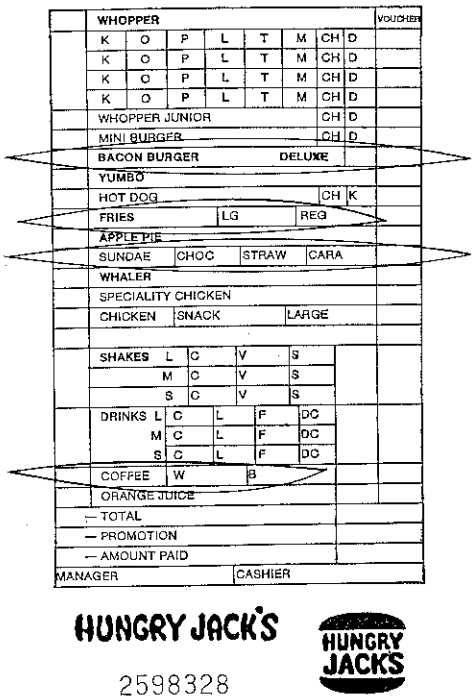
Chapter 1:

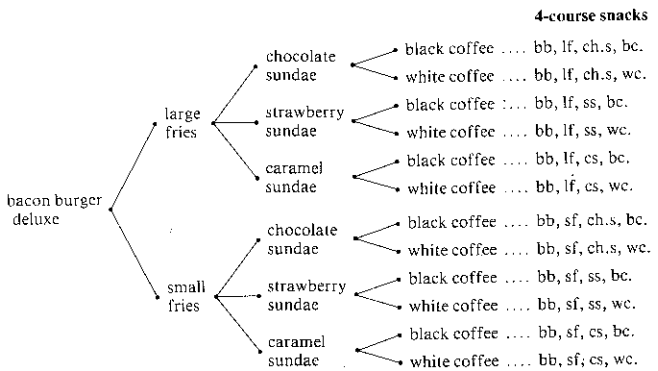
The Multiplication Principle

It's time for a snack! A regular Hungry Jacks customer **always** has a Bacon Burger Deluxe, then fries followed by a sundae, and a coffee to finish. When he is feeling very hungry he has a large serve of fries instead of a small serve. Usually he has a strawberry sundae but sometimes tries a different flavour. Occasionally he has milk in his coffee.



Our regular customer wonders how many different 4-course snacks are possible.

All of the possible 4-course snacks can be listed in a **tree diagram.**



Starting with the Bacon Burger Deluxe and ending with the coffee, there are **12** different branches in the tree diagram and each of these represents a different 4-course snack.

There are 12 different snacks in the list.

By examining the tree diagram we can make a rule for counting the number of possible snacks.

Notice that, after the Bacon Burger Deluxe, there are: two choices of fries and then, with each of these, 3 choices of sundaes.

This makes $2 \times 3 = 6$ different 4-course snacks so far!

Then **with each of these 6**, there are 2 choices of coffee.

This makes $6 \times 2 = 12$ different 4-course snacks or $2 \times 3 \times 2 = 12$ different 4-course snacks.

Multiplication Principle

To find the number of ways of performing a task:

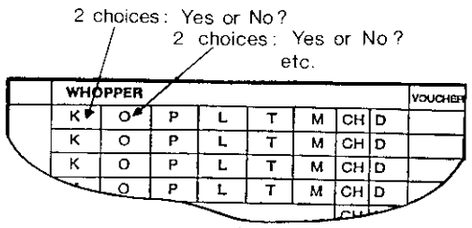
- Break the task into stages;
- count the number of choices at stage 1 and after one of these choices has been taken, count the number of choices at stage 2 and so on;
- multiply together, the number of choices at each stage.

Example 1.

Hungry Jacks' biggest seller is the Whopper. It is a sesame seed bun with a meat pattie inside. As well, there are 8 extra fillings: Ketchup, Onions, Pickles, Lettuce, Tomato, Mayonnaise, Cheese, Double Meat.

Many people have the lot, but any (or all) of the extra fillings can be left out to create a "custom-built" Whopper.

How many different "custom-built" Whoppers are possible?

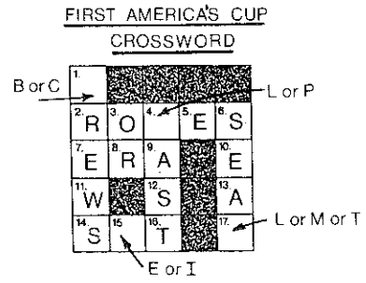


Solution:

Number of "custom-built" Whoppers $= 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$

2 choices of ketchup: take it or leave it

Example 2.



Across:

2. The first race got off to a bad start because some crew members had trouble with their . . . . . .
7. The inaugural race was the beginning of an . . . . . in yachting.
14. Thomas Lipton crossed the line first because, in the early stage of the race, he got a good . . . . . .

Down:

1. Thomas Lipton attributed his successes to his fine . . . . . .
3. Otherwise
4. The winner of the challenge was decided on the . . . . . event.
6. Progress in the race was hampered by the presence of a troublesome . . .

Complete the crossword and find out how many different solutions are possible.

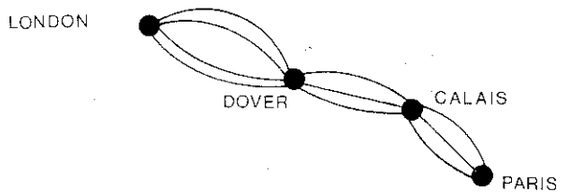
Solution:

Number of solutions $= 2 \times 2 \times 2 \times 3 = 24$

2 choices for box 1
2 choices for box 4
2 choices for box 15
3 choices for box 17

Example 3.

A tourist in London wishes to travel to Paris via Dover and Calais.



She has a choice of hire car, bus, train or hitchhike from London to Dover; ferry, hovercraft or swim from Dover to Calais and either bus, train, or cycle on her final stage of the journey.

a. In how many different ways can she elect to complete the journey from:

i. London to Dover? ii. Dover to Calais? iii. Calais to Paris?
iv. London to Calais? v. London to Paris?

b. After spending two weeks in Paris she decides to return to London via Calais and Dover but is too exhausted to cycle or swim. In how many different ways can she elect to make the return trip?

Solution:

a. i. Number of ways from London to Dover = 4
ii. Number of ways from Dover to Calais = 3
iii. Number of ways from Calais to Paris = 3
iv. Number of ways from London to Calais = $4 \times 3 = 12$
v. Number of ways from London to Paris = $4 \times 3 \times 3 = 36$

b. Number of possible selections for the return trip = $2 \times 2 \times 4 = 16$

train or bus
ferry or hovercraft
hire car or bus or train or hitchhike

Exercise 1: The Multiplication Principle

1. A teacher selects one student from a class of 20 to hand out a worksheet at the beginning of a lesson. At the same time another student is selected to collect an assignment. In how many ways may the two students be selected?

2. A student has a choice of 5 foreign languages and 6 science subjects. In how many ways may a particular student choose to study:
   a. one foreign language and one science subject?
   b. Indonesian and one science subject?

3. If a die and a coin are tossed simultaneously, determine the number of possible outcomes if the die is:
   a. cubical.
   b. octahedral.

4. A young girl has a Barbie doll. She has 4 different jackets, 3 different dresses and 6 different pairs of shoes. In how many different ways can the girl dress her doll using a jacket, a dress and a pair of shoes?

5. A florist has 5 different varieties of cut flowers, 3 types of greenery and 4 different colours of wrapping paper. He makes up small bouquets consisting of 1 variety of flower, 1 type of greenery and wrapped in 1 piece of paper. How many different bouquets are possible?

6. A pastrycook makes birthday cakes. He makes 2 types of base: sponge or madeira; 4 types of filling: jam, lemon sauce, mock cream or real cream; and 2 types of icing: royale icing or butter icing. How many different birthday cakes can he offer a customer?

7. To cope with varying weather conditions an airline pilot has a choice of 3 routes from Melbourne to Singapore; 5 routes from Singapore to Bahrain; 3 routes from Bahrain to London. How many different ways can the pilot plan his trip from Melbourne to London?

8. In London there are 5 morning newspapers, 3 evening newspapers and 4 weekend newspapers. If a Londoner decides to order one morning, one evening and one weekend newspaper, how many different choices of 3 newspapers has he?

9. How many different 5-course meals are possible if a customer can choose from a menu which offers 2 appetisers, 4 soups, 3 entrees, 7 main courses and 4 desserts?

10. Call signs for most radio stations in Australia consist of a non-zero digit followed by two letters. How many different call signs are possible?

11. To manufacture turned table legs, a length of timber must undergo four processes: rough sawing, machine turning, hand sanding and spray painting. The factory has 3 saw benches, 5 lathes, 4 sanding benches and 3 paint bays. Through how many different routes may a piece of timber pass before it becomes a finished table leg?

12. A hardware department advertises a special in its sale. For $19.99 a customer can purchase a 1-litre tin of gutter paint, a paint brush, a paint scraper, a sheet of glass paper and a bottle of paint brush cleaner. The customer can select from 5 different colours of paint, 2 different sizes of brushes, 2 different shaped scrapers, 6 different grades of paper and has a choice of 2 different brands of paint brush cleaner. How many different sales are possible?

13. How many different 7-digit telephone numbers are possible provided that no number may start with zero?

14. A security firm offers the facility of a security safety box for its customers. The customer is given a security number which, when entered into the code reader, will open his safety box. The code consists of three letters followed by four digits and finishes with letters SDB. The only letters not used in the code are O and I. Determine how many different codes can be developed by the security firm.

15. An "outdoor living centre" store advertises BBQs as a super special. The customer can assemble his own portable BBQ by selecting a burner system, a portable cylinder size and the type of stand for the BBQ. The burners which are available are either 2, 3 or 4 jets; the cylinders are either 1·5, 4·5 or 9·5 kg and the BBQ stands can either be with 4 wheels, 2 wheels or free standing (no wheels). If the store wishes to display all of the possible BBQs, how many different BBQ sets would be in the display?

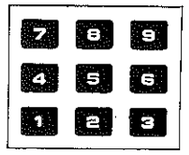
16. What is the maximum number of different car number plates that a country town can issue if the prefix is AB and is followed by 4 digits?

17. A motor vehicle distributor specialises in new 4-wheel-drive vehicles. As a special on the purchase of a new standard 4-wheel drive, a customer has the choice of: a brass

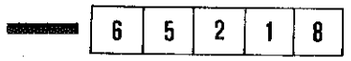
towball or a chrome-plated towball; and a standard fuel tank or long-range tanks; and a bull bar or not; and a roof rack or not; and a tool kit or an extra wheel. How many differently assembled 4-wheel-drive vehicles are possible?

18. A secretary leaves a message for her boss to contact Pete on the third floor. The boss reads the message but cannot remember Pete's extension number. He realises that the extension numbers for the third floor commence with a 3 followed by four digits. Rather than ask for help he decides to try and contact Pete by dialling a 3 then a selection of four digits from the digits 0, 1, 2, 3, ..., 9. How many possible calls may he have to make before selecting the correct extension number?

19. A young child randomly presses the counting numbers on a calculator. An 8-digit number is the largest that the calculator can display. How many different 8-digit numbers is it possible for the child to punch in?



20. A cyclist purchases a bicycle security chain with a combination lock. The lock can be opened when the five small circular discs are rotated to a particular position and are opposite a mark on the lock. Each disc consists of ten digits, 0, 1, 2, 3, ..., 9.

e.g. 

On returning home with the purchase the cyclist notices the security chain lock is locked and the combination code which usually appears on the package has been erased. He accepts his poor buy and decides to try to break the combination. How many possible codes may he have to try?

21. The telephone numbers in a district exchange consists of the digits 321 followed by four digits. How many telephone numbers are possible in this district if:

a. no restrictions apply?
b. the numbers from 321 0000 to 321 0500 cannot be used?

22. There are five roads between A and B and six roads between B and C.

a. In how many ways can one drive from A to C via B?
b. In how many ways can one complete the round trip: A to C via B then back again via B?

23. A woman has a choice of five different routes that she may take from her home to her place of employment. In how many ways could she travel to work and back on a particular day if:

a. she selects a different route each way?
b. she selects the same route each way?
c. no restrictions apply?

24. A secretary has 2 letters to post and 6 letter boxes available at a post office. In how many ways may the letters be posted if:

a. no restrictions apply?
b. both letters are deposited in the same box?
c. different boxes are used to post the letters?

25. To paint a bedroom, an interior decorator decides on blue walls, white ceilings and white woodwork. The colour choice for walls is Duck-egg Blue, Azure Blue, Smokey Blue, Blue Mist or Wedgewood Blue. The colour choice for ceilings and woodwork is Brilliant White, Antique White, Broken White or Offwhite. How many different colour schemes are possible if:

a. the ceiling and woodwork are to be the same colour?
b. the ceiling and woodwork are to be different colours?

26. A telephone company offers the following selection guide for household subscribers:

Colour:	Red, white, grey, black, brown
Model Type:	Desk, wall-mounted
Dial Type:	Circular, press-button (2 types—console and handpiece)

a. In how many different ways may a subscriber select a telephone?
b. How many different red telephones may a subscriber select?
c. How many different circular-dial white telephones could a subscriber choose?

27. A child's Crazy Animals book consists of three separate fold-over sections consisting of 8 different heads, 8 different bodies and 9 different leg-and-feet shapes.

a. How many different crazy animals may a child create?
b. How many different crazy animals may be created having one particular body shape?

28. A police "identi-kit" box for men contains:

6 different face shapes	8 different hair lines
4 different eyebrow styles	10 different eye shapes
10 different noses	5 different moustaches
8 different sets of lips	5 different beards





a. How many different identi-kit pictures can be made up?
b. If a suspect is masked from the nose downward (obscuring his nose, moustache, lips, beard and face shape), how many different identi-kit pictures of a masked suspect can be made up?

29. In a local tavern, dinners are made up for a main course from steak, fish or chicken; from baked, mashed or fried potatoes; and from mixed vegetables or salad.

a. Find, by means of a tree diagram, the number of different dinners possible.
b. How many different dinners contain fish?
c. How many different dinners contain fish and salad?

30. A travel agent offers a choice of 5 different down-town city tours, 6 different tours to the surrounding hills and 4 different coastal tours. How many different tours are possible for a tourist who wishes to see:

a. the city itself and the surrounding hills?
b. the coastline and the hills regions?
c. a specific down-town tour, a specific hills tour and any coastal tour?

31. A young couple are selecting their kitchen bench space for their new home and have decided that the Formica tops should be in a yellow tone, the cupboards and drawers a woodgrain veneer, the tile surrounds in brown, the taps and power points in white. The builder offers a choice from:

6 different Formica tops	4 different woodgrain veneers
5 different tiles in brown	4 different tap styles.

a. In how many ways may they select their kitchen design?
b. They have made up their minds that the cupboards and drawers will be in woodgrain pine. How many options have they now in completing their selection?

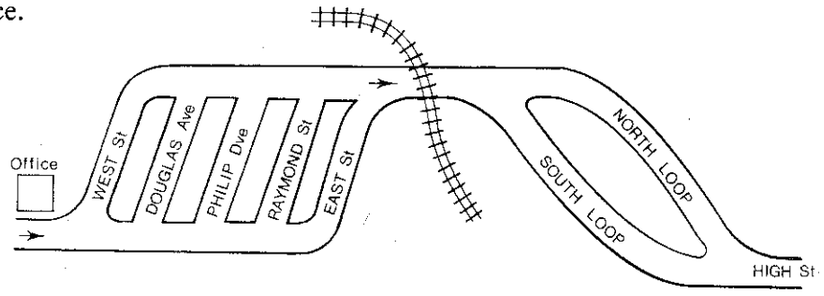
32. The New Zealand Broadcasting Commission produced a children's television serial consisting of eight episodes. At the end of each episode the audience was invited to write in and choose one of two story lines for the next episode. Naturally, there was no choice for episode 1. The writer made sure that, whichever story line was chosen, the episode ended with a pre-determined scene.

a. How many different complete stories were possible?
b. After the serial was over the NZBC transmitted a complete story made up of episode 1 and the unused story lines of the other episodes. How many different complete stories are possible, made up this way?

33. A computer programme is written to create ordered pairs, (x,y). A number x is selected at random from the set $\{0, 1, 2, \ldots, 99\}$ and then a number y is selected at random from the set $\{0, 1, 2, \ldots, 9\}$.

a. How many different ordered pairs are possible?
b. How many different ordered triples (x,y,z) can be made if z must be selected from the set $\{0, 1, 2, \ldots, 999\}$?

34. An office worker travels from the office to High Street each evening. She chooses her route to avoid traffic congestion and, naturally never goes back towards the office.



a. How many different routes could she take to High Street?
b. If Douglas Avenue, East Street and North Loop are blocked with traffic, how many different routes are possible from the office to High Street?

35. How many 3-digit numbers less than 700 can be formed using only the non-zero digits if:

a. repeated digits are permitted?
b. repeated digits are not permitted?
c. the first and second digit must be the same and no further repetition is permitted?

36. Numbers with 4 digits are to be formed from the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, or 9. No number may start with a zero but digits may be repeated.

a. How many of these numbers are there?
b. How many of these numbers are greater than 4 999?
c. How many of these numbers are greater than 9 000 and even?

37. Four-digit numerals may be formed using the digits 1, 3, 5, 7 and 9 such that no digit can be used more than once. How many of these four-digit numerals:

a. may be formed?
b. begin with 7?
c. end in a 3 or 5?
d. are less than 597?

38. In how many ways can 4 different Chinese cooking books and 1 Mexican cooking book be displayed on a shelf at a food convention if a salesman has 7 different Chinese and 3 different Mexican cooking books at his disposal and the Mexican cooking book must be in the middle of the display?

39. A bicycle with Derailleur gears has 2 sprockets (toothed wheels) on the pedal shaft. Either of these can be paired with any of 5 different sprockets on the rear axle by jumping the connecting chain. Each pairing of sprockets gives a different speed.

a. How many different speeds are possible on this bicycle?
b. How many sprockets would you guess to be on the pedal shaft and the axle shaft on a bicycle with:
   i. 5 speeds?
   ii. 3 speeds?

40. There are 4 corridors in a hospital which lead from the main emergency entrance to the examination room and 3 corridors which lead from the examination room to the X-ray department. Determine how many different paths a hospital orderly can take in wheeling a patient:

a. from the main emergency entrance to the X-ray department via the examination room.
b. from the main emergency entrance to the X-ray department via the examination room and return,
   i. if it is possible to return via the same corridor.
   ii. if the corridors used on the return trip must differ from those used in getting to the X-ray department.

41. The local newsagent's card display contains 80 different humorous birthday cards, 75 different traditional birthday cards, 45 different anniversary cards and 12 different get-well cards. In how many different ways can a customer select:

a. a humorous birthday card and an anniversary card?
b. a traditional birthday card, an anniversary card and a get-well card?
c. a birthday card and a get-well card?

42. A section of a small country library contains 85 different fictional war novels, 42 different historical war novels, 25 different home-handyman books and 44 different western yarns. In how many different ways may a borrower select:

a. an historical war novel?
b. a fictional war novel and a home-handyman book?
c. a western yarn, a home-handyman book and an historical novel?
d. a war novel and a western yarn?

43. A customer at Hungry Jacks decides to order a four-course meal. The meal the customer orders is a double meat miniburger with cheese, a large fries, an apple pie and a large diet coke. How many different orders could a customer order if he decides on:

a. a double meat miniburger with cheese, fries, an apple pie and a diet coke?
b. a double meat miniburger with cheese, a regular serve of fries, an apple pie and a diet coke?
c. a double meat miniburger with cheese, fries, an apple pie and a large diet coke?
d. a double meat miniburger with cheese, fries, an apple pie and a small diet coke?

Refer to the docket illustrated at the beginning.

44. In a multiple-choice test there are 10 questions. Each question has 5 alternatives and the candidate must choose 1 alternative for each question. A candidate decides to answer all questions even if she must guess. How many different ways are there of answering the test if:

a. she answers all questions by guessing?
b. she is certain about 5 questions and guesses the rest?
c. she eliminates 2 alternatives from each question and then guesses?

45. A tennis club is running a doubles competition. There are 143 women and 126 men available to play. How many different mixed doubles pairs can be formed?

46. The Victorian Football League's six games for a particular weekend were:

Hawthorn	v	Footscray
Sydney Swans	v	Essendon
North Melbourne	v	Collingwood
Fitzroy	v	Carlton
St Kilda	v	Geelong
Melbourne	v	Richmond

Entrants in a football tipping competition had to predict the winner of each game and write their six "winners" on a card. From their list of winners, entrants were to predict which team would get the highest score. Also, from the losers they had to predict which team would get the lowest score.

a. How many different cards were possible?
b. How many different cards could have Hawthorn, Melbourne and Essendon predicted as winners?

47. A car manufacturer offers several options on a number of different features, as shown in the following table:

Feature	Choices
Size	4 cyl., 6 cyl., V-8
Transmission	manual, automatic
Brakes	power drum, power disc, regular drum, regular disc
Colour	blue, brown, red, green, white, gold
Sound	standard radio, AM-FM radio, stereo cassette
Steering	regular, power
Air Conditioning	yes, no
Interior	vinyl, material
Tyres	radial, steel radial, conventional

a. How many different new cars may a customer order?
b. A customer insists on a red automatic V-8 with steel radial tyres. How many different new cars may he order?

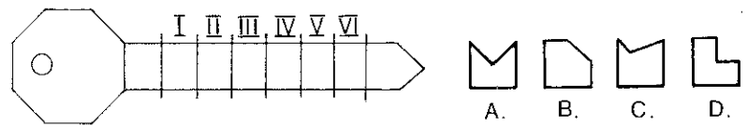
48. A steakhouse offers as a menu:

Entree	Steak
Prawn and Bacon Kebab	Fillet
Squid Rings	Porterhouse
Lamb Kebab	Rump
	T-Bone

Steak is served with French Fries or Jacket-Baked Potatoes and Salad or Vegetables.

a. In how many different ways is it possible to order a 2-course meal?
b. How many different 2-course meal orders could begin with a Prawn and Bacon Kebab?
c. If a particular customer is allergic to seafood, how many different ways may this person order a 2-course meal?
d. Steaks may be ordered to be cooked Rare, Medium-Rare or Well-Done and with an option of Mushroom Sauce or House Special Steak Sauce. In how many different ways is it now possible to place an order for a 2-course meal:
   i. to contain the Rump Steak, Well-Done?
   ii. having a sauce on the steak?
   iii. having mushroom sauce on a steak which must not be Well-Done?

49. A key has six parts, I, II, III, IV, V, VI. There are four patterns A, B, C, or D available for each part.



a. How many different key designs are possible?
b. How many different key designs are possible if A must ocupy position I?
c. How many different keys can be designed if, where possible, each pattern may be reversed to create a different pattern?

50. A teacher wishes to make up simple subtractions using the digits 5, 6, 7, 8, 9.

For example, $6 - 5 = ?$ $8 - 5 = ?$ $5 - 5 = ?$ $8 - 9 = ?$

Draw a square array (subtraction table) which gives all of the answers.

a. How many different examples are possible?
b. How many different answers are possible?
c. How many different examples and how many different answers are possible if he wishes to use the same digits for addition examples?

51. A well known brand of calculator has ten single-digit buttons and ten function buttons. If a digit button is pressed, followed by a function button, an answer is displayed.

a. How many answers can be produced in this way?
b. How many answers can be produced if a 2-digit number is punched in before a function button is pressed?
c. A digit button is pressed, followed by an "operator" button ($\times$, $\div$, $+$, or $-$), followed by a digit button and then the "equals" button and an answer is displayed.

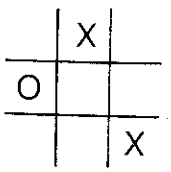
   i. How many different sequences of this type are possible?
   ii. How many of the sequences would display an answer which is not a number?

   Hint: [5] [÷] [0] [=] gives E, an error message.

52. A large store has 5 departments on the upper floor and 5 on the ground floor. There are 4 trainee shop assistants to be assigned to the departments. In how many different ways may they be assigned if:

a. no department is to have more than one trainee?
b. they must all be assigned to the ground floor and no department is to have more than one trainee?
c. they must all be assigned to the ground floor?
d. they must all be assigned to the same floor?
e. two particular trainees must be assigned to the same floor and no department is to have more than 1 trainee?

53. In the game of noughts (O) and crosses (X) there are nine squares in which players may place their marks. The players take turns and the first player to complete a line of three marks (horizontally, vertically, or diagonally) is the winner. In this game the player who starts always uses a cross.



How many different patterns are possible:

a. after 2 turns: that is, 1 turn for each player?
b. after 3 turns? See the diagram.
c. after 4 turns: that is, 2 turns for each player?
d. if the "cross" player, who starts, wins on the 5th turn?

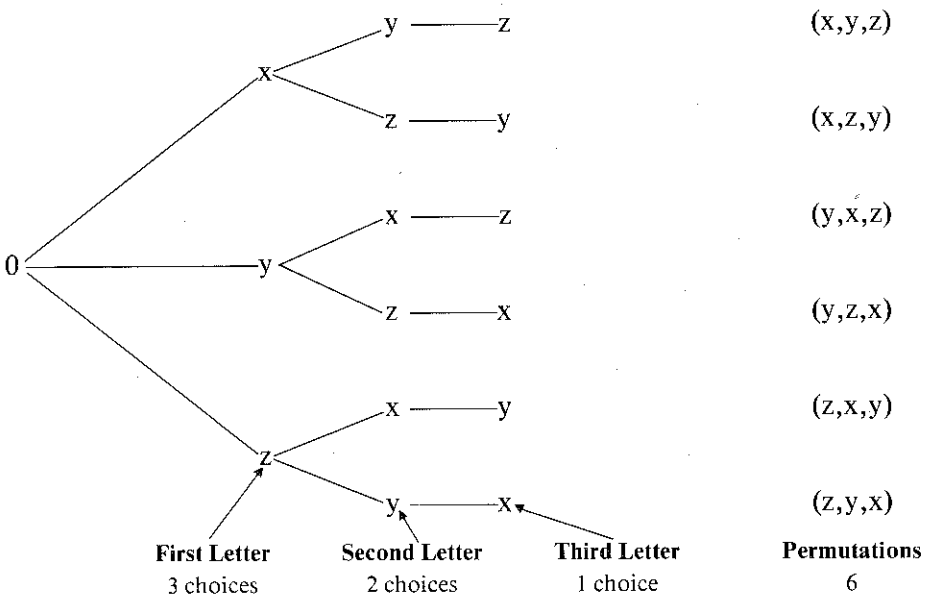
Chapter 2:

Permutations

A permutation is an arrangement of things in a definite order. For example the letters x, y and z can be arranged (permuted) in 6 different ways:

(x,y,z) (x,z,y) (y,x,z) (y,z,x) (z,x,y) (z,y,x)

A systematic way of listing these permutations is by means of a tree diagram.



If we are interested only in counting the permutations, the fundamental counting principle can be used. The number of permutations is the product of the choices at each stage.

First Letter		Second Letter		Third Letter		Permutations
3	×	2	×	1	=	6

PERMUTATIONS AND FACTORIALS

Example 1.

In a television competition viewers were asked to rank, in order of merit, 8 spectacular cricket catches. How many different orders were possible?

Solution:

The number of orders = 8 × 7 × 6 × 5 × 4 × 3 × 2 × 1 = 40 320 $= 8!$

There are 8 choices for the best catch and 7 remaining choices for the second best catch and so on. The **Multiplication Principle** is used to calculate the number of different orders.

A mathematical shorthand notation for representing the product
$8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$ is 8! and is read as "eight factorial".

If the competition had required viewers to rank in order of merit, n spectacular cricket catches instead of 8, then there would have been n! different orders possible.

Permutation Notation, ${}^nP_n$

The number of different ways of arranging n different objects using all of them is denoted by:

$$ {}^nP_n = n! $$

Factorial Notation

The product of consecutive factors decreasing from the number n to 1 is written as n!

that is, $n \times (n - 1) \times (n - 2) \times \ldots \times 3 \times 2 \times 1 = n!$

Example 2.

Express each of the following in factorial notation:

a. $5 \times 4 \times 3 \times 2 \times 1$ 5!
b. $52 \times 51 \times 50 \times \ldots \times 2 \times 1$ 52!
c. $11 \times 10!$ 11!
d. $7 \times 6 \times 5$ $7! \div 4! = \frac{7!}{4!}$

Solution:

a. $5 \times 4 \times 3 \times 2 \times 1 = 5!$
b. $52 \times 51 \times 50 \times \ldots \times 2 \times 1 = 52!$
c. $11 \times 10! = 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 11!$
d. $7 \times 6 \times 5 \times \frac{4!}{4!} = \frac{7!}{4!}$

Example 3.

Express each of the following as the product of consecutive factors:

a. 4! b. $\frac{8!}{5!}$ c. $\frac{n!}{(n - 2)!}$

Solution:

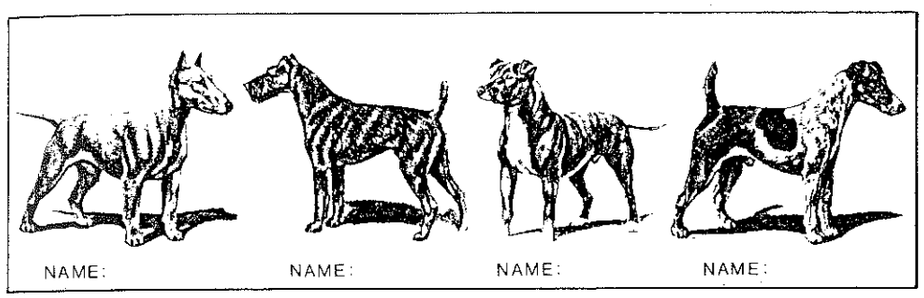
a. $4! = 4 \times 3 \times 2 \times 1$

b. $\frac{8!}{5!} = \frac{8.7.6.\cancel{5.4.3.2.1}}{\cancel{5.4.3.2.1}} = 8.7.6$

c. $\frac{n!}{(n - 2)!} = \frac{n(n - 1)(n - 2)(n - 3) \ldots 3.2.1}{(n - 2)(n - 3)(n - 4) \ldots 3.2.1} = n(n - 1)$

Example 4.

In a competition, the contestants must match the names: Spot, Rover, Tippy and Boy to the pictures:



How many different ways can these names be assigned?

Solution:

Number of ways $= 4 \times 3 \times 2 \times 1 = 4! = 24$

This is the number of different ways of permuting (arranging in order) the four names underneath the pictures.

Example 5.

A small child had 5 building blocks each of which was a different colour. She placed 3 of these in a stack. How many different-looking stacks could she have built?

Solution:

Number of stacks $= 5 \times 4 \times 3 = 60$

She had 5 choices for the bottom block and then she had 4 choices for the middle block and then she had 3 choices for the top block. If she had had n blocks instead of 5, she could have built $n(n - 1)(n - 2)$ different-looking stacks of 3 blocks.

Permutation Notation, ${}^nP_r$

The number of different ways of arranging r objects from a set of n different objects is denoted by:

$$ {}^nP_r = n(n-1)(n-2)\ldots(n-r+1) $$

This product of consecutive factors can be expressed in factorial notation as:

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

Exercise 2A: Permutations and Factorials

1. Evaluate each of the following:

a. $3!$ b. $11!$ c. ${}^4P_4$

d. $6.5!$ e. $8! - 7!$ f. $\frac{12!}{2!10!}$

g. $\frac{8!}{7!}$ h. $\frac{10! - 6!}{4!}$ i. $3.4! + 2.4!$

j. $4.(3!)!$

2. Express each of the following in factorial notation:

a. $3 \times 2 \times 1$ b. $1 \times 2 \times 3 \times 4 \times 5$ c. $12.11.10.9$

d. $17.16.15$ e. ${}^7P_7$ f. $25.16.9.4.1$

g. $n(n-1)(n-2)(n-3)$ h. $\frac{8.7.6}{3.2.1}$ i. $\frac{52.51.50.49.48}{5.4.3.2.1}$

j. $(n+2)$

3. Express each of the following as the product of consecutive factors:

a. $5!$ b. $\frac{8!}{4!}$ c. $\frac{12!}{9!}$

d. $\frac{n!}{(n-3)!}$ e. $\frac{(r+1)!}{(r-1)!}$ f. ${}^5P_5$

4. Given $10! = 3\,628\,800$, find the value of:

a. $9!$ $10! \div 10 = 362880$ b. $11!$ $10! \times 11 = 39916800$

5. If there are seven horses in a race, in how many different ways can they cross the finishing line, assuming no dead heats for any position? $7! = 5040$

6. In how many different orders can the names of 6 candidates be listed on a ballot paper for a local council election? $6! = 720$

7. A mechanic buys a set of 15 chrome-plated spanners; each spanner is made to fit a special sized nut; each of the spanners is different in length. In how many different ways can the mechanic display the spanners on the specially prepared workshop tool board? $15!$

8. A competition involves matching the faces of 5 well known musicians with the 5 different instruments they are noted for playing. How many different entries would need to be submitted to guarantee a correct entry by a person knowing nothing about famous musicians? $5! = 120$

9. How many different 3-digit numbers may be formed using only the digits 5, 7 and 9 where repeated digits are not permitted? $3! = 6$

10. Four friends enter a bar together and find that there are four stools vacant at the bar. In how many different ways can they seat themselves at the bar? $4! = 24$

PERMUTATION FORMULA AND APPLICATIONS

Example 6.

Verify that $n(n-1)(n-2)\ldots(n-r+1) = \frac{n!}{(n-r)!}$

where $n = 7$ and $r = 3$.

Solution:

L.H.S., $7.6.5 = 210$

R.H.S., $\frac{7!}{(7-3)!} = \frac{7!}{4!} = \frac{7.6.5.4.3.2.1}{4.3.2.1} = 7.6.5 = 210$

Example 7.

Evaluate each of the following:

a. ${}^8P_5$ $\frac{8!}{3!} = 6720$ b. ${}^6P_6$ $\frac{6!}{0!} = 6! = 720$

Solution:

a. ${}^8P_5 = \frac{8!}{(8-5)!} = \frac{8!}{3!} = 8.7.6.5.4. = 6720$

b. ${}^6P_6 = 6! = 720$

Alternatively, using ${}^nP_r = \frac{n!}{(n-r)!}$ we find that,

$${}^6P_6 = \frac{6!}{(6-6)!} = \frac{6!}{0!} = \frac{720}{0!}$$

If we know that (by another method) the answer is 720, we need a definition for 0!

Define: $0! = 1$

Example 8.

A person has 6 books in his briefcase but not room for all on a shelf. He calculated that there were $\frac{6!}{2!}$ possible arrangements. How many books did he place on the shelf?

Solution:

$${}^6P_r = \frac{6!}{2!} = \frac{6!}{(6-4)!} = \frac{6!}{(6-r)!}$$

$\therefore \quad r = 4$

Hence, he placed 4 books on the shelf.

Example 9.

A student entering high school has to choose 4 optional subjects from a list of 12 and write them in order of preference on his application form. In how many different ways may this student complete the option selection form?

Solution:

Number of selections for first preference	$= 12$
Number of selections for second preference	$= 11$
Number of selections for third preference	$= 10$
Number of selections for fourth preference	$= 9$
By the Multiplication Principle, the total number of ways	$= 12.11.10.9$
	$= 11\,880$

The student may complete the option selection form in 11 880 different ways.

Example 10.

A hospital builds 12 new single rooms for patients. In how many different ways can they allocate the rooms if:

a. 12 patients apply? b. 13 patients apply? c. 7 patients apply?

Solution:

a. Number of ways $= 12.11.10.9.8.7.6.5.4.3.2.1 = 12! = 479\,001\,600$
There are 12 choices of **room** for the first patient, 11 choices for the second and so on.

b. Number of ways $= 13.12.11.10.9.8.7.6.5.4.3.2 = 6\,277\,020\,800$
There are 13 choices of **patient** for the first room, 12 choices of patient for the second room and so on.

c. Number of ways $= 12.11.10.9.8.7.6 = 3\,991\,680$
There are 12 choices of room for the first patient, 11 choices for the second patient and so on, until there are 6 choices of room for the last patient.

Example 11.

Find the integer n if ${}^nP_{n-7} = 720$

Solution:

$${}^nP_{n-7} = 720$$

$$\text{i.e.} \quad \frac{n!}{(n-(n-7))!} = 720$$

$$\text{i.e.} \quad \frac{n!}{7!} = 720$$

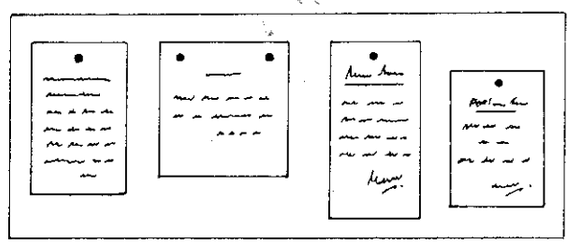
$$\text{i.e.} \quad n! = 720 \times 7!$$

$$\text{i.e.} \quad n! = 10 \times 9 \times 8 \times 7!$$

$$\text{i.e.} \quad n = 10$$

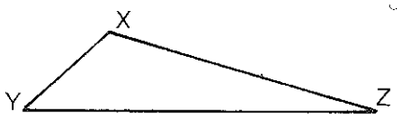
Exercise 2B: Permutation Formula and Applications

1. Five ladies leave their identical coats in a checkroom at a theatre. If they all lose their check tickets, in how many different ways is it possible for their coats to be assigned to them on leaving?
2. In how many different ways could the four different notices be arranged across the noticeboard?



3. How many different ways are there of naming this triangle? For example, XYZ is a name. List all of the names.

3.2.1 = 6



XYZ YZX ZXY
XZY YXZ ZYX

4. In how many different ways can the four symbols *, @, N and $ be arranged in a row? Each symbol is used only once in each arrangement. 4! = 24

5. How many different 6-digit numbers can be made by permuting all of the digits: 1, 3, 5, 8, 6, 4? Repetitions of the digits are not allowed. 6! = 720

6. Simplify:

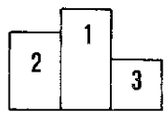
a. $\frac{7!}{3!}$ 840 b. $\frac{6!}{4!}$ 30 c. $\frac{7!}{4!} \cdot \frac{6!}{4!}$ 6300 d. ${}^5P_3$ $\frac{5!}{2!}$ = 60 e. ${}^{52}P_2$ $\frac{52!}{50!}$ = 2652 = 51.52

7. A family owns 4 cars but their driveway is only one car wide. At night they park them in the driveway one behind the other. How many different permutations are possible? 4! = 24

8. How many permutations of the letters of the word *ARGUMENT* are there if all letters are used? 8! = 40320

9. How many different "words" can be made from four of the following letters: *MEDICO*? Letters may not be used more than once in each word. 6.5.4.3 = 360

10. Eight athletes are to run in the final of the 100-metre sprint. To meet a newspaper deadline, a sports photographer wishes to take photographs of all the possible place-getters on the victory dais before the race!



8.7.6 = 336

How many different photographs should he take to cover all the possibilities?

11. *CLEAN RIDS BOY more than a little*

This was a clue in a cryptic crossword and the person doing the crossword decided that the 12-letter answer was an anagram of the capital letters.

a. How many anagrams of the capital letters are possible? 12! = 479001600

Totally irrelevant

The correct anagram was a word which means "more than a little".

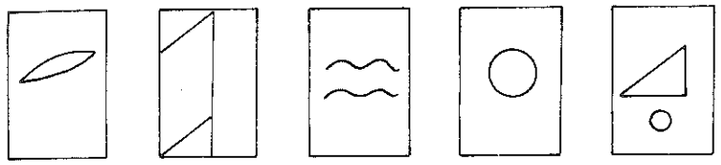
b. What was the word? CONSIDERABLY

12. Five people share a house. Each week they prepare a roster for cleaning, cooking and shopping. How many different rosters are possible?

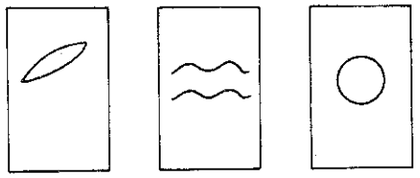
5.4.3 = 60

13. A teacher has 25 children in his class. When he calls the roll in the morning it is usually in alphabetical order. In how many different orders can he call the roll if he decides not to use alphabetical order? Do not simplify your answer. 25! - 1

14. A fortune teller predicts your future by asking you to choose 3 of these cards and put them in a row.



For example:



suggests "wealth"

but the reverse suggests "poverty". How many different "futures" are possible? 5.4.3 = 60

15. In how many ways can the positions of president, vice-president, secretary and treasurer be filled from a nominated committee of nine? 9.8.7.6 = 3024

16. a. Can you "untangle" the following capital cities?

G D I U E N B R H	=	EDINBURGH
S P A R I	=	? PARIS
O M E R	=	? ROME
B I S L O N	=	? LISBON
S W O T G A N H I N	=	? WASHINGTON
O P I N A S G E R	=	? SINGAPORE

b. In how many different ways can you "tangle" the following capital cities?

i. DELHI 5! = 120, 5! - 1 = 119 ii. BUDAPEST 8! = 40320, 8! - 1 = 40319 iii. PRAGUE 6! = 720, 6! - 1 = 719

17. A newspaper held a competition in which 4 photographs of radio personalities had to be matched to 4 photographs of their feet! Each entry cost 10 cents. How much would it cost if you wished to submit one entry for each possibility? 4! × 10¢ = $2.40

18. There are 24 desks in a classroom. The desks are in 4 rows of six, facing the teacher's table at the front. There are 16 children in the class. In how many different ways may they arrange themselves if:

a. there is no restriction? ${}^{24}P_{16} = \frac{24!}{8!}$
b. the back 4 desks must be left vacant? ${}^{20}P_{16} = \frac{20!}{4!}$
c. the back 8 desks must be left vacant? ${}^{16}P_{16} = \frac{16!}{0!} = 16!$

Leave your answers in factorials.

19. A Number-One-Method cake contains flour, butter, sugar, eggs and milk. An inexperienced cook knows the ingredients but has forgotten the order in which they are to go into the mixing bowl.

a. In how many different orders may he put the ingredients into the bowl?
b. If he remembers that the eggs and milk are put in together, in how many different orders may he put the ingredients into the bowl?

20. An overworked executive writes down twenty jobs which must be completed before the holidays. He asks his secretary to make a list of just 5 of these in order of priority.

a. How many different lists is it possible for the secretary to make?

He tells his secretary that one particular job is urgent and must be at the top of the list.

b. How many different lists is it possible for the secretary to make with this restriction?

21. In an election there are 9 candidates.

a. In how many different orders may their names be listed on the ballot paper?

J. ALBERT	☐	M. FREEMAN	☐	P. KING	☐
P. BLACK	☐	A. HOLT	☐	C. LORD	☐
R. DAWSON	☐	S. JONES	☐	T. WHITE	☐

In the optional-preferential system of voting, voters are asked to number as many boxes as they wish, to indicate their order of preference for candidates. In how many different ways can the above ballot paper be filled in if the voter decides to number:

b. i. all nine boxes? ii. exactly three boxes?

22. How many different 5-digit numbers can be formed from the digits 7, 8 and 9? Repetition of digits will be necessary.

23. Numbers are to be formed from the digits 5, 6 and 7.

a. How many 2-digit numbers can be formed if digits may not be repeated?
b. How many 3-digit numbers can be formed if digits may not be repeated?
c. How many 4-digit numbers can be formed? Some repetition of digits will be necessary.

24. How many 4-digit numbers can be formed from the digits 1, 2, 5, 7 and 9 if:

a. repetitions are allowed?
b. repetitions are not allowed?

25. The following small tiles [7] [9] [1] [4] are to be used to make numbers.

For example, [9] [7] [1]

How many different:

a. four-digit numbers are possible? b. one-digit numbers are possible?
c. two-digit numbers are possible? d. numbers do NOT contain three digits?

26. A map of Australia is divided into 8 regions (viz. 6 states and 2 territories). In how many different ways can the map be coloured if:

a. there are 10 different colours available and each region must be a different colour?
b. there are 10 different colours available, both territories must be the same colour but each state must be a different colour?

27. Prove that:

a. $^nP_r + r\left(^nP_{r-1}\right) = {}^{n+1}P_r$

b. $^{n+1}P_{r+1} = (n + 1)^nP_r$

28. Find the numbers n and r if:

a. $^nP_4 = 61 \times 60 \times 59 \times 58$ b. $^{19}P_r = 19 \times 18 \times 17 \times 16 \times 15$

c. $^nP_2 = 132$ d. $^nP_3 = 720$

e. $^nP_n = 120$ f. $^nP_r = \frac{17!}{9!}$

g. $^nP_r = 20$ h. $^nP_r = 990$

APPLYING PERMUTATIONS

Example 12.

a. How many different arrangements are possible using all the letters of the word *TRIANGLE*?
b. How many of these arrangements begin with the letter *T* followed by *R*?
c. How many of these arrangements have the letter *T* following *R*?
d. How many of these arrangements have the letters *T* and *R* adjacent?
e. How many of these arrangements do not have the letters *T* and *R* adjacent?

Solution:

a. Number of arrangements = 8! = 40 320

b. Number of arrangements = 1.1.6! = 720
There is 1 choice for the first letter, 1 choice for the second letter, 6 choices for the third letter, 5 choices for the fourth letter and so on.

c. Number of arrangements = 7! = 5 040
Consider a composite "letter" *(RT)* as a unit and each of the other letters as units. This gives 7 units altogether.

d. Number of arrangements = 7! × 2 = 10 080
If *T* and *R* are adjacent then the composite "letter" may be *RT* or *TR*. Therefore, the answer in part (c) is doubled.

e. Number of arrangements = 40 320 − 10 080 = 30 240
The number of arrangements where T and R are not adjacent is equal to the total number of arrangements of all of the letters less the number of arrangements where T and R are adjacent.

Example 13.

Using each of the digits 1, 2, 3, 4, 5, 6, 7, 8 once only, 4-digit numbers are formed: e.g. 1 238, 7 658, and 1 435.

a. How many different 4-digit numbers can be formed? $^8P_4 = \frac{8!}{4!} = 1680$

b. How many of the 4-digit numbers are less than 5 000? $4.7.6.5 = 890$

c. How many of the 4-digit numbers are even numbers? $8.7.6.4 = 840$

Solution:

a. Number of 4-digit numbers = 8.7.6.5 = 1 680

b. Number of 4-digit numbers = 4.7.6.5 = 840
There are four choices for the front digit. It must be under 5. There are seven choices for the second digit and so on.

c. Number of 4-digit numbers = 4.7.6.5 = 840
Even numbers have 2, 4, 6 or 8 as their final digit and the first stage is to determine the final digit; there are 4 choices for the first stage. For the second stage we move to the front digit and there are 7 choices. For the second digit there are 6 choices and for the third digit, 5 choices.

Exercise 2C: Applying Permutations

1. How many different 4-digit numbers less than 6 000 can be made from the digits 7, 6, 4 and 2? The digits may not be used more than once. $2.3.2.1 = 12$

2. How many different 5-digit numbers can be formed from 0, 1, 2, 3, 4, 5, 6, 7, 8, 9? Digits are to be used once only and zero may not be used in the front of a number. $9.9.8.7.6 = 27216$

3. A second-hand car yard has 6 cars and 2 camper-vans to be displayed in a row along its footpath frontage.
   a. How many different arrangements of all 8 vehicles are possible? $8! = 40320$
   b. How many different arrangements are possible if a camper-van is to be on each end of the display? $2.1.6! = 1440$

4. A swimming squad has 8 swimmers. The first, second, third and fourth swimmers for a relay team must be decided. In how many different ways may this be done if:
   a. there are no restrictions? $^8P_4 = 1680$
   b. one particular swimmer must swim fourth? $^7P_3 = 210$
   c. also, a fifth swimmer must be named as a reserve? $^8P_5 = 6720$

5. There is a row of 12 parking bays, but only 7 cars are parked in them.
   a. How many different possible arrangements of cars are possible? $^{12}P_7 = 3991680$
   b. If one of the cars belongs to the manager and he always parks in the reserved bay, how many different arrangements of cars are possible? $^{11}P_6 = 332640$

6. There are 11 men in a cricket team. How many different batting orders of 10 men are possible? Three particular batsmen always bat before any other players. How many different batting orders are possible in this case? $3!8! = 241920$

7. How many 3-digit even numbers can be formed with the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 if repetition of digits is:
   a. not allowed? $8.7.4 = 224$ b. allowed? $9.9.4 = 324$

8. How many 6-digit numbers, which are multiples of 5, can be formed from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9? Repetitions of digits are not allowed. $^8P_5 .1 = 6720$

9. Four-digit numbers can be made up by permuting all of the digits 5, 6, 1 and 3. Digits may be used only once in each permutation. 6135 6513 6351 6531 6153 6315
   a. List all of these numbers which are greater than 6 000. $1.3! = 6$
   b. How many of these numbers would be less than 6 000? $3.3! = 18$

10. In how many ways can the letters of the word *PERMUTE* be arranged if both *E*s stay in the same place: second and seventh? $1.1.5! = 120$

11. A crossword fan has one word to complete.

			E					E

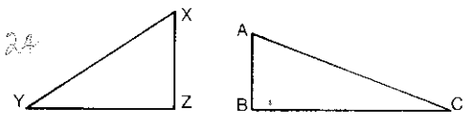
The two Es are fixed in place and she decides that the remaining seven letters must be from the list: S I A M B R L T or N. $^9P_7 = 181440$
   a. How many different ways are there of arranging 7 different letters from the list?
   b. The missing word means "unhappy". What is it? MISERABLE

12. There are 7 hooks in a row on a wall and 5 paintings to hang on them. In how many different ways can the paintings be arranged? How many of these arrangements leave the middle hook vacant? $^7P_5 = 2520$; $6! = 720$ (assume blank is one painting)

13. In how many different ways can 5 oil paintings of different value be arranged along a wall if:
   a. the most valuable is placed in the middle of the display? $4.3.1.2.1 = 24$
   b. the two least valuable paintings are placed at the ends of the display? $2.3.2.1 = 12$

14. A girl collects model cars and now has 15 of them. She has room on her top shelf for 6 and on her bottom shelf for 5. The rest cannot go on display. How many different arrangements are possible for:
   a. the top shelf? $^{15}P_6 = 3603600$
   b. the top and bottom shelves? $^{15}P_{11} = \frac{15!}{4!}$

15. The following two triangles are equal in area.



How many different ways are there of writing this true sentence: "Triangle **XYZ** has the same area as triangle **ABC**", if only the letters in heavy type may be rearranged?

16. In an experiment on short-term memory, subjects are asked to read a list of 10 nonsense syllables. These are presented one at a time on cards.

RAQ	*TES*	*FIL*	*POZ*	*NUR*
BAC	*CEY*	*GIC*	*HUS*	*KOH*

After a short fixed interval of time each each subject is asked to write down all of the nonsense syllables in the correct order.

a. How many **correct** orders are possible?
b. How many different cards consisting of a consonant, followed by a vowel, followed by a different consonant could be prepared?

To take away the effect of order of presentation, the syllables are presented to each subject in a different order.

c. In how many different orders can the cards be presented?
d. Which two cards do you guess that nearly all subjects recall correctly?

17. The comics page of the paper consists of 5 comic strips and one large advertisement. The comic strips and the advertisement each occupy the full width of the page.

a. In how many different ways can the page be set out?
b. In how many different ways can the page be set out if the advertisement must be at the top?
c. In how many ways can the page be set out if all of the comics must be together?

18. Consider the symbols: *X 8 T I 2 4*

a. i. For fun, write down a permutation of all of the symbols so that a sentence is formed when they are read from left to right.
   ii. How many permutations of all six symbols are possible?
b. i. For fun, write down a permutation of four of these symbols so that a sentence is formed when they are read from left to right.
   ii. How many different permutations of **these** four symbols are possible?
c. A computer programme is designed to print out all of the possible arrangements of any four symbols chosen from this list. How many different arrangements will it print?
d. i. For fun, arrange these symbols to form a sentence: *M 5 T 2 U X C*
   ii. How many arrangements of these symbols have *M* and *T* together in that order?

19. In how many ways can 7 men and 6 women arrange themselves in a row if men and women must alternate?

20. The mayor, the town clerk and 10 councillors attend a council meeting. In how many different ways may they arrange themselves around the table if:

a. the mayor always sits at the head of the table?
b. the mayor always sits at the head of the table and the town clerk sits on his right?
c. the mayor always sits at the head of the table and the town clerk sits next to him?

21. A photograph of the seven school prefects is to be taken for the school magazine. In how many different ways may the photographer order the seven students if the photograph is to be taken with:

a. all the prefects in a row?
b. four prefects standing and three sitting?
c. the head boy and head girl seated and the remaining five prefects standing in a row behind them?

22. A model train enthusiast has 2 locomotives: a steam locomotive and diesel locomotive; 8 different flat-top wagons; 3 different passenger carriages and a guardsvan. Determine in how many different ways a train may be coupled containing:

a. a steam locomotive, 8 flat-top wagons and a guardsvan.
b. a diesel locomotive, 8 flat-top wagons, 3 passenger carriages and the guardsvan.
c. two locomotives, 3 passenger carriages and the guardsvan.
d. a diesel locomotive in the front, followed by the 8 flat-top wagons and the guardsvan at the end.

23. In how many different ways can 3 men and 4 women sit in a row if:

a. there are no restrictions?
b. the men are to sit together?

24. The State Emergency Service has a list of 8 volunteers: Alice, Bob, Carol, David, Ellen, Fred, Gwen and Harry. The secretary is to roster one volunteer to the telephone switchboard for each day of the Easter holiday period: Good Friday, Easter Saturday, Easter Sunday and Easter Monday. No volunteer may be rostered for more than one day.

a. How many different rosters are possible?
b. How many of these rosters would have Alice working on Easter Monday?
c. How many of these rosters would not have Alice working on Easter Monday?
d. How many different rosters are possible if it was decided to have a man and a woman on the switchboard for each day?

25. In a mixed-doubles tennis match, one man with one woman forms a team. How many different mixed-doubles teams are possible if there are:

a. 6 men and 6 women?
b. 8 men and 6 women?

26. Five men and four women enter a ballroom dancing championship. How many different pairs (a man and a woman) can be formed? At the last moment, x **extra** women entered the championship and it was calculated that 2 520 different pairs were possible. Find x.

Hint: Find consecutive factors of 2 520.

27. Two pistol-shooting pairs are to be selected from 5 pistol shooters. A first and second shooter must be nominated for each pair. In how many different ways can:

a. an A-pair and a B-pair be selected? $5,4,3,2 = 120$ b. two pairs be selected?

28. Three sets of twins stand in a row for a photograph. There is a boy and a girl in each set. In how many different ways could they arrange themselves if: $(2.2.2)\times 3! = 48$

$6! = 720$ $3!\times 2 = 12$

a. there was no restriction? b. each twin pair must not be separated?
c. each twin pair must not be separated and boys and girls must alternate?

29. There are 4 French books, 3 German books and 2 Italian books to be placed on a shelf. In how many different ways can they be arranged if:

a. there are no restrictions? $9! = 362880$
b. the French books must be on the left, the German books in the middle, and the Italian books on the right? $4!\,3!\,2! = 288$
c. the French books must be together, the German books must be together, and the Italian books must be together? $(4!\,3!\,2!)\times 3! = 1728$

30. Three married couples attend the theatre. In how many different ways can they seat themselves in a block of six vacant seats if: $(2.2.2).3! = 48$

$6! = 720$ $(3!3!)\times 2 = 72$

a. there are no restrictions? b. each man decides to sit next to his wife?
c. men and women alternate? d. the three men sit together? $3!\times 3!\times 4 = 144$

31. A computer is programmed to list all of the permutations of the 8 letters of the word *COMPUTER*.

a. How many permutations are there? $8! = 40320$

How many of these permutations:

b. start with *C*? $1.7! = 5040$
c. start with a consonant? $5.7! = 25200$
d. start and end with a consonant? $5.6!.4 = 14400$
e. end with the letters *ER* in that order? $6!\times 1 = 720$
f. end with the letters *E* and *R* in either order? $6\times 2 = 1440$
g. have the letters *E* and *R* together? $(7! = 5040)\times 2 = 10080$ [ER or RE]

32. Arnold Schoenberg (1874–1951) invented the "tone row" technique for composing music. In these compositions he used all 12 notes of the musical scale:

C, C#, D, Eb, E, F, F#, G, Ab, A, Bb, B

To start a composition he set down a permutation of all 12 notes: a tone row.

a. How many different tone rows are possible? $12! = 479001600$

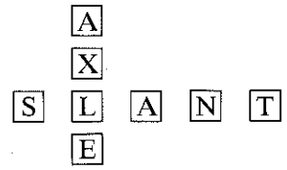
How many of these tone rows:

b. start with *C* and end with *Bb*? $1.10!.1 = 3628800$
c. start with *C* and have *C#* second? $1.1.10! = 3628800$
d. have *C* followed directly by *C#* somewhere in the row? $11! = 39916800$
e. have *C* and *C#* together somewhere in the row? $11!\times 2 = 79833600$

33. Words are to be formed from the letters of the word *EDUCATION*. Letters are to be used once only and each word will contain all 9 letters. How many different words:

a. commence with the letter *E* and end with the letter *N*? $1.7!.1 = 5040$
b. have the letter *D* following the letter *E*? $8! = 40320$
c. have the letter *D* adjacent to the letter *E*? $8!\times 2 = 80640$
d. have all the vowels together and all the consonants together? $(5!4!)\times 2 = 5760$
e. have all the vowels together? $5!\times 4!\times 5 = 14400$
f. commence with the letter *T*, followed directly by the letter *E*, and the last two letters *C* and *D* in either order? $1.1.5!.2.1 = 240$
g. commence with the letter *A* and end with the letter *A*? none, i.e. 0

34. In the word game SCRABBLE each player has 7 randomly-selected tiles with letters printed on them. The players must make dictionary words on a playing board by building on letters already in place.



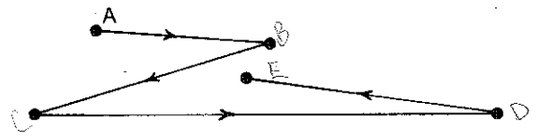
Players must think of as many arrangements of their letters as possible before they make a word. Unfortunately, not every arrangement makes a dictionary word. The first player has [A] [B] [R] [E] [X] [U] [L]. How many different arrangements are possible using:

a. all 7 letters? $7! = 5040$ b. 4 out of the 7 letters? $^7P_4 = 7!\div 3! = 840$

A second player holds [B] [S] [A] [T] [M] [N] [U] and she decides to build horizontally on the [L]. One possibility is shown in the diagram. How many different arrangements are possible if she decides to:

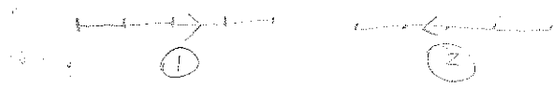
c. start with [L] and use 4 of her own letters? $^7P_4 = 840$
d. use the [L] in the second place and use 4 of her own letters? $^7P_4 = 840$
e. use the [L] in any place and use 4 of her own letters? 840×5, ie $^7P_4\times 5 = 4200$
f. start with [L] and use 4 of her own letters including her [S] in the last place? $^6P_3 = 120$

35. The diagram below shows one straight-line pathway connecting 5 dots.

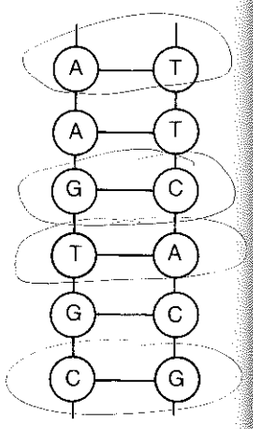


A pathway must start at one dot, end at another and pass through every dot exactly once.

a. How many different pathways are possible? $5! = 120$
b. How many different pathways start at dot A? $1.4! = 24$
c. How many different pathways do not start at dot A? $120 - 24 = 96$
d. How many different pathways end with dot A? $4!\times 1 = 24$
e. How many different pathways have dot A neither at the start nor at the end? $120 - 24 - 24 = 72$
f. It is possible to draw 5 dots with only two different pathways. Draw 5 dots with this property.



36. Deoxyribonucleic acid (DNA) is a long complex molecule found in the nucleus of all living cells. Its structure dictates inherited characteristics: whether a rose will be red, whether a girl will be tall, and so on. The molecule looks like a very long twisted ladder. Each rung of the ladder is called a nucleotide and is made by pairing two bases. There are 4 bases: adenine (A), guanine (G), cytosine (C) and thymine (T). The diagram shows that A can only be paired with T, and G can only be paired with C, and vice versa. GC CG AT TA



It is the **sequence** of the nucleotides which determines inherited characteristics.

a. How many different sequences of 6 nucleotides are possible? (Assume AT is different from TA and that GC is different from CG). $4^6 = 4096$

b. How many different sequences of n nucleotides are possible? 4^n

c. What is the minimum number of nucleotides necessary to give 1 000 000 different sequences? $4^x \geqslant 1000000 \therefore x = 10, 4^{10} = 1\,048\,576$

37. How many different 3-permutations can be formed from:

a. a 5-set? ${}^5P_3 = 60$

b. a 3-set? ${}^3P_3 = 3! = 6$

c. i. an n-set? What are the restrictions on n? ${}^nP_3 = \frac{n!}{(n-3)!}$ where $n \geqslant 3$ and an integer

ii. Use the information in part i. to find a positive integral root of $n^3 - 3n^2 + 2n - 990 = 0$. $n(n-2)(n-1) = 990$ $11 \cdot 10 \cdot 9 = 990 \therefore n = 11$

38. Show ${}^{n+1}P_{r+1} - {}^nP_r = n.\left({}^nP_r\right)$ where n and r are positive integers and $n \geqslant r$.

39. An integer, r, is greater than 0 but less than 9. ${}^{10}P_{r+2} = \frac{10!}{(8-r)!}$

a. How many (r + 2)-permutations are there from a 10-set?

b. There are 12 times more (r + 2)-permutations than r-permutations from a 10-set. Find r. ${}^{10}P_r = 12 \times {}^{10}P_{r+2} \therefore r = 6$

40. A child has a set of 9 blocks. They are identical cubes except that they are all different colours.

a. In how many different orders can she arrange them in a row? $9! = 362880$

b. If she wishes to make a stack of just 5 of the blocks, how many different arrangements are possible? ${}^9P_5 = 9! \div 4! = 15120$

41. A red die is placed in front of you on a table so that one face points towards you.

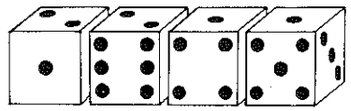


a. i. How many different faces could be uppermost? 6

ii. With a particular one of these faces uppermost, how many different faces could point towards you? 4 (not top or bottom)

iii. How many different looking ways could a die be placed on the table so that one of the faces points towards you? $6 \times 4 = 24$

A white die, a green die, and a black die are placed in a row with the red die so that one face of each die points towards you.

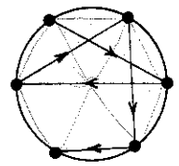


b. i. In how many different ways could the colours be permuted? $4! = 24$

ii. In how many different looking ways could you arrange the 4 dice in a row? $6 \times 4! = 7962624$

42. There are 6 points evenly spaced around the circumference of a circle. A straight-line pathway starts at any point and finishes at any other point. Every point is included once in the pathway.

a. How many different pathways are possible? $6! = 720$



If the radius of the circle is 1 unit,

b. what is the length of a shortest pathway?

c. how many different shortest pathways are there?

PERMUTATIONS OF THINGS NOT ALL DIFFERENT

There are **six** permutations of the letters *SEA*:

SEA *SAE* *AES* *EAS* *ASE* *ESA*

However, if we replace *A* with another *E*:

SEE *SEE* *EES* *EES* *ESE* *ESE*

there are only **three** distinct permutations.

Example 14.

How many distinct permutations are there of all of the letters *S E E M*? $\frac{4!}{2!} = 12$

Solution:

Number of permutations $= \frac{4!}{2!} = 12$

If we pretend, at first, that one of the *E*s is an *A* (*SEAM* instead of *SEEM*), then there are 4! permutations.

But we divide by 2! because *EA* and *AE* are really just *EE*.

Example 15.

Three $1 coins and two 10c coins are arranged in a pile. How many different-looking stacks are possible?



Solution:

Number of piles $= \frac{5!}{3!2!} = 10$

Pretend that all the coins are distinct: 5! permutations. Then divide by 2! because of the two 10c coins and by 3! because of the three $1 coins.

Permutations of Things not all Different

The number of distinct permutations of n things of which r_1 things are alike, another r_2 things are alike, ..., is given by:

$$\frac{n!}{r_1!\, r_2!\, r_3! \ldots}$$

Exercise 2D: Permutations of Things not all Different

1. A baby has 4 red blocks, 2 white blocks and 1 green block. How many different-looking vertical stacks can the baby build? How many of these stacks have the two white blocks together?
2. A stamp collector purchases 5 identical 30-cent stamps and 2 different 45-cent stamps. In how many different ways could he mount these stamps in an album?
3. A box of groceries contains 3 identical brands of tomato soup, 3 containers of the same brand of margarine, a packet of bacon and 4 different packages of cold meat. In how many different orders can a householder place the grocery items on a bench?
4. A book display unit has room for 12 books. A bookseller selects 8 different best sellers to display and fills the remaining positions with 4 copies of a new release. In how many different ways may he display the books?
5. How many distinct permutations of all of the digits 1, 1, 1, 3, 3 and 5 are possible?
6. Determine the number of different words which can be formed by permuting the letters of the word:
   a. *STATISTICS.* b. *CALCULUS.* c. *CORRECT.*
7. There are 20 men in the chorus of an opera. Half of them must wear long baggy pants; one-quarter, long straight pants; and the rest, short pants. In how many different ways may the pants be allocated to the chorus?

8. *TAA'S AIRMAN was confused as Columbus' flagship came into view.*

   This was a clue in a cryptic crossword and the person doing the crossword decided that the 10-letter answer was an anagram of the capital letters.
   a. How many different anagrams are possible?
   b. What was the answer?

9. A family of eight consists of a father, a mother and six children of which the two eldest are identical twins. They line up in a row for a family photograph. How many different-looking photographs are possible if you cannot tell one twin from the other?

10. The prime factors of 18 000 are $2^4 \times 3^2 \times 5^3$. One way of expressing this (in full) is $2 \times 3 \times 3 \times 2 \times 2 \times 5 \times 2 \times 5 \times 5$.
    a. In how many different ways may 18 000 be expressed, in full, as a product of primes?
    b. In how many different ways may 3 087 be expressed, in full, as a product of primes?

11. A strand of deoxyribonucleic acid (DNA) consists of the following sequence of bases:

    A A G G C A A A T T T

    a. How many different sequences can be made with these 11 bases?
    b. If base *C* is replaced with another *T*, how many sequences are possible?

12. Twelve students are each to receive a book prize. In how many different ways can the prizes be awarded if:
    a. all the books are different?
    b. all the books are the same?
    c. six books are identical of one type and six books are identical of another type?

13. A soccer team has 11 players. There are 3 identical long-sleeved jumpers, 4 identical short-sleeved jumpers and 4 identical sleeveless jumpers to be worn by the team.
    a. In how many different ways can the jumpers be allocated?
    b. If each jumper had a different number sewn on the back, how many different ways could the jumpers be allocated?

14. A car accident witness cannot recall the number plate of a car accurately. He can remember that the first three characters were letters X, R and A but he cannot remember the order. Also he remembers that the next three characters were the digits 6, 6 and 5 but he cannot remember the order. How many different number plates fit this description?

15. A long corridor has 12 doors. A painter decides to paint one-third of them white, one-third light green and one-third dark green.
    a. How many different colour schemes can he choose for the doors?
    b. If he discovers 3 more doors after finishing the 12 doors, how many different colour schemes for the whole corridor are now possible?

16. a. How many different 11-letter words can be formed from the word *CALCULATORS*?

b. How many different 11-letter words can be formed from the word *CALCULATORS* which:

i. begin with a *T* and end with an *R*?

ii. have an *S* in the middle?

17. In the 1983 America's Cup, *Australia II* won 4 races and *Liberty* won 3 races. *Australia II* won the cup! The first yacht to win 4 races wins the cup.

Let W denote "wins a race" and $\overline{W}$ denote "loses a race".

W W W W , W $\overline{W}$ W W W , W $\overline{W}$ W $\overline{W}$ W W , $\overline{W}$ $\overline{W}$ $\overline{W}$ W W W W are all examples of possible cup-winning sequences.

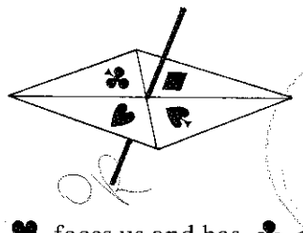
a. Which sequence represents *Australia II*'s 1983 cup win?

b. How many different sequences could lead to a cup win in:

i. the sixth race? ii. the seventh race?

18. A man makes a pile of 10 coins; some are \$1 coins and the rest are 10c coins. He calculates that there are 120 distinct piles he could make. If he has less than \$7, how many of each coin has he?

CIRCULAR PERMUTATIONS: ARRANGEMENTS IN A CIRCLE

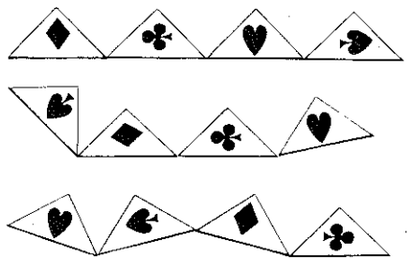


The four suits (♥ ♣ ♠ ♦) are to be positioned around a spinner.

Notice that ♥ faces us and has ♣ on the left, ♠ on the right and ♦ opposite. This is just one of many circular permutations. How many different ways can the suits be positioned around the spinner?

If we keep the four suits in the same position and rotate the spinner clockwise through 90°, we still have the same arrangement of the suits relative to each other.

If we cut and stretch **this** spinner, the possible arrangements in a row are:



For each circular permutation of the suits there are 4 different arrangements in a row.



∴ The number of arrangements of 4 objects in a circle

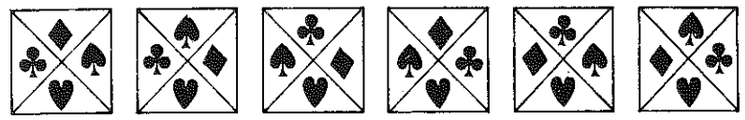
= (number of arrangements of 4 objects in a row) ÷ 4

$= \frac{4!}{4}$

$= 3!$

$= 6$

These are:



Circular Permutations

The number of circular permutations of all n different objects $= \frac{n!}{n} = (n-1)!$

Example 16.

A "chocolate" wheel has 8 equal sectors. Each sector is to be painted a different colour and there are 8 colours available. How many different-looking chocolate wheels are possible?

Solution:

Number of wheels $= \frac{8!}{8} = 7! = 5\,040$

Example 17.

Snow White arranges the Seven Dwarfs around a may-pole.

a. In how many ways can she do this if Sleepy and Dopey are to be together?

b. In how many ways can the Seven Dwarfs be arranged if Sleepy and Dopey must not be together?

Solution:

a. Number of arrangements $= \frac{2.6!}{6} = 240$

Thinking of Sleepy and Dopey as one unit, there are now 6 distinct units to be arranged around the may-pole. Sleepy and Dopey may switch positions but still stay together. There are hence $\frac{6!}{6}$ circular permutations of the units and 2 different ways Sleepy and Dopey can arrange themselves as a unit.

b. Number of arrangements $= \frac{7!}{7} - \frac{2.6!}{6} = 480$

There are $\frac{7!}{7}$ circular permutations of the Seven Dwarfs without any restrictions. By subtracting the number of circular permutations where Sleepy and Dopey are together the result will be the required number of circular permutations where Sleepy and Dopey are separated.

Example 18.

At a cattle show 5 bulls and 5 cows are paraded in a circle for judging.

a. In how many different ways may the cattle be arranged if the cows and bulls alternate?
b. In how many different ways may the cattle be arranged if the cows and bulls alternate and if 2 particular cows and 1 particular bull, from the same stud, must be separated?

Solution:

a. Number of arrangements $= \frac{5!\ 5!}{5} = 2\,880$

There are $\frac{5!}{5}$ circular permutations of the bulls and 5! arrangements of the cows in the alternate spaces.

b. Number of arrangements $= \frac{5! \times 3 \times 2 \times 3!}{5} = 864$

There are $\frac{5!}{5}$ circular permutations of the bulls, 3 choices of cow for the space to the left of the particular bull, 2 choices of cow for the space to the right of the particular bull and 3! arrangements of the remaining cows.

Exercise 2E: Circular Permutations

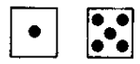
1. An astrologer wishes to arrange the 12 signs of the Zodiac on a circular star chart:



In how many different ways can he arrange them if he ignores the usual sequence?

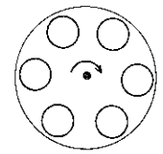
2. There are 9 German Shepherd dogs to be paraded for judging in a circle. In how many different ways can they be arranged?

3. A circular gaming table is divided into equal sectors each of which corresponds to one of the totals from rolling a pair of dice.

For example: 

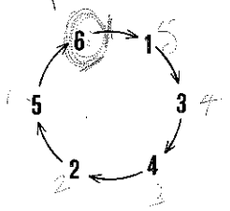
In how many different ways may the totals be arranged?

4. A wheel containing six different-coloured filters rotates in front of a spotlight. The colours are red, orange, green, blue, yellow and pink. How many different colour cycles are possible?



5. A car has 6 cylinders. The wiring can be arranged so that the cylinders fire in a particular cycle.

For example:

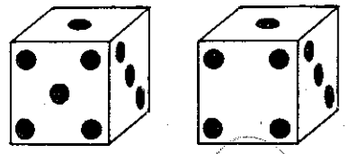


a. How many different cycles are possible?
b. If in another car there are 6 possible firing cycles, how many cylinders has it?

6. King Arthur and 12 of his knights sit at the round table.
a. In how many different ways can they arrange themselves?
b. In how many different ways can they be arranged if King Arthur's throne is in a set position at the table?

7. A key ring holds 12 keys. In how many different ways can the keys be arranged on the ring, given that the ring can be turned over?

8. a. Which one of the dice is not "regular"? Why?



Given that opposite faces of a die must add up to seven, answer the following questions.

b. i. What is the minimum number of faces that must be seen before the numbers on all faces are known?
ii. In how many different ways may the numbers one, two and three be arranged in a circle?
iii. Hence, in how many different ways can regular dice be numbered?

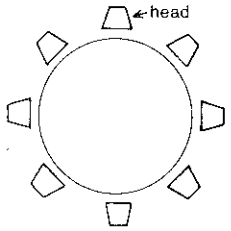
9. In how many ways can 8 people be arranged in a circle if 2 particular people must always:

a. be together? b. be separated?

10. In how many ways can 5 boys and 3 girls be arranged in:

a. a row if the girls must always stand together?
b. a circle if the girls must always stand together?

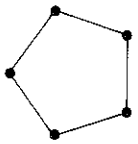
11. Four married couples are to be seated around a round table.



Each woman will sit next to her husband. In how many different ways can they arrange themselves if:

a. a woman must sit at the "head of the table"?
b. either a man or a woman may sit at the "head of the table"?
c. there is no "head of the table"?

12. a. In how many different ways may the vertices of a regular pentagon be named with the letters A, B, C, D and E?

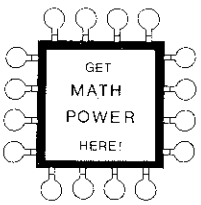


b. In how many different ways can the vertices be named if the pentagon's name must be ABCDE?

c. In how many different ways can vertices be named if a pentagon's name must be ABCDE and if all of its sides are of different lengths?

13. Mother and father and 4 children stand in a circle. In how many ways can they arrange themselves if mother and father stand opposite each other?

14. An illuminated sign is square and has 4 light bulbs protruding from each side: 16 bulbs flashing on and off. The bulbs can be wired so that they flash one at a time for about one-tenth of a second.



a. How many different cycles of illumination are possible?

To get a wheel effect the bulbs are wired so that they flash on and off in numerical order.

b. How many different wheel-effect cycles are possible?

To get a dazzling effect the bulbs on each side are wired to come on together, one side at a time. Now, there are 4 circuits instead of 16 and they can switch on in any order.

c. How many different dazzling-effect cycles are possible?

15. At an Australian Premiers' Conference, 6 premiers each with 2 advisors attended a planning session. In how many ways could they arrange themselves about a round table if each premier must sit between his two advisors?

Chapter 3:

Combinations

There are 5 coins in a set:

$$\{5c,\ 10c,\ 20c,\ 50c,\ \$1\}$$

How many different sums of money can be made by choosing three coins? Here is the list.

{ 5c, 10c, 20c}	35c
{ 5c, 10c, 50c}	65c
{ 5c, 10c, $1 }	$1.15
{ 5c, 20c, 50c}	75c
{ 5c, 20c, $1 }	$1.25
{ 5c, 50c, $1 }	$1.55
{10c, 20c, 50c}	80c
{10c, 20c, $1 }	$1.30
{10c, 50c, $1 }	$1.60
{20c, 50c, $1 }	$1.70

There are 10 different combinations of coins. They are listed as subsets because the order does not matter.

Combination

A **combination** is a selection of things in which order is not important.

To find the number of combinations without making a list, we start with the fundamental counting principle. There are $5 \times 4 \times 3 = 60$ different ways of **arranging** three coins out of the five, which includes for example,

5 10 20 5 20 10 10 20 5 10 5 20 20 5 10 20 10 5

But these $3 \times 2 \times 1 = 6$ different arrangements represent just one combination of the same three coins. Therefore, to determine the number of combinations we must divide the total number of arrangements by 6. That is, the number of combinations of 3 things selected from 5 different things is:

$$\frac{{}^5P_3}{{}^3P_3} = \frac{5 \times 4 \times 3}{3 \times 2 \times 1} = 10$$

$5P_3 \div 3!$

COMBINATION FORMULA AND APPLICATIONS

Example 1.

There are 9 questions on an exam paper and a candidate must answer 7. How many different combinations of 7 questions are possible?

Solution:

$$\text{Number of combinations} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3}{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} = 36$$

Combination Notation, ${}^nC_r$

${}^nC_r$ is a notation which means "the number of combinations of r things selected from n different things".

$\binom{n}{r}$ is an alternative notation.

Example 2.

Evaluate: a. ${}^{13}C_5$ b. $\binom{20}{2}$

Solution:

a. $${}^{13}C_5 = \frac{13 \times 12 \times 11 \times 10 \times 9}{5 \times 4 \times 3 \times 2 \times 1} = 1\,287$$

b. $$\binom{20}{2} = \frac{20 \times 19}{2 \times 1} = 190$$

Example 3.

Express $\binom{15}{4}$ in factorial notation.

Solution:

$$\binom{15}{4} = \frac{15 \times 14 \times 13 \times 12}{4 \times 3 \times 2 \times 1}$$

$$= \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(4 \times 3 \times 2 \times 1) \times (11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1)}$$

$$= \frac{15!}{4!\,11!}$$

Factorial Notation for Combinations, ${}^nC_r$

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} \quad \text{for integers } n \geqslant r \geqslant 0$$

Exercise 3A: Combination Formula and Applications

1. Express the following in factorial notation:

a. ${}^5C_2$ b. ${}^{10}C_8$ c. ${}^{20}C_9$ d. ${}^{15}C_3$

2. How many swimming teams of 5 swimmers can be chosen from a squad of 20?

3. In how many different ways is it possible to select a sample of 3 eggs for a freshness test from a carton of a dozen eggs?

4. In how many different ways may a laboratory assistant select three rats from a group of 15 rats for testing the same drug?

5. A large chain store has 9 branches. In how many different ways can the management select two of these branches to take part in a special promotion offer?

6. Evaluate:

a. $\binom{10}{2}$ b. $\binom{10}{8}$ c. $\binom{12}{9}$ d. $\binom{12}{3}$

e. ${}^4C_3$ f. ${}^4C_4$ g. ${}^5C_0$ h. $\binom{52}{5}$

Hint: 0! = 1

7. Determine the number of ways in which a social worker can select six of nine families on her "must visit" list?

8. In a 100-question objective test, John got 4 questions wrong. How many different sets of questions could he have got wrong?

9. Eleven points lie on a circle. How many different inscribed octagons is it possible to form using these points as vertices?

10. A hospital report reveals that an accident victim broke 2 fingers. How many different injuries fit this description? A thumb is not a finger in this problem.

11. Twelve rowers are members of a squad which is training for the final team for the "eights" at a forthcoming regatta. In how many ways can the coach select the final eight to contest the regatta?

12. In how many ways can a coach select the first 5 players to start a basketball match from a squad of 9 players, if we assume that all 9 can play in any of the 5 starting positions?

13. Seven small holes are drilled in a piece of steel such that no three holes are collinear. An apprentice metalworker is asked to scribe lines on the metal sheet connecting any two holes. Determine how many lines are possible?

14. How many different:

a. poker hands (5 cards) can be dealt from a pack of 52 cards?

b. cribbage hands (4 cards) can be dealt from a pack of 52 cards?

15. Eight points lie in a plane such that no three are collinear.

a. How many different lines can be formed containing any 2 points?

b. How many different triangles can be formed using the points as vertices?.

c. How many different quadrilaterals can be formed using the points as vertices?

16. Show:

a. i. $^{10}C_8 = {}^{10}C_2$ ii. $^{6}C_1 = {}^{6}C_5$

b. i. $\binom{5}{2} = \binom{5}{3}$ ii. $^{7}C_4 = \binom{7}{3}$

17. Six holes have been dug but only 4 trees are available.



a. How many different ways can you choose holes which will remain empty if you plant all 4 trees?

b. If you plant all trees, how many different selections of holes are possible?

18. Seven people are to travel by plane but there are only 5 seats left on the early flight.

a. How many different selections can be made of 5 people to travel on the early flight?

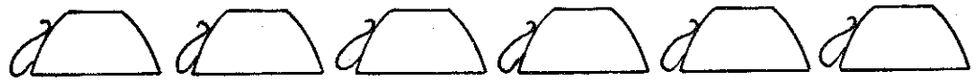
b. How many different selections can be made of 2 people to travel on a later flight?

19. On joining a video club a new member is offered 3 free video movies out of a special list of 5.

a. How many different combinations of 3 films are possible?

b. A customer decides to select his free films by indicating the 2 video films he will **not** choose. In how many ways can this be done?

20. There are three identical golden eggs hidden under these cups.



You may keep all of the golden eggs if you lift up the 3 cups concealing the eggs in just three trials.

a. How many different ways may the eggs be concealed?

b. An extra golden egg is added. Would you be more likely to win 4 golden eggs in just four trials?

21. Eight women try out for a running team but only 6 are to be chosen. One selector decides to choose the team by eliminating 2 runners.

a. In how many different ways may she choose the 2 to be eliminated?

Another selector decides to choose six directly.

b. In how many different ways may she choose 6 for inclusion in the team?

22. Find the integer r such that:

a. $\binom{5}{r} = 1$ b. $\binom{12}{r} = \frac{12!}{3!\,9!}$ c. $\binom{n}{r} = \frac{n!}{2!\,(n-2)!}$

d. $^{11}C_r = \frac{11 \times 10 \times 9}{3 \times 2 \times 1}$ e. $^{12}C_r = 66$ f. $\binom{10}{r} = 210$

Hint: There are two possible answers for each part.

23. There are 7 novels, 5 biographies and 6 mathematics books on a shelf. How many different selections of 3 books may be made if:

a. they are all to be novels? b. they are all to be biographies?

c. they are all to be mathematics books? d. they must not be novels?

24. How many 6-man indoor hockey teams can be selected from a squad of 13 players if:

a. the coach disregards the positions played by the members of the squad?

b. the coach includes the only goalie in the squad in each team?

25. How many different bridge hands (13 cards) are there in a deck of 52 cards? How many of these bridge hands contain no jacks, no queens, no kings, and no aces? Express your answers in factorials.

26. A fishing enthusiast has the choice of 3 rod types: fibreglass, cane (finished), cane (rough); 4 different brands of reels and 3 different brands of fishing line. In how many different ways can he:

a. select an assembled rod which includes a rod, reel and line?

b. select an assembled rod if he decides to purchase a fibreglass rod?

27. How many different hands of 5 cards dealt from a deck of 52 cards:

a. are all spades?

b. are all red cards?

c. contain neither jacks, queens nor kings?

28. A student is asked to select any five digits from the digits: 1, 2, 3, 4, 5, 6, 7, 8, 9. How many unordered selections can the student make if:

a. there are no restrictions?

b. the digit 9 cannot be selected?

c. the digits 2 and 3 must be included?

29. Consider the set, {1, 2, 3, 4, 5, 6, 7, 8, 9}. How many different subsets of four numbers:

a. consist of all odd numbers?
b. consist of all prime numbers?
c. consist of prime numbers or odd numbers?
d. consist of multiples of 3?

30. A set A = {p, q, r, 8, 9, y, 6, z, †, #, $, §}. How many different subsets of set A:

a. contain 3 elements?
b. contain 3 elements which are all letters?
c. contain 3 elements which are all digits?
d. contain 3 elements which are neither digits nor letters?

31. An opinion poll company has a list of 200 voters put into categories.

	Liberal	Labor	Swinging Voters	Total
Males:	24	35	42	101
Females:	30	31	38	99
Total:	54	66	80	200

The company conducts a "straw" poll by contacting 25 voters. How many different sets of 25 voters are all:

a. Liberal?
b. Swinging Voters?
c. Females?
d. Female Swinging Voters?

Express your answers in the form $\binom{n}{r}$

32. An electric circuit contains five switches which can be either "on" or "off". Let, for example, (0, 0, 0, 1, 1) represent a state in which the first three switches are "off" and the last two are "on".

a. How many different states have:
   i. three switches "off" and two switches "on"?
   ii. four switches "off" and one switch "on"?
b. How many different states are there?

33. A part of a computer screen is divided into 16 square regions.

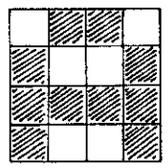


Figure 1

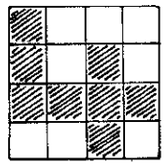


Figure 2

The programmer wishes to create patterns by shading in squares. How many different patterns are possible if:

a. ten squares are to be shaded? (figure 1)
b. eight squares are to be shaded? (figure 2)
c. all 16 squares are to be shaded?

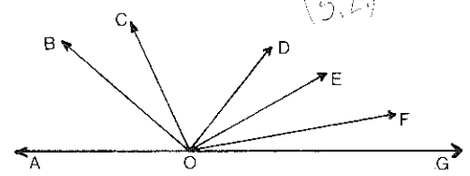
34. A manager has 3 vacancies to be filled and his personnel manager has narrowed the number of applicants to a short list of 6. In how many ways can the manager fill the positions if:

a. the 3 positions are all similar?
b. the 3 positions are all different?

35. An internal airline company offers a stop-over holiday package where a choice of 5 towns is to be made from a brochure containing 11 holiday towns. In how many ways may a tourist select a package if:

a. the order of stop-overs is important?
b. the order of stop-overs is unimportant?

36. AOG is a straight line.



How many angles less than 180° are determined by the rays in the diagram?

37. There are 12 girls in a junior hockey team and three are to be awarded players' prizes. In how many different ways can the prizes be awarded if:

a. all the prizes are the same?
b. all the prizes are different?

38. Find n:

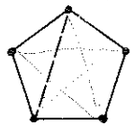
a. $\binom{n}{4} = \frac{12!}{4!\,8!}$
b. $\binom{n}{r} = \frac{7 \times 6 \times 5}{3 \times 2 \times 1}$
c. $\binom{n}{3} = 220$
d. $\binom{n}{17} = 1$
e. ${}^{n}C_{n-1} = 52$
f. $\binom{n}{7} = \binom{n}{4}$

39. A professor is asked to select 3 students from the mathematics class. He replies that there are 120 different selections. How many students were in the mathematics class?

40. In an examination a candidate must answer 5 questions. He calculates that there are 21 possible combinations of five questions. How many questions were on the examination paper?

41. A section of an art exhibition includes water colour paintings by the same artist. On entry to the gallery, the public is asked to nominate (on a form) the six which are most appealing. The forms are sorted into 28 different piles, each pile representing a different selection of paintings. How many different water colour paintings were exhibited?

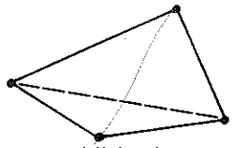
42.



pentagon



hexagon



quadrilateral

The diagram shows three examples of convex polygons. The dots represent vertices, the lines represent sides and the broken lines represent diagonals. How many different diagonals are possible in a polygon with n sides? Hint: Copy and complete the following table.

Polygon	Number of sides	Number of lines joining any two vertices	Number of diagonals
Triangle	3	$\binom{3}{2} = 3$	$\binom{3}{2} - 3 = 0$
Quadrilateral	4	$\binom{4}{2}$	$\binom{4}{2} - 4 = 2$
Pentagon	5	$\binom{5}{2}$	$\binom{5}{2} - 5$
Hexagon	6	$\binom{6}{2}$	$\binom{6}{2} - 6$
Heptagon	7	$\binom{7}{2}$	$\binom{7}{2} - 7$
Octagon	8	$\binom{8}{2}$	$\binom{8}{2} - 8$
n-gon	n	$\binom{n}{2}$	$\binom{n}{2} - n$

COMBINATIONS USING THE MULTIPLICATION PRINCIPLE

Nine people nominate for the debating team.



Two women and 3 men are to be selected from this group. How many different debating teams are possible?

Break the task into two stages: select the women **and then** the men. Using the multiplication principle:

$$\text{The number of debating teams} = \binom{5}{2} \times \binom{4}{3}$$
$$= \frac{5.4}{2.1} \times \frac{4}{1}$$
$$= 10 \times 4$$
$$= 40$$

Example 4.

Wild Bill Hickock was playing poker when he was shot. He was holding 5 cards: the "dead man's hand". It consisted of 2 aces, 2 eights and 1 other card that was neither an ace nor an eight. How many different poker hands fit this description?

Solution:

$$\text{Number of hands} = \binom{4}{2}\binom{4}{2}\binom{44}{1} = 1\,584$$

4 aces, choose 2
4 eights, choose 2
44 other cards, choose 1

Example 5.

There are 17 women and 12 men in a chess club. The club enters a team of 5 in the national championships. How many different teams can be selected if:

a. the club champion and the runner-up must be included?

b. three particular women are unavailable?

Solution:

a. $$\text{Number of different teams} = \binom{1}{1}\binom{1}{1}\binom{27}{3} = \binom{27}{3} = 2\,925$$

1 champion, choose 1
1 runner-up, choose 1
27 other members, choose 3

b. $$\text{Number of different teams} = \binom{3}{0}\binom{26}{5} = \binom{26}{5} = 65\,780$$

3 unavailable women, choose 0
26 available people, choose 5

Example 6.

Twenty tourists on a package tour are to catch taxis from the hotel to the airport. There are 5 taxis waiting in a queue outside the hotel. Each taxi will carry 4 passengers. In how many different ways can the tourists be allocated to the taxis?

Solution:

Number of different allocations $= \binom{20}{4}\binom{16}{4}\binom{12}{4}\binom{8}{4}\binom{4}{4} \approx 3{\cdot}0554 \times 10^{11}$

20 tourists, choose 4 for the first taxi

16 remaining tourists, choose 4 for the second taxi

and so on.

Exercise 3B: Combinations using the Multiplication Principle

1. Evaluate:

a. $\binom{7}{3}\binom{2}{2}$ b. $\binom{6}{4}\binom{4}{2}$ c. $\binom{13}{3}\binom{13}{2}\binom{13}{1}$

d. $\binom{1}{1}\binom{15}{4}\binom{15}{4}$ e. ${}^3C_3 \times {}^5C_2$ f. ${}^7C_0 \times {}^8C_4 \times {}^8C_3$

2. A student has a choice of 3 foreign languages and 6 sciences. How many different sets of choices has she if she must have 2 foreign languages and 3 sciences?

3. In how many ways can a set of 3 boys and 3 girls be selected from a set of 5 boys and 6 girls?

4. A committee of 3 students and 2 teachers is to be formed from a group consisting of 5 teachers and 20 students. How many different committees are possible?

5. A winner at a quiz competition is asked to select as her prize any two of the seven wrapped packages with a box of chocolates from the seven different boxes available. How many different prizes are possible?

6. Five letters are to be chosen by selecting three letters from the word *DRAWING* and two from the word *HELP*. How many different unordered selections can be made?

7. In a class there are five A-grade students, seven B-grade students and twelve C-grade students. How many different combinations of 6 students include 2 from each grade?

8. A sale of paintings includes 9 oil paintings and 13 water-colour paintings. In how many different ways can a customer select two oil paintings and three water-colour paintings?

9. In how many different ways can a reader select three books from the library shelves if she wishes to select two romance novels and a travel book when the library has 52 romance novels and 14 travel books?

10. A jury of 12 is to be selected from 30 nominated people.

a. How many different juries can be selected?

b. How many different juries can be formed which will include two particular people?

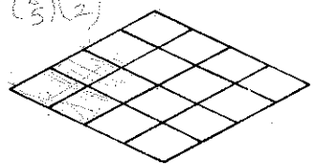
11. Determine how many different poker hands (5 cards) contain 3 clubs and 2 diamonds.

12. A kitchen-hand of a restaurant is instructed to prepare a fruit salad consisting of 5 different fruits. There are 8 different fruits available. The menu states that fruit salad and icecream is the only dessert available, however the customer has a choice of 4 different flavours of icecream.

a. How many different varieties of fruit salad can be prepared?

b. How many different orders of dessert are possible if a customer chooses his own 5 fruits with a little of each of 2 flavours of icecream?

13. How many different parallelograms can be found in the diagram?



14. From 8 Malaysians, 6 Vietnamese and 4 Indonesians, how many different sets of 6 people consist of:

a. two from each country?

b. four Malaysians and 2 non-Malaysians?

15. A newsagent displays 5 different grades of black and red felt pens: extra wide, wide, medium, fine and extra fine. In how many different ways may a customer choose:

a. two felt pens?

b. two pens, a black and a red?

c. a black medium pen and a red pen?

16. A portable electronic printer is powered by 4 small batteries. When the owner removed the used batteries from the printer he accidentally mixed them with the 4 new batteries which were identical in appearance. In how many different ways can the person select 4 batteries consisting of:

a. all new batteries? b. two new and two old? c. three new and one old?

17. A building contractor purchases 6 blocks of land. He decides to build 3 houses of different styles on 3 of the blocks. The draftsman prepares plans of 5 different styles of houses which are all acceptable to the builder. Taking into account both the location of a house and its style, how many different combinations are available to the builder?

18. In a school calendar there are 13 weeks in first term, 14 weeks in second term and 13 weeks in third term. The principal decides to choose 2 weeks in each term for parent interviews. How many different combinations of weeks are possible?

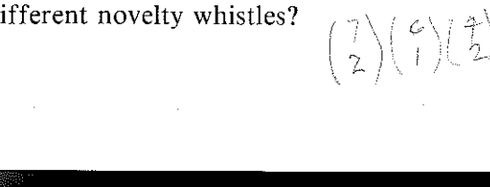
19. A class room has 4 rows with 7 desks in each row. There are 16 students in the class and they seat themselves, 4 to each row. How many different combinations of seats may they occupy?

20. To produce a new computer program to teach English a team of six people is required: 3 foreign language experts, 2 English teachers and 1 computer programmer. Six foreign language experts, 12 English teachers, and 4 computer programmers apply to be on the team. How many different teams of 6 people are possible?

21. A first year technical course in computer programming requires a full-time student to enrol in six subjects. Two of these subjects are compulsory and the student must select two elective units from five offered and two enrichment units from the three offered. How many different selections are available for a full-time student?

22. A Hockey Council is to have 5 members. There are to be 2 members from each of the A and B grade teams. There are 13 players in each team. The other council member is to be selected from the 6-member organizing committee. How many different councils are possible?

23. At the beginning of a battle there are 13 positions left on the castle rampart but only 4 archers and 3 pikemen available.



If the 4 archers have identical fighting strengths and the 3 pikemen have identical fighting strengths, in how many different ways may they be deployed along the rampart as a fighting force?

24. A competition called PICK-A-BRICK requires an entrant to select 10 numbers. By placing a cross through the selected numbers, the entrant must select one from row 1, two from row 2, three from row 3 and four from row 4.

		1	2				Row 1
	3	4	5	6			Row 2
7	8	9	10	11	12		Row 3
13	14	15	16	17	18	19	20 Row 4

How many different ways could an entry form be completed?

25. In preparation for a small child's birthday party a mother decides to present each of the guests with a parcel containing two different-coloured balloons, a party hat, two different face masks and two novelty whistles. If the mother has an ample supply of each gift, how many different combination parcels can she make if there are 7 different-coloured balloons, 4 different face masks, 6 different party hats and 5 different novelty whistles?

26. There are 20 women and 10 men in a room. How many different groups of 6 people from this room will consist of:

a. all women? b. all men? c. three women and three men?

27. A, B and C are disjoint sets:

$A = \{1, 2, 3, 4\}$ $B = \{5, 7, 9, 11, 13\}$ $C = \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \frac{1}{11}\}$

How many different sets of 6 elements can be formed by:

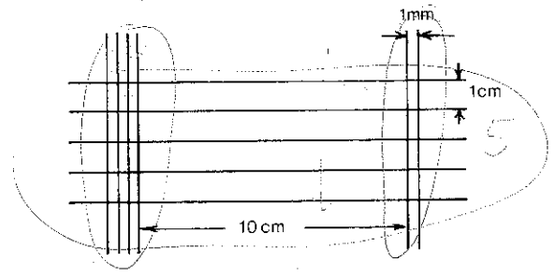
a. choosing 2 elements from each set?
b. choosing 4 elements from B and 2 from C?
c. choosing 3 elements from A and the other 3 from the union of B with C?

28. Five cards are drawn from a standard deck of 52 cards. How many different sets of 5 cards consist of:

a. 2 clubs and 3 hearts? b. 3 kings and 2 queens?
c. 2 clubs, 2 hearts and 1 spade? d. a ten, a jack, a queen, a king and an ace?
e. 2 red cards and 3 black cards? f. the ace of spades and 4 non-aces?
g. a pair of threes, a pair of jacks, and a seven?

29. In hiring a research team, the director of the research programme has to choose 3 analytical chemists from 7 applicants, 2 biologists from 5 applicants, 6 laboratory assistants from 10 applicants and 2 computer operators from 6 applicants. Determine the total number of ways in which the director can choose his team.

30. There are 5 horizontal lines crossed by 6 vertical lines. The distances between the lines are indicated on the diagram.



How many different rectangles, with areas greater than 9 square centimetres, can be formed?

31. How many different five-card poker hands contain:

a. all face cards (court cards)? Jacks, queens and kings are face cards.
b. three face cards and two non-face cards?

32. A hockey squad of 17 players contains 8 specialist forwards, 4 specialist halfbacks, 3 specialist fullbacks, and 2 players who can only play as a goalie.

a. How many different 11-man hockey teams can be nominated if no regard is given to the position the player is to occupy?

b. How many different 11-man hockey teams can be selected containing:
   i. five forwards, three halfbacks, two fullbacks and a goalie?
   ii. six forwards, two halfbacks, two fullbacks and a goalie?

33. Determine how many different bridge hands (13 cards) contain 4 hearts, 3 diamonds, 5 clubs and a single spade.

34. How many different poker hands (5 cards) consist of:

a. three jacks and 2 sevens?
b. three jacks and any pair (2 sevens, 2 queens, etc.)?
c. three of a kind (3 jacks, 3 nines, etc.) and any pair (2 sevens, 2 queens, etc.)?

35. Some patients in a speech clinic are nominated for a special training programme. They fall into the following categories:

	4 years and under	Between 4 and 12 years	12 years and over
Hearing impaired	12	29	14
Hearing not impaired	8	12	14

Twelve patients are to be selected.

How many different selection are possible if:

a. there must be 4 from each age group?
b. the programme is suitable only for children under 12 years of age or whose hearing is not impaired?
c. irrespective of age, there must be equal numbers of hearing-impaired and hearing-unimpaired subjects?

Do not simplify the answers.

36. A bridge player is dealt the following hand of 13 cards from a standard deck: 9 spades, 3 hearts and 1 club.

a. How many different bridge hands fit this description?
b. How many different bridge hands contain:
   i. no diamonds?
   ii. 10 diamonds and 3 clubs?
   iii. exactly 10 diamonds?

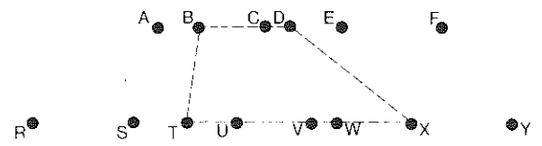
37. A swimming squad consists of 4 from Zone A, 5 from Zone B and 6 from Zone C. A team of 6 is to be selected to represent the state at the National Titles.

a. Determine the number of possible teams that could be selected.
b. Determine the number of possible teams that could be selected if 2 swimmers from each zone must be included.

38. A committee of 3 boys and 3 girls is to be selected from a group of 7 boys and 7 girls. Find how many different committees can be formed if:

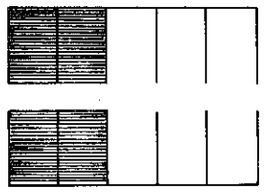
a. a particular girl must be chosen.
b. a particular boy and a particular girl must be excluded.

39. Quadrilaterals are to be formed using (as vertices) 2 dots from the top row and 2 dots from the bottom row.



a. How many different quadrilaterals may be formed?
b. How many of these quadrilaterals:
   i. do not have B as a vertex?
   ii. do not have B as a vertex nor contain B in the uppermost side?

40. There are ten parking bays for a new building and four of them are covered.



Two companies rent the building and are to have five bays each. In how many different ways may:

a. the bays be allocated?
b. the bays be allocated if they must get two covered bays each?

41. A man has 8 different paintings. He decides to hang 2 on each of the 4 walls of his living room. In how many different ways can he allocate the paintings to the walls?

42. There are 30 pigeon-holes in the office. Fifteen of these are to be reserved for the accounts section and 13 for the despatch section. The rest are spare. In how many different ways can the pigeon-holes be selected?

43. Eight men and 8 women are to be subjects in a nutrition test. They are to be split into 2 equal-sized groups. The experimental group is to be given a fat-free diet and the control group is to continue its normal diet. How many different splits are possible if:

a. the sex of the subjects is disregarded?
b. equal numbers of men and women must be in each group?

44. A row of 2 ticks and 8 crosses $X\,X\,X \checkmark X\,X \checkmark X\,X\,X$ represents one student's performance on a true-false test.

a. How many different rows contain:
   i. 2 ticks and 8 crosses?
   ii. 5 ticks and 5 crosses?
b. If you answered the paper by guessing every answer, are you more likely to get 2 correct or 5 correct? Why?

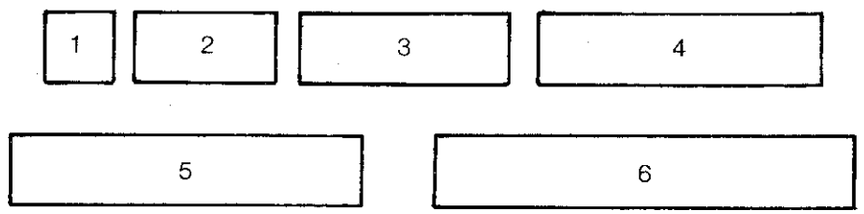
45. Prove, that for all integers n:

a. $\binom{n}{4}\binom{n-4}{3} = \binom{n}{3}\binom{n-3}{4}$, $n \geqslant 4$

b. $\binom{n}{r}\binom{n-r}{2} = \binom{n}{2}\binom{n-2}{r}$, $n \geqslant r \geqslant 2$

46. Justify that there are 624 possible poker hands of 5 cards which contain exactly four of a kind; that is, 4 aces, 4 sevens, 4 kings, etc.

47.



The wooden blocks are 1, 2, 3, 4, 5 or 6 centimetres long, respectively. Blocks are placed end to end to make one "long block".

a. How many "long blocks" of different lengths can be formed by using 3 blocks?
b. How many of these "long blocks" are more than 9 centimetres in length?

Hint: List some possibilities.

48. Five noughts and 4 crosses could fill a noughts and crosses grid:

o	o	x
x	o	o
x	x	o

o	x	o
o	o	x
o	x	x

o	o	x
x	x	o
o	o	x

a. How many different patterns of 5 noughts and 4 crosses are possible?
b. How many different patterns consist of 3 noughts and 3 crosses?
c. How many different patterns consist of 4 noughts and 3 crosses?

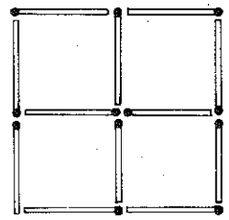
49. There are 6 tennis players: Alf, Bill, Chris, David, Ernie and Fred. They are to be formed into three doubles teams, 2 players in each team.

a. If Alf is always paired with Bill, **list** all of the possible doubles teams.
b. How many different partners may Alf have if he is not necessarily paired with Bill?

Hence,

c. how many different ways may they be formed into 3 doubles teams if there are no restrictions?

50. Make 2 squares by removing a pair of matches. Do not re-arrange the others.



a. How many different pairs of matches could be removed to form 2 squares?
b. If you removed pairs of matches at random, what is the largest number of different trials possible before forming 2 squares?

51. There are 4 football games each weekend and 2 umpires are needed for each game. There are 12 umpires available for next weekend's games. How many different ways can the umpires be allocated to next weekend's games?

52. A triad is sounded when 3 musical notes are played together.

C, C#, D, E♭, E, F, F#, G, A♭, A, B♭, B

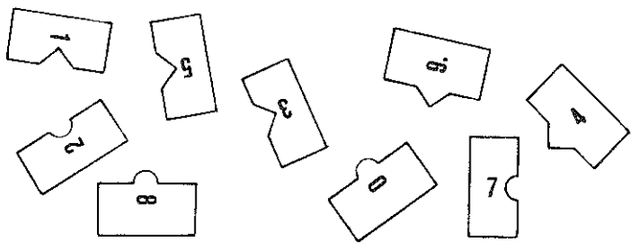
low pitch ↑ ↑ high pitch

a. How many different triads can be formed from the given musical scale?
b. How many of these triads contain $A\flat$?
c. How many of these triads have $A\flat$ as the note of highest pitch?
d. How many of these triads have $A\flat$ as the note of lowest pitch?
e. How many of these triads have $A\flat$ as the note of middle pitch?

Chapter 4:

The Addition Principle and Miscellaneous Counting

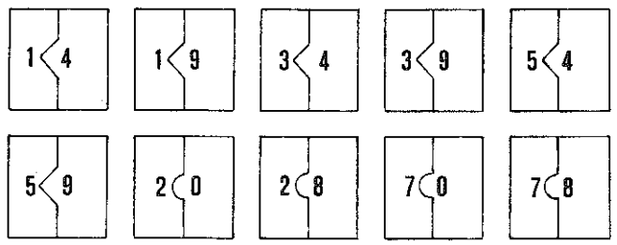
Two number-tile sets have been mixed up and only nine tiles are left.



The tiles are to be fitted together to build two-digit numbers. How many different two-digit numbers can be built?

To build numbers we join a pair of pointed tiles OR a pair of rounded tiles.

The set of all possible two-digit numbers is:



There are 10 different two-digit numbers.

The set of 10 different two-digit numbers is the union of two mutually exclusive subsets, namely:

Subset A: the two-digit numbers built using the "pointed" tiles. There are 6 of these.
Subset B: the two-digit numbers built using the "rounded" tiles. There are 4 of these.

Therefore, the total number of two-digit numbers

= the number of two-digit numbers which are "pointed" or "rounded"
= 6 + 4
= 10

Addition Principle

If the set of all possible outcomes of performing a task can be separated into two **mutually exclusive** subsets, the first of which has m outcomes and the second n outcomes, then the total number of ways of performing the task is $m + n$.

PERMUTATIONS USING THE ADDITION PRINCIPLE

Example 1.

Before the birth of their first child, a couple lists all of their preferences for first and second names.

	First Name	Second Name
Boys' Names	Richard Harold Thomas Simon	Jonathan Glenn Benjamin
Girls' Names	Anne-Marie Leanne Kate	Susan Pippa Dawn

In how many different ways can they name their first child?

Solution:

The number of names
= the number of ways of naming EITHER a boy OR a girl
= 4.3 + 3.3
= 12 + 9
= 21

We use the addition principle here because the task (to name a child) leads to two mutually exclusive subsets:

EITHER a pair of boys' names OR a pair of girls' names.

The multiplication principle is then used to count the number of different ways of naming the boy. There are 4 choices for first name **and** then 3 choices for second name. Hence there are $4 \times 3 = 12$ ways of naming the boy. Similarly, there are 9 different ways of naming the girl.

Example 2.

How many different numbers greater than 400 can be formed from the digits 3, 5, 6, 8 if repetition of digits is not allowed?

Solution:

Number of numbers greater than 400

= number of numbers with EITHER three digits OR four digits

= 3.3.2 + 4.3.2.1

= 18 + 24

= 42

Exercise 4A: Permutations using the Addition Principle

1. In how many different ways can the three rear seats of a car be filled, either all by boys or all by girls, if there are 5 boys and 3 girls wishing to be transported?

2. How many words consisting of 3 or 4 letters can be formed using letters from the word *FORMULA*?

3. How many words consisting of 4 or 5 letters can be formed using letters from the word *THEOREM* if the *E*s must be excluded from all words?

4. a. Use a tree diagram to list all of the numbers greater than 50 that may be formed from the tiles: [2] [5] [6]. Repetitions are impossible.
b. How many different numbers may be formed? There is no restriction on the size of the number.

5. How many 2-digit or 4-digit numbers can be formed from the digits 1, 3, 5, 7, 9 if:

a. repetition of digits is not allowed?
b. repetition of digits is allowed?

6. How many different 4-digit even numbers may be formed from the digits 2, 3, 4, 8 if:

a. digits may be repeated?
b. digits may not be repeated and the number must be greater than 5 000?
c. digits may not be repeated and the number must be greater than 2 000?
d. digits may not be repeated and the number must be greater than 2 500?

7. How many words consisting of at least 2 letters can be formed using letters from the word *MIXED*?

8. Six men stand together in a line. In how many ways is it possible for them to position themselves if:

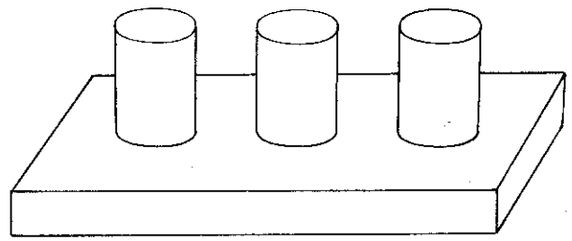
a. no restrictions apply?
b. a particular man, B, must occupy the last position?

c. A and B must stand next to each other?
d. A is to stand to the right of, and adjacent to, B?
e. A is to stand to the right of B?

9. Six candidates are nominated for a local council election. In how many different ways can a voter number a voting slip if:

a. he places a digit from 1, 2, 3, 4, 5, 6 against each candidate's name?
b. the voting is 'optional preferential'? i.e. the voter numbers the candidates in order of preference but has the option of leaving some candidates out.

10. Three sockets for light globes are mounted in a row on a board.



Coloured globes are to be used to send signals at night. There are available: 3 red, 3 blue and 3 yellow globes, all of which are identical except for colour.

a. How many different 1-globe signals can be sent if the person receiving the signal cannot tell in which socket the globe is placed?
b. How many different 2-globe signals can be sent if the person receiving the signal cannot tell which sockets have been used but is able to tell if the globes are close together or far apart using:
i. globes of the same colour? ii. globes which differ in colour?
c. How many different 2-globe signals can be sent?
d. How many different 3-globe signals can be sent using:
i. globes of the same colour?
ii. globes all differing in colour?
iii. globes, two of one colour and one of another colour?
e. How many different 3-globe signals can be sent?
f. How many different signals can be sent using at least one globe?

11. There are 6 roads connecting towns A and B, and 3 roads connecting towns B and C. Three direct routes exist between towns A and C. In how many different ways could a motorist drive:

a. from town A to town C via town B?
b. from town A to town C via town B, and return via town B?
c. from town A to town C and return without travelling on the same road more than once?

12. A typist, when responding to customers' letters, always includes a reference number at the top of her letters. These reference numbers contain either four, five or six digits, none of which commences with a zero; or three digits, all of which commence with a zero. Determine the number of possible 3, 4, 5 or 6 digit reference numbers which may be generated if repetition of digits is allowed.

13. A golfer has 4 hats, 3 caps, 5 shirts and 6 skirts. All articles of clothing are different from each other. How many different outfits consisting of a skirt, a shirt and one item of head gear can she select?

14. How many different "words" of at least four letters can be formed from the letters of the word *SOFTWARE*?

15. A digital watch displays a 3-digit or a 4-digit number.



How many different numbers are possible?

16. Numbers are to be formed using the tiles: [8] [5] [6] [9] . Repetition of digits is impossible.

a. How many different three-digit numbers are possible?
b. How many different numbers of at least three digits are possible?
c. How many different four-digit even numbers are possible?
d. How many different four-digit even numbers greater than 6000 are possible?

17. Determine the number of different six-letter "words" which can be formed from the letters of the word *MATRIX*. How many of these words:

a. end with an *R*?
b. commence with an *R*?
c. begin or end with an *R*?
d. begin with a *T* or end with an *M*?

18. A squash player has a choice of either a pink outfit or a blue outfit. She has 3 pink skirts, 2 blue skirts, 1 pink shirt, 3 blue shirts, 2 pairs of pink socks, 2 pairs of blue socks, and 3 pairs of white socks which can be worn with either outfit.

a. How many different squash outfits are possible?
b. If the pink shirt is lost, how many different outfits are possible?
c. If she finds her pink shirt but decides not to wear white socks with a pink outfit, how many different outfits are possible?

19. In front of you on a table there is a ten-cent coin, a twenty-cent coin, a fifty-cent coin and a one-dollar coin. After you put on a blindfold you may touch two of the coins but the first one you touch will be picked up before you touch the second. If the second coin you touch is of higher value than the first you may keep the coins.

a. How many different sequences will enable you to keep the coins?
b. How many different sequences are there altogether?

20. A train consists of two locomotives and eight flat cars, all of different weights. The locomotives have different powers and a train is made up with both locomotives at the front or, the more powerful one at the front with the less powerful one at the rear. Taking into account the weight of the flat cars and the power of the locomotives, how many different trains are possible?

21. A dog breeder's method of naming pups in a litter is to use the kennel name first, followed by a selected name to identify the litter, and a third name to identify the pup, depending on its sex. The breeder selects the following set of names:

	Kennel Name	Litter Name	Other Name
Dog Names	Bardic	Kind	Sir Boy Man Prince King
Bitch Names	Bardic	Kind	Miss Madam Girl Princess Queen

a. In how many different ways can the breeder allocate names to a litter of:
   i. 4 bitches?
   ii. 2 bitches and 2 dogs?
b. If, before the birth of 4 pups, the breeder wished to list all of the possible pedigree names (e.g. Bardic Kind Princess) how many different names could he have on his list?

22. To operate an automatic teller machine the user must first insert his plastic card and then key in his personal identification number (P.I.N.). The P.I.N. is a secret four-digit number. A user has almost forgotten his P.I.N. but can recall that it contains the digits *4* and *3* in that order but not necessarily together.

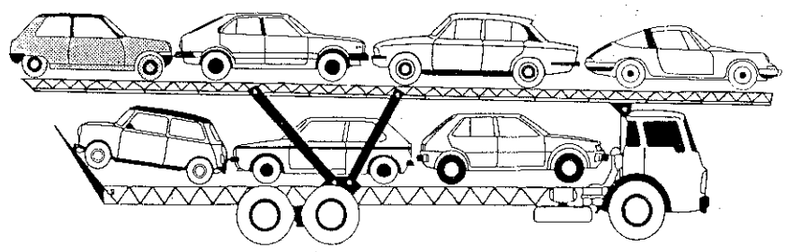
a. List all of the possible P.I.N.s which satisfy this recollection; use question marks (*?*) for the missing digits. For example, *?4?3*.
b. How many different P.I.N.s are of the form *4??3*?
c. How many different P.I.N.s contain the digits *4* and *3* in that order?

23. In a horse race there are 10 horses. To win a TRIFECTA, a punter must nominate the first, second and third horse in the correct order. To win a FORECAST QUINELLA a punter must nominate the first and second horse in the correct order.

a. If a punter wishes to bet on all possible TRIFECTAS and all possible FORECAST QUINELLAS, how many different bets must he take?
b. A punter is "certain" that a particular horse will be either first or second. Taking this into account, how many different bets must he take if he wishes to bet on all possible TRIFECTAS and FORECAST QUINELLAS?

24. Six boys and 4 girls stay at a holiday camp. There is a 6-bed and an 8-bed dormitory available. In how many different ways may the beds be allocated if boys and girls must sleep in separate dormitories?

25. A double-decker auto transporter has room for three sedans on the bottom deck and four sedans on the top deck.



Sometimes vans are transported but these take the space of two sedans.

a. In how many different ways can a full load be arranged which consists of:
   i. only sedans? ii. only vans?
b. In how many different ways can a load be arranged which consists of:
   i. 1 van and 5 sedans? ii. 2 vans and 3 sedans?

26. A shop owner has 3 spaces to fill on a display shelf. He decides to fill the spaces with either all different boxes of toffees or all different boxes of chocolates. If he has 5 different varieties of toffees and 7 different varieties of chocolates in his store-room, in how many different ways can he fill the display shelf?

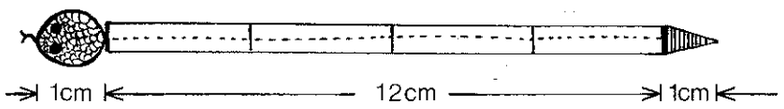
27. How many 2-digit or 3-digit numbers are there such that:

a. they are odd numbers?
b. they are divisible by 5?
c. they contain no two digits alike?
d. they contain no two digits alike and are divisible by 5?
e. they contain no two digits alike and are odd?

28. To make toy snakes, beads are threaded onto a piece of string. A large purple bead represents the head of the snake and a small black cone-shaped bead, the tip of the tail. The body of the snake is made of thin cylindrical beads which are available in unlimited numbers in a variety of colours and lengths:

3 cm long — red, green, orange, white, blue
4 cm long — blue, yellow, red.

The body of the snake (excluding the head and the tip of the tail) is to be 12 centimetres long.



How many different-looking snakes can be made if:

a. a colour may not be used more than once and only 4-centimetre cylindrical beads may be used?
b. a colour may not be used more than once but there is no other restriction?
c. colours may be used more than once but two cylindrical beads of the same colour may not be placed together?

29. How many numbers less than 3 000 can be formed from the digits 1, 2, 3, 4, 5, 6 if repeated digits are not permitted?

30. How many 3-digit numbers have exactly 2 digits alike?

31. The menu at a small restaurant is:

First Course	Second Course	Third Course
Prawns	Fish Mornay	Icecream
Oysters	Chicken	Fruit
Olives	Beef	

A customer may select one item, at most, from each course.

a. How many different 3-course meals are possible?
b. How many different 3-course meals are possible if sea-food may not be taken on both the first and second courses?
c. How many different meals of at least one course are possible?

COMBINATIONS USING THE ADDITION PRINCIPLE

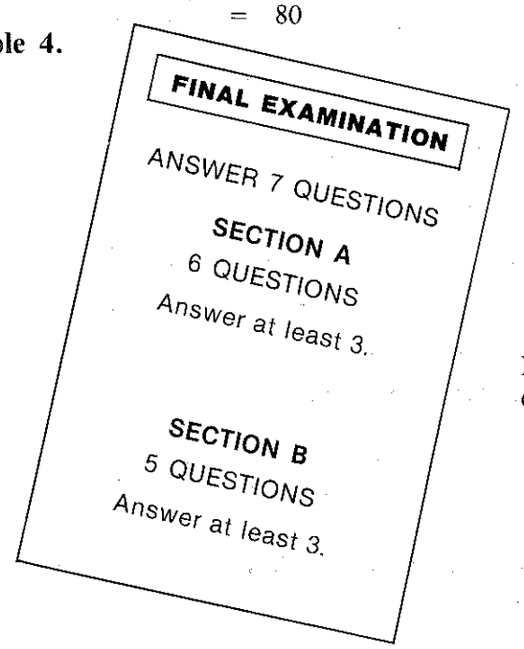
Example 3.

For her birthday, a girl receives a gift voucher redeemable at a music shop. When she gets to the shop she finds 8 cassettes and 5 records to her liking. With the voucher she can afford **either** 4 cassettes **or** 3 records. How many different combinations are available to her?

Solution:

$$\text{Number of combinations} = \binom{8}{4} + \binom{5}{3}$$
$$= 70 + 10$$
$$= 80$$

Example 4.



How many different selections of questions are possible?

Solution:

The candidate may answer:

EITHER 4 questions from Section A and 3 from B
OR 3 questions from Section A and 4 from B.

Using the addition principle:

$$\text{Number of selections} = \binom{6}{4}\binom{5}{3} + \binom{6}{3}\binom{5}{4}$$
$$= 15 \cdot 10 + 20 \cdot 5$$
$$= 150 + 100$$
$$= 250$$

Example 5.

A four-man golf team is to be selected from 5 Australians and 4 Americans. How many different teams contain:

a. no Americans? b. at least one American?

Solution:

a. Number of teams containing no Americans

$$= \binom{4}{0}\binom{5}{4} \quad \text{(No Americans AND 4 Australians)}$$
$$= 5$$

b. Number of teams containing at least one American

= number of teams containing EITHER

1 American and 3 Australians	OR	2 Americans and 2 Australians	OR	3 Americans and 1 Australian	OR	4 Americans and 0 Australians
$\binom{4}{1}\binom{5}{3}$	+	$\binom{4}{2}\binom{5}{2}$	+	$\binom{4}{3}\binom{5}{1}$	+	$\binom{4}{4}\binom{5}{0}$
40	+	60	+	20	+	1

$= 121$

Alternatively:

Number of teams containing at least one American

= Total number of teams — Number of teams containing NO Americans

$$= \binom{9}{4} - \binom{4}{0}\binom{5}{4}$$
$$= 126 - 5$$
$$= 121$$

Exercise 4B: Combinations using the Addition Principle

1. A committee of at least 4 and at most 6 is to be formed from a group of 12 people. How many different committees are possible?

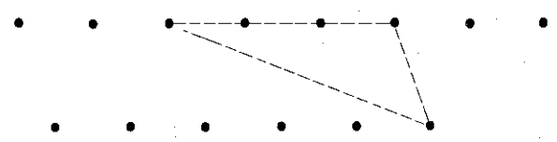
2. In how many ways can a student select one or more novels from five recommended novels?

3. An artist has tubes of red, green, blue, yellow, brown and purple paint. New colours are made by mixing together equal quantities of paint from these tubes. How many different colours are possible if the artist does not use up all of the paint from any one of the tubes?

4. A person has a 1c, 2c, 5c, 10c, 20c, 50c and a $1 coin. How many different sums of money can the person form using these coins?

5. How many different fruit-juice cordials can be prepared from apples, passionfruit, pineapples, peaches, pears, oranges and grapes, if a customer may select one or more fruits?

6. A beam balance has the following weights: 5 g, 15 g, 25 g, 50 g and 100 g.
   a. How many different combinations of weights are possible?
   b. How many of these combinations come to more than 20 grams?

7. 

There are 8 dots in the top row and 6 in the bottom row. Triangles are formed by using 3 dots as vertices. How many different triangles are possible?

8. A committee of six people is to be selected from 6 men and 5 women. How many different committees of at least 3 women can be formed?

9. A committee of six is formed from 8 Australian and 5 Indonesian students. In how many ways this committee be selected if:
   a. no restrictions apply?
   b. it must contain 2 Indonesian students?
   c. it must contain at least 2 Indonesian students?

10. How many different juries of twelve may be chosen from 10 men and 9 women if:
    a. no restrictions apply?
    b. there are equal numbers of men and women to serve?
    c. no more than 4 women are to serve?

11. A six-person trade delegation is to be chosen from 12 business people and 4 politicians. How many different delegations can be selected to include::
    a. 4 business people?
    b. no politicians?
    c. at least 2 politicians?

12. An international mountain climbing team consisting of nine climbers is to be chosen from 7 Austrians, 5 Swiss and 4 Frenchmen. In how many ways can this team be formed so as to contain:

a. an equal number of each nationality?
b. at least 3 Frenchmen?
c. at the most 2 Austrians?

13. A school has 8 boy and 8 girl prefects. A sub-committee of five is selected to organize the school ball. How many different committees can be selected if:

a. no restrictions are made?
b. 3 boys and 2 girls are required?
c. 3 boys and 2 girls are required and the head boy and head girl must be included?
d. at least one girl must be included?

14. A student must select a set of 16 pieces of poetry to study for the final examinations from an anthology of 24 poems, which includes 8 sonnets. In how many ways can this be done if:

a. exactly 5 sonnets are selected? b. at least 6 sonnets are selected?

15. A batch of ten computer chips is known to contain three defective chips. A sample of three chips is taken from the batch. How many selections of three chips are possible?

a. with no restrictions?
b. which contain three defective chips?
c. which contain at least two defective chips?
d. which contain at most one defective chip?

16. How many 5-card hands from a 52-card pack contain:

a. exactly 3 aces? b. at most 2 aces?

17. Show that:

a. $\binom{8}{5} + \binom{8}{4}\binom{7}{1} + \binom{8}{3}\binom{7}{2} + \binom{8}{2}\binom{7}{3} + \binom{8}{1}\binom{7}{4} + \binom{7}{5} = \binom{15}{5}$

b. $\left({}^3C_0\right)^2 + \left({}^3C_1\right)^2 + \left({}^3C_2\right)^2 + \left({}^3C_3\right)^2 = {}^6C_3$

c. $\left({}^8C_0\right)^2 + \left({}^8C_1\right)^2 + \left({}^8C_2\right)^2 + \ldots + \left({}^8C_8\right)^2 = {}^{16}C_8$

d. ${}^{m+n}C_2 = {}^mC_2 + {}^nC_2 + {}^mC_1 . {}^nC_1$

18. A student is to attempt 8 out of 10 questions in an exam.

a. How many choices has the student?
b. How many choices has the student if the first 3 questions are compulsory?
c. How many choices has the student if at least 4 of the first 5 must be attempted?

19. An examination paper consists of two sections:

7 questions in Section A and 5 questions in Section B.

In how many different ways can a candidate select his questions if:

a. he must answer 5 questions from Section A and 3 questions from Section B?
b. he must answer 8 questions with at least 3 from each section?

20. In a 12-question examination paper there are 4 questions on trigonometry. In how many ways may a candidate select 8 questions if:

a. no restrictions apply?
b. no trigonometry questions are to be attempted?
c. exactly 3 trigonometry questions are to be attempted?
d. at most 2 trigonometry questions are to be attempted?
e. at least 3 trigonometry questions are to be attempted?

21. A private school offers the following options:

Woodwork, Metalwork, French, Art, Music and Technical Drawing.

In how many ways can a four-subject option course be chosen if:

a. no restrictions are made?
b. students choosing Technical Drawing must choose either Woodwork or Metalwork, but not both?

22. A college debating team (consisting of three members) is to be chosen from eleven students, seven of which are females. In how many ways can this team be selected if:

a. the team must include exactly two males?
b. the team must include exactly two females?
c. the team must include at the most two females?

23. A club contains 8 women and 12 men. How many 6-member committees can be formed if:

a. there are no restrictions?
b. the committee must contain more women than men?
c. the committee must contain more men than women?

Hint: Single-sex committees are permitted.

24. Six men and 4 women go on a canoeing trip. They have a 4-person canoe and six individual kayaks. How many different unordered selections of 4 people can be made to travel in the 4-person canoe if:

a. there are no restrictions?
b. there must be 2 men and 2 women?
c. there must be 3 men and 1 woman?
d. one particular woman must travel in it?
e. two particular men refuse to be in it together?

25. A lady gives a dinner party for 6 guests. In how many ways may they be selected from 8 friends if:

a. there are no restrictions?
b. if 2 of her friends will only attend together?
c. if 2 of her friends will not attend together?

26. How many different committees of 4 people may be selected from 10 people if:

a. no restrictions apply?
b. 2 particular people must be included?
c. 2 of the people refuse to serve on the committee together?

27. A woman has 12 close friends. How many different combinations of 5 of them may she invite for dinner if:

a. two of them are married and will only attend together?
b. two of them have been divorced and will not attend together?

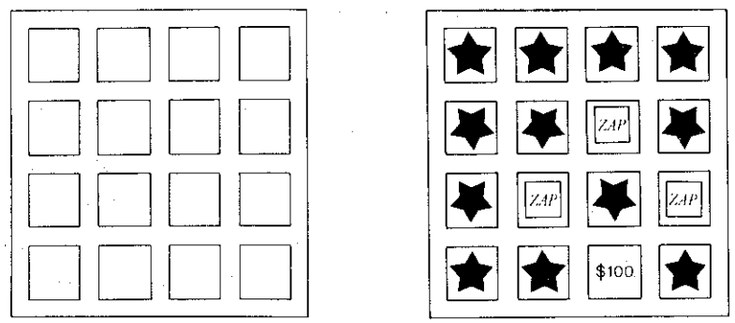
28. A carton of one dozen eggs contains 3 underweight eggs. Four eggs are selected from this carton. How many different combinations of four eggs contain:

a. 3 underweight eggs and 1 correct-weight egg?
b. no underweight eggs?
c. 2 underweight eggs exactly?
d. exactly 1 underweight egg?

All underweight eggs are between 1 and 2 grams less than the correct weight.

e. How many different selections of 4 eggs would be at least 2 grams underweight?

29. A new "scratch-and-win" game is called *$100-ZAP.*



There are 16 squares on each card: a *$100* square, three *ZAP*s and 12 stars. You must uncover exactly eleven squares. To win $100 you must uncover the *$100* square but you must not uncover a *ZAP*. How many different combinations of eleven squares:

a. are there altogether?
b. will win you $100?
c. include *$100* and at least one *ZAP*?
d. include neither *$100* nor a *ZAP*?

30. Four students are to be selected from a class of 10 girls and 15 boys. How many different selections do not contain 2 girls and 2 boys?

31. There are 7 Australians, 4 Malaysians and 2 Indonesians in a room. How many different combinations of 3 people from this room contain people from different countries?

32. A man makes a list of 17 people to play bridge. There are 4 people at a table for each game. Consequently, he cannot invite all seventeen. Remembering that he wishes to play himself, how many different combinations of people can he invite?

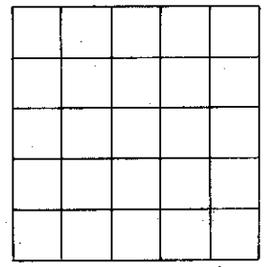
33. Six people wish to catch a taxi. A taxi driver will take 4 people at most so two taxis must be used. In how many ways can they split themselves into 2 groups if:

a. 3 people are to travel in each taxi?
b. either 3 people are to travel in each taxi or 4 people are to travel in one of the taxis?

34. In how many ways can a group of 8 people subdivide into two unequally-sized groups?

35. A supermarket manager is offered a special deal from a food wholesaler who wishes to promote a new product (Product B). There are 3 varieties of Product B and 9 varieties of the old product (Product A). For every 3 varieties of Product A he takes, he will be given any 1 variety of Product B. If he decides to try at least 1 variety of Product B, how many different combinations are available?

36. Six equally-spaced horizontal lines are crossed by six equally-spaced vertical lines to form a 5×5 square.



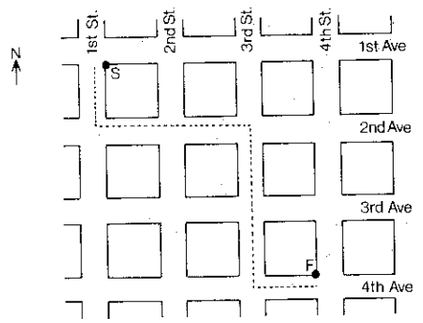
a. How many different rectangles (including squares) are formed?
b. How many different squares are formed?
Hint: Count the 1×1 squares, then 2×2 squares, and so on.
c. How many rectangles which are not squares are formed?

37. Three people approach 4 turnstiles at a football ground. Ignoring the identities of the people and the order in which they enter, how many different ways may they enter the ground if:

a. they all enter through the same turnstile?
b. they all enter through different turnstiles?
c. two enter by one turnstile and the other enters by a different one?
d. they may choose any turnstile(s)?

COUNTING USING ARRAYS

Example 6.

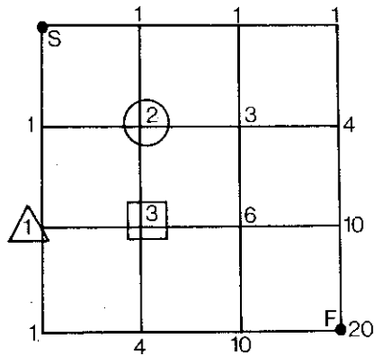


A shopper wishes to walk from a shop (S) on the corner of 1st Avenue and 1st Street to her flat (F) on the corner of 4th Avenue and 4th Street. How many different routes are possible if she never walks north (↑) or west (←)?

Solution:

The number of different routes to any intersection is given by the following array:

For example:



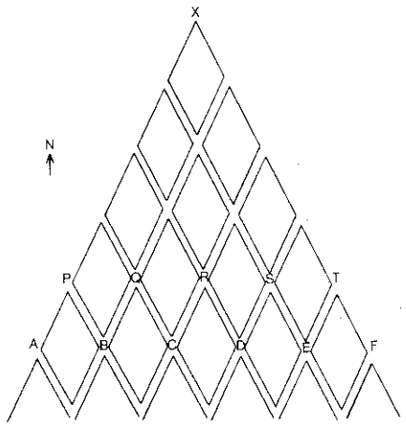
To reach the intersection ◯ of 2nd Street and 2nd Avenue there are two routes, namely, →↓ or ↓→. To reach the intersection △ of 1st Street and 3rd Avenue there is only 1 route.

Therefore, there are ② + ⚠ = 3 possible routes to reach the intersection □ of 2nd Street and 3rd Avenue.

To obtain any number in the grid, add together the number to the north of it and the number to the west of it. There are, clearly, 20 different routes she may take.

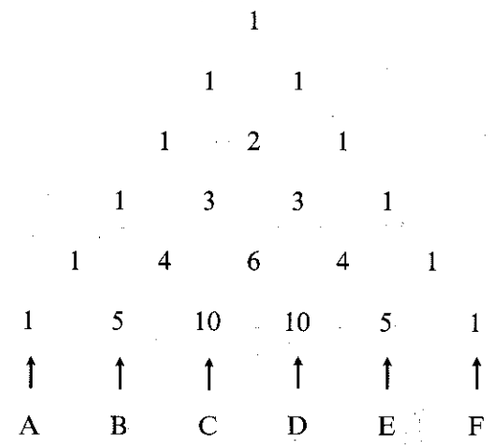
Example 7.

How many different southerly routes are there from X to each of A, B, C, D, E and F?



Solution:

The array is:



There is only 1 route to A. There are 5 routes to B; 10 routes to C; 10 routes to D; 5 routes to E; and 1 route to F.

This array is called PASCAL'S TRIANGLE after Blaise Pascal (1623–1662).

Example 8.

For the map in the previous example:

a. How many blocks do you need to travel to get to each of P, Q, R, S and T?

b. How many times do you travel a block in the S.E. direction (↘) to reach each of P, Q, R, S, and T?

c. Evaluate $\binom{4}{0}$, $\binom{4}{1}$, $\binom{4}{2}$, $\binom{4}{3}$ and $\binom{4}{4}$ and compare your result with the corresponding line in Pascal's Triangle.

d. Express the number of different routes from X to each of A, B, C, D, E and F in combinatorial notation.

Solution:

a. 4,4,4,4,4.

b. 0,1,2,3,4.

c. 1,4,6,4,1.

d. $\binom{5}{0}$, $\binom{5}{1}$, $\binom{5}{2}$, $\binom{5}{3}$, $\binom{5}{4}$, $\binom{5}{5}$.

Example 9.

Determine the sum of the first 100 counting numbers.

Solution:

Sum is $1 + 2 + 3 + 4 + \ldots + 96 + 97 + 98 + 99 + 100$

$= (1 + 100) + (2 + 99) + (3 + 98) + \ldots + (50 + 51)$

$= 101 + 101 + 101 + \ldots + 101$

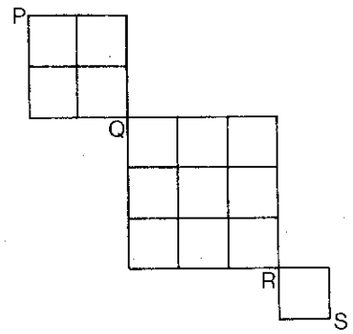
$= 50 \times 101$

$= 5\,050$

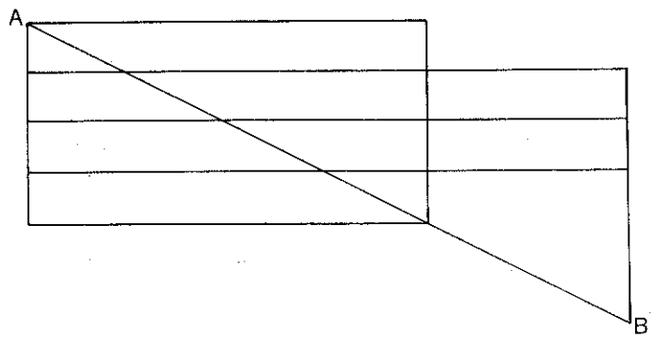
This method of counting is attributed to Carl Gauss (1777–1855).

Exercise 4C: Counting using Arrays

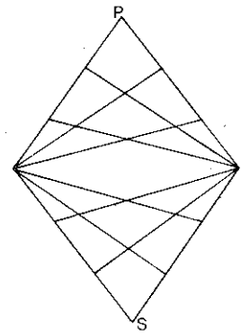
1. Counting paths that go to the right or downwards only, how many different paths are possible from P to S?



2. How many different routes from A to B are possible, if travel in a westerly or northerly direction is not permitted?



3. Counting paths that go downwards only, how many different paths are possible from P to S?



4. Simplify:

a. $1 + 2 + 3 + 4 + \ldots + 50$
b. $1 + 2 + 3 + \ldots + 1\,000$
c. $1 + 3 + 5 + 7 + 9 + \ldots + 51$
d. $1 + 3 + 5 + 7 + 9 + \ldots + 51 + 53$
e. $2 + 4 + 6 + 8 + \ldots + 1\,000$

Exercise 4D: Miscellaneous Counting Problems

1. In each of the following, state whether combinations or permutations are being counted:

a. The number of groups of 4 people from a group of 12 people.
b. The number of 13-card bridge hands from a 52-pack.
c. The number of ways of choosing any 6 questions from 10 in an examination paper.
d. The number of ways of selecting 6 Lotto numbers from the 40 available numbers.
e. The number of all 4-letter words from 5 letters.
f. The number of ways of placing guests around an oval dining table.

2. A class competition attracts 42 different but correct entries. The first three drawn are to receive prizes of \$10, \$5 and \$1, respectively. How many different allocations of prizes are possible?

3. There are 23 burrows in a rabbit warren but a rabbit trapper has only 4 traps. He sets all of his traps in the entrance of 4 different burrows. How many choices of 4 burrows has he?

4. If a fishing club has a membership of 12, how many different 3-man committees can be formed?

5. If there are 9 horses which start in a steeplechase, in how many different orders can they cross the finishing line, if two of the horses fail to complete the race? Assume no dead heats.

6. A squad of 25 state footballers has 6 recognised centre-line players. How many different centre-lines of 3 players can be selected and positioned from the 6 players available?

7. Evaluate:

a. $\frac{8!8!}{4!4!}$ b. $(4!)^{2!}$

8. A set of prime numbers A = {2, 3, 5, 7, 11, 13, 17}. How many distinct products can be formed by multiplying together any two different numbers from the set?

9. In how many ways is it possible to select three balls from a bag containing 12 differently-numbered balls if they are selected:

a. one at a time, replacing each ball before the next is drawn?
b. one at a time, without replacing any ball before the next is drawn?
c. all in one group?

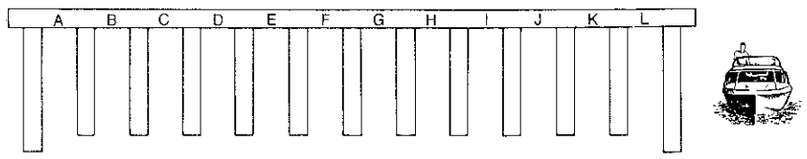
10. A set of 3-digit random numbers is to be made by choosing three of the balls:



Typical numbers are 363, 051, and 994. The balls are to be drawn from a box without looking. How many different random numbers are possible if:

a. they are drawn one at a time, replacing each ball before the next is drawn?
b. they are drawn one at a time, without replacing any ball before the next is drawn?

11. A new marina has 12 berths for boats. There are 15 applicants for the new berths.



a. How many different allocations of berths are possible? For example:

Berth	Name
A	F. JONES
B	P. SMITH

b. To make the allocations, the marina controller put all of the names in a "hat" and, as a first step, threw away the first 3 drawn out. How many different selections of 3 "throw-away" names are possible?

12. In a shipment of 12 cars, 2 are known to have defective brakes. How many different selections of 4 cars are possible if:

a. a dealer is unaware of the fault?
b. a dealer knows of the fault and wishes to choose 4 roadworthy cars?

13. There are 4 different streets that a motorist may take from his home to the freeway, and 3 different streets he may take from the end of the freeway to his place of work.

a. How many different routes can the motorist take from his home to work via the freeway?
b. How many routes are possible for the motorist to complete a return trip: that is, home to work via the freeway and back home via the freeway?

14. An examination paper consists of 10 questions of which candidates are required to attempt any 6. In how many ways can the 6 questions be selected?

15. A musical chord is made of 5 different notes played together. Consider a musical scale of 12 notes:

C, C#, D, E♭, E, F, F#, G, G#, A, B♭, B.

low pitch ↑ ↑ high pitch

a. How many different 5-note chords can be made from this scale?
b. How many of these 5-note chords include *G#*?
c. How many of these 5-note chords include *G#* as the:
i. lowest note? ii. highest note? iii. middle note?

16. In how many ways can six different brands of calculators and one microcomputer be arranged on a shelf at an electronics show, if the salesman has nine different brands of calculators and four different brands of microcomputers at his disposal and decides that a microcomputer is to be in the middle of the display?

17. The FUN-ABROAD travel agency wishes to promote 2 package tours from 5 offered by Cathay Airlines. In how many ways can the agency choose 2 tours?

18. A witness to a car accident can remember only the first and last characters of a number plate.

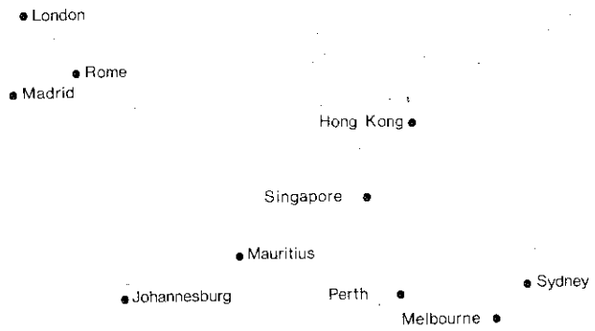
6					7

Each of the second and third characters is a letter; the remaining two characters are each digits. How many different car registrations fit this description?

19. How many distinct "words" can be formed by scrambling the letters of the word:

a. *MANGO*? b. *PINEAPPLE*?

20. An International Airline offers flights between any two of these cities:



a. Determine how many different single-stage flights must be scheduled.
b. Determine how many different single-stage flights must be scheduled to depart from Singapore.

21. How many different permutations of the nine letters of the word *CHEMISTRY* are there if:

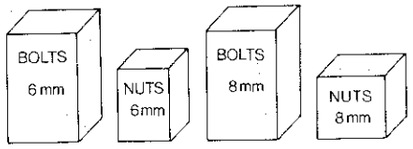
a. a vowel must be placed at the beginning and end of the word?
b. if the *C* must occupy the first position, the *I* the middle position, and the *Y* the end position?

22. A class consisting of 9 boys and 10 girls has to select a debating team of four members. They decide that it should contain 2 boys and 2 girls. In how many ways can they select:

a. the 2 boys? b. the 2 girls? c. the team?

23. If a girl selects 2 chocolates from a box containing 10 chocolates, and her friend selects 8 chocolates from another box also containing 10 chocolates, who has the greater choice? Assume that the order of selection does not matter.

24.



In how many different ways could a hardware assistant display the four packets on a shelf if:

a. no restrictions apply?
b. he wishes to put the bolts together and the nuts together?
c. he wishes to put the bolts and nuts of the same size together?

25. The price of an Asian Tour includes four stop-overs. A prospective customer is asked to make a selection of any four of eight cities nominated. In how many ways can a tourist plan an Asian tour with four stop-overs of his choice if:

a. the order of the stop-overs does not matter?
b. the order of the stop-overs does matter?

26. A tennis tournament attracts 4 professionals who become the top "seeds". There are 20 other entrants as well. The tournament organizers decide to allocate the players to 4 different groups. There is to be a "seed" in each group. In how many different ways may the players be allocated?

27. A large furniture manufacturing company has a policy of always having upon display one sample item of every product they manufacture.

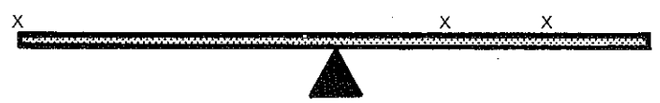
a. They manufacture 6 different styles of dining-room chairs and customers can have a choice of 3 types of wood and 10 types of fabric. How many chairs do they have on display?
b. If a customer wants a chair made out of one particular wood, how many choices of chair does the customer have?

28. How many different juries of twelve may be selected from 9 women and 7 men if:

a. the foreman of the jury must be a man?
b. exactly 6 women and 6 men are to serve?

29. An examination paper requires students to answer 4 of 6 questions from Section A and 6 of 10 questions from Section B. In how many different ways may a student select the 10 questions to be answered?

30. A see-saw is horizontal and balanced in the middle.



It is 6 metres long and there are three crosses chalked on it. One is at the left-hand end and the others are on the right-hand side, 1 metre and 2 metres from the middle. Three sand bags weighing 3 kg, 5 kg and 6 kg are to be placed on the see-saw, one on each cross.

a. How many arrangements are possible if the see-saw is to remain horizontal?
b. What is the maximum number of arrangements that could be tried before finding the one that left the see-saw horizontal?

31. Three people arrive in a city for a conference, and on arrival are assigned to hotels. The organizers have 4 hotels on their list. How many different allocations are possible if:

a. there are no restrictions?
b. a given pair must stay at the same hotel?
c. they all stay at different hotels?
d. one person insists on being accommodated at the hotel nearest the conference centre?

32. A teacher has 8 different animal-shaped erasers to award among 8 students as prizes. In how many ways may these items be awarded if:

a. no restrictions apply?
b. each student is to receive one prize only?

33. How many four-digit numbers less than 6 000 can be formed from the digits 0, 1, 2, 3, 4, 5, 7, 8 if:

a. repeated digits are permitted? b. repeated digits are not permitted?

34. A person playing poker was dealt five cards: 3 clubs and 2 spades. How many different poker hands fit this description?

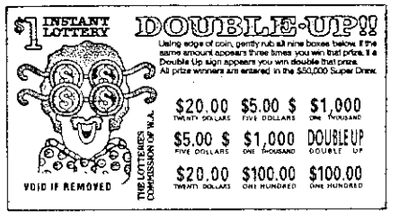
35. How many selections of 3 cards from a standard 52-card pack are of the same suit?

36. Eight greyhounds are listed for a race. In how many different ways could the first 3 places be filled:

a. if dead heats were disregarded?
b. if the favourite is certain (!) to win?

37. Seven people leave home to travel to the local football ground. They have a small car which can only seat 4 people, a motorbike with a pillion seat and a bicycle. In how many different ways may they divide themselves into the 3 groups to travel to the football ground?

38. A W.A. Lotteries Commission's **DOUBLE-UP** scratch-and-win ticket has 9 rectangles printed, with selections from *$2*, *$5*, *$20*, *$50*, *$100*, *$1 000*, *$10 000*, or *DOUBLE UP*.



← a losing ticket !

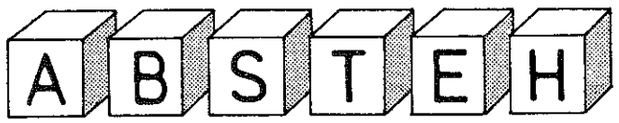
A winning ticket has the same amount appearing three times and the winner collects that amount. If DOUBLE UP is uncovered as well, the prize is doubled.

a. How many different ways are there of arranging the amounts (including DOUBLE UP) shown on the losing ticket?
b. To win $20 000 a person must find *$10 000* three times as well as DOUBLE UP. In how many different ways can these four be distributed on a ticket?

39. Nine identically-sized advertisements (6 cm × 6 cm) are to be placed in a section of a newspaper. The editor allows an area of 18 cm × 18 cm for the advertisements.

a. In how many different ways could the advertisements be displayed in the area allotted?
b. If the centre position must be a particular advertisement, in how many different ways could the remaining advertisements be displayed?

40. A child has a set of 6 alphabet blocks.



There is one letter per block but every block is different.

a. A child places 3 of these blocks in a row on the floor in front of him. How many different permutations are possible?
b. How many different permutations of all 6 blocks are possible?
c. Another child has similar blocks but 3 of these bear the letter B and three bear the letter L. How many different permutations of all 6 of these blocks are possible?

41. How many 5-card hands from a 52-card pack contain:

a. exactly 2 aces? b. 2 aces and a queen? c. 3 spades and a diamond?

42. A boy has 8 trucks, 11 cars and 5 tractors in his toy collection. He wishes to take 2 of each to school for part of a display on transport. How many different subsets of vehicles may he select?

43. There are four T-model Fords and three Packards in an old car line-up.

a. How many different permutations of these 7 old cars are there?
b. In how many ways can they be lined up if a Packard is to occupy first, fourth and last position?
c. In how many ways can they be lined up if a T-model Ford is to occupy first position and makes are to alternate for the remaining six positions?

44. Find the number of 13-card hands from a standard 52-card pack which contain 4 spades, 4 hearts, 3 clubs and 2 diamonds.

45. In a geometry test, each student may select 2 theorems to prove from five, and 3 constructions to complete from four. In how many different ways can a student select the five questions he is required to answer in this test?

46. How many teams of 3 can be chosen from a squad of 6 if:

a. both the oldest and youngest are excluded?
b. the youngest is excluded?

47. A couple celebrating their wedding anniversary dine out at a Chinese restaurant. They decide beforehand that they will each have a 4-course meal and that they will order something different from each other at each course and share the dishes. The menu offers 8 appetisers, 10 entrees, 14 main courses and 12 desserts.

a. In how many different ways may they select their meal?

Neither of them eats seafood and half of the dishes in each of the first 3 courses contain seafood.

b. In how many ways may they select their meal?

48. a. A $1 coin, a 50c coin and a 20c coin are tossed. How many different outcomes are possible?
b. If all three coins were $1 coins, how many different outcomes are now possible?

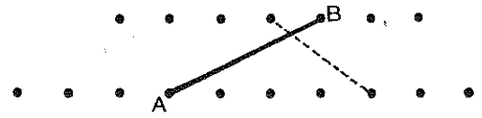
49. Find the integer n if:

a. ${}^{n}P_{n-4} = 210$ b. ${}^{n}P_{n-9} = 17\,160$

50. "Words" are to be formed from the letters P, A, N and S. How many words without repeated letters contain:

a. three letters? b. four letters? c. at least three letters?

51.



A line segment is drawn by joining a dot in the top row with a dot in the bottom row. How many different line segments cut the line segment AB but not at its end-points?

52. How many 3-letter words can be formed from the letters of the word *GRAPH* if the words must:

a. begin with an A and end with a P?
b. begin with an A or end with a P?

53. A push-button radio has 5 buttons but there are 7 radio stations to be assigned.

a. In how many different ways may the stations be assigned to buttons if as many stations as possible must be assigned?

In a remote area only 3 particular stations can be received.

b. In how many ways can these stations be assigned to the buttons if all stations must be assigned at least once and no button may be left unassigned?

54. In the 1984 Olympic Games there was a dead heat for the gold medal in the women's 100 m breast-stroke. There were 8 competitors in this race. In how many possible orders could the swimmers have finished?

55. Prove that for integers n and r, $n \geqslant r \geqslant 2$:

$$n(n-1)\binom{n-2}{r-2} = r(r-1)\binom{n}{r}$$

56. A football coach names a team of 21 players on Thursday night without allocating the positions they will play in the match on Saturday.

a. The coach decides to name the 18 players to take the field just prior to the match. How many different combinations are possible if positions are ignored?
b. Of the 21 players, ten are contract players and must play in each match. How many different combinations of 18 players are possible in this case?

57. Joe's Burger Bar sells hamburgers with cheese, lettuce, tomato, mayonnaise, pickle or sauce. How many different hamburgers can be prepared using any three of the extras?

58. At a meeting, after everyone had shaken hands once with everyone else, it was found that 780 handshakes had taken place. How many people were at the meeting?

59. A storekeeper has space on the shelf for 6 cans. She wishes to display 3 different varieties of soup, 2 different varieties of vegetables and 1 type of corn from a stock of 9 soup varieties, 5 types of vegetables and 3 types of canned corn. In how many different ways may she fill the display shelf?

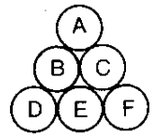
60. Find x if $(x!)^2 - 4x! - 12 = 0$.

61. A horse-owner has a stable with 12 stalls. She has 5 horses to stable. In how many ways can the horses be assigned to stalls?

62. Committees of four are to be chosen from a list of seven women. Each is allotted a letter A, B, C, D, E, F or G. How many different committees can be formed if:

a. there is no restriction? b. A must be included? c. A is included but G is not?
d. A and B are included? e. A and B are excluded?

63. Six discs marked A, B, C, D, E or F are arranged to form a triangle which may be viewed from any side.



a. How many different arrangements are possible?
b. How many different arrangements have A on a corner?

64. How many numbers greater than 10 can be formed from the digits 1, 2, 3, 7 if no digit can be used more than once in any particular number?

65. Four identical ping-pong balls are to be distributed into 4 different boxes. How many different distributions are possible?

66. How many different "words" consisting of 2 consonants and 2 vowels from the word *FACETIOUS*, are possible?

67. In a magic square the sum of the rows, column and diagonals is the same. For example:

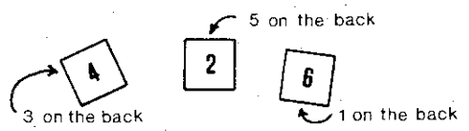
8	3	4
1	5	9
6	7	2

The magic number for this square is 15. How many different magic squares are possible using each of the digits 1, 2, 3, 4, 5, 6, 7, 8 and 9?

68. One digit from the list, 1, 2, 3, 4, 5, 6, was printed on each side of a card. No digit was used more than once. Three identical cards were used and, besides the digit, it was impossible to tell the front from the back.

a. Determine how many different-looking cards could have been made?

The cards were:



b. How many different three-digit numbers could be formed by arranging the cards in a row?

69. From a group of 8 boys and 5 girls, in how many ways can 4 boys and 2 girls be arranged:

a. in a row?
b. in a row if boys are to be together?
c. in a circle?

70. Six people are to be elected to the council. There are 10 nominations.

a. How many different councils may be formed?

The mayor and deputy mayor are members of any council of six.

b. How many different councils may be formed taking into account the positions of mayor and deputy mayor?

71. An 8-letter word in a crossword is known to contain three *T*s.

a. How many different ways could the three *T*s fit into the word?
b. If, apart from three *T*s, the word was known to contain two *E*s, how many different ways could the three *T*s and the two *E*s fit into the word?
c. Hence, how many different ways can the letters of the word *TOTTERED* be arranged?

72. The sum of five variables p, q, r, s and t may be written for example, as:

$$p + q + r + s + t \quad \text{or} \quad q + p + r + t + s$$

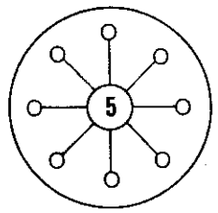
a. How many different ways are there of expressing the sum of these five variables without introducing extra $+$, $-$, $\times$, or $\div$ signs?
b. How many different ways are there of expressing the product of the variables a, b and c without using any multiplication signs?
c. How many different ways are there of re-writing the expressions:

i. $abc - xyz$; ii. $abc + xyz$;

without changing their implied value and without introducing extra $+$, $-$, $\times$ or $\div$ signs?

73. There are 9 different books of which 4 are fiction, 3 are science fiction and 2 are biographies. How many different selections of 5 books contain at least one of each type?

74.

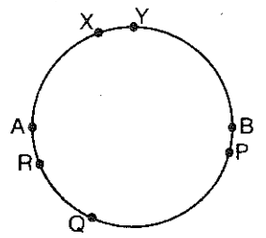


a. How many different arrangements of each of the digits 1, 2, 3, 4, 5, 6, 7, 8 and 9 can be made in the circles above such that the sum of any line is 15?
b. How many different circular permutations are possible?

75. A painter has 10 tins of paint, all of different colours. He decides to make new colours by mixing together a little paint from any two of his original tins. He decides to try all of the combinations. How many different colours will he have when he has finished?

76. In how many ways can a dozen different stamps be divided into 3 groups of four?

77.



AB is the diameter of a circle. Three of the points A, X, Y, B, P, Q and R are chosen to define a triangle.

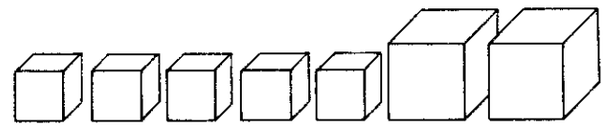
a. How many different triangles are possible?
b. How many different ways are there of naming each triangle?
c. How many different triangles' names are possible?

No two points, except A and B, determine a diameter.

c. How many different right-angled triangles are possible?

78. In how many different ways can a group of 10 people pair off into 5 groups?

79. There are five 2-centimetre cubes and two 3-centimetre cubes. They are all different from each other in colour.



A vertical pile of cubes exactly 10 centimetres high is to be built.

a. Taking into account only the colour of the cubes,
   i. how many different-looking piles containing no large cubes are possible?
   ii. how many different-looking piles containing exactly one large cube are possible?
   iii. how many different-looking piles are possible?
b. If all of the cubes were painted black, how many different-looking piles are possible taking into account only the size of the cubes?

80. An urn contains 10 balls, all identical except for colour. Seven are red and 3 are white. How many different-looking combinations of 4 balls are possible?

81. A newspaper Bingo card is a rectangular array of numbers selected from:

1, 2, 3, ..., 99, 100.

The rest of the squares are left blank.

GAME : 590

1	10	21			51	62			92
4			34	46		64	70	89	
	15	22			54	65	77		98

a. How many different selections of eighteen numbers are possible? Leave your answer in factorials.

An examination of the card shows that the first column always contains one, two or three numbers selected from: 1, 2, 3, 4, 5, 6, 7, 8, 9.

b. How many selections of three numbers are possible for the first column?
c. How many selections of one, two or three numbers are possible for the first column?

The second column from the left always contains one, two or three numbers selected from 10, 11, 12, ..., 19.

d. How many different selections of one, two or three numbers are possible for the second column?

The third column from the left always contains one, two or three numbers selected from 20, 21, 22, ..., 29. The remaining columns are filled in a similar manner.

e. How many different selections of eighteen numbers are possible for a Bingo card filled in this way? Leave your answer in factors.

82. There are 15 people in a group booking for a theatre.

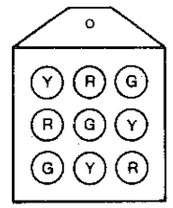
a. They are to be split into two groups: ten to go on one night and five to go on another. In how many different ways can the split be made if:
   i. the larger group is to go on the first night?
   ii. it does not matter which group goes on the first night?
b. In how many different ways can the people be split into three groups of 9, 4 and 2 people if it does not matter on which night they attend?

83. If ${}^{20}C_{r-10} = {}^{20}C_r$ evaluate:

a. r b. ${}^{r}C_{12}$ c. ${}^{18}C_r$

84. Ten people wish to travel in two cars but only five of them can drive. In how many different ways may they split into two groups of 5 if there must be at least one driver in each group?

85. Nine coloured marbles: 3 red, 3 green and 3 yellow, are to be arranged in a three by three array on a tray.

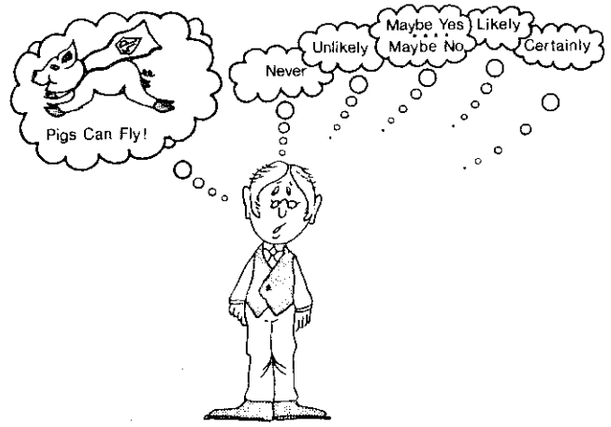


There is to be exactly one ball of each colour in each row and each column. Ignore the diagonals.

a. How many different patterns are possible with a red marble in the middle?
b. How many different patterns are possible?

Chapter 5:

Beginning Probability



What is the probability of each of the following events occurring? Use the terms: never, unlikely, maybe yes—maybe no, likely, certainly.

- You will get younger.
- You will get married.
- The next person you see will be a male.
- You win the lottery the next time you buy a ticket.
- You will live to the age of 60 years.
- You will see a U.F.O.
- You will meet the Prime Minister.
- You have a brain.
- You will pass your first driving test.

A job of a mathematician is to measure probability, to replace words such as "likely", "never", etc. with numbers.

EXPERIMENTAL PROBABILITY

One way to measure the probability of an event is to conduct an experiment. For example, if a coin is flipped high in the air, what is the probability that it will come down "heads"?

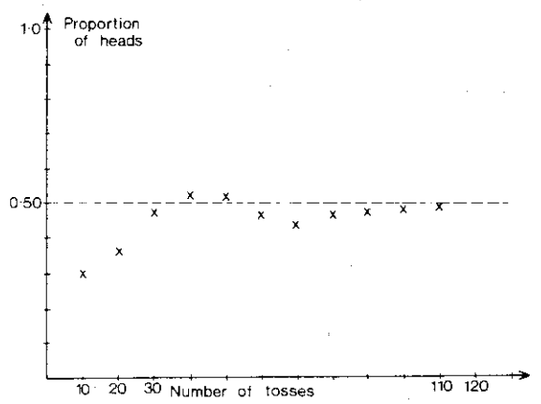
If your answer is 50%; 1 chance in 2; $\frac{1}{2}$; or 50–50 you are measuring this probability correctly. An experiment will confirm this.

A coin was tossed repeatedly and a record kept:

Number of tosses	Number of heads	Proportion of heads
10	3	0·300
20	7	0·350
30	14	0·477
40	21	0·525
50	26	0·520
60	28	0·467
70	31	0·443
80	37	0·463
90	43	0·478
100	48	0·480
110	54	0·491
120	?	?
130	?	?

Continue the table and the following graph by performing 20 more **trials**; that is, toss a coin 20 more times. A few words of caution before starting:

- the authors' coin should be used but make do with one of your own!
- whichever coin is used, make sure that the outcome on each trial is **random**. This means that, even though the possible outcomes (head or tail) are known, the experiment must be performed in such a way that the outcome is not predetermined by the way in which the coin is tossed. No cheating; a high spinning toss should do the trick!



Apparently, as the number of tosses increases, the proportion of heads gets closer to 0·50. This limiting value, 0·50, is taken as the probability of obtaining a head on a single toss.

The probability of a head on a single toss is approximately equal to the proportion of heads in a very large number of tosses.

Measuring Probability by an Experiment

The probability of a success in a single trial is given by:

$$\text{P(success in a single trial)} \approx \frac{\text{number of successes}}{\text{number of trials}}$$

provided that:

i. the outcome of each trial was random, and
ii. there was a large number of trials.

Example 1.

A sample of 100 "One Litre" bottles of a well known soft drink was chosen at random and the contents were measure to the nearest 10 millilitres. The results were:

Millilitres:	980	990	1 000	1 010
Number of bottles:	2	15	80	3

Using this information, estimate the probability that the next "One Litre" bottle of this soft drink will contain:

a. between 995 ml and 1 005 ml.

b. less than 995 ml.

Solution:

a. P(bottle contains between 995 ml and 1 005 ml) $\approx \frac{80}{100} = 0{\cdot}8$

b. P(bottle contains less than 995 ml) $\approx \frac{15 + 2}{100} = 0{\cdot}17$

Example 2.

A bag contains 6 discs, identical except for colour, some are red and some are white. Four discs are selected at random from the bag. All four discs are drawn simultaneously and all are replaced before the next draw. The colours are recorded:

1st selection:	2 red	2 white
2nd selection:	3 red	1 white
3rd selection:	4 red	0 white
4th selection:	2 red	2 white
5th selection:	4 red	0 white
6th selection:	2 red	2 white
7th selection:	3 red	1 white
8th selection:	2 red	2 white
9th selection:	2 red	2 white
10th selection:	3 red	1 white

What are the most likely numbers of red and white discs in the bag?

Solution:

For all 10 selections there were 27 red : 13 white
$\approx$ 2·08 : 1
$\approx$ 4 red : 2 white

Example 3.

In a game of "two-up", 2 coins were tossed simultaneously. An observer kept a record of the outcomes of 500 trials:

2 heads	121 times HH
1 head and 1 tail	253 times HT
2 tails	126 times TT
	500

If the result is 1 head and 1 tail, no bets are paid.

a. Use the recorded data to estimate the probability that on the next trial no bets are paid. P(HT) = $\frac{253}{500}$ = 0·506.

If the result is 2 heads, the "spinner" keeps tossing.

b. Use the recorded data to estimate the probability that, after the next trial, the "spinner" keeps tossing. P(HH) = $\frac{121}{500}$ = 0·242

Solution:

a. If no bets are paid the result must be 1 head and 1 tail.

$$\text{P(1 head and 1 tail)} \approx \frac{253}{500} = 0{\cdot}506$$

b. If the spinner is to keep tossing the result must be 2 heads.

$$\text{P(2 heads)} \approx \frac{121}{500} = 0{\cdot}242$$

Exercise 5A: Experimental Probability

1. The contents of 120 packets of seeds were counted and the results recorded as shown:

Number of Seeds:	9	10	11	12	13	14	15
Number of Packets:	2	5	31	48	29	4	1

Total = 120.

On this information, what is the probability that the next packet of the same type of seed would:

a. contain less than 12 seeds?
b. contain more than 12 seeds?
c. contain 11, 12 or 13 seeds?

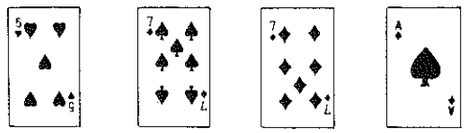
2. A gambler suspected a die of being loaded, tossed it 12 times, and obtained:

Outcome :	⚀	⚁	⚂	⚃	⚄	⚅
Frequency :	4	0	1	3	3	1

He decided that the die was loaded because "a fair die should give equal proportions of each number". Do you agree with his conclusion? Why?

3. A coin was pushed from a table and allowed to fall on to the floor. This was repeated 176 times and 89 heads were obtained. Therefore, the probability of a head on the next trial is approximately equal to 0·51. Comment on this result.

4. A man "cut" a standard pack of 52 cards four times. His results were:



He said to himself, "The proportion of clubs is 0 out of 4. Therefore I estimate that the probability of getting a club on my next cut is zero!" Do you agree? Why?

5. A pallet of 288 paving bricks was selected at random and the length of each brick was measured to the nearest millimetre. The recorded lengths were:

Length (mm):	227	228	229	230	231	232	233
Number of bricks:	3	12	31	209	18	9	6

If a paving brick from the same batch is selected at random, what is the probability that this brick will measure:

a. 230 mm? b. less than 229 mm? c. more than 230 mm?

6. A die is rolled 200 times. After each roll, the score appearing is recorded. The results were:

Number on die:	1	2	3	4	5	6
Frequency:	36	28	27	36	35	38

Using this information, estimate the probability that on the next roll of the same die:

a. a four will occur? b. an odd number will occur?
c. an even number will occur? d. a number less than 4 will appear?

7. Three dice are rolled together three hundred times. After each roll, the total score obtained is recorded. A summary of this experiment follows:

Total Score:	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Frequency:	1	3	10	13	17	28	31	40	40	31	34	24	17	7	3	1

Using this data, estimate the probability that on the next roll of the three dice the total score will be:

a. nine? b. less than seven?
c. more than twelve? d. greater than eight but less than fourteen?
e. even? f. odd?

8. A gambler conducts an experiment which involves drawing a card from a well-shuffled standard pack of cards. He conducts 440 such trials and records the results of his experiment as shown:

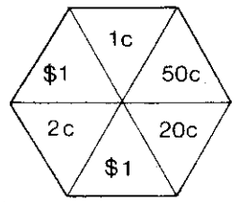
Suit:	Hearts	Diamonds	Spades	Clubs
Frequency:	101	118	109	112

After he had finished he noticed that 2 cards were missing.

a. If the gambler chooses one more card from the depleted pack estimate the probability that the card chosen will be:
i. a heart. ii. a diamond. iii. a spade.
iv. a club. v. a black suit. vi. a red suit.

b. If the 2 missing cards were from the same suit, guess the suit? Why?

9. A hexagonal spinner as shown in the diagram is spun 300 times. The results of the 300 spins is shown in the following table:



Result:	1c	2c	20c	50c	$1
Frequency:	53	47	51	52	97

If the spinner is spun again, use the results in the table to determine the approximate probability of the result being:

a. 2 cents. b. 50 cents. c. less than 50 cents. d. $1.

10. A large bin contains 2 000 newly manufactured pencils. A sample of 150 pencils is selected and the length of each pencil is measured to the nearest millimetre. The lengths of the pencils of this sample are recorded as:

Length (mm):	174	175	176	177	178	179	180
Number of pencils:	2	3	15	109	12	6	3

Determine, using the results in the table, the probability that a pencil selected from this bin and then measured will be:

a. 177 millimetres. b. less than 178 millimetres. c. more than 175 millimetres.

11. A social worker interested in "three children families" collected the following information:

Subject: 3-children families
Survey Conducted: 31/12/84–13/2/85
Number in Survey: 526

Composition of families:	3 boys	2 boys/1 girl	2 girls/1 boy	3 girls
Number in survey:	66	193	208	59

Determine, from the results presented in the survey, the probability that the next 3-child family surveyed consists of:

a. 3 boys. b. 3 girls. c. 2 girls and 1 boy. d. 2 boys and 1 girl.

12. Four coins were tossed simultaneously. The number of heads obtained was recorded. A total of 50 similar trials was conducted and the results were as shown:

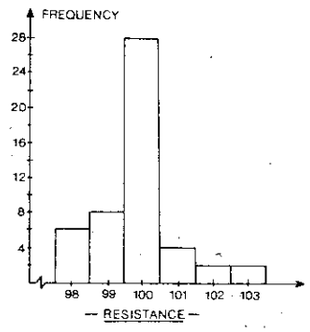
Number of heads	**Tally marks**	**Frequency**
0	||	2
1	𝍸 𝍸 ||	12
2	𝍸 𝍸 𝍸 ||||	19
3	𝍸 𝍸 ||||	14
4	|||	3

Using this information, approximate the probability that on the next toss of the same four coins the result will be:

a. 1 head only. b. exactly 3 heads. c. exactly 2 heads.
d. 4 tails. e. at most, 3 heads.

13. A sample of fifty 100-ohm resistors was selected at random and tested. Measurement was made to the nearest ohm. The results of the testing programme were given by means of the frequency graph:

50



Using the results shown by the graph, determine the probability that the next 100-ohm resistor selected and tested will have a resistance of:

a. more than 100·5 ohms. b. less than 99·5 ohms.

14. A drawing pin was tossed 100 times. The number of times it came to rest point-downwards was recorded:

Number of tosses:	10	20	30	40	50	60	70	80	90	100
Number of times:	4	6	12	?	15	?	22	25	?	26
Proportion of times:	0·40	0·30	?	0·375	?	0·30	?	?	0·28	?

a. Fill in the missing entries on the table and use your results to estimate the probability that your drawing pin will land point-downwards on a single random toss? 26/100
b. Is it equally likely to land point-downwards or point-upwards?

15. A self-service petrol station has eight bowsers. Six dispense super petrol and the others regular. The proprietor is concerned about congestion on Saturday mornings. He decides to record the number of cars at each bowser during the next Saturday.

Super-6 Regular-2

	SOUTH SIDE				WEST SIDE			
Bowser:	1	2	3*	4	5	6	7	8*
Number of cars:	60	62	20	50	21	12	4	1
	S	S	R	S	S	S	S	R

* Bowsers 3 and 8 dispense regular petrol and the others dispense super petrol.

Total = 230

Use proportions to estimate the probability that, on the following Saturday morning, a randomly selected car:

a. uses bowser 4. b. takes super-grade petrol.
c. does not take super-grade petrol. d. uses a WEST SIDE bowser.
e. uses a SOUTH SIDE bowser.

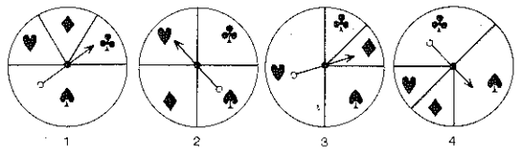
16. A sample of eighty "335 gram net" packets of a popular cake mix was chosen at random. The contents of each packet was weighed to the nearest gram. The recorded weights were:

Weight (grams):	332	333	334	335	336	337	338
Number of packets:	2	6	15	39	12	5	1

Using this information, estimate the probability that the next "335 gram net" packet of this brand of cake mix will weigh:

a. more than the advertised net weight.
b. less than 334 grams.
c. within $2\frac{1}{2}$ grams of the advertised net weight.

17. Each needle of the spinners was flicked 100 times.

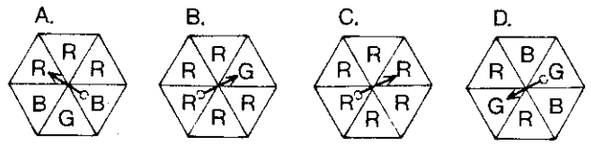


The results were:

	Spinner	Hearts	Diamonds	Spades	Clubs
A.	?	10	14	62	14
B.	?	42	17	29	12
C.	?	14	9	40	37
D.	?	30	23	25	22

Fill in the number of the spinner in the results table.

18.

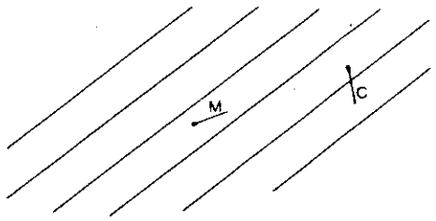


The arrow was spun 50 times for each hexagon. The arrow stopped on a colour and this was recorded:

I	Colour:	R	B	G
	Number of times the arrow stopped:	42	0	8
II	Colour:	R	B	G
	Number of times the arrow stopped:	24	16	10
III	Colour:	R	B	G
	Number of times the arrow stopped:	20	16	14
IV	Colour:	R	B	G
	Number of times the arrow stopped:	50	0	0

a. Match the results with the spinners.
b. Use the results for spinner C to measure the probability of a certain event.

19. Parallel lines are drawn on a very large sheet of paper.



The distance between the lines is exactly the length of a needle. The needle is tossed randomly onto the paper. It either crosses a line (C) or misses a line (M). The results for 300 tosses were 201 "crosses" and 99 "misses".

a. Estimate the probability that on the next toss the needle will cross a line.
b. This famous experiment was performed by the Count de Buffon (1707–1788). He calculated, by an analytic method, that the probability should be $\frac{2}{\pi}$. Does your estimate confirm or deny this?

20. An ordinary coin, a twisted coin, and a double-headed coin were tossed 70 times each.

Coin A:

HHHTT THTHH HTTTH THHTT HTHHH TTHTH HTHTH
THTHT THTTH HTTHH HHHTT HTTHT HHHHT THTTT

Coin B:

TTHTH TTTHH THTHH TTTTH TTHHH TTHTT THTHH
THTHT TTHTH HTHHH TTHHH THHTT TTTHT TTHTT

Coin C:

HHHHH HHHHH HHHHH HHHHH HHHHH HHHHH HHHHH
HHHHH HHHHH HHHHH HHHHH HHHHH HHHHH HHHHH

a. Either Coin A or coin B was the twisted one. Which one was it? Why?
b. Which of coins A, B or C was a double-headed coin?
c. For the double-headed coin:
   i. what is the proportion of heads?
   ii. what is the proportion of tails?

   Hence,

   iii. what is the probability of a certain event?
   iv. what is the probability of an impossible event?

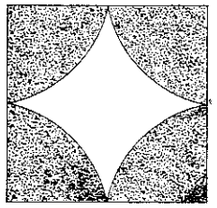
21. In a survey, a sample of 190 people was asked to answer "yes" or "no" to a particular question. Those who did not answer were recorded as "undecided".

	Yes	No	Undecided	Total
Men:	53	27	4	84
Women:	65	29	12	106
Total:	118	56	16	190

The same question was put to a randomly selected member of the population from which this sample was drawn. Use relative frequencies to estimate the probability that this person:

a. would answer yes.
b. would be a woman.
c. would not answer yes.
d. would be a man and would answer no.
e. would be a man or would not answer.

22. Flies land on a square measuring 2 units by 2 units.



The shaded portion is made up of 4 quadrants of a circle each with a radius of 1 unit. The record shows that 329 flies landed on the shaded portion and 97 landed on the unshaded portion.

a. What proportion of flies landed on the shaded portion?
b. What proportion of the total area is shaded?
c. By comparing the answers to part a. and part b., what may be said about the way the flies chose their landing points?
d. Write a formula for approximating π, which uses the number of flies landing on various parts of the square.

23. An industrial chemist adds a strengthening chemical agent to a concrete batch. Twenty concrete brick samples are made and each brick is subjected to the same pressure test. If the brick breaks a *Y* is recorded, if not, *N*.

Y N N N N N N N Y N Y N N N Y N N Y Y N

If another similar brick is made from the identical concrete batch and then subjected to the same pressure test, is it more likely to break or not break? Explain.

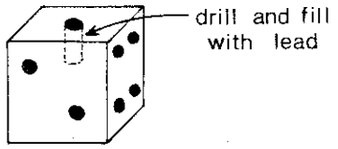
24. A life insurance company provided the following data relating to the mortality rate on a sample of 200 000 people.

Alive at Age 12	**Alive at Age 21**	**Alive at Age 40**
200 000	186 750	170 500

Estimate the probability that a child aged 12:

a. will die during the next 9 years.
b. will reach the age of 40.

25. If a dice player wishes to cheat, he "loads" the die. Assume he wishes to throw as many sixes as possible.

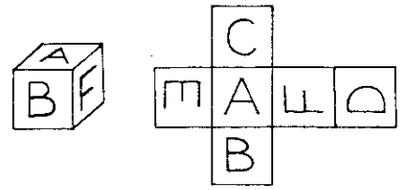


Health warning: People who load a die in this way risk severe injury from other players. A die was thrown 300 times with the following results:

33521	16361	22116	11226	54361	32531
13542	52151	24613	65261	56554	11441
11611	13121	63655	43212	61646	13414
45665	31154	45122	26424	24354	45552
43643	25523	44413	66563	11535	12366
11131	22423	55442	66263	34625	42533
13311	26145	61541	44413	24515	52224
24242	25654	31122	64536	66244	32345
55635	53235	66211	26664	16236	65461
36121	56465	15541	52631	42331	13422

a. Prepare a table which shows the proportion for each outcome.
b. Would you guess that this die was loaded to enhance the probability of a six? Why?

ASSIGNING PROBABILITIES



A child's alphabet block is a cube. She rolls it along the floor and the uppermost face shows a vowel (*A* or *E*) when it stops. What is the probability of this happening?

There are 2 ways of measuring the probability:

a. Experimentally.

Roll it many times and then the proportion of vowels is an estimate of the probability of a vowel on a single roll. For example,

$$\text{P(vowel on a single roll)} \approx \frac{131 \text{ vowels}}{400 \text{ rolls}} \approx \frac{1}{3}$$

b. Assigning.

Without doing an experiment we may assign the answer, $\frac{1}{3}$

There are 6 possible outcomes $\{A, B, C, D, E, F\}$, of which 2 are vowels, $\{A, E\}$. We say, the probability of a vowel in a single roll is $\frac{2}{6} = \frac{1}{3}$.

$$\text{That is, P(vowel)} = \frac{\text{number of vowels}}{\text{number of possible letters}}$$

Example 4.

A fair die is rolled once.

a. List the set, S, of all the possible outcomes.

b. List the set, E, of all of the outcomes which are prime numbers.

c. What is the probability of getting a prime number on a single random roll?

Solution:

a. $S = \{1, 2, 3, 4, 5, 6\}$

b. $E = \{2, 3, 5\}$

c. $P(E) = \frac{3}{6} = \frac{1}{2}$

Sample Space, S

The sample space is the set of all possible outcomes.

Sample Point

A sample point is one of these outcomes.

Event, E

An event is one or more sample points: i.e., a subset of the sample space.

n(E) represents the number of sample points in the event E.

Example 5.

There are 4 yellow and 3 white ping pong balls in a bag. They are all the same size. One ball is selected at random. What is the probability that it was white?

Solution:

$$S = \{Y_1, Y_2, Y_3, Y_4, W_1, W_2, W_3\} \qquad E = \{W_1, W_2, W_3\}$$

$$P(\text{white}) = P(E) = \frac{n(E)}{n(S)} = \frac{3}{7}$$

Each ping pong ball in the bag has an **equally likely** chance of being selected.

This method of assigning probabilities is valid only if all outcomes are equally likely. Therefore, S = {Y,W} was rejected because Y is more likely than W.

Assigned Probability

$$P(\text{event E}) = \frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}} = \frac{n(E)}{n(S)}$$

provided:

i. all outcomes are equally likely, and

ii. outcomes are random.

Example 6.

A pack consists of 10 cards, numbered 1, 2, 3, ..., 10. The cards are shuffled and one card is chosen.

a. List the sample space.

b. Determine the probability of the events A, B, C, and D where:

i. A is the event, "a 7".

ii. B is the event, "an even number".

iii. C is the event, "a number greater than 11".

iv. D is the event, "a number less than 11".

Solution:

a. $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

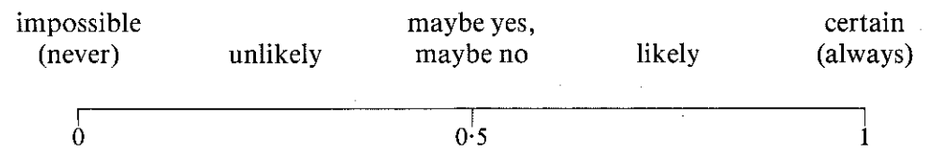
b. i. $P(A) = \frac{1}{10}$.

ii. $P(B) = \frac{5}{10} = \frac{1}{2}$.

iii. $P(C) = \frac{0}{10} = 0$.

iv. $P(D) = \frac{10}{10} = 1$.

The assigned probabilities can be placed on a scale ranging from 0 to 1.



That is, $0 \leqslant P(E) \leqslant 1$.

Exercise 5B: Assigning Probabilities

1. A person buys a ticket in a raffle. There are 1 000 tickets in the raffle, numbered from 1 to 1 000, inclusive. What is the probability that the person holding ticket number *496* wins?

2. There are 88 keys on a grand piano and 36 of these are black. A key is chosen at random. What is the probability that it is white?

3. A container holds 3 green, 6 red, 3 blue and 8 white cubes. One cube is selected at random. What is the probability that it is:

a. a green cube? b. a cube which is not red?
c. a green or a blue cube?

4. A single element is chosen at random from the 7-element set S.

$$S = \{\$, *, b, 17, 9, c, ;\}.$$

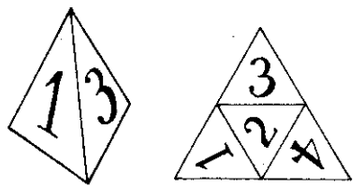
What is the probability that it is:

a. an asterisk? b. a letter? c. not a semi-colon?
d. a number? e. a number greater than 20?

5. One disc is selected from a container holding 3 white discs, 6 black discs and 11 red discs. Determine the probability that the disc selected is:

a. white. b. black. c. red.
d. not red. e. not white.

6. A four-sided die is in the shape of a regular tetrahedron.



After it has been tossed only 3 faces are visible. What is the probability that after a random toss:

a. a 4 will not be visible?
b. a 2 will be visible?
c. the visible faces add up to an even number?
d. the visible faces add up to 5 or less?

7. An octahedral (8-sided) die is rolled once. The sides are numbered 1, 2, 3, 4, 5, 6, 7, and 8.

a. What is the probability of getting:
i. a four? ii. a number less than 7? iii. an even number?
b. What assumptions did you make to obtain these probabilities?

8. An urn contains 3 red marbles, 4 blue marbles, 2 yellow marbles and 1 black marble. One marble is drawn from the urn. What is the probability that the marble drawn is:

a. yellow? b. red? c. black? d. blue?

9. A dodecahedral die, whose faces show the numbers 1, 2, 3, ..., 12, is rolled. What is the probability that the number 3 is rolled?

10. A letter is chosen at random from the alphabet. What is the probability that:

a. it is a Z? b. it is a consonant? c. it is an X, Y, or Z?

11. A motorist knows that there are traffic lights around the corner. There is a red light, an amber light and a green light. He decides that since there are three lights, his probability of encountering a red light is $\frac{1}{3}$. Comment on his reasoning.



12. Let $S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, the set of digits. What is the probability that a randomly chosen digit would be:

a. 5? b. prime? c. odd?
d. odd and prime? e. 5 or 8?

13. A coin is tossed once. What is the probability that the toss results in:

a. a head? b. a tail?

14. At a quiz evening, 75 tickets numbered 1 to 75 are sold in a raffle for a prize of "a meal for four" at a popular restaurant. What is the probability that:

a. the ticket number *48* is drawn as the winning ticket?
b. the winning ticket bears a number greater than 31 and less than 43?

15. A card is drawn from a standard pack of cards. What is the probability that the card drawn is:

a. a spade? b. a club? c. a five of clubs?
d. a jack of diamonds? e. a four? f. an ace?

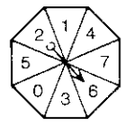
16. Each of the letters of the word *PROBABILITY* are printed on equal-sized pieces of paper and then placed in a container. One piece of paper is selected. What is the probability that the piece of paper selected contains the letter:

a. *P*? b. *O*? c. *L*? d. *T*? e. *I*? f. *B*?

17. From a standard pack of cards all the red cards are removed. From the remaining cards, a card is drawn. What is the probability that the card drawn is:

a. the seven of spades? b. the five of clubs? c. a six?
d. not the seven of spades? e. a black card?

18. A spinner is in the shape of a regular octagon.



It is spun once. What is the probability of obtaining a 6?

19. It is known that a child will celebrate her birthday in the next week. What is the probability the child's birthday is on:

a. Wednesday? b. Saturday? c. a weekday? d. the weekend?

20. A roulette wheel has 37 numbers: *0, 1, 2, 3, 4, 5, ..., 36*. The zero is green, half of the rest of the numbers are red and the others black. A ball is thrown onto the spinning wheel and comes to rest on one of the numbers. What is the probability that the number is:

a. red? b. green? c. odd? d. *23*? e. not zero?

21. On a table, there were the following coins: 1 cent, 2 cent, 5 cent, 10 cent, 20 cent, 50 cent, 1 dollar. A blind man was asked to choose one to keep. What is the probability that he chose the 1 dollar coin?

22. A motorist stops because of a flat tyre. What is the probability that the flat tyre is:

a. a front tyre? b. the rear passenger-side tyre?

23. When a regular 6-sided die is thrown, which is more likely: "getting a six" or "getting an odd number"?

24. A bag contains 75 red discs and 25 yellow discs. If selection is made without looking, is the selection more likely to be a red or a yellow disc? Why?

25. An ordinary cubical die is made out of plastic. It was left in a hot place and became distorted:



"The probability of obtaining a six in a single random toss is $\frac{1}{6}$."

Do you agree with this statement? Why?

26. There are 6 green and 7 red apples in a box. Tom chooses one at random and eats it.

a. What is the probability that it was red?

Tom's apple was red. Then Dick chose one at random and ate it.

b. What is the probability that it was red?

Dick's was red, too. Then, Harry chose one at random and ate it.

c. What is the probability that it was green?

27. Suppose one card is selected at random from a standard pack of 52 cards.

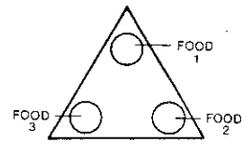
a. What colour card is most likely to be drawn?
b. Is the card selected more likely to be a Jack or a diamond?

28. A tray holds 5 pens. Two are black, 2 are blue and the other is red. The tray is accidentally knocked onto the floor and all of the pens fall out except for one. Assuming that the pens are equally likely to fall out, find the probability that:

a. the red pen is left in the tray. b. the red pen falls out.
c. a blue pen is left in the tray. d. a blue pen falls out.

29. Food companies wish to test the flavour, texture, colour, juiciness or taste of a new product before putting it on the market. The companies are often concerned to find out whether their new product is different from one of their old products. Tests to determine this are called sensory evaluation tests and there are 2 types:

A: **The triangle test.**



Three coded food samples are presented to a taster. Two are identical (old product). The other is the new product.

a. What is the probability that a taster, who cannot pick the odd sample, takes a lucky guess and gets it right?

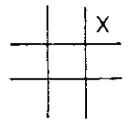
B: **The duo-trio test.**

The taster is given a sample of new product first, followed by 2 test samples: one of the old product and one of the new.

b. What is the probability that a taster, who guesses, correctly identifies the new product in the test sample?

30. A car is parked by the kerb and the parking inspector notes that the valve on the left-rear tyre is closest to "4 o'clock". The car parks in the same spot on the next day. What is the probability that the valve of the same wheel is closest to "4 o'clock"?

31. In a game of noughts and crosses, C goes first and uses a cross.

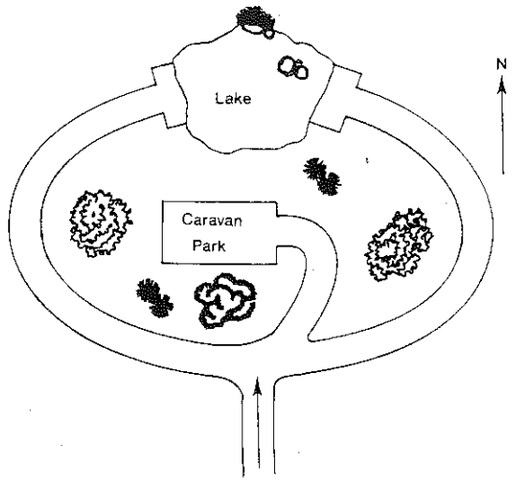


a. i. What is the probability that he would start in the top right-hand corner?
ii. What assumptions did you make to arrive at this probability?

Assume C used the top right-hand corner on his first turn.

b. What is the probability that the player who uses noughts chose a square next to the cross on the next turn?

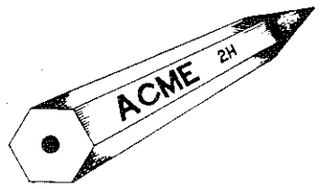
32.



A motorist travelling north on this road (see arrow) reaches a three-way intersection. He does not know which road to take and decides to choose his road at random.

a. Describe, briefly, how he may use a regular die to choose his road.
b. If he chose his road in a random fashion, what is the probability that:
   i. he takes a road to the lake?
   ii. he takes a road to the caravan park?

33.



The cross-section of a pencil is in the shape of a regular hexagon. On one of the faces, the brand is printed. The pencil is rolled across the table. What is the probability that the face with the brand comes to rest:

a. uppermost? b. on the table? c. uppermost or on the table?

34. Match some of the numbers: 0, $-0{\cdot}4$, $0{\cdot}94$, 1, $0{\cdot}004$, $0{\cdot}5$, $3{\cdot}1$ to the following probability statements:

A: the probability that it will get darker at night.
B: the probability that a man aged 51 will live until 55.
C: the probability that the next flat tyre on a car will be at the rear.
D: the probability that it will rain in the Sahara desert next year.
E: the probability that a peacock lays an egg.

35. There are 8 men and 7 women in Room A and 9 men and 11 women in Room B. A person is to be chosen at random. Which is the more likely event:

E_1, "a woman is chosen from Room A"?
E_2, "a woman is chosen from Room B after all the people from Room A have joined those in Room B"?

36. There are red, green and blue beads in a bag and altogether there are less than 15 but more than 10. The probability that a randomly selected bead is red is $\frac{1}{2}$; the probability that it is green is $\frac{1}{3}$; and the probability that it is blue is $\frac{1}{6}$. How many beads are in the bag?

37. An ace is removed from a standard deck of cards. One card is to be chosen at random from the remaining cards. By looking at the card which has already been removed, a mathematician decides that the probability of choosing a red card next is $\frac{26}{51}$ and the probability of choosing a club next is $\frac{13}{51}$. What card had been removed?

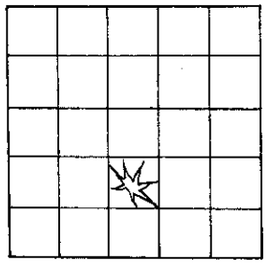
38. Each of the following probability statements is determined by either:

A: estimating the probability with the proportion of successes over a large number of trials; or
B: assigning probabilities by assuming equally likely random outcomes; or
C: "guesswork", that is, somebody's judgment.

Classify each of the following as A, B, or C. Some have 2 classifications.

a. There is a 4% chance of rain on Christmas Day.
b. The probability of drawing an ace is $\frac{1}{13}$.
c. The chance of a "meltdown" in a nuclear reactor is practically zero.
d. There is a 56% chance of recovery from a particular type of cancer.
e. The probability of throwing a six with a loaded die is 0.24.
f. The marathon record is almost certain to be broken at the next Olympic Games.

39. A cricket batsman deflected a ball through this window accidentally.



The ball smashed one of the panes of glass. All of the 25 panes are the same size. They are square and make a square window when put together.

a. What is the probability that the pane which was smashed was:
   i. in the centre? ii. in a corner?
   iii. in the top row? iv. not along the edge of the window?
b. After the window had been repaired, a child threw a stone at it and smashed a pane. What is the probability that the centre pane was smashed?

40. There are a dozen coffee mugs in the cupboard. Some are white and the others are blue. The probability that a randomly selected mug is blue is $\frac{1}{3}$. How many white mugs are there?

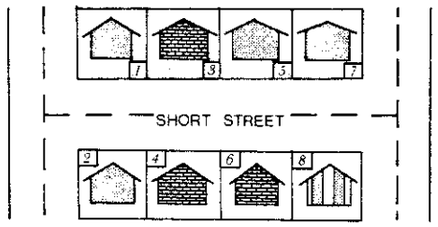
41. The Smith family consists of:

Bill Smith, the father, aged 42 years;
Mary Smith, the mother, aged 41 years;
Tom Smith, a son, aged 16 years;
Betty Smith, a daughter, aged 12 years;
Fred Smith, a son, aged 11 years; and
Colin Smith, a son, aged 6 years.

a. A single member of this family is chosen at random. What is the probability that:
   i. it is Tom?
   ii. she is a female?
   iii. he is a male?
   iv. it is not the father?
   v. she is the sister of someone in this family?
   vi. he is the brother of someone in this family?

b. Describe a method by which you might make sure that the choice of a single family member was random.

42.



There are 4 houses on each side of Short Street. A postman has a letter addressed to Short Street but with no name on it. The postman delivers the letter by guessing the house. What is the probability that he delivers it:

a. to the correct house?
b. to the wrong house?
c. to the wrong side of the street?
d. to a corner house on the wrong side of the street?

43. Jane, who is 2 years old, is an only-child. She has a mother called Betty who has a mother called Daisy. One of these three females is chosen at random. What is the probability that:

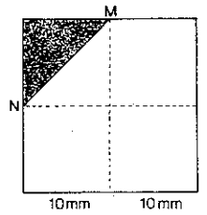
a. she is Jane?
b. she is a grandmother?
c. she is a mother?
d. she is someone's daughter?

44. Suppose each of the 8 letters of the words *TOP POOLS* are placed on separate cards and the pack is then shuffled. If a card is selected at random, find the probability that it was:

a. the letter *P*.
b. a vowel.
c. a letter in the word *POLICE*.
d. not a letter in the word *POLICE*.
e. a letter in the word *HOSPITAL*.

PROBABILITY BY COMPARING QUANTITIES

A point is selected at random in the square. M and N are mid-points of sides.



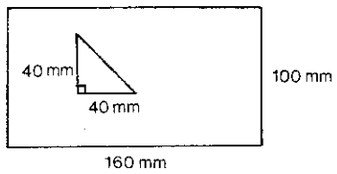
What is the probability that the point lies within the shaded region?

Intuitively, the probability is $\frac{1}{8}$. We may assign probabilities by comparing like measureable quantities: that is, a ratio may be formed between two areas, two volumes, two lengths, etc. This ratio is taken as the probability.

$$\text{P(shaded region)} = \frac{\text{Area of shaded portion}}{\text{total area}} = \frac{50\,\text{mm}^2}{400\,\text{mm}^2} = \frac{1}{8}$$

Exercise 5C: Probability by Comparing Quantities

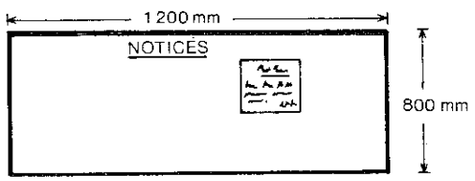
1. A triangle of base 40 mm and of height 40 mm is somewhere within a rectangle whose dimensions are 160 mm by 100 mm.



A point is chosen at random from within the rectangle. Find the probability that it also lies within the triangle.

2. A rectangular table measures 1·8 metres by 1 metre. A fly lands on the table. A man hits blindly at the fly using a fly swat which is also rectangular. The fly swat measures 80 mm × 100 mm. What is the probability that he hits the fly?

3.

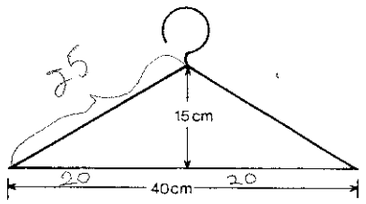


A rectangular notice (180 mm × 240 mm) is affixed to a rectangular notice board (1 200 mm × 800 mm). A spare drawing pin is stuck into the notice board in a randomly selected position. What is the probability that the pin is stuck into:

a. the notice?
b. the board, missing the notice?

4. A plane left the airport on an unknown bearing. What is the probability that it was in the minor sector between a bearing of 012° (i.e., N 12°E) and 120° (i.e. S 60°E)?

5.



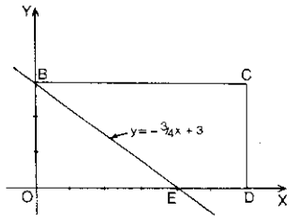
A wire coat hanger is in the shape of an isosceles triangle with a hook on top. A piece of cotton is tied onto the hanger (not the hook). What is the probability that it is within 10 cm of the base of the hook?

6. A man agrees to arrive for an appointment between 6.00 p.m. and 6.15 p.m. The man honours his agreement but chooses his arrival time randomly. What is the probability that he arrives after 6.10 p.m.?

7. A dart board is circular with a radius of 230 mm. Four playing cards, each measuring 90 mm × 55 mm, are stuck to it. A blindfolded person hits the board with one dart. Find the probability that the dart did not pierce a card.

8. A small fish swims in a "rectangular" aquarium. The aquarium is 1 metre long and 0·4 metres wide. The water is 0·5 metres deep. Assume that the fish swims about randomly. What is the probability that, at any instant, it is:
   a. within 5 centimetres of the surface?
   b. nearer the front left-hand bottom corner than any other bottom corner?

9.

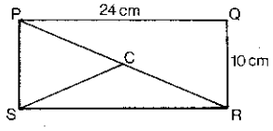


OBCD is a rectangle in a Cartesian plane. The co-ordinates of O, B, C, and D are (0,0), (0,3), (6,3) and (6,0), respectively. A line $y = -\frac{3}{4}x + 3$ divides the rectangle into two regions. A point is selected at random from within the rectangle.
   a. What is the probability that it is in:
      i. the smaller region? ii. the larger region?
   b. A point is selected at random in the line segment BE. What is the probability that it is closer to B than E?

10. A point is selected at random inside a circle of radius 4 cm. Determine the probability that the point is closer to the centre than it is to the circumference.

11. An X-ray detected a minute flaw, the size of a pin point, in a manufactured ball bearing. Determine the probability that the flaw was situated closer to the centre of the ball bearing than the surface.

12.



PQRS is a rectangle. C is the centre. A point is selected at random in one of the line segments PQ, QR, RS, SP, PR or CS. What is the probability that it is closer to S than R?

SAMPLE SPACES—LISTS AND DIAGRAMS

Assigning probabilities is sometimes difficult because a set of random outcomes may be large and hard to imagine. To assist us in determining the outcomes we may use a tree diagram, a Venn diagram, a grid or a list.

Example 7.

A three-digit number is to be formed using the digits, 1, 8 and 9. No digit may be used more than once. Find the probability the number 918 was formed.

Solution:

Listing:

{189, 198, 819, 891, 918, 981} i.e., n(S) = 6

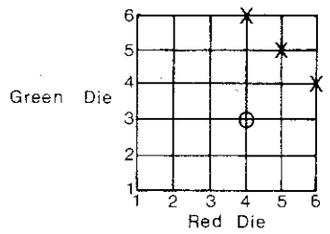
Hence, $P(918) = \frac{1}{6}$

Example 8.

A red and a green die are rolled together. Determine the probability of obtaining:
a. a 4 on the red die and a 3 on the green die.
b. a total of 10.

Solution:

Grid:



a. The outcome, "4 on the red die and 3 on green die", is one of the 36 possible outcomes. That is, n(S) = 36.

Hence, P(4 red, 3 green) = $\frac{1}{36}$

b. The outcome, "a total of 10", can be formed in 3 possible ways: (6 red, 4 green), (5 red, 5 green), (4 red, 6 green).

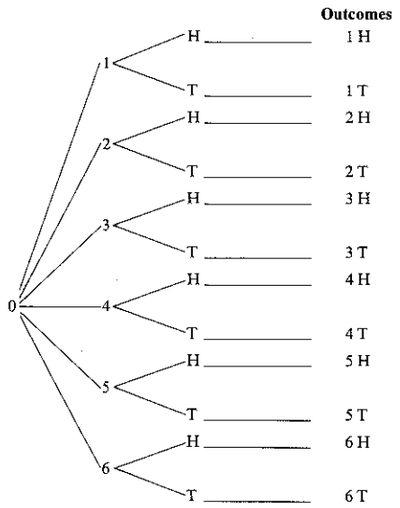
Hence, P(score 10) = $\frac{3}{36} = \frac{1}{12}$

Example 9.

A coin is tossed and a die is rolled at the same time. Determine the probability of obtaining a 4 on the die and a head on the coin.

Solution:

Tree diagram:



i.e., $n(S) = 12$.

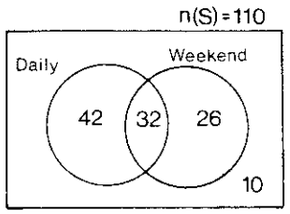
Hence, $P(4,H) = \frac{1}{12}$.

Example 10.

A survey was carried out in a new development area to gain data on home-delivered newspapers. One hundred and ten homes were selected at random and the occupants asked whether they had the daily paper or the weekend paper home delivered. 74 received the daily paper; 58 received the weekend paper; 26 received only the weekend paper and 10 received no paper at all. Determine the probability that the last home visited in this survey received both the daily and weekend papers.

Solution:

Venn diagram:



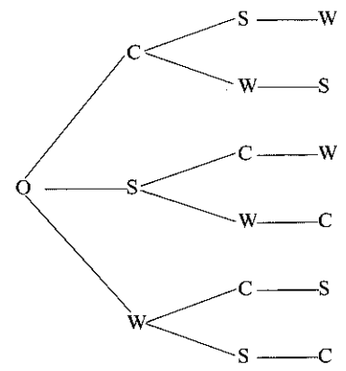
P(both types of newspaper) $= \frac{32}{110} = \frac{16}{55}$

Example 11.

A car dealer has an end-of-model sell out. He reduces the price on 3 vehicles: a red coupe, a white sedan and a white station wagon. Assume that the cars are equally attractive to a purchaser and all sell. Determine the probability that:

a. the coupe sells first and the sedan next.
b. the station wagon sells last.
c. a white vehicle is the last to sell.

Solution:



a. P(coupe first, sedan next) $= \frac{1}{6}$

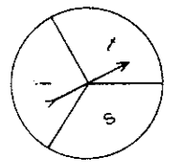
b. P(wagon last) $= \frac{2}{6} = \frac{1}{3}$

c. P(white vehicle last) $= \frac{4}{6} = \frac{2}{3}$

Exercise 5D: Sample Spaces—Lists and Diagrams

1. The letters A, T, and N are placed on separate cards. A kindergarten student is asked to arrange the cards in any order so as to make a three-letter "word". Determine the probability that the child arranges the cards:
   a. in the order A T N.
   b. with the vowel in the middle.
   c. in such a manner that the word formed is the name of an insect.

2. The spinner is spun twice to spell out a 2-letter word.



   a. List all of the possible 2-letter words.
   b. What is the probability that the word is in the Oxford English Dictionary?

3. A strand of electrical cable contains three wires: one green (earth), one black (neutral), and one red (active). It is wired to a power point. Assuming the person has no knowledge of electrical wiring, what is the probability that the wiring is correct?

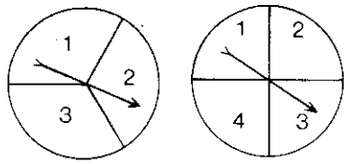
4. A target consists of three concentric rings, a bullseye, an inner and an outer. An archer is just about to shoot at the target. He decides that since there are three regions, he has 1 chance in 3 of hitting the bullseye. Do you agree with his decision? Why?

5. In a National Lottery there are many tickets. A person who has done a little probability theory reasons, "There are two types of tickets, winners and losers. Therefore, my chance of winning a prize is $\frac{1}{2}$". Comment on this reasoning.

6. A coin is to be flipped 3 times. Use a tree diagram to determine all of the possible sequences of heads and tails.

a. Determine the chance of obtaining:
   i. three heads in a row.
   ii. a tail, then a head, then a tail.
   iii. three tails in a row.
b. Use the tree diagram to assist you in determining the probability of:
   i. three boys in a row in a 3-child family.
   ii. two boys and then one girl in a 3-child family.
   iii. two boys and one girl in a 3-child family.

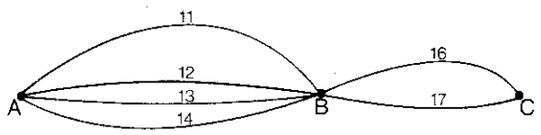
7. The spinners are spun simultaneously.



Calculate the probability that:

a. the arrows stop on the same number.
b. the arrows stop on two numbers which add up to 5 or more.
c. each arrow stops on an odd number.
d. at least one arrow stops on a 2.

8. Four highways connect town A to town B and two highways connect town B to town C.



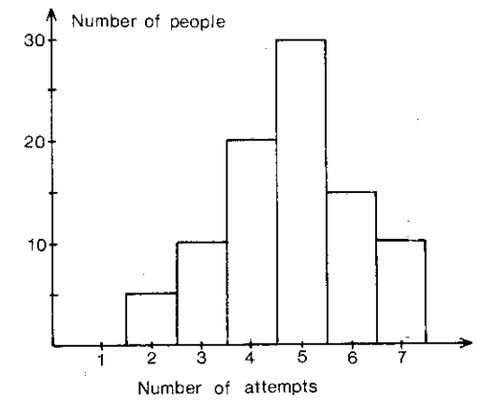
A motorist chooses his highways at random. Determine the probability that the motorist, wishing to travel from town A to town C, will:

a. use highway 11 then highway 16.
b. use odd-numbered highways.
c. use an even-numbered highway to B and then an odd-numbered highway to C.

9. A teacher instructs a student to write the three fractions $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{3}$ in any order across the page.

a. List all the possible orders which could be written from left to right across a page.
b. Determine the probability that the student wrote them in the order:
   i. $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{3}$
   ii. $\frac{3}{4}$, $\frac{1}{3}$, $\frac{1}{2}$
c. Determine the probability that the first two fractions have a sum greater than one.

10. An observer above the Mirror Maze at the Royal Show recorded the number of attempts taken by 90 people to find the correct way through the maze. His findings are recorded in the histogram below.



Use this data to answer the following questions:

a. What is the probability that a person from this sample will need exactly 5 attempts?
b. What is the probability that a person from this sample will need exactly 3 or 7 attempts?
c. What is the probability that a person from this sample will need at least 4 attempts?
d. What is the probability that a person from this sample will succeed on the first attempt?
e. The observer stated that from this sample the probability of success on or before the fourth attempt was equal to 0·5. Was this a true assessment for the above sample?

11. There are three parking bays A, B and C in front of a shop. They are equally popular with customers and two bays are already filled by a Volkswagen and a Ford.

a. Use a diagram to describe all of the possible parking arrangements.
b. Determine the probability that:
   i. the Volkswagen is in bay A.
   ii. the Volkswagen is in either bay A or bay B.
   iii. the Ford is in bay A and the Volkswagen is in bay B.
   iv. the Ford is not in bay B.
   v. the two cars are side by side.

12. A group of 50 girls from a school entered an eisteddfod. Twenty-eight played the piano, 25 sang and 15 neither sang nor played the piano. If a girl is chosen at random from the group, find the probability that she:

a. played the piano.
b. played the piano and sang.
c. played the piano or sang, but not both.

13. John, Peter and Susan queue to buy their lunch at the canteen. Use an appropriate diagram to assist you to determine the probability that:

a. John is at the front of the queue.
b. Susan is between the two boys.
c. Peter is next to Susan.

14. The hot plates of an electric stove can be set on "Off", "Hi" or "Low". A small child plays with the controls of the two hot plates on the front of a stove and leaves them randomly set.

a. List all of the possible combinations for these two controls.
b. Determine the probability that:
   i. both plates are left on "Hi".
   ii. at least one plate is left on.
   iii. the plates are set on something different.

15. Mr and Mrs Brown agree, initially, on the three possible Christian names of Peter, James and David for their newly-born son. On registering the birth of their son, only two different Christian names were given.

a. Determine the probability that their son's Christian names were registered as:
   i. Peter James.
   ii. David Peter.
b. Determine the probability that the registered name of their son:
   i. included the name James.
   ii. did not include the name James.

16. An examination paper consists of two sections: Section A, questions 1 and 2; and Section B, questions 3 and 4. Candidates are instructed to attempt two questions only, one from each section.

a. List all the possible orders a candidate may take in answering the questions and complying with the instructions.
b. Determine the probability that a student attempted:
   i. questions 1 and 3.
   ii. questions 2 and 3.
   iii. question 2 first then question 4.

17. Father, mother and son each have a car. The cars are red, white and green, but not necessarily in respective order. Complete the following table which shows all the possible permutations.

Father	Mother	Son
Red	White	Green
Red	Green	White
Green	Red	White
Green	?	?
White	?	?
White	?	?

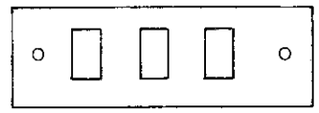
What is the probability that:

a. Mother had the red car?
b. Father's was red, Mother's white and the son's green?
c. the son's car was not green?
d. the son's car was not green and the mother's car was not white?

18. Each of the digits 3, 4, 5, and 6 is written on separate cards. The cards are shuffled and two are selected at random.

a. List the sample space.
b. Determine the probability that the sum of the two cards is odd.

19. A bank of three switches is installed on a wall in the kitchen.



One switch operates the ceiling fan, another, the exhaust fan and a third, a dome light situated above the sink. A person working in the kitchen asks for all the switches to be turned on. Assuming that the switches are turned on at random and one at a time, determine the probability that the switches are turned on in the order of:

a. ceiling fan, exhaust fan and then the light.
b. the light and then a fan.
c. a fan, a light and then a fan.

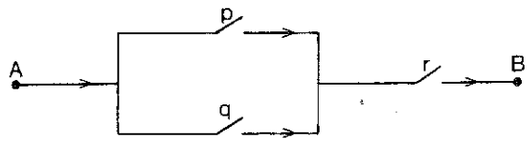
20. The last three digits of a bank account is known to contain a *4*, a *7*, and a *5*.

a. Determine the probability that the order of the three last digits is *4 5 7*.
b. Determine the probability that the last digit is:
   i. a *5*.
   ii. not a *7*.

21. A farmer has 3 chickens, Henny, Penny and Jenny. They are each about to lay an egg but there is only room for one chicken at a time on the nest. Assume that the order in which they lay eggs is random.

a. List all of the possible orders.
b. What is the probability that:
   i. Henny lays an egg first?
   ii. Penny is last?
   iii. Jenny is not last?

22. Part of an electrical circuit AB contains 3 switches, p, q, and r.



Let O represent "switch open" and C represent "switch closed".

(O, C, O) represents switch p open, switch q closed, and switch r open.

a. Using ordered triples, list all the possible states of the circuit.

Each switch is randomly open 50% of the time and operates independently of the other switches.

b. Find the probability that, at any instant:
   i. all switches are open.
   ii. an electric current could flow from A to B.
   iii. exactly 2 switches are open.

23. Of the 800 students enrolled at a Senior College, 532 are enrolled in Mathematics, 468 are enrolled in Chemistry and 380 are enrolled in both subjects. Use a Venn diagram to determine the number of students who are not enrolled in either course. What is the probability that a student chosen at random from this College population:

a. does Mathematics only? b. does both Chemistry and Mathematics?
c. does Chemistry only? d. does not study either Mathematics or Chemistry?

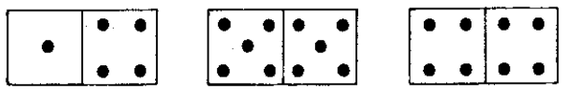
24. Out of a class of 30 students who are proficient at water skiing or ice skating, 16 can ski and 23 can skate. One of these students is selected at random. What is the probability that the student can do both?

25. From the base camp, a cub is given instructions which will get him to the barbecue spot where they are to rendezvous.

"... 1 kilometre North, 1 kilometre East, 1 kilometre South ... the order in which you follow the directions, is your choice ...". Determine the probability that the cub will:

a. follow the directions as written.
b. return to the camp before reaching the barbecue spot.
c. visit the barbecue spot twice.

26. These three dominoes are placed end to end on the table.



a. List all of the possible outcomes.
b. Determine the probability that:
   i. all of the 4s are together. ii. a 5 is not next to a 4.
   iii. the numbers are in ascending order from left to right.

27. The four aces from a pack of cards are face down on a table. One is turned face up and then a second one.

a. Use a tree diagram to describe all of the possible sequences. How many are there?
b. Hence, find the probability that:
   i. the ace of diamonds is the second card turned over.
   ii. the ace of clubs is not revealed.
   iii. the two aces are the same colour.

28. A pizza parlour offers three sizes of pizzas, small (S), medium (M), family (F), with the choice of pepperoni (P), ham (H) and capsicum (C). List the sample space and hence, give the probability that the next customer selects:

a. a family-size pizza with capsicum.
b. a pepperoni pizza other than a family-size.

29. The three cards, 3 of hearts, 4 of diamonds and 5 of spades are shuffled and then placed face down on a table. Each of the cards are then turned over one at a time. Determine the probability that:

a. the cards are turned over in ascending numerical order.
b. the first two cards are red.
c. the first two cards sum to an even number.
d. the middle card is a 3.
e. the last card is numerically larger than the middle card.

30. There are three children in the Kelly family. Find the probability that:

a. they are all girls. b. only one is a boy. c. the eldest is a girl.
d. the eldest is a boy and the youngest is a boy. e. at least two are boys.

31. A bag contains two white balls and three red balls. A ball is drawn and placed on a table. If a second ball is then selected, find the probability that the following events occurred:

a. A white ball was followed by a red ball.
b. Two white balls were chosen.
c. One white ball was chosen.
d. At least one white ball was chosen.
e. Three red balls were chosen.

32. Two sisters both promise to bake their father a cake for his birthday. They each can bake chocolate cakes, fruit cakes or madeira cakes. Independently and at random, the sisters choose a type of cake to bake. What is the probability that:

a. the sisters present the same type of cake?
b. the elder sister bakes a madeira and the younger sister bakes a chocolate cake?
c. neither sister bakes a fruit cake?
d. the father is presented with a chocolate cake and a madeira cake?

33. A survey conducted at a local supermarket revealed that of the 200 customers interviewed: 142 used product A, 145 used product B, 126 used product C, 101 used products A and C, 106 used products B and C, 113 used products A and B, and 96 used all three. Eleven customers indicated that they did not use A or B or C. Determine the probability that a customer selected at random from the 200 interviewed:

a. does not use any of the three products.
b. uses product A only.
c. uses product C only.
d. uses A and B only.
e. uses one product only.

34. Each face of a regular tetrahedral die is an equilateral triangle with faces numbered 1, 2, 3, 4. This is tossed with a regular cubical die and the two faces which land on the table are noted. What is the probability that the tetrahedral face has more spots in contact with the table than the cubical face?

35. Of a sample of 148 teenagers who enjoy listening to music, 84 listened to cassettes, 84 listened to records and 80 listened to live music. Sixteen teenagers listened to cassettes only, 20 listened to records only and 40 listened to both cassettes and records. Eight from the sample listened to all three sources of music. Determine the probability that a teenager selected at random from this sample listened to:

a. cassettes only.
b. live music only.
c. one source of music only.
d. cassettes and live music, but not records.

36. Two fair dice, one red and one green, are rolled. Find the probability that:

a. both dice show the same number.
b. both dice show an odd number.
c. the green die shows a number greater than 4.
d. the green die shows a number greater than 4 and the red die shows an even number.
e. the number showing on the green die is greater than the number showing on the red die.
f. one die shows a number less than 3 and the other die shows an odd number.

37. There are 3 switches in a line and each is left either "up" or "down". Let U represent "up", let D represent "down", and let (U,D,D) represent the state where the first switch is up and the others are down.

a. List all of the possible outcomes.

If each switch is set at random what is the probability that:

b. all switches are "up"?
c. exactly two switches are "down"?

38. A menu consists of an entree (prawn cocktail, seafood cocktail or squid), a main course (beef or pork), and dessert (chocolate mousse or fresh fruit salad). A person selects a three-course meal at random. What is the probability that this person:

a. had prawn cocktail, beef and fresh fruit salad?
b. had a meal which included pork?
c. ended the meal with chocolate mousse?

39. A bag is known to contain 2 blue and 2 yellow discs. Two discs are selected at random.

a. List the possible sample space.
b. Determine the probability that the two discs drawn are the same colour.

40. In the Indian game of Tong, two players simultaneously show their right hands with either one, two or three fingers extended. Assuming that each player is equally likely to extend one, two or three fingers, what is the probability that:

a. the total number of fingers extended is odd?
b. both players extend the same number of fingers?
c. both players extend a different number of fingers?

To win at this game, players take alternate turns nominating one of the outcomes, "odd total" or "even total".

d. What call would be the best to make? Why?

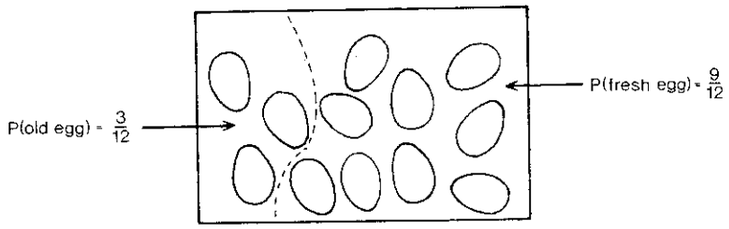
Chapter 6:

Applying Probability: Expectation and Odds

COMPLEMENTARY EVENTS

A method used to test the freshness of eggs is to place them in a sink full of water. Fresh eggs remain on the bottom of the sink and old eggs float.

Twelve eggs are put into a sink and three float. Suppose the water leaks from the sink and an egg is then chosen at random.



Clearly, $P(\text{old egg}) + P(\text{fresh egg}) = 1$.

There are only two types of eggs in the sink: old eggs and fresh eggs.

The event, "choosing a fresh egg (F)" and the event, "choosing an old egg ($\overline{F}$)" are complementary events.

$n(F) = 9$; $n(\overline{F}) = 3$ and $n(S) = 12$.

We note: $n(F) + n(\overline{F}) = n(S)$

Hence with $n(S) \neq 0$ we have: $\dfrac{n(F)}{n(S)} + \dfrac{n(\overline{F})}{n(S)} = \dfrac{n(S)}{n(S)}$

i.e., $P(F) + P(\overline{F}) = 1$

Complementary Events

If E and $\overline{E}$ are two mutually exclusive events which together fill a sample space, then:

$$P(E) + P(\overline{E}) = 1$$

i.e., $P(\overline{E}) = 1 - P(E)$

Example 1.

A new drug designed to reduce blood pressure was administered to 90 people. After the treatment period, the blood pressure of each person was recorded. For the purpose of comparison, another 90 people were checked for blood pressure even though they received no treatment.

BLOOD PRESSURE

	Significant increase	No change	Significant decrease	Total
Drug:	6	28	56	90
No treatment:	32	29	29	90
Total:	38	57	85	180

If one person was selected at random from the 180, what is the probability that this person:

a. showed a significant change in blood pressure?

b. was not one of the drug group which showed a significant decrease in blood pressure?

Solution:

a. $P(\text{significant change}) = 1 - P(\text{no change})$

$= 1 - \dfrac{57}{180}$

$= \dfrac{123}{180}$

$= \dfrac{41}{60}$

b. P(not drug group, significant decrease)

$= 1 - P(\text{drug group and significant decrease})$

$= 1 - \dfrac{56}{180}$

$= \dfrac{124}{180}$

$= \dfrac{31}{45}$

Exercise 6A: Complementary Events

1. A box contains 12 biros, two of which are known not to work. If one is selected at random, determine the probability that:

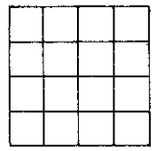
   a. it will work.

   b. it will not work.

2. A bag of 24 oranges is known to contain 3 which are rotten. Determine the probability of selecting:

a. a rotten orange. b. an orange which is not rotten.

3.



A person is asked to place a tick in one of the squares. Determine the probability that the person:

a. ticks a corner square. b. does not tick a corner square.

4. A student is asked to select an integer greater than 3 but less than 40. Determine the probability that the integer selected is not divisible by 3.

5. Each of the letters of the word *EXPERIMENTAL* is written on separate cards and the cards shuffled. One card is selected at random. Determine the probability that the card drawn contains a letter which:

a. is a consonant. b. is a vowel.
c. is not a vowel. d. is not a consonant.

6. One card is drawn from a standard pack of 52 cards. Determine the probability that the card drawn was:

a. not a red card. b. not a king. c. not the ten of clubs.

7. If two dice are rolled, determine the probability that:

a. the combined score is not greater than 11.
b. neither die shows a 5.

8. A bag contains 5 blue marbles, 3 white marbles and 7 red marbles. One marble is selected at random from the bag. Determine the probability that the marble drawn was:

a. white. b. red. c. blue.
d. not blue. e. not red. f. not white.

9. Two one-dollar coins are tossed. Determine the probability that:

a. no tails occur. b. at least one tail occurs.

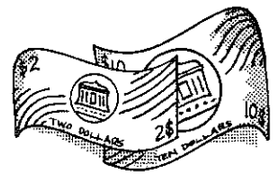
10. If the probability that an event will not occur is $\frac{4}{N}$, then which one of the alternatives is the probability that it will occur?

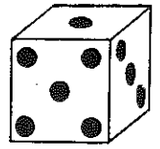
A. $\frac{N}{4}$ B. $\frac{4}{N} - 1$ C. $1 - \frac{N}{4}$ D. $\frac{N-4}{N}$ E. none of these.

11. Six schoolboys have caps which all look the same, but they are all different sizes. The caps get mixed up and one of the boys chooses a cap at random. The probability that it is too big is $\frac{1}{2}$. What is the probability that it is too small?

EXPECTATION

Would you pay \$2 to play this game?





If you throw a one, you receive \$10, otherwise you lose your \$2. We are interested in the variable, "number of dollars gained". There are two possible values of this variable: a net gain of \$8 or a loss of \$2. The corresponding probabilities are shown in a **probability distribution table**.

Variable, x_i	**Probability, p_i**
\$8	$\frac{1}{6}$
$-$\$2	$\frac{5}{6}$

In the long run, you would make a net gain of \$8 on one-sixth of the throws. This is an average return of one-sixth of \$8, that is, \$1.33 per throw. On the losing side, you could expect to lose \$2 on five-sixths of throws. This is an average loss of \$1.67. A negative is used to indicate a loss.

The probability distribution table may be extended to show these products.

Variable, x_i	**Probability, p_i**	**Product, x_ip_i**
\$8	$\frac{1}{6}$	$\$8 \times \frac{1}{6} \approx \$1.33$
$-$\$2	$\frac{5}{6}$	$-\$2 \times \frac{5}{6} \approx -\$1.67$
Total:	1	$-$\$0.34

A player should expect to lose 34 cents per game, in the long run. This amount is the **expected value** of the variable and we write:

$$E(X) = -\$0.34$$

Would you pay \$2 to play the game? Not likely!

Expected Value

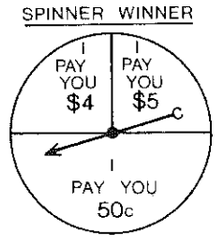
The expected value of a random variable X is given by:

$$E(X) = x_1p_1 + x_2p_2 + x_3p_3 + \ldots + x_np_n$$

provided that $x_1, x_2, \ldots, x_n$ are mutually exclusive events which fill the sample space.

Example 2.

To play this game you must put up a bet of \$1, \$2, \$3, or \$4. What is the most you should pay for a single spin if you want the game to be in your favour?



Solution:

Let X represent "the number of dollars gained".

	Variable, x_i	Probability, p_i	Product, x_ip_i
	\$0.50	$\frac{1}{2}$	\$0.25
	\$4.00	$\frac{1}{4}$	\$1.00
	\$5.00	$\frac{1}{4}$	\$1.25
Total:		1	\$2.50

$$E(X) = \$2.50$$

If you base your judgment on the expected value, you should pay no more than \$2.

Example 3.

A baker makes four special pies each day. Each pie costs \$1.40 to make and is sold for \$4.40. Pies may be sold only on the day in which they are baked. He sells 0, 1, 2, 3 or 4 pies each day with respective probabilities $\frac{1}{20}, \frac{1}{20}, \frac{1}{5}, \frac{1}{5}, \frac{1}{2}$.

a. List profits on the sale of 0, 1, 2, 3, or 4 pies and calculate the expected profit.

b. How much profit could she expect to make in seven days of trading?

Cost = 4 × \$1·40 = \$5·60

Profit = No. pies sold × \$4·40 − \$5·60

Solution:

a.

No. of pies sold	Profit, x_i	Probability, p_i	Product, x_ip_i
0	−\$5.60	$\frac{1}{20}$	−\$0.28
1	−\$1.20	$\frac{1}{20}$	−\$0.06
2	\$3.20	$\frac{1}{5}$	\$0.64
3	\$7.60	$\frac{1}{5}$	\$1.52
4	\$12.00	$\frac{1}{2}$	\$6.00
Total:		1	\$7.82

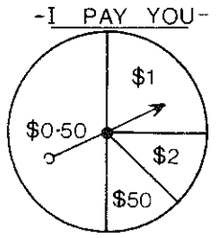
b. Expected amount in seven days of trading = 7 × \$7.82 = \$54.74

Exercise 6B: Expectation

1. An importer of cane furniture is offered a shipment of furniture for \$20 000. The associated probabilities that he will be able to sell the shipment for \$24 000, \$22 000, \$20 000 or \$18 000 are 0·22, 0·41, 0·19 and 0·18, respectively. Determine the expected value of the profit.

2. A master builder tenders for a housing construction which could yield a profit of \$40 000 with an associated probability of 0·8. However, because of strikes and late delivery of supplies, the construction could yield a loss of \$18 000 with an associated probability of 0·2. Determine the builder's expectation.

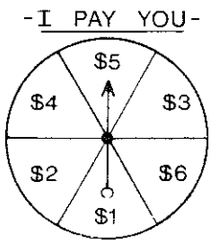
3. Would it be worth paying \$5 to play this game? Why?



4. A lucky dip box contains 25 wrapped packages. Ten of the parcels are valued at 20 cents each; another ten, at 60 cents each; and the remaining five at \$1 each. For the chance of selecting a package would you pay:

   a. 20 cents? b. 50 cents?

5. Would it be worth paying \$4 to play this game? Why?

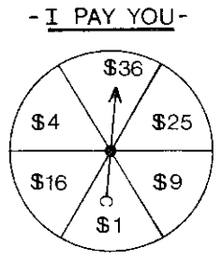


6. The probability that a person will sell her yacht for a profit of \$4 000 is 0·12; the probability of a profit of \$2 000 is 0·28; the probability that she will break even is 0·45. The probability that the sale will result in a loss of \$2 000 is 0·15. Determine the expected value of the profit.

7. When a car is stopped at a police road block the probabilities that there will be 1, 2, 3, 4, or 5 people in a car are 0·16, 0·48, 0·19, 0·12 and 0·05, respectively. Determine the expected number of *passengers* per car.

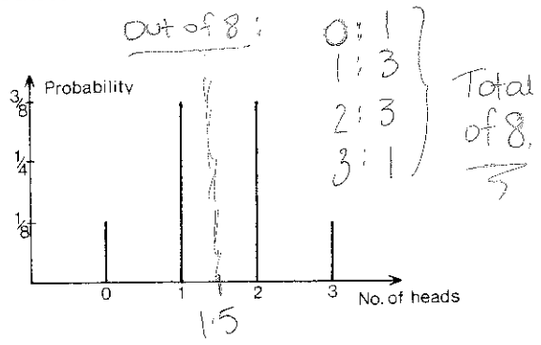
8. An amateur sporting club decides to run a non-profit gambling night for its members. One game involves rolling a fair die. The entry fee for each roll is \$4. If a one is rolled the club pays out \$1; if a two, \$2; if a three, \$3; if a four, \$4; if a five, \$6; and if a six, \$8 is paid out. Is the game a fair game? Why?

9. Would it be worth paying \$15 to play this game. Why?



10. A gambling game involves rolling a fair die. If a prime number occurs the gambler receives that number of dollars. However, if a non-prime number occurs he must pay the house that number of dollars. Determine the expected value. Is the game in favour of the house or in favour of the gambler? Why?

11. The probability distribution for the number of heads when three coins are tossed is graphed.



a. Find the average number of heads in a very large number of trials.
b. In a gambling game, a punter was offered \$30 for three heads, \$20 for two heads, \$10 for one head and nothing for no heads. The bet was \$15. Calculate the expected net return for the punter.

12. A random variable X may take the values 1, 2, 3, 6, or 12 with probabilities $\frac{2}{15}$, $\frac{2}{15}$, $\frac{2}{15}$, $\frac{1}{5}$, and $\frac{2}{5}$, respectively. Find the expected value, E(X).

13. The "Numbers" racket was a favourite of American criminals. "Numbers" is an illegal gambling game in which a player selects any of one thousand 3-digit numbers such as 401, 080, or 179. He pays \$2 to the "runner". A winning number is selected in an honest and random manner. For example, the last three digits of the total number of share transactions on the Chicago stock exchange were often used. Any player who selects the correct numbers collects \$1 000. What is the expected value of this game?

14. A batch of computer chips is known to contain 10% defectives. Each minute of production, a random sample of 3 chips is taken. The probabilities of finding 1, 2, or 3 defectives are 0·243, 0·027 and 0·001, respectively.

a. What is the probability of finding no defectives in a random sample?
b. In a very large number of samples, what would be the average of the number of defectives?
c. The defective chips are set aside when they are detected. How many hours should it take to accumulate 100 defective chips?

15. A company which manufactures car batteries guarantees them for 2 years. It promises to replace any battery which fails within the guarantee period. The batteries cost \$30 to manufacture and sell for \$60. Records show that 7% of the manufactured batteries fail within the guarantee period. In the long run, what is the company's expected return per battery?

16. The bank notes, \$2, \$5, \$10, \$20, \$50, and \$100, are sealed in separate envelopes. What is the most a gambler should pay to be allowed to select:

a. one envelope at random? b. two envelopes at random?

17. A pair of fair dice is tossed. Let X represent "the total number of spots on the uppermost faces". List all the possible values of X and their corresponding probabilities to determine the expected value of X.

18. In the game of KENO, a barrel of 80 marbles is used. The marbles are numbered 1, 2, 3, ..., 80. In ONE-SPOT KENO, the player pays \$5 and punches a single hole through the card on the number of his choice.

ONE-SPOT KENO

1	*2*	*3*	*4*	*5*	*6*	*7*	*8*
9	*10*	*11*	*12*	*13*	*14*	*15*	*16*
17	*18*	*19*	*20*	*21*	*22*	*23*	*24*
25	*26*	*27*	*28*	*29*	*30*	*31*	*32*
33	*34*	*35*	*36*	*37*	*38*	*39*	*40*
41	*42*	*43*	*44*	*45*	*46*	*47*	*48*
49	*50*	*51*	*52*	*53*	*54*	*55*	*56*
57	*58*	*59*	*60*	*61*	*62*	*63*	*64*
65	*66*	*67*	*68*	*69*	◯	*71*	*72*
73	*74*	*75*	*76*	*77*	*78*	*79*	*80*

Punch ONE number only.

The casino selects 20 numbers from the barrel at random. If the number punched out by the player is among the 20 selected, the player is paid \$15.

a. Find the expected net winnings.
b. If the game was completely fair the expected net winnings should be zero. How much should the player pay for a fair game of ONE-SPOT KENO?
c. If 500 cards were sold for any one game, how much profit could the casino expect to make?

19. A random variable Y can take the values 0, 1, 2, 3, or 4.

a. Construct a probability distribution for Y given that:

$$P(Y>0) = 0{\cdot}9, \quad P(Y>1) = 0{\cdot}6, \quad P(Y = 1) = P(Y = 2), \quad \text{and}$$
$$P(Y = 0) = P(Y = 4).$$

b. Find the expected value of Y.

20. In a game of American roulette there are eighteen red numbers, eighteen black numbers, a zero (green) and a double zero (also green).

The casino offers "evens" for betting on *red* or on *black*. That is, if you bet \$1 on a red number and it comes up, they will return your dollar and give you another \$1.

a. Find the expected net gain (to the nearest cent) if you bet \$1 on *red* in this game.
b. The casino offers 35 to 1 for bets on a single number. Find the expected net gain (to the nearest cent) if you bet \$1 on number *23* in this game.
c. Which is more favourable to you: betting on *red* or betting on *23*? Why?

21. A television repair company charges for each fifteen minutes spent on a repair. The charge is a flat fee of \$15 and then \$9 for each fifteen minutes. The company records for a typical week's repairs show:

No. of 15-minute periods:	1	2	3	4	5	6	7	8
No. of repairs:	20	15	19	10	3	0	0	3

Use relative frequencies as estimates of probabilities to find:

a. the expected number of fifteen-minute periods for a television repair.
b. the company's expected income for next week's television repairs.

ODDS

"Odds" are quoted in many places: on the racing pages of a newspaper or in gambling games. "Odds" is an alternative way of expressing probabilities.

Odds in Favour, Odds Against

The odds in favour of an event E is the ratio of the number of favourable outcomes to the number of unfavourable outcomes.

$$\text{Odds in favour of E} = n(E) : n(\overline{E})$$

The reverse ratio gives the odds against E.

$$\text{Odds against E} = n(\overline{E}) : n(E)$$

Example 4.

A single-numbered disc is drawn from a bag containing eight identical-shaped discs numbered 1 to 8.

a. List all the possible outcomes.
b. Determine:
   i. the odds in favour of selecting a disc whose number is less than 3.
   ii. the odds against selecting a disc whose number is less than 3.

Solution:

a. $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$
b. The favourable outcomes, $E = \{1, 2\}$ and
   the unfavourable outcomes, $\overline{E} = \{3, 4, 5, 6, 7, 8\}$.
   i. Odds in favour of E $= 2:6 = 1:3$
   ii. Odds against E $= 6:2 = 3:1$

Example 5.

A bookmaker at the horse races quotes the odds for eleven horses in a race.

Double Bay	33 : 1	White Knight	7 : 2
Johnston Town	20 : 1	Domino	33 : 1
Clearday	50 : 1	Flying Fox	2 : 1 ON
Waterboy	EVENS	Whatsisname	25 : 1
Big Booma	33 : 1	Mysterious	16 : 1
Try Again	100 : 1		

a. How much is a bookmaker's payout for a \$1 win bet if the winner is:
   i. Double Bay? ii. Waterboy? iii. Flying Fox? iv. White Knight?

b. Express, as probabilities, the odds quoted on:
   i. Clearday. ii. Flying Fox. iii. White Knight. iv. Waterboy.

Solution:

a. i. Payout for Double Bay = \$33 + \$1 = \$34
   ii. Payout for Waterboy = \$1 + \$1 = \$2
   iii. Payout for Flying Fox = \$0.50 + \$1 = \$1.50
   Note: "2:1 ON" means that for each \$2 bet on Flying Fox, the possible winnings are only \$1.
   iv. Payout for White Knight = \$3.50 + \$1 = \$4.50

b. i. $P(\text{Clearday}) = \frac{1}{50+1} = \frac{1}{51}$ ii. $P(\text{Flying Fox}) = \frac{2}{2+1} = \frac{2}{3}$

   iii. $P(\text{White Knight}) = \frac{2}{7+2} = \frac{2}{9}$ iv. $P(\text{Waterboy}) = \frac{1}{1+1} = \frac{1}{2}$

Exercise 6C: Odds

1. What are the odds of obtaining a three in one roll of a fair die?

2. On the average, a football team wins 3 out of every 4 games played. What are the odds in favour of this team winning its next game?

3. If the odds that *Honest Trier* will win the third race are 3:2, what is the probability of *Honest Trier* winning the race?

4. *No Comparison* has won 32 races from 40 starts. Assuming no changes in the competition against this horse, what odds might be quoted for his next race?

5. A horse, *Geraldton Bay*, is said to have a probability of 0·2 of winning. Express this probability as:
   a. a percentage.
   b. an odds against the horse winning.
   c. an odds in favour of the horse winning.

6. The probability that it will rain on June 14th is $\frac{1}{5}$.
   a. What is the probability that it will not rain on June 14th?
   b. What are the odds in favour of it raining on June 14th?
   c. What are the odds against rain on June 14th?

7. If a red die and a green die are rolled together, determine the odds:
   a. in favour of both dice showing the same number.
   b. against the number showing on the green die being greater than the number showing on the red die.

8. The probability that an event E occurs is $\frac{2}{5}$.
   a. What are the odds in favour of E occurring?
   The probability that an event F occurs is p.
   b. What are the odds in favour of F occurring?

9. A bag contains black and white balls. The odds against a randomly selected ball being white are 2:1. If five black balls were removed, the odds that a randomly selected ball would be white are EVEN. How many balls were in the bag originally?

10. The odds in favour of being caught behind a bus on the freeway when driving to work are 7:3. For the next occasion that you drive to work on the freeway determine:
   a. the probability of being caught behind a bus.
   b. the probability of not being caught behind a bus.
   c. the odds against being caught behind a bus.

11. If the probability of sleeping through an alarm clock after a late night is $\frac{1}{8}$, determine the odds that, after your next late night, the alarm clock will not awaken you.

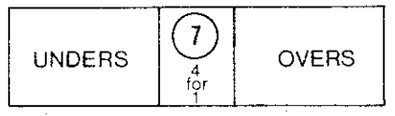
12. The odds against a particular athlete clearing 2 metres in high jump are 5:1. Determine the probability that on his next attempt at this height he:
   a. clears the bar. b. dislodges the bar.

13. A musical preference survey taken at random from 40 people in the street revealed, that 14 preferred rock, 23 preferred country and western and 3 preferred classical music. What are the odds that a person selected at random from this group:
   a. preferred rock music? b. did not prefer country and western?

14. Determine the odds in favour of a family of 3 children consisting of 2 girls and 1 boy.

15. A bag contains 2 red balls, 2 white balls and 5 yellow balls. If a ball is selected at random from the bag, determine:
   a. the odds that it is white.
   b. the odds against it being yellow.
   c. the odds that it is yellow.

16. In the gambling game of "Unders and Overs" participants place their bets on the board.



Two dice are rolled and if the total score is less than 7 (*Unders*) or greater than 7 (*Overs*) "even money" is paid out. That is, if you bet \$5 on the *Overs* and the result

is over 7, then you collect \$10: the \$5 you bet plus the \$5 you win. If the result is *7* then *4 for 1* is paid out. Determine the odds in favour of:

a. *Unders.* b. *Overs.* c. *7.*

Why do you think "The House" offers odds of *4 for 1* on a seven?

17. Weather bureau records show that the temperature on Christmas Day has exceeded 35 °C on fifteen occasions over the last sixty years in the town of Warmington.

a. What is the probability that this year, Christmas Day will be hotter than 35 °C at Warmington?
b. What are the odds that it will be cooler than 35 °C on Christmas Day at Warmington this year?
c. One Warmington citizen offered \$10 to your \$3 as a bet that the temperature this Christmas Day will not exceed 35 °C. Whom do the odds favour in this bet?

18. Seeds are collected for a new hybrid sweet pea plant. The seeds are expected to produce red, pink, or white flowers. Red is the most popular colour. However, a randomly selected seed is just as likely to produce white flowers as red. It is twice as likely to produce pink flowers as red.

a. What are the odds that a randomly selected seed will produce red flowers?
b. What is the probability that a randomly selected seed will produce pink flowers?

19. Some calculators break down while under the guarantee. There are three causes: manufacturing error, rough handling, or excessive heat. Past experience has indicated that the cause of the breakdown is twice as likely to be rough handling as manufacturing error. However, it is three times as likely to be caused by manufacturing error than excessive heat. If a calculator breaks down while under the guarantee, determine the probability that it is caused by:

a. manufacturing error. b. rough handling. c. excessive heat.

20. There are 33 discs in a bag. Some are red, some are white and the others are green. A single disc is to be drawn at random. The odds against it being red are 9 : 2. The odds against it being green are 5 : 6.

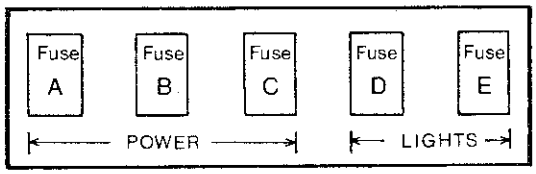
a. How many discs of each colour are in the bag?
b. What are the odds against it being white?

Chapter 7:

Conditional Probability, Independence and Laws

CONDITIONAL PROBABILITY

One of these fuses "blows" and there is an electricity failure in the house.



What is the probability that it was fuse A? Answer, $\frac{1}{5}$

Just before you start checking the fuses, somebody from inside the house calls out, "All of the lights are still on!" Now, what is the probability that it was fuse A?

The clue is, that it must be a power fuse. The number of possibilities is reduced from 5 to 3. Thus the probability that it is fuse A, **given** that it is a power fuse, is $\frac{1}{3}$.

This is a conditional probability because we are given a clue. We write,

$$P(\text{fuse A} \mid \text{power fuse}) = \frac{1}{3}$$

The vertical line means "given".

Example 1.

To test a new drug, laboratory rats are given lethal doses. All of the rats in column L required less than 100 mg of the drug before they died but those in column G required more. The rats are classified according to sex, as well.

	L less than 100 mg	G greater than 100 mg	Total
Male (M):	23	29	52
Female (F):	11	67	78
Total:	34	96	130

Autopsies are conducted on some of the rats.

a. Given that a randomly selected rat was a female, what is the probability that the lethal dose was less than 100 mg.

b. What is the probability that a randomly selected rat was a male given that the lethal dose was greater than 100 mg.

Solution:

a. P(less than 100 mg | female) = P(L | F) = $\frac{11}{78}$ (n(L and F) / n(F))

b. P(male | greater than 100 mg) = P(M | G) = $\frac{29}{96}$ (n(M and G) / n(G))

Conditional Probability

The probability that event A occurs, given that event B has already occurred is given by:

$$P(A \mid B) = \frac{n(A \cap B)}{n(B)}$$

It is sometimes more convenient to express this law as:

$$P(A \mid B) = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(B)}{n(S)}} = \frac{P(A \cap B)}{P(B)}$$

INDEPENDENT EVENTS

A club runs two types of unusual raffles, a BACKWARDS raffle and a REPLACEMENT raffle.

In a BACKWARDS raffle there are six prizes. The first ticket drawn out wins the booby prize and the sixth ticket drawn out wins $10 000. There are 10 tickets and they are drawn one at a time. What is the probability that your ticket comes out:

a. first?

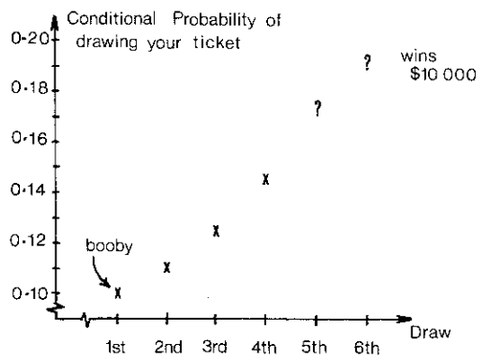
b. second, given that it was not the first one?

c. third, given that it was not drawn previously?

d. fourth, given that it was not drawn previously?

The probabilities in parts a, b, c, and d are shown in the graph.

e. Complete the graph to show the conditional probabilities for the fifth and the sixth draws.



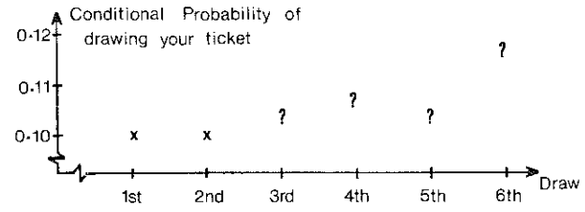
In a REPLACEMENT raffle you may win more than one prize because each ticket is put back after it has been drawn.

f. What is the probability that your ticket comes out first?

g. What is the probability that your ticket comes out second, given that it was not the first one drawn?

The probabilities for parts f and g are shown in the graph.

h. Complete the graph to show the conditional probabilities for the third, fourth, fifth and sixth draws.



In the REPLACEMENT raffle all of the probabilities are $\frac{1}{10}$. We say that the results of the draws are **independent** because the outcome of the first draw has no effect on probabilities associated with the following draws.

Independent Events

Two events, A and B are independent of each other if the probability that one of them occurs is not influenced by the occurrence or non-occurrence of the other.

That is, $P(A) = P(A \mid B)$ or $P(B) = P(B \mid A)$

Example 2.

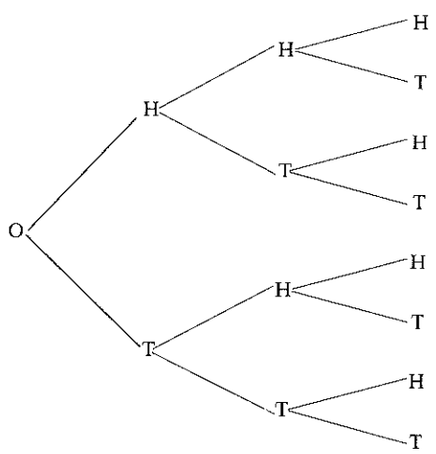
Three fair coins are tossed.

Let the event A be "all tails or all heads".
Let the event B be "at most 2 heads", ie 0 heads, 1 head, 2 heads
Let the event C be "at least 2 heads", ie 2 heads, 3 heads

Which of the pairs (A,B), (A,C), and (B,C) are independent?

(A,B) –

Solution:



	Event
HHH	A,C
HHT	B,C
HTH	B,C
HTT	B
THH	B,C
THT	B
TTH	B
TTT	A,B

Now $P(A \mid B) = \frac{1}{7}$ and $P(A) = \frac{2}{8} = \frac{1}{4}$.
Therefore A and B are NOT independent.

Now $P(A \mid C) = \frac{1}{4}$ and $P(A) = \frac{1}{4}$.
Therefore A and C are independent.

Now $P(B \mid C) = \frac{3}{4}$ and $P(B) = \frac{7}{8}$.
Therefore B and C are NOT independent.

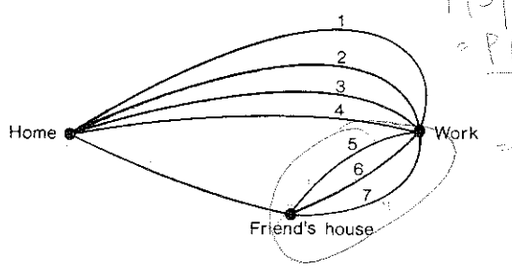
That is, (A,C) is the only independent pair.

Exercise 7A: Conditional Probability and Independence

1. A bag contains 5 yellow, 3 red and 7 blue discs. A disc is selected at random, its colour noted and then another disc is drawn. Determine the probability of selecting:

a. a yellow disc on the second draw if a red was selected first.
b. a yellow disc on the second draw if a blue was selected first.
c. a yellow disc on the second draw if a non-yellow was selected first.

2. A man travels to work by any of four different routes which he chooses at random. Occasionally he may detour to pick up a friend and from the friend's house he may choose any of three different routes at random. Given that he picked up his friend, what is the probability that he drove along route 5?



3. A dish contains 10 jellybeans and 5 mints. The dish is passed around and children take them one at a time. Assume they choose randomly. Determine the probability that:

a. the first child takes a jellybean.
b. the second child takes a jellybean, given that the first child took a jellybean.
c. the third child takes a mint, given that the first two children took jellybeans.

4.

Jones	Smith	Williamson
Harris	Johns	Taylor

A secretary delivers five identical memos to the staff pigeon holes. Determine the probability that Mr Taylor collected his memo first if the memo was written by Mr Harris.

5. The letters of the word *PURCHASE* are arranged in a line in a random order. Determine the probability that:

a. the last letter of the word formed is an *A* if the first two letters are *PU*.
b. the last letter of the word formed is a *P* given that the first letter of the word is a vowel.

6. A novel contains 300 pages. Determine the probability that a reader is on page 186 if it is known she is beyond the half-way mark.

7. Which of the following pairs of events are independent? Answer "Yes" or "No".

a. a person's height; a person's weight.
b. a person's height; a person's income.
c. colour of hair; weight.
d. sex of driver; ability of driver.
e. age of driver; probability of an accident.
f. getting a three on the first roll of a die; getting a three on the second roll.
g. a head on a coin; a three on a die.
h. hours of golf practice; golf score.

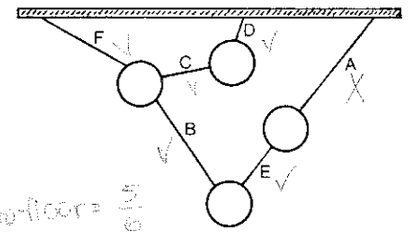
8. Each of the following newspaper extracts contain one or two probability statements. Mostly they are conditional probability statements. Express them in the form $P(A \mid B) =$ constant.

a. *ALCOHOL AND ROAD DEATHS:* Nine percent of car accidents attended by Police involve a fatality. Of the fatal accidents, 80% are related to alcohol consumption.
b. *LIFE ASSURANCE:* 72% of men aged 60 years live until the age of 65 years at least.
c. *BALLET:* "Without a long warm-up before the performance, I guarantee that a dancer will sustain an injury. Our dancers warm up before every performance. There are 124 in the company and we have had only 5 serious injuries".

d. *DRUGS:* "Dr Nahas said that a survey conducted in New York State revealed that 28 per cent of the daily users of marijuana among high school seniors will go on and experiment with stronger, more destructive drugs, while only one per cent of the kids who never use marijuana will go directly to the use of those more destructive drugs."

e. *SMOKING:* A study just released from Boston University in the U.S., indicated that women who smoke tobacco during pregnancy have a five times greater risk of having children with a much lower than average birth weight.

9. Four identical rings are suspended from the ceiling as shown in the diagram:



One of the strings is cut. After the cut, the bottom ring is still closest to the floor. What is the probability that:

a. string A was cut?

b. string B was cut?

10. Western Australia has three television stations: two commercial stations, Channel 7 and Channel 9, and a non-commercial station, Channel 2. Determine the probability that a television viewer is watching:

a. Channel 7.

b. Channel 9 if the viewer always tunes to a commercial station.

11. A motorist "blows" a tyre on a car. Determine the probability that the tyre which burst was:

a. a rear tyre.

b. a front tyre.

c. the rear tyre on the driver's side, if the motorist experienced no problems with the steering of the car.

12. A die is rolled. Determine the probability that a 3 is uppermost if it is known that:

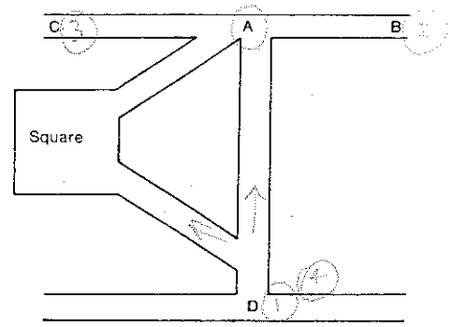
a. an odd number is uppermost.

b. a number less than 5 is uppermost.

c. an even number is uppermost.

13. Determine the probability that the second ball drawn in a forty-ball Lotto draw will be *7* if the first number drawn was *14*.

14.



a. A tourist is walking about this village. She arrives at A. What is the probability that:

i. she came from B?

ii. she came from C?

iii. she came from D via the square?

iv. she came directly from D?

Now she tells you that she did not see the Square.

b. Recalculate the probabilities in part a.

15. Responses to a questionnaire about the banning of analgesic drug commercials on T.V. were as follows:

	For	Against	No Opinion	
Men:	27	19	6	52
Women:	28	16	4	48
	55	35	10	100

What is the probability that any person selected at random from this sample:

a. is a woman?

b. is a woman, given that the person was in favour of banning the commercial?

c. is a man, given that the person had no opinion either way?

16. A and B are events in a sample space, S.

a. If $P(A \cap B) = 0{\cdot}48$, $P(A) = 0{\cdot}6$ and $P(B) = 0{\cdot}8$ find:

i. $P(A \mid B)$.

ii. $P(B \mid A)$.

b. Are A, B independent events? Why?

17. Find the probability of drawing a jack from a well-shuffled pack of 52 cards if:

a. all of the spades have been removed.

b. all of the kings have been removed.

c. two jacks have been removed.

18. There are 100,000 tickets in the lottery and you have bought one.

a. What is the probability that you will win first prize?

b. If you do not win first prize, what is the probability that you won second prize?

c. If you do not win first or second prizes, what is the probability that you won third prize?

19. In Reno, Nevarda (U.S.A.), on 15th May, 1981, a U.S.A. record was set on a roulette table.* The number *17* came up fourteen times in a row on a particular wheel.

a. Given that the number *17* had come up fourteen times, what is the probability it comes up on the fifteenth time?

b. What is the probability that the number *17* comes up on the sixteenth spin?

* An American roulette wheel has a zero, a double zero, and the digits 1 to 36.

20. A standard pack of 52 cards contains 4 aces and 4 kings. A card is dealt from the pack. Then, another one is dealt without replacing the first.

Let A_1 represent the event: "an ace is dealt first";
let A_2 represent the event: "an ace is dealt second";
let K_1 represent the event: "a king is dealt first"; and so on.

a. Calculate the following probabilities:
i. $P(A_1)$ ii. $P(A_2|A_1)$ iii. $P(K_1)$ iv. $P(K_2|A_1)$

b. Calculate these probabilities again. This time, the first card was replaced and the pack shuffled before the second one was dealt.

21. Jonathan chose two letters at random from the word *TONE*. Both letters appear in his name. Determine the probability that:

a. they were *T* and *N*. b. *O* was included.

22. If M represents the event that you pass your next mathematics exam and H represents the event that you do all of your mathematics homework, express each of the following in words:

a. P(M). b. P(M | H). c. P($\overline{M}$ | $\overline{H}$).

These probabilities are 0·4, 0·95 and 0·9, respectively.

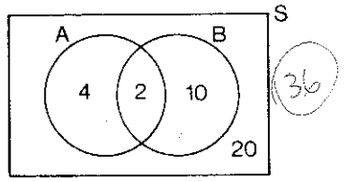
d. Are M and H independent events? Support your answer by quoting probabilities.

23. A single number was eliminated from this set: {6, 7, 8, 9, 10}. After the number was eliminated, a second was chosen at random. Determine the probability that:

a. the second was 6 if the eliminated number was 7.
b. the eliminated number was 6 if the second number was 10.
c. the eliminated number was even, if the second number was 10.
d. the eliminated number was even, if the the second number was odd.

Hint: List all of the possibilities.

24. This is the Venn diagram for events A and B in a probability space, S.



The number of equally likely sample points in each event is given. Use conditional probabilities to determine whether or not:

a. A and B are independent. b. A and $\overline{B}$ are independent.
c. $\overline{A}$ and $\overline{B}$ are independent.

25. A person holding a photograph of two children, points to it and says, "He looks like Uncle Harry!" Given this clue, what is the probability that, in the photograph:

a. both children are boys?
b. the elder child is a girl?
c. the younger child is a boy?

26. A pair of fair dice is tossed. One is red and the other is green. Define events A, B and C by:

A: green die shows more spots than red.
B: total number of spots is an even number.
C: exactly one die shows five spots.

Which pairs of events (A,B), (A,C) or (B,C) are independent? Why?

27. There are two evening newspapers, the *Advertiser* and the *Bugle*. The *Bugle* has the larger sales. The question is: "Does the fact that a customer is female, influence which paper is bought?" Perhaps newspaper preference is independent of sex. A random sample of 100 customers was tabulated.

Number of Sales	*Advertiser*	*Bugle*	**Total**
Male:	18	42	60
Female:	12	28	40
Total:	30	70	100

a. What proportion of this sample bought:
i. The *Bugle*? ii. The *Advertiser*?

b. What proportion of females from this sample bought:
i. The *Bugle*? ii. The *Advertiser*?

Express these proportions as conditional probabilities.

c. Based on this evidence, what is the answer to the question?

28. If two dice are rolled together, determine the probability that:

a. exactly one of the numbers showing is a multiple of three.
b. the sum of the uppermost numbers showing is eight, given that exactly one of the numbers is a multiple of three.

29. A market survey is designed to test the appeal of a new toilet soap package. A stratified sample of 140 people is asked to rate the package on a 5-point scale:

A, B, C, D, E

very unattractive ——— very attractive

The number of responses in each category is tabulated in the following table:

	Rating				
	A	**B**	**C**	**D**	**E**
Over 25 years:	0	6	7	53	4
Under 25 years:	3	9	15	35	8

a. What proportion of respondents gave the package a rating of C, D, or E?
b. What proportion of the "over 25" age group gave the package a rating of C, D, or E?
c. On the basis of this evidence, would you say that, in this sample, rating response is independent of age?

The 140 people are re-classified according to sex.

	Rating				
	A	**B**	**C**	**D**	**E**
Male:	2	14	11	28	11
Female:	1	1	11	60	1

d. What proportion of females gave the package a rating of C, D, or E?
e. On the basis of this evidence, would you say that, in this sample, rating response is independent of sex?

30. A primary school class of 30 students went on an end-of-term picnic to a stream in the hills. Eight students fell into the water, twelve skinned their knees and fifteen returned without mishap.

a. What is the probability that a child who got wet did not get skinned knees?
b. What is the probability that a child with skinned knees also fell into the stream?

31. There are four cakes on a plate: a vanilla slice, a custard pie, and two chocolate eclairs. A man ate two cakes. What is the probability that the first cake he ate was a chocolate eclair if:

a. a custard pie was left after he had eaten his two cakes?
b. he ate a chocolate eclair second?

32. A family has three children. What is the probability that the family has two boys given that the youngest child is a girl?

33. This newspaper clipping gives 1981 Australian census figures: the number of Australians of each religious denomination. The number of Australians without a religion is not given.

Our most popular religion is Anglican

FOR those curious about the number of people in Australia's church denominations the figures are disclosed in the 1981 Commonwealth Census.

There were 3,810,769 Anglicans and 3,785,505 Catholics.

Membership of the smaller denominations are: Armenian Apostolic 3582; Baptist 190,259; Bretheren 21,489; Church of Christ 89,424; Congregational 23,017; Jehovah's Witness 51,815; Latter Day Saints 32,444; Lutheran 199,760; Methodist (including Wesley) 490,767; Orthodox 421,281; Pentecostal 72,148; Presbyterian 637,818; Salvation Army 71,570; Seventh Day Adventist 47,747; Uniting Church 712,609.

Non-Christian: Buddhist 35,073; Hebrew 62,126; Muslim 76,792.

Express (correct to two decimal places) each of the following proportions as conditional probabilities:

a. the proportion of Anglicans.
b. the proportion of Baptists.
c. the proportion of Baptists in religions which are neither Catholic nor Anglican.
d. the proportion of Muslims in the non-Christian religious group.

THE MULTIPLICATION LAW FOR PROBABILITIES

Half of the people entering a jeans shop buy a pair of jeans (J).

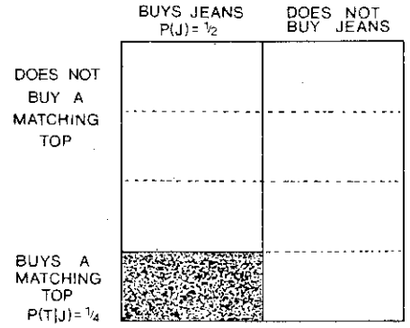
That is, $P(J) = \frac{1}{2}$

Of the ones who buy jeans, 25% buy a matching top (T).

That is, $P(T \mid J) = \frac{1}{4}$

What is the probability that the next person entering the shop buys jeans AND a matching top?

that is, $P(J \cap T) = ?$



Probability of buying jeans AND a matching top is $\frac{1}{8}$

That is, $P(J \cap T) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

($P(J)$, $P(T \mid J)$)

Multiplication Law

For events A and B in a sample space, S, the probability of A AND B both happening is given by:

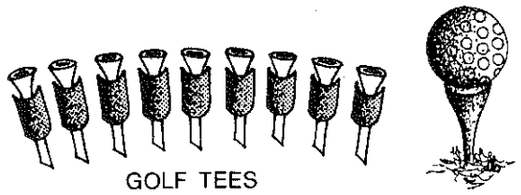
$$P(A \cap B) = P(A) \times P(B \mid A)$$

If A and B are independent events the law simplifies to:

$$P(A \cap B) = P(A) \times P(B)$$

Example 3.

A golfer had nine golf tees displayed in a holder on her golf bag. Three of the tees were red, two were black and the remainder were yellow.



GOLF TEES

She selected tees at random for each drive and returned unbroken ones to the holder. She broke only the tee used on her first drive. Determine the probability that the golfer used:

a. a red tee first and then a yellow tee.

b. a red tee and then a yellow tee and then a black tee on her third drive.

Let R represent: "red tee on first drive".
Let Y represent: "yellow tee on second drive".
Let B represent: "black tee on third drive".

Solution:

a. $P(R \cap Y) = P(R).P(Y|R) = \frac{3}{9} \times \frac{4}{8} = \frac{1}{6}$

b. $P(R \cap Y \cap B) = P(R).P(Y|R).P(B|Y \cap R) = \frac{3}{9} \times \frac{4}{8} \times \frac{2}{8} = \frac{1}{24}$

Example 4.

A certain hybrid of rat has short hair and a straight tail. If these hybrids are cross-bred there is a 25% chance that any offspring will have long hair instead of short hair. Quite independently, there is a 25% chance that any offspring will have a curly tail instead of a straight tail. If one of the offspring is randomly selected, determine the probability that it will have:

a. long hair and a curly tail.
b. long hair and a straight tail.
c. short hair and a curly tail.
d. short hair and a straight tail.

Solution:

a. $P(\text{long} \cap \text{curly}) = P(\text{long}).P(\text{curly}) = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$

b. $P(\text{long} \cap \text{straight}) = P(\text{long}).P(\text{straight}) = \frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$

c. $P(\text{short} \cap \text{curly}) = P(\text{short}).P(\text{curly}) = \frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$

d. $P(\text{short} \cap \text{straight}) = P(\text{short}).P(\text{straight}) = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$

Example 5.

In a flashing display sign, two electric circuits control the lights.



There is one circuit for the bulbs around the edge and another for the neon tubes in the middle. At any instant the probability that the circuits are ON is given by:

		Light bulbs	
		ON	**OFF**
Neon tubes	**ON:**	0·42	0·28
	OFF:	0·18	0·12

Are the circuits operating independently?

Solution:

P(Neon tube is ON) = P(N) = 0·42 + 0·28 = 0·7
P(Light bulbs are ON) = P(L) = 0·42 + 0·18 = 0·6

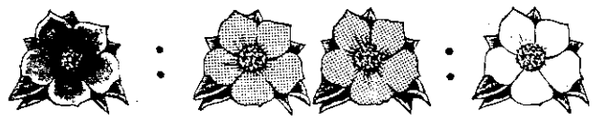
Now, $P(N \cap L) = 0{\cdot}42$, from the table.

But $P(N) \times P(L) = 0{\cdot}42$.

Therefore N and L are independent events which suggests that the circuits are operating independently.

Example 6.

The flowers of first-cross Four O'Clock plants are pink. Seeds from these produce either red-flowered, pink-flowered, or white-flowered plants in the ratio 1 : 2 : 1



red : pink : white

a. What is the probability that a seed selected at random does not produce a red-flowered plant?

b. What is the probability that no red-flowered plants are produced from a planting of:

i. 2 seeds? ii. 3 seeds? iii. 4 seeds? iv. 5 seeds?

c. If you wanted to be almost certain (that is, more than 99% sure) of producing at least one red-flowered plant, what is the minimum number of seeds you should plant?

Solution:

Let R_i represent the event that the ith plant is red-flowered.

a. $P(\overline{R}_1) = \frac{2+1}{1+2+1} = \frac{3}{4}$

b. i. $P(\overline{R}_1 \cap \overline{R}_2) = P(\overline{R}_1) \times P(\overline{R}_2) = (\frac{3}{4})^2 = 0{\cdot}5625$

ii. $P(\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_3) = (\frac{3}{4})^3 \approx 0{\cdot}4219$

iii. $P(\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_3 \cap \overline{R}_4) = (\frac{3}{4})^4 \approx 0{\cdot}3164$

iv. $P(\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_3 \cap \overline{R}_4 \cap \overline{R}_5) = (\frac{3}{4})^5 \approx 0{\cdot}2373$

c. We require the chance of no reds to be under 1%.

That is, $P(\overline{R}_1 \cap \overline{R}_2 \cap \overline{R}_3 \cap \ldots \cap \overline{R}_n) < 0{\cdot}01$

That is, $(\frac{3}{4})^n < 0{\cdot}01$

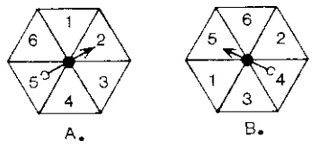
Now, $(\frac{3}{4})^{16} \approx 0{\cdot}010025$. Check on your calculator.

Therefore 17 is the minimum number of seeds that should be planted to be almost certain of producing at least one red-flowered plant.

Exercise 7B: The Multiplication Law for Probabilities

1. The probability that a hardware salesman sells eight tins of paint on a Saturday morning is 0·6 and that he sells three bundles of garden stakes is 0·4. What is the probability, that on a Saturday morning, a salesman sells eight tins of paint and three bundles of garden stakes?

2. Both spinners, A and B, are spun once, simultaneously.



a. Determine the probability of:

i. a 3 on spinner A.

ii. a combined score of 6.

iii. a 3 on spinner A and a combined score of 6.

b. T represents: "a 3 on spinner A"; and R represents: "a combined score of 6".

Are the events T and R independent events? Justify your answer.

3. If it is cloudy in September, the probability that it rains is 0·7. In September, an average of 12 out of 30 days are cloudy. What is the probability that September 23rd next year is cloudy and raining?

4. In the game of Monopoly two dice are used. A player "Goes to Jail" if he throws three doubles in a row. What is the probability of going to jail by this method?

5. An athlete is to compete in two events, the 400 metres and the 1 500 metres. The chances of the athlete winning these events are 0·2 and 0·85, respectively. What is the probability that the athlete:

a. wins both events?

b. wins the 400 metres and loses the 1 500 metres?

c. loses the 400 metres and wins the 1 500 metres?

d. loses both events?

6. The probability that Peter James graduates is 0·6 and the probability it will rain on June 22nd is 0·25. What is the probability that:

a. Peter James graduates and it rains on June 22nd?

b. Peter James does not graduate and it rains on June 22nd?

c. Peter James graduates but it does not rain on June 22nd?

d. Peter James does not graduate and it does not rain on June 22nd?

7. A and B are two independent events with respective probabilities of 0·3 and 0·8.

a. Copy and complete each of the following:

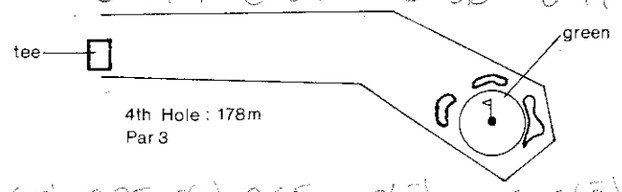
i. $P(A \cap B) = ?$ ii. $P(\overline{A} \cap B) = ?$

iii. $P(A \cap \overline{B}) = ?$ iv. $P(\overline{A} \cap \overline{B}) = ?$

b. Determine the value of:

$$P(A \cap B) + P(\overline{A} \cap B) + P(A \cap \overline{B}) + P(\overline{A} \cap \overline{B})$$

8.



Two golfers, John and Peter, tee off at the 4th hole. The probability that John's tee shot hits the green is 0·85 and that Peter's tee shot hits the green is 0·65. Determine the probability that:

a. both golfers' tee shots land on the green.

b. both golfers' tee shots do not land on the green.

c. Peter's tee shot misses the green and John's tee shot lands on the green.

d. Peter's tee shot lands on the green and John's tee shot misses the green.

9. Two events R and S are known to be independent events. Determine P(S) and P(R) if $P(\overline{S}) = 0{\cdot}6$ and $P(\overline{R} \cap \overline{S}) = 0{\cdot}42$.

10. A communication satellite has an incorrect orbit. Space-shuttle astronauts have the task of correcting the orbit. The task has three main stages:

I. extend the shuttle's mechanical arm.
II. an astronaut leaves the shuttle and effects repairs to the satellite's power system.
III. the astronaut returns to the shuttle and the satellite is moved into the correct orbit by a controlled firing of the satellite's rockets.

The probabilities that these three main stages are successful are $\frac{9}{10}$, $\frac{3}{4}$ and $\frac{2}{3}$, respectively. What is the probability that the satellite's orbit is corrected?

11. If the independent probabilities that Bill and Helen will be alive in 25 years' time are 0·25 and 0·3, respectively, determine the probability that, in 25 years' time:

a. Bill and Helen will be alive.
b. only Bill will be alive.
c. neither of them will be alive.
d. at least one of them will be alive.

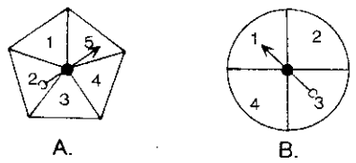
12. The probability that an aeroplane is delayed for take off due to bad weather is 0·14. The probability that a passenger will be allocated to a window seat is 0·2. Determine the probability of:

a. the aeroplane not being delayed for take off because of bad weather and a passenger not having a window seat.
b. the aeroplane being delayed for take off because of bad weather and a passenger having a window seat.

13. Three weight lifters (Bill, Reno and Paul) enter a weight lifting competition. The chances that Bill, Reno and Paul will be successful at lifting the qualifying weight are 0·8, 0·7 and 0·75, respectively. Determine the probability that:

a. all are successful on their first attempt at lifting the weight.
b. Bill and Reno are successful but Paul fails to lift the weight.
c. only Paul is successful in lifting the weight.

14. Spinner A and spinner B are spun simultaneously.



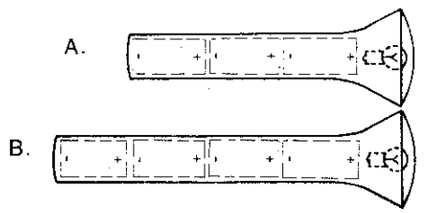
Determine the probability of obtaining:

a. a three with Spinner A.
b. a four with spinner B.
c. a three on spinner A and a four on spinner B.

15. In a survey, seven out of ten tertiary graduates said that if they had the opportunity again they would have chosen a different degree. Only 32% of students who enter a tertiary institution graduate. What is the probability that a randomly chosen person entering a tertiary institution graduates and is satisfied with his or her degree?

16. The independent probabilities that two dogs will live for 12 years are 0·25 and 0·15, respectively. Determine the probability that both dogs will live to the age of 12 years.

17. One per cent of new batteries of a particular brand are faulty. In a torch every battery must be good or the torch will not light. Each of these torches is filled with randomly selected new batteries of this brand. Which is more likely to light? Why?



18. Two extra red threes are added to a standard pack of 52 cards. A card is drawn, its colour noted, and then it is replaced before a second card is drawn. Determine the probability of:

a. a red card being drawn first then a five, second.
b. the first card not being red and then the second card not being a 5.

19. A box contains fifteen coloured discs: 6 blue, 5 white and 4 red. A disc is selected at random, its colour noted, and then it is replaced before another disc is drawn.

a. Determine the probability that:
   i. the first disc drawn is red.
   ii. the second disc drawn is blue.
   iii. the first disc drawn is red and the second disc drawn is blue.
b. What is the probability that the first disc drawn is not red and the second disc drawn is not blue?

20. An athlete is to compete in two events, the long jump and the triple jump. The long jump is her first event and the probability that she will win is 0·2. If she wins this event, the probability that she will win the triple jump is 0·65, but if she loses, the probability is only 0·55. What is the probability that she will win:

a. both events?
b. the second event only?

21. From past weather records, a particular city is known to have, on average, 21 rainy days in the month of June. This city experiences 80% or more cloud coverage for 24 days of the same month.

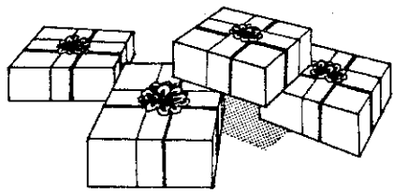
a. Determine:
   i. P(rainy day).
   ii. P(cloudy day).
b. Explain why P(cloudy day and rainy day) is not equal 0·56.

22. Terry, Ray and Jim are given a mathematical problem to solve. Each student works on the problem independently. The probabilities of each being able to solve the problem are 0·95, 0·65 and 0·75, respectively. Determine the probability that:

a. Terry and Jim are the only two to solve the problem.
b. Terry is the only one to solve the problem.
c. no student solves the problem.
d. all three solve the problem.
e. the problem is solved by at least one person.

23. The odds against "Johnston Town" winning Race 1 are 4:1. The odds in favour of an international touring soccer team winning its first match are 1:2. What is the probability that Johnston Town wins Race 1 and the touring soccer team wins their first match?

24. A lucky dip contains four identical packages: three are empty and one contains a $100 note.



Four people take turns at removing and opening a package.

a. Calculate the probability that the fourth person gets the $100 prize.
b. Is it better to have first, second, third, or fourth try at removing a package? Why?

Another lucky dip is the same, except that there are only two empty packages. One contains a $100 note and another contains a $50 note.

c. Calculate the probability that the first person gets $50 and the last person gets $100.

25. In a tennis match a particular player has an 80% chance of winning a game in which he serves but only a 40% chance of winning a game in which he does not serve. There are six games in a set and the players take turns in serving. Assume that the result of any game is independent of the results of any previous games. Determine the probability that this player:

a. wins all six games in a set.
b. loses all six games in a set.
c. wins at least one and loses at least one game in a set.

26. A person who has never tried a crossword puzzle before has a probability, $p = 0{\cdot}05$, of solving one. However, this person can only improve by solving puzzles. The probability of solving a kth puzzle is given by:

$$p_k = 0{\cdot}1 + 0{\cdot}9p_{k-1} \text{ where } k = 2,3,4,5\ldots$$

a. Complete this probability distribution table to show the probabilities of solving:
i. a fourth puzzle. ii. a fifth puzzle. iii. a sixth puzzle.

k	p_k
1	0·05
2	0·145
3	0·2305
4	?
5	?
6	?
7	0·495131
8	?

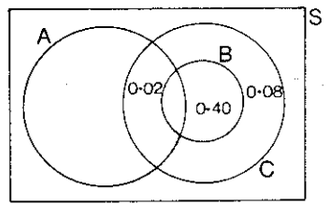
b. What is the probability that this person would NOT solve an eighth puzzle?
c. How many puzzles would this person need to solve before she had a greater than a 50–50 chance of solving the next one?
d. Explain, by referring to the equation, why p_k is less than 1.
e. Show that the probability q_k of NOT solving a kth puzzle is given by:

$$q_k = 0{\cdot}9q_{k-1}$$

27. A production line puts out a very large number of tyres. Eight tyres in one hundred are defective. A quality controller takes a random sample from the production line.

a. What is the minimum number of tyres that should be in the sample so that there is greater than a 95% chance of detecting a defective tyre?
b. He took a sample of this number of tyres and, detecting no defective tyres, concluded that the production line must be putting out less than 8% defective tyres. Could he be wrong? Explain.

28. There are three non-zero events A, B and C in a probability space, S. Their probabilities are 0·5, 0·8 and 0·9, respectively. Probabilities of compound events associated with A, B and C are illustrated in the Venn diagram.



a. Write down the probabilities of the following compound events:
i. $A \cap B$. ii. $A \cap C$. iii. $B \cap C$. iv. $A \cap \overline{C}$.
b. Which of the following pairs of events are independent: (A,B), (A,C), (B,C), or $(A,\overline{B})$?

THE ADDITION LAW FOR PROBABILITIES

Example 7.

A woman works part time on Wednesday, Thursday and Friday mornings. Her name is on the hospital waiting list for elective surgery. The hospital said that it would phone on a week day.

a. What is the probability that the hospital telephones while she is at work?

As well, she plays tennis all day Monday.

b. What is the probability that the hospital telephones on her tennis day?

c. What is the probability that the hospital telephones while she is out?

Solution:

	Mon.	Tues.	Wed.	Thurs.	Fri.
Morning:	Tennis	—	Work	Work	Work
Afternoon:	Tennis	—	—	—	—

a. P(at work) $= P(W) = \frac{3}{10}$.

b. P(at tennis) $= P(T) = \frac{1}{5}$.

c. P(out) $= P(\text{at work OR at tennis})$
$= P(W \cup T)$
$= P(W) + P(T)$
$= \frac{3}{10} + \frac{1}{5}$
$= \frac{1}{2}$

Addition Law for Mutually Exclusive Events

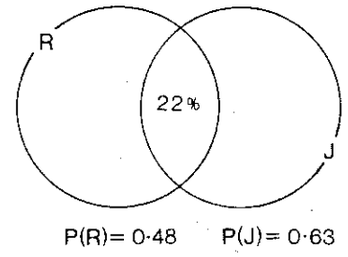
If two events A and B are mutually exclusive, the probability that A OR B happens is given by:

$$P(A \cup B) = P(A) + P(B)$$

Example 8.

A computer check on all premature babies born at a particular hospital showed that 48% had respiratory problems (R), 63% were jaundiced (J) and 22% had both problems ($R \cap J$). What is the probability that a new-born premature baby will have respiratory problems or be jaundiced?

Solution:



We use a more general form of the Addition Law when events are not mutually exclusive. So that babies with both problems are not counted twice, we use:

$$P(R \cup J) = 0{\cdot}63 + 0{\cdot}48 - 0{\cdot}22 = 0{\cdot}89$$

Addition Law

For events A and B the probability that A OR B happens is given by:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example 9.

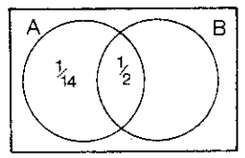
a. If $P(A) = \frac{4}{7}$, $P(B) = \frac{5}{7}$ and $P(A \cup B) = \frac{11}{14}$, find:

i. $P(A \cap B)$ ii. $P(A \mid \overline{B})$

b. Are the events A and B,

i. independent? ii. mutually exclusive?

Solution:



a. i. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

i.e., $P(A \cap B) = P(A) + P(B) - P(A \cup B)$
$= \frac{4}{7} + \frac{5}{7} - \frac{11}{14}$
$= \frac{1}{2}$

ii. $P(A \mid \overline{B}) = \frac{P(A \cap \overline{B})}{P(\overline{B})}$

now, $P(A \cap \overline{B}) = P(A) - P(A \cap B)$

$= \frac{4}{7} - \frac{1}{2}$

$= \frac{1}{14}$

$\therefore \quad P(A \mid \overline{B}) = \frac{P(A \cap \overline{B})}{P(\overline{B})} = \frac{\frac{1}{14}}{\frac{2}{7}} = \frac{1}{4}$

b. i. A and B are not independent, since

$$P(A).\ P(B) \neq P(A \cap B)$$

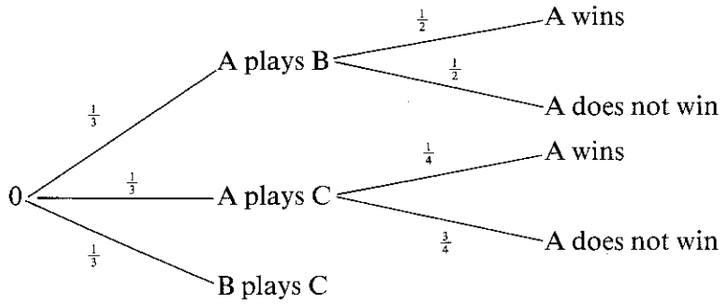
ii. A and B are not mutually exclusive, since

$$P(A \cap B) \neq 0.$$

Example 10.

Three baseball teams A, B and C reach the final round. Past performances indicate that A and B are evenly matched but team C has a 75% chance of beating either of the others. Lots are drawn to see which pair of teams play the first match. What is the probability that A wins the first match?

Solution:



P(A wins) = P(A plays B and wins OR A plays C and wins)

$= \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{4}$

$= \frac{1}{4}$

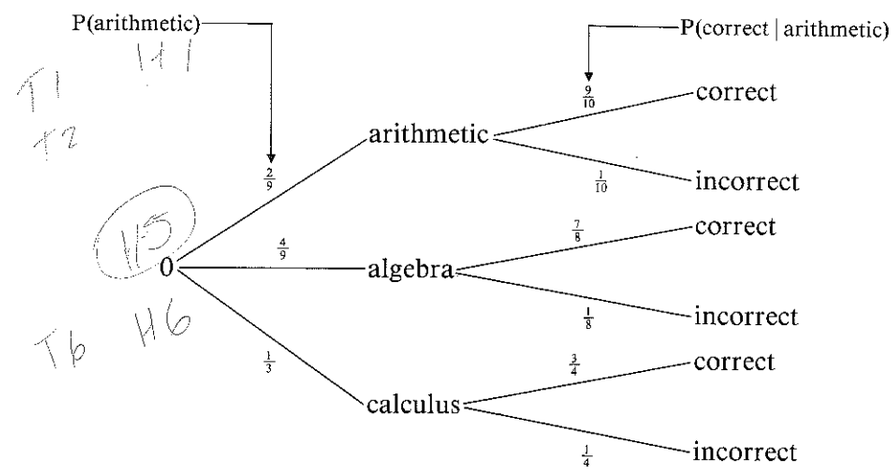
Example 11.

A student elects to solve a problem about compound interest using either arithmetic methods, algebraic methods or calculus methods with corresponding probabilities of $\frac{2}{9}$, $\frac{4}{9}$ and $\frac{1}{3}$. With each of these methods, the respective probabilities of obtaining the correct answer are $\frac{9}{10}$, $\frac{7}{8}$ and $\frac{3}{4}$. Determine the probability that the student:

a. obtains the correct answer.

b. used algebra, if the answer is known to be incorrect.

Solution:



a. P(correct) = P(choosing arithmetic and being correct OR choosing algebra and being correct)

$= \frac{2}{9} \times \frac{9}{10} + \frac{4}{9} \times \frac{7}{8} + \frac{1}{3} \times \frac{3}{4}$

$= \frac{2}{10} + \frac{7}{18} + \frac{1}{4}$

$= \frac{151}{180}$

b. $P(\text{algebra} \mid \text{incorrect}) = \frac{P(\text{algebra and incorrect})}{P(\text{incorrect})}$

$= \frac{\frac{4}{9} \times \frac{1}{8}}{1 - \frac{151}{180}}$

$= \frac{10}{29}$

Exercise 7C: The Addition Law for Probabilities

1. A digit is selected at random from the set D = {1,2,3,4,5,6,7,8,9}. Determine the probability that the digit selected is odd or is less than 5.

2. A coin and a die are tossed simultaneously. Determine the probability of:
   a. a head or a five. b. a tail or a three.

3. A and B are two events in a probability space, S. If S = {a,b,c,d,e,f,g}, A = {a,b,c} and B = {b,c,d,e,f} and we assume that each sample point has an equal probability of being selected, determine:
   a. $P(A \cap B)$. b. $P(A \cup B)$.

4. A and B are two events such that $P(B) = 0{\cdot}4$, $P(A \cap B) = 0{\cdot}32$ and $P(A \cup B) = 0{\cdot}86$. Find:

a. $P(A)$. b. $P(B \mid A)$.

5. A coin is tossed and a card is drawn from a standard pack of 52 cards. Find the probability of getting:

a. a head and a king. b. a head or a king.

6. If A and B are two mutually exclusive events where $P(A) = 0{\cdot}48$ and $P(B) = 0{\cdot}36$, determine:

a. $P(A \cup B)$. b. $P(A \cap B)$. c. $P(A')$. d. $P(B')$. e. $P(A \cap B')$.

Note: A′ is the complement of A.

7. An integer is chosen at random from the first twenty-five positive integers. Determine the probability that the integer chosen is:

a. divisible by 2 and 3. b. divisible by 2 or 3.

8. A class contains thirty students, 12 boys and 18 girls. One third of the boys and one third of the girls have blue eyes. Determine the probability that a student selected at random is a girl or has blue eyes.

9. If $P(R) = \frac{1}{3}$, $P(S) = \frac{1}{2}$ and $P(R \cap S) = \frac{1}{4}$, determine:

a. $P(R \mid S)$. b. $P(S \mid R)$. c. $P(R \cup S)$.

10. A classroom teacher, at the commencement of her Maths activity lesson asked her 32 students two questions.

Question 1: How many people require a compass? Fifteen students required a compass.

Question 2: How many people require a protractor? Twenty students required a protractor.

Four students in the class had their own compass and protractor. If a student was selected at random from this class, what is the probability that the student required:

a. a compass only? b. a protractor only?

c. a compass and a protractor? d. a compass or a protractor?

11. Two playing cards are selected from a standard pack of 52 cards. The first card selected is replaced before the second card is selected. Using the Addition Law of Probability, determine the probability that at least one of the cards selected is a diamond.

12. A and B are two events such that $P(A) = 0{\cdot}7$, $P(B) = 0{\cdot}3$ and $P(A \cup B) = 0{\cdot}9$. Find:

a. $P(A \cap B)$. b. $P(A \mid B)$. c. $P(B \mid A)$.

13. A standard pack of 52 cards is shuffled and a card is drawn at random. Determine the probability that the card drawn is:

a. a ten or a red card. b. a king or a club.

c. a court card or a black card.

14. If M and N are two independent events such that $P(M) = 0{\cdot}85$ and $P(M \cup N) = 0{\cdot}95$, determine $P(N)$.

15. Two dice are rolled. Determine the probability of:

a. a 3 with the first die and a 4 with the second die.
b. a 3 and a 4.

16. A card is selected at random from a standard pack of 52 cards. Determine the probability that the card drawn is:

a. either less than five or a spade.
b. neither less than five nor a spade.

An ace counts as "one" in this problem.

17. Two cards are drawn in succession from a standard pack of 52 cards. Determine the probability of:

a. a jack first, followed by a seven.
b. a jack and a seven.

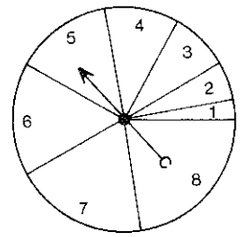
The first card is not replaced before the second is drawn.

18. A box contains 3 red, 2 blue and 4 black marbles. Two marbles are drawn in succession. The first marble drawn is not replaced. Find the probability that:

a. the first marble is red.
b. both marbles are red.
c. the first marble is red and the second is black.
d. one marble is black and the other is blue.
e. at least one marble is red.
f. neither marble is black.
g. a blue marble is drawn last.
h. the marbles differ in colour.

19. The probability of buying a television set which has a faulty picture tube is 0·08 and independently, the probability of one having a faulty colour control is 0·12. Determine the probability of purchasing a television set which has at least one of these faults.

20. The size of the angle in each sector in this circle is directly proportional to the number in it.



The pointer is spun.

a. Determine the probability that the pointer will stop on:
   i. 5. ii. 5 or 8.

The pointer is spun twice.

b. Determine the probability that the pointer will stop on 5 or 8 each time.

21. In a year of high summer rainfall a wheat crop has a probability of $\frac{1}{4}$ of being affected by fungus. In a year when the summer rainfall is not high, the probability is reduced to $\frac{1}{9}$. High summer rainfall occurs, on average, one year in ten.

Let H represent the random event: "high summer rainfall".
Let F represent the random event: "affected by fungus".

Calculate:

a. $P(F \cap H)$. b. $P(H' \mid F)$. c. $P(H \mid F')$.

22. The chance of receiving a payout from poker machine P is $\frac{1}{8}$ and from poker machine Q is $\frac{1}{12}$. Determine the probability that the next person to play these machines has a win on the first attempt. Assume that this person is just as likely to choose machine P as machine Q.

23. To commence a darts game each player shoots for the bulls-eye. Closest to the "bull" starts the game. The probability that Jack hits the bulls-eye is 0·75 and that Peter hits the bulls-eye is 0·85. Determine the probability that:

a. both players hit the bulls-eye. b. Jack or Peter hits the bulls-eye.

24. In a recent Gallup Poll it was found that 3 out of 5 people were in favour of a certain proposal, 1 out of 5 was against, and 1 out of 5 was undecided. Imagine three randomly selected people being interviewed by the poll taker. Determine the probability that:

a. the first person was in favour of the proposal or was undecided.
b. the first person was in favour of the proposal and the other two were against.
c. one of them was in favour of the proposal and the other two were against.

25. A study was conducted on 100 motorists who committed a traffic violation whilst entering a city. The traffic violations were categorised as either "not serious" or "serious". The time of the day was recorded as either "morning" or "afternoon". A summary of the results of the survey is recorded in the table:

	Morning	**Afternoon**
Not serious:	42	24
Serious:	28	6

If one of these 100 motorists was selected at random, what is the probability that this motorist committed:

a. a serious traffic violation?
b. a serious traffic violation during the morning?
c. a not-serious traffic violation or was guilty of a traffic offence in the morning?

26. The owners of a large passenger liner employ 300 stewards. The following table summarises the types of stewards and the length of service with the shipping line.

	Cabin steward	**Dining-room steward**	**Drinks steward**
Less than one year:	90	20	20
1 to 4 years:	60	20	10
More than 4 years:	40	30	10

If a steward is selected at random, what is the probability that:

a. he is a cabin steward?
b. he has less than one year's service with the shipping line?
c. he is a dining-room steward or has more than 4 years' service?
d. he is a drinks steward or has between 1 to 4 years' service?
e. he does not have less than 1 year's service?

27. A Malaysian student studying in Perth, Australia, is ten times more likely to go overseas after exams than to stay in Perth. However, he is three times more likely to go to the eastern states of Australia than to go overseas. What is the probability that:

a. he will stay in Perth after the exams?
b. he will stay in Australia after the exams?

Of the Malaysian students who did not stay in Perth,

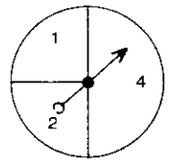
c. what proportion could you expect to have gone overseas?

28. A dart player has three darts to throw at the bulls-eye. He stops when (and if) he hits the bulls-eye. The probability that he hits it with his first dart is $\frac{1}{50}$. If he misses with the first dart, the probability that he hits it with the second dart is $\frac{1}{35}$. If he misses with the first two darts, the probability that he hits it with the third is $\frac{1}{17}$. What is the probability that he hits the bulls-eye?

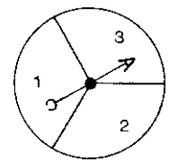
29. A farmer driving home during shearing time has two possible routes: the left route or the right route. The probability of being slowed down by sheep on the left route is $\frac{2}{3}$ and on the right route is $\frac{2}{5}$. Determine the probability of not being slowed down in reaching home if the choice of routes is random.

30. A rocket is designed to put a satellite into orbit. It is a three-stage rocket and all three stages must "fire" for it to attain the correct orbit. The probabilities that the three stages "fire" are 0·95, 0·9 and 0·8, respectively. What is the probability that it does not attain the correct orbit?

31. The pointers on these two spinners are spun simultaneously.



A.



B.

Determine the probability of getting:

a. a 2 on A.
b. a 2 on B.
c. a 2 on A and a 2 on B.
d. a 2 on A or a 2 on B.
e. the same number on each spinner.
f. a larger number on A than on B.

If the sum of the numbers was four,

g. what is the probability that A showed 2 and B showed 2?

32. An ordinary six-sided die is tossed four times. The die is assumed to be fair. Calculate the probabilities of the following events. Express your answers in fraction form.

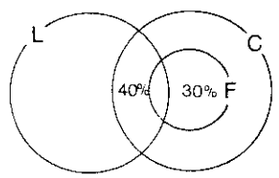
A: "the numbers 2, 3, 4 and 5 in that order."
B: "the numbers 2, 3, 4 and 5 in any order."
C: "the four numbers are all sixes."
D: "the four numbers are all the same."
E: "the numbers are 6, 6, 6 and 3 in any order."
F: "there are three numbers the same and one different."

33. A fair coin is tossed repeatedly. What is the probability of getting the *second* head:

a. on the second toss?
b. on the third toss?
c. on the fourth toss?
d. on the *n*th toss?

34. Three 10-cent coins are placed in a container. One of the coins has two heads. A coin is selected at random and tossed twice. A head results from each toss. Determine the probability that the coin tossed was the two-headed coin.

35. Customers book into a garage for a lubrication, (L), or an oil change, (C). The mechanic will only instal new oil filters, (F), for customers who request an oil change. Fifty per cent of all customers have a new filter installed. Sixty per cent have a lubrication and an oil change, and of these, one-third buy a new oil filter. Only 4% of customers request a lubrication only. Complete the Venn diagram.



Hence, determine:

a. the probability that a randomly selected customer has an oil change only.
b. the probability that a customer who has an oil change also has a lubrication.
c. whether L and C are independent events.
d. whether L and F are independent events.

36. Bag A contains 2 black jellybeans and 4 red ones. Bag B contains 3 black jellybeans and 9 red ones. One of these bags is chosen at random. Then, a jellybean is removed without looking. What is the probability that:

a. Bag A was chosen and a black jellybean was removed?
b. Bag B was chosen and a black jellybean was removed?
c. a black jellybean was removed?

After the first jellybean was removed a second one was chosen from the other bag.

d. What is the probability that they were both black?

37. Alf and Betty are both divers. The probability that Alf gets a high score on his first dive is 0·7, while the probability that Betty gets a high score on her first dive is 0·9. The probability that Alf gets a high score on each of his two dives is 0·56. Find the probability that:

a. both Alf and Betty get a high score on their first dive.
b. Alf or Betty or both get a high score on their first dive.
c. Alf gets a high score for his second dive if he gets a high score for his first dive.

38. Two random variables X and Y are independent. X can take any of the values 0, 1, 2, 5, or 10 and Y can take any one of the values 0, 1, 2, 3, or 4. The associated probabilities are given by the distributions:

x	:	0	1	2	5	10
P(X = x)	:	0·3	0·2	0·3	0·1	0·1

y	:	0	1	2	3	4
P(Y = y)	:	0·5	0·1	0·1	0·1	0·2

Find:

a. $P(X \geqslant 2)$
b. $P(X = 5 \cap Y = 4)$
c. $P(X = 5 \cup Y = 4)$
d. $P(X + Y = 14)$
e. $P(X - Y = 2)$

39. A calculator is faulty. The reciprocal button $\boxed{1/x}$ works correctly only 70% of the time. When it is not working, the number which is in the display remains in the display after the reciprocal button has been pressed. A *16* is entered on the display. What is the probability that:

a. the display shows *0·0625* after pressing the reciprocal button once?
b. the display shows *0·0625* after pressing the reciprocal button twice?

40. Three-digit numbers are formed from the set of digits {1,2,3,5}. The numbers formed must be greater than 100 and less than 400. No digit can be repeated. If one of these three-digit numbers is selected at random, what is the probability that:

a. the number is odd?
b. the number is not divisible by 3?
c. the number is divisible by 3 or 5?

41. In a new clothing factory 70% of the machinists are inexperienced, 20% have had a little experience, and the rest are experts. The manager calculates that (respectively) 20%, 50%, and 90% of the garments on which they are working will pass the first inspection.

a. Calculate the probability that a randomly selected garment was made by an inexperienced machinist and did not pass the first inspection.

Determine the probability that a randomly selected machinist:

b. will be an expert or will have had a little experience.
c. will be an inexperienced machinist whose next garment will pass the first inspection.
d. will have his or her next garment pass the first inspection.

If the last garment did not pass the last inspection,

e. what is the probability that the machinist was either an expert or had had little experience?

42. When a regular tetrahedron is dropped on to the floor, three of its four faces are visible when viewed from above.



P and Q are each regular tetrahedra. P has three faces painted red and one white; Q has one face painted blue and three white. They are thrown on to the floor. Determine the probability of being able to see:

a. three red and three white faces.
b. four white faces.
c. exactly two faces of each colour.

43. There are three bags set out in front of a blindfolded person. Bag A contains 6 red, 4 green and 2 yellow discs; bag B contains 2 red, 2 green and 5 yellow discs; and bag C contains 3 red, 2 green and 3 yellow discs. The bags are randomly arranged and the person is led to the table and required to select a disc at random.

a. Determine the probability that the person:
   i. selected a red disc from bag A. ii. selected a yellow disc.
b. If the person was asked to select any two discs at random, find the probability that they are both red.
c. The person was handed bag A and bag B and asked to select one disc from each. Find the probability that a red and a green disc were selected.

44. A post-graduate student has applied for two scholarships: one at an American university and the other at an English university. He reckons that the probability of being offered at least one of them is 0·8. His professor commented, "You stand a better chance from an American university. Anyhow, there's a 30% chance that you will be offered both because they decide independently."

a. By taking into account the student's reckoning and his professor's comments, find the probability that the student will be offered a scholarship in:
   i. the American university. ii. the English university.
b. The student took a scholarship to an English university. What is the probability that he was not offered one from America?

45. A standard pack of 52 cards is divided into separate and equal piles. One of the piles contains 8 hearts, 5 spades, 7 clubs and 6 diamonds. A bystander is asked to select two cards. He is told he can select one from each pile or two from the same pile. What is the probability that the person will select two spades? Give your answer correct to 4 decimal places.

46. A boy challenges his father to a table tennis match. They decide to play up to nine games and the overall winner is the first one to win five games. The boy calculates that the odds against beating his father are 2:1 in any game.

a. What is the probability that the boy will win the first game?
b. What is the probability that the boy will win the first four games and the father will win the fifth game?

After five games the score was: Boy, 4; Father, 1.

c. What is the probability associated with this score?
d. What is the probability that the boy:
   i. wins the sixth game?
   ii. loses the sixth but wins the seventh game?
   iii. loses the next two games but wins the eighth?
   iv. loses the next three games but wins the last?
   v. goes on from the score (Boy, 4; Father, 1) to win the match?

47. Last January, in London, 18 out of 31 days were below freezing. Using these statistics as a forecast for next January, calculate the probability that in London next year:

a. January 8th will be below freezing.
b. January 8th and January 20th will both be below freezing.
c. January 8th will not be below freezing.
d. neither January 8th nor January 20th will be below freezing.
e. January 8th or January 20th or both days will be below freezing.

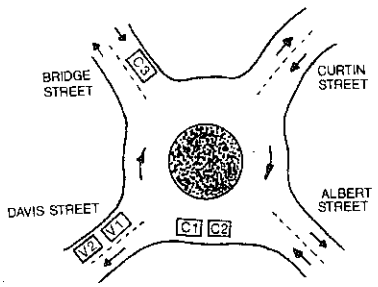
There are four Sundays in next year's January.

f. What is the probability that exactly one of these will be below freezing?

48. In a game of quoits the player is required to throw five rope rings in succession on to the spike. The probability that John is successful with his first attempt in a game of quoits is 0·6. The probability of a success on his second, third, fourth, and fifth attempts are 0·65, 0·70, 0·75 and 0·8, respectively. Determine the probability that John is successful:

a. on his first two attempts.
b. on his first three attempts.
c. with at least four of his shots.

49. An aerial photograph is taken of a traffic roundabout.



There are two vans, V_1 and V_2, and a car, C_3, about to enter the roundabout. Two cars, C_1 and C_2, are on the roundabout. There are several assumptions concerning the behaviour of the vehicles on the roundabout:

A: their exits are chosen independently and are equiprobable;
B: vehicles do not enter and exit via the same street;
C: vehicles do not go past their chosen exit.

Take into account these assumptions and determine the probability that:

a. van V_1 exits via Albert Street.
b. both vans exit via Albert Street.
c. one van exits via Albert Street and the other via Curtin Street.
d. the three cars exit via the same street.
e. at least one vehicle exits via Bridge Street.
f. all vehicles exited via Curtin Street, if it was known that all five used the same exit.

It is not known from which streets the cars C_1 and C_2 entered the roundabout.

g. List all possible entry street and exit street pairs. For example, (B,A) represents "enter via Bridge Street and exit via Albert Street".

Cross out all of the pairs which violate the assumptions for cars C_1 and C_2.

h. What is the probability that both cars entered via Curtin Street and will exit via Bridge Street?

50. At the singles tennis championships four people A, B, C and D reach the semi-finals. Two semi-final games will be played and the two winners will play in the final. The names are drawn from a hat to decide who plays whom in the semi-finals.

a. What is the probability that A plays against B in a semi-final?

Hint: List all the possibilities.

C is the top seed (the best player) and A is the poorest. The percentages of wins in previous matches are:

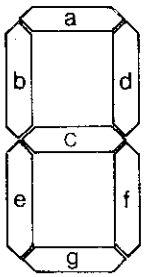
B wins against A; 50%	C wins against A; 90%
D wins against A; 60%	C wins against B; 70%
D wins against B; 40%	D wins against C; 30%

b. What is the probability that A plays B in a semi-final and A wins?
c. What is the probability that A wins a semi-final match?
d. What is the probability that A plays C in the final match and wins?

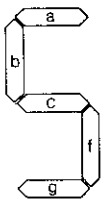
Chapter 8:

Probability Using Counting Techniques

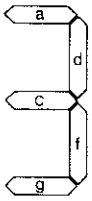
The digit *8* on a calculator is composed of seven separate liquid crystals.



Because the batteries are low, two randomly determined crystals fail to operate. If the *8* button is pressed, determine the probability that the digit displayed is a five. There are many possibilities.



d and e fail



b and e fail

Each possibility can be represented by a combination of two letters. There are $\binom{7}{2} = 21$ possible combinations. Only one combination (d with e) makes a *5*.

$$\text{Hence,} \quad P(5) = \frac{1}{21}$$

Example 1.

In the Australian Soccer Pools gambling game, a player must choose 6 numbers from the following array of 36 numbers.





Six "correct" numbers are determined by the results of soccer matches. A seventh number, called the supplementary number, is determined similarly. To win a prize a player must match 4, 5, or 6 of these numbers according to the following list:

1st division prize:	6 correct numbers.
2nd division prize:	5 correct numbers + supplementary.
3rd division prize:	5 correct numbers.
4th division prize:	4 correct numbers.
5th division prize:	3 correct numbers + supplementary.

A player can win a prize in only one division.

a. How many different combinations of 6 numbers are possible?

b. A player selects 6 numbers by guessing. Calculate the probability of winning a 4th division prize.

Solution:

a. There are $\binom{36}{6} = 1\,947\,792$ different combinations of six numbers.

b. To win a 4th division prize, exactly 4 of the 6 correct numbers are required. The other 2 numbers must come from the remaining 30.

	Correct numbers		Others
	6		30
	↓		↓
4th division requires:	4	AND	2

Therefore, the number of 4th division possibilities is:

$$\binom{6}{4}\binom{30}{2} = 6\,525$$

Thus the probability of a 4th division prize is:

$$\frac{\binom{6}{4}\binom{30}{2}}{\binom{36}{6}} = \frac{6\,525}{1\,947\,792} \approx 1 \text{ chance in } 300$$

Example 2.

A computer programme prints, in random order, all of the possible permutations of the letters *S*, *P*, *T*, and *O*.

a. What is the probability that the first permutation printed is a word from the list:

SPOT, *STOP*, *TOPS*, *OPTS*, *POTS*, *POST*?

b. What is the probability that the first permutation starts or finishes with *S*?

Solution:

a. There are 4! = 24 possible permutations but only 6 in the list.

$$\therefore \quad P(\text{first permutation is from list}) = \frac{6}{24} = \frac{1}{4}$$

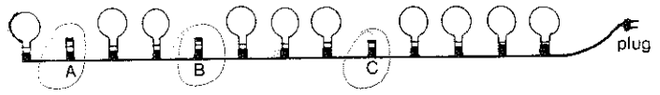
b. $$P(\text{first permutation starts or finishes with S}) = \frac{1.3.2.1 + 3.2.1.1}{4!}$$

$$= \frac{6 + 6}{24}$$

$$= \frac{1}{2}$$

Example 3.

A set of Christmas-tree lights has three empty sockets.



A householder has a supply of 4 red, 3 green and 2 yellow globes. Without looking, he picks up three globes. What is the probability that he chose:

a. all red globes?

b. all green globes?

c. three globes of the same colour?

d. three globes not all the same colour?

e. each globe of a different colour?

Solution:

a.

	Red	Green	Yellow
Available:	4	3	2
	↓	↓	↓
Require:	3	0	0

$$P(\text{all red}) = \frac{\binom{4}{3}\binom{3}{0}\binom{2}{0}}{\binom{9}{3}} = \frac{\binom{4}{3}}{\binom{9}{3}} = \frac{1}{21}$$

b.
$$P(\text{all green}) = \frac{\binom{4}{0}\binom{3}{3}\binom{2}{0}}{\binom{9}{3}} = \frac{\binom{3}{3}}{\binom{9}{3}} = \frac{1}{84}$$

c. P(all same colour) = P(all red OR all green) $= \frac{1}{21} + \frac{1}{84} = \frac{5}{84}$

d. P(not all the same colour) $= 1 - \frac{5}{84} = \frac{79}{84}$

e.

	Red	Green	Yellow
Available:	4	3	2
	↓	↓	↓
Require:	1	1	1

$$\text{P(each of a different colour)} = \frac{\binom{4}{1}\binom{3}{1}\binom{2}{1}}{\binom{9}{3}} = \frac{2}{7}$$

Example 4.

A bridge hand consists of 13 cards dealt from a standard pack of 52 cards. Determine the probability that a player is dealt:

a. 5 hearts, 4 clubs, 3 spades and 1 diamond.

b. 5 of one suit, 4 from another suit, 3 of a third suit and 1 of the remaining suit.

Solution:

a.

	Hearts	Clubs	Spades	Diamonds
Available:	13	13	13	13
	↓	↓	↓	↓
Require:	5	4	3	1

$$\text{P(5H, 4C, 3S and 1D)} = \frac{\binom{13}{5}\binom{13}{4}\binom{13}{3}\binom{13}{1}}{\binom{52}{13}} \approx 0{\cdot}00539 \approx 1 \text{ chance in } 186$$

b. Here are some possibilities:

	Hearts	Clubs	Spades	Diamonds
Available:	13	13	13	13
	↓	↓	↓	↓
Require:	5	4	3	1

OR

	Clubs	Hearts	Diamonds	Spades
Available:	13	13	13	13
	↓	↓	↓	↓
Require:	5	4	3	1

OR

etc.

The only difference between these possibilities is the arrangements of the suits. There are $4! = 24$ different arrangements.

P(5 of 1 suit, 4 of another, 3 of a third, 1 of the remaining)

$$= \frac{\binom{13}{5}\binom{13}{4}\binom{13}{3}\binom{13}{1}}{\binom{52}{13}} \times 4! \approx 0{\cdot}13 \approx 1 \text{ chance in } 8$$

Exercise 8: Probability Using Counting Techniques

1. Four children take turns to play a new computer game. Their father puts their names into a hat and makes a draw to determine who goes first, who goes second, who goes third and who goes fourth. What is the probability that they take their turns in alphabetical order?

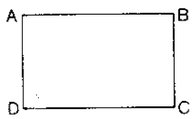
2. A small child is given five "letter cards", namely, *A*, *P*, *C*, *S* and *E*. The child arranges them in a line. What is the probability that the child makes the word *PACES*?

3. There are four filters (or spreaders) in the aerator of a tap. After cleaning, they are to be put back. If they are not put back in the correct order the water squirts out sideways. If they are put back in a random order, what is the probability that the water squirts out sideways the next time the tap is turned on?

4. The letters of the word *PERTH* are arranged in a random order. Calculate the probability that they will not be in alphabetical order.

5. Two war novels and three other novels are arranged at random on a shelf. What is the probability that the two war novels occupy the left and right hand ends?

6.

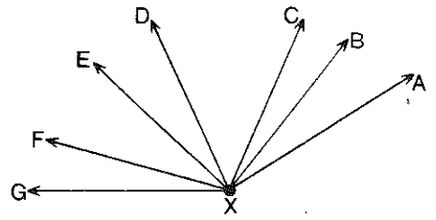


If two vertices of this rectangle are chosen at random, what is the probability that they form:

a. a side? b. a diagonal?

7. Five different veteran cars line up one behind the other at a motor show. What is the probability that the T-model Ford and the M.G. sports car occupy the end positions?

8. The diagram shows seven rays with a common end point X.



a. How many different angles less than 180° are determined by the rays in the diagram?

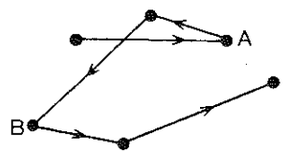
Two rays are selected at random. Find the probability that the rays determine:

b. angle EXB.

c. angle EXB or angle EXA.

9. Seven men and eight women arrange themselves in a row, taking up positions at random. What is the probability that the men and women occupy alternate positions? Express your answer as a fraction.

10. The diagram shows one straight-line pathway connecting six dots.



A pathway must start at one dot, end at another and pass through every dot exactly once.

a. How many different pathways are possible?

If a pathway is chosen randomly, what is the probability that:

b. it starts at A and finishes at B?

c. A is followed immediately by B somewhere in the pathway?

11. To win a first-division prize in Lotto, a player must nominate six correct numbers out of forty. A group of ten office workers decide to form a syndicate and to enter a Lotto System-10 coupon. In Lotto System-10, ten numbers are crossed instead of six as with a standard entry.

a. Show that the probability of winning a first-division prize with System-10 is 210 times greater than with a single entry of six numbers.

b. If a single entry for Lotto costs 25 cents, how much should it cost to enter a Lotto System-10 coupon?

12. The feature race at a greyhound meeting attracts eight starters. A punter decides to place six quinella bets. To win a quinella he must predict the first two placegetters in either order. His selections are 1 and 4; 2 and 4; 3 and 4; 2 and 3; 3 and 5; 2 and 5. If each of the greyhounds is equally likely to win, determine the probability that the punter held a winning quinella ticket.

13. a. How many different arrangements of four letters can be made from the following letters: *ABSENT*?

b. If a 4-letter arrangement is made in a random fashion, what is the probability that:

i. it is *SENT*? ii. it starts with *A*? iii. it contains no vowels?

14. Two different letters are chosen at random from the alphabet. Determine the probability that they are:

a. both vowels. b. both consonants.

c. one vowel and one consonant. d. *T* and a vowel.

15.

HOUSE OF REPRESENTATIVES

Mark your ballot paper 1, 2, 3, and 4 in order of preference.

THOMAS, Bert	Liberal	☐
ALLEN, Peter	N.D.P.	☐
GREGORY, Clive	Labor	☐
LEWIS, Anne	Independent	☐

a. What is the probability that a ballot paper which has been filled in randomly gives Thomas 1, Allen 3, Gregory 4 and Lewis 2?

b. What is the probability that Lewis received second preference from a person who was known to give the Labor Party his first preference?

16. In a police line-up of eight men, 7 are selected at random from the street. This sample includes two brothers. Find the probability that the two brothers stand side by side in the line-up.

17. A relay team of four is to be selected from a group of six girls: Diana, Beverley, Glenda, Anne-Marie, Leanne and Kate. What is the probability that Leanne starts and Anne-Marie finishes?

18. In a class there are twenty-four students: 12 boys and 12 girls. Five are grade A, seven are grade B, and twelve are grade C. There are four girls in each grade. Six students are chosen at random. What is the probability that:

a. there are two from each grade?

b. there are equal numbers of boys and girls?

c. there are three boys from grade C?

19. A jury of twelve is to be selected from 11 men and 19 women. The jurors are chosen at random. What is the probability that, on this jury:

a. there are 6 men and 6 women?

b. two particular men are chosen?

c. the jury consists entirely of women?

If the jury consisted of 6 men and 6 women,

d. what is the probability that two particular men were chosen?

Do not simplify your answers.

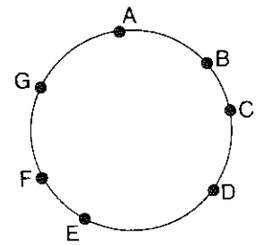
20. Three married couples wish to attend a concert. They book a row of six seats. When they reach the concert they choose their seats at random. Calculate the probability that:

a. each woman sits next to her husband.
b. the three men sit together and the three women sit together.
c. the three women sit together.

21. From 5 mathematicians and 6 physicists, a committee of 3 mathematicians and 2 physicists is to be formed. Determine the probability that:

a. two particular mathematicians are on the committee.
b. one particular physicist is not on the committee.

22. Seven points A, B, C, D, E, F and G lie on the circumference of a circle.



a. How many different triangles (which use the points as vertices) may be formed?
b. One of these triangles is selected at random. What is the probability that:
   i. the triangle is AGC?
   ii. the triangle is not AGC?
   iii. the triangle does not contain the point A?

23. A lawn mower repair store takes in thirteen mowers. Nine of these mowers require only minor repairs and four require major repairs. The service manager makes a list of the mowers to determine the order in which they are to be repaired.

a. How many different lists are possible if:
   i. any order is satisfactory?
   ii. the four major repairs must be at the bottom of the list?

   Express your answers in factorials.

To help train a new repair man, four mowers are selected at random.

b. How many different selections of four mowers can be made?
c. What is the probability that:
   i. none of the four mowers requires major repairs?
   ii. one requires major repairs and the other three require minor repairs?

24. There are eight different musical notes in a certain scale:

doh, *re*, *me*, *fah*, *soh*, *la*, *te*, *doh'*

A triad is a 3-note chord: that is, three notes played simultaneously.

a. How many different triads are possible?

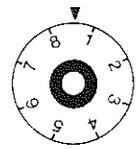
If three different notes are selected randomly, find the probability that:

b. the triad is *doh*, *me* and *soh*.
c. the triad contains *fah*.
d. the triad contains at least two consecutive notes.

25. A householder buys a set of coloured party lights. There are ten globe sockets on the electrical cable. The buyer is provided with sixteen coloured globes: 5 blue, 3 red, 4 yellow, 3 green and 1 orange. The householder chooses the globes randomly. Determine the probability that the householder selects:

a. 3 blue, 3 red, 2 yellow, 1 green and 1 orange globe.
b. only red, yellow and green globes.

26. A safe has a combination lock with eight numbers.



To open the safe, four *different* numbers must be presented to the pointer in the correct order. After the first number, the rotations must alternate between clockwise *C* and anticlockwise *A*, starting with clockwise. The "combination" which opens the safe is *4*; C, *7*; *A*, *8*; *C*, *5*. A burglar wishes to open the safe but has no idea of the "combination", except that he knows it is a four-number lock with alternating rotations. He does not know the direction of the first rotation.

a. How many "combinations" are possible?
b. What is the probability that the burglar gets the safe open on the first try?
c. Another burglar, who is familiar with the type of safe, knows only that the combination uses the numbers *4*, *7*, *5* and *8*. What is the probability that he gets it open on the first try?

27. There are twelve people in an audition for two singing awards. Three are men. The twelve draw lots to decide the order in which they will sing.

a. How many different orders are possible?
b. How many different orders have all of the men singing after all of the women?
c. What is the probability that all of the men sing after all of the women?

There is a "short list" of 4 women and 2 men for the two awards. The judges cannot agree and decide to draw two names out of a hat. Determine the probability that:

d. no men receive awards.
e. no women receive awards.
f. at least one man receives an award.

28. The letters of the word *PROBABILITY* are arranged at random. Find the probability that:

a. the *B*s are together.

b. the *B*s are separated.

29. Three cards are drawn from a well-shuffled pack of 52 cards. Find the probability that:

a. two are clubs and one is a diamond.

b. all cards are of one suit.

c. two are queens and one is a jack.

d. at least two are kings.

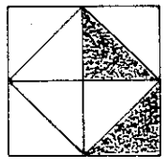
30. The Perth Cup has a field of eighteen horses, 12 of which are local horses, 5 are from the eastern states and the other horse is from New Zealand.

a. What is the probability that the first three placegetters are horses from the eastern states?

b. What is the probability the first three placegetters are horses from Western Australia?

c. Determine the probability that the quinella (1st and 2nd places) is filled by local horses.

31. A child selects two small triangles at random from the square and colours them.



a. What is the probability that the two shaded portions:

i. form a square?

ii. form a single triangle?

iii. form neither a square nor a single triangle?

Another child selects two small triangles with a common side.

b. What is the probability that the coloured portions, when taken together, form a square or a triangle?

32. A relay team of four runners is to be chosen from a training squad of eight runners. Answer the following questions on the assumption that all eight are of equal ability and have the same chance of being selected.

a. How many different teams could be selected?

b. Suppose that you are one of the eight.

i. How many of the different teams could you be in?

ii. What is the probability that you will be chosen?

c. Suppose that your friend is also one of the eight. What is the probability that you AND your friend are in the team?

d. What is the probability of you OR your friend being in the team?

e. Given that the team chosen is one that your friend is in, what is the probability that you will be in it also?

f. Once the team has been chosen, how many different running orders are possible?

33. Determine the probability that a poker hand of 5 cards contains exactly 3 kings or exactly 4 jacks.

34. A boy has the following coins: 1c; 2c; 5c; 10c; 20c; 50c; \$1. He selects three coins at random. What is the probability that:

a. the \$1 coin is selected?

b. the sum of money is 80 cents?

c. the sum of money is 35 cents or less?

d. the \$1 coin or the 50 cent coin is included?

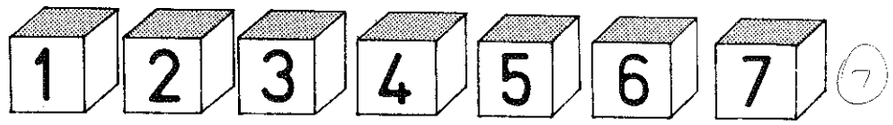
35. A set X is determined by choosing two numbers at random from A = {1,2,3,4}. A set Y is determined by choosing two numbers at random from B = {0,1,2,3,5,10}. Find the probability that:

a. $X \cup Y = \{1,2,5,10\}$

b. $X \cup Y = \{1,2,10\}$

36. A poker player in a "school" of five, won the "pot" with a full house: 3 kings and 2 jacks. Determine the odds against a player on the next deal being dealt a full house: that is, any three of a kind and any one pair.

37.



A small child is playing with this set of number blocks and arranges three of them in a row. Find the probability that the child put together:

a. an odd number.

b. a number greater than 400.

c. a number greater than 660.

d. a number less than 350.

e. a number greater than 800.

38. From 8 male and 7 female students, five representatives were selected. If the selections were made randomly, determine the probability that:

a. the representatives were all male.

b. the representatives were all female.

c. there was at least one representative of each sex.

d. there were 3 females and 2 males.

e. there were more females than males.

39. What is the approximate odds against a bridge hand of 13 cards containing:

a. all black or all red cards?

b. all black cards or all hearts?

40. A space team of four is to be chosen from a group of eight astronauts, 5 of whom are men and 3 of whom are women. Calculate the probability that the team selected contains at least two women astronauts.

41. A horse race has a field of twelve starters. A punter wishes to place six Trifecta bets. In a Trifecta the punter must pick the finishing order of the first three horses. His selections are (1,2,4), (1,4,2), (2,4,1), (2,1,4), (4,2,1) and (4,1,2). Determine the odds against a punter holding a winning Trifecta ticket if it is assumed that each horse has an equal chance of winning the race.

42. Six men and five women nominate for a ballroom dancing championship. One of the men is married to one of the women. The rest are unmarried. Names are drawn out of a hat to make 5 couples (a man and a woman). Determine the probability that:

a. the married couple is included in the list of 10 people.
b. the married couple dance together.

43. Eight Olympic Games swimmers compete in a heat of the 100 metres women's breaststroke. There are 2 Americans and 2 Japanese in this heat. The lanes are assigned by ballot.

a. In how many different ways can the lanes be assigned?
b. What is the probability that the 2 Americans are together?
c. What is the probability that the 2 Japanese are together?
d. What is the probability that the 2 Americans are together and the 2 Japanese are together?
e. What is the probability that the 2 Americans are together or the 2 Japanese are together?
f. If the 2 Americans were together, what is the probability that the 2 Japanese were also together?

44. A women's swimming squad has seven members. The first, second, third and fourth swimmers for a relay team are to be nominated.

a. How many different relay teams are possible?

If the nominations are done randomly, what is the probability that:

b. the oldest member of the squad swims first?
c. the oldest member of the squad is not nominated?
d. the oldest member of the squad swims first or fourth?

If it is known that the oldest member of the squad was nominated,

e. what is the probability that she swam first or fourth?

45. Bob, Chung, Roger, Avril and Anne share a house. Each week three people are rostered for cleaning, cooking and shopping. For the first roster they draw names from a hat and allocate the jobs.

a. How many different sets of three names are possible?
b. How many different rosters are possible?

Determine the probability that, on this roster,

c. Chung does the cleaning, Avril does the cooking, and Anne does the shopping.
d. Avril does the cleaning and Anne does the cooking.
e. Roger is left off.
f. Roger does the shopping or is left off.

46. A plant disease expert wishes to sample wheat plants in a rectangular field. She divides the field into seventy squares, so that there are 7 rows and 10 columns of squares. She wishes to take a random sample of ten squares.

a. How many different random samples are possible?
b. What is the probability that there is one square in every column?

Do not simplify either answer.

47. Determine the odds against a 5-card poker hand containing:

a. four jacks. b. any four of a kind.

48. A student's locker contains four science books, three social studies books and one mathematics book. Two books are taken out at random. Find the probability that:

a. two social studies books were taken.
b. two science books were taken.
c. a mathematics book was taken followed by a science book.
d. a science book was taken followed by a mathematics book.

49. There are twelve members of a tee-ball squad, 6 boys and 6 girls. The coach must choose nine for the team. Imagine that the coach chooses his team randomly. Find the probability that he chooses:

a. all of the boys.
b. all of the girls.
c. 4 boys and 5 girls.

Only one of the girls has a brother in the squad.

d. What is the probability that the brother and sister were both included in the team?

The coach chose 4 boys and 5 girls and wrote down the order in which they were to bat. Imagine that the coach did this randomly. Find the probability that the batting order:

e. starts with the 5 girls.
f. has boys and girls alternating.

50. Peter, Richard, Tom and Alan play the card game, bridge. Richard is Tom's partner and Peter is Alan's partner. Each player is dealt 13 cards from a standard pack of 52 cards.

a. If Tom holds one ace, the ace of spades, what is the probability that his partner has the other three aces?
b. If Alan and Peter have 9 clubs between them, what is the probability that either Richard or Tom has the remaining 4 clubs?

51. A selection of three novels is to be randomly chosen from 4 Western, 3 Science Fiction and 5 Fiction novels. Find the probability that:

a. the three are of different types.
b. the three are the same type.
c. one particular Western is in the selection.

52. Triangles are formed using three of these dots as vertices.



The dots lie in three straight rows. How many different triangles:

a. have a vertex in each row?
b. have 2 vertices in the top row?
c. have 2 vertices in the middle row?
d. have 2 vertices in the bottom row?
e. are there altogether?

A triangle is chosen at random. What is the probability that:

f. it has a vertex in each row?
g. it does not have a vertex in the top row?
h. it has 2 vertices in the middle row, given that it has no vertex in the top row?

53. A jury of twelve is to be selected from 12 men and 5 women. Find the probability that:

a. exactly three women are selected.
b. no more than two women are selected.

54. A table is to seat eight people. Determine the probability that, if 4 men and 4 women seat themselves randomly about the table, men and women alternate.

55. Supermarket trolleys are stacked one behind the other for storage.



If the basket of a trolley is too big, the trolley gets jammed with the one in front of it. However if the basket is too small, it gets jammed with the one behind it. No jamming occurs when trolleys of the correct size are stacked together. In a stack of ten trolleys, eight are the correct size, one is too big and one is too small.

a. How many different arrangements of ten trolleys are possible?

The trolleys are stacked in a random order. Next morning the first ten customers take a trolley each until the stack is depleted.

b. What is the probability that:
   i. no customer has trouble with a jammed trolley?
   ii. one customer has trouble with a jammed trolley?
   iii. two customers have trouble with a jammed trolley?

56. To win a prize at an amusement park, six ping-pong balls are placed into the mouth of a clown's head. The balls exit randomly into any of six available channels. What is the probability that, after the six balls have been used, exactly one channel remains unoccupied?

57. A milk crate has space for 20 bottles. Sixteen bottles of cows' milk and four bottles of goats' milk are randomly placed in the crate. If a customer selected four bottles of milk at random from this crate, find the probability that:

a. all four contained cows' milk.
b. at least one bottle contained goats' milk.
c. all four contained cows' milk, given that the first bottle contained cows' milk.

58. A dentist is to examine eighteen six-year-olds, 10 of them boys. She wishes to select six as a random sample for a research project. Determine the probability that the sample consists of:

a. six boys.
b. three boys and three girls.
c. at least one of each sex.

On average, 25% of six-year-old girls and 20% of six-year-old boys have tooth decay. The dentist's sample consisted of three boys and three girls.

d. What is the probability that at least one child had tooth decay?
e. What is the probability that at least one boy and at least one girl had tooth decay?

In the light of these probabilities, it was decided to take a larger random sample consisting of equal numbers of boys and girls. The dentist wished to be better than 95% sure of getting at least one girl and at least one boy with tooth decay.

f. What is the minimum number of children that should be in the sample?

59. A box contains twelve light bulbs in packets. Five have bayonet fittings and seven have screw fittings. Three are chosen at random and removed from the box. Determine the probability that:

a. all three bulbs have bayonet fittings.
b. the three bulbs include at least one of each kind of fitting.
c. the three bulbs all have the same kind of fitting.
d. all three light bulbs had the same kind of fitting, given that the first two had the same kind of fitting.

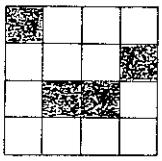
60. Of the first eleven people interviewed by an opinion poll company, 3 were in favour of a certain proposal, 4 were against it, and 4 were non-commital.

a. Determine the probability that the first three people interviewed were against the proposal.

Three people were randomly selected from the eleven. Determine the probability that:

b. all three were against the proposal.
c. one was in favour of the proposal, one was against it, and one was non-committal.
d. at least one was not against the proposal.
e. all three were in favour of the proposal, given that at least one was not against it.

61. Four small squares (1 centimetre by 1 centimetre) are selected at random from this 4-centimetre by 4-centimetre square.



Find the probability that the four squares:

a. form a diagonal.
b. form a 2-centimetre by 2-centimetre square.
c. include at least one square from the top row.
d. do not form a complete row or column or diagonal.
e. include one square in each row and column.

The first square selected was in a corner.

f. What is the probability that none of the others was in a corner?

62. A very large consignment of apples consists of 60% A grade, 30% B grade, and 10% C grade, all mixed together.

a. If 10 apples are chosen at random, what is the probability that:
   i. they are all A grade?
   ii. exactly two are A grade?
   iii. at least one is A grade?
   iv. they are not all the same grade?
b. What is the most likely number of B grade apples in the sample?

Consignments of apples should contain no C grade apples.

c. What is the minimum number of apples which should be inspected so that there are at least nine chances in ten of detecting one or more C grade apples?
d. In the sample of 10 the inspector found four C grade apples. He decided that the consignment did not contain 10% C grade. Do you agree? What is the chance that he is wrong?

63. A baseball team consists of twelve players but only 9 can be on the field at one time. Assume that the players are all of equal ability and can take the field in any position.

a. How many different ways can the coach allocate 9 players to first, second, third and home bases and the five other field positions? $^{12}P_9 = 79833600$

The coach allocates 9 of the twelve players at random. $^{12}P_9$

b. What is the probability that the captain is on first base and the vice-captain is on third base? $1.1.\ ^{10}P_7 = 604800 \therefore P = 604800/79833600 = 0.00768$
c. What is the probability that the captain and the vice-captain occupy two of the four bases?

Four members of the team are left-handers.

d. What is the probability that the four left-handers are selected?
e. Given that the four left-handers take the field, what is the probability that they occupy the bases?

Chapter 9:

The Binomial Distribution

A coin has been damaged and is not flat.



The probability that it comes up heads on a single toss is unknown, say p. The probability that it comes up tails on a single toss is also unknown, say q, which is equal to $1 - p$. What is the probability that, if you toss this coin 3 times, you get:

0 heads? 1 head? 2 heads? 3 heads?

The probability of getting 0 heads is simply, the probability of getting a tail on the first toss, a tail second, and a tail on the last toss. That is, $q \times q \times q = q^3$.

Similarly, the probability of getting 3 heads is $p \times p \times p = p^3$.

Now, to get 1 head we must also get 2 tails. The probability of a head on the first toss, followed by 2 tails is $p \times q \times q = q^2p$.

However, the single head may come anywhere in the sequence. It can occur on the first toss, the second toss or the third toss. Therefore, the probability of getting exactly one head is $3q^2p$. Similarly, the probability of getting 2 heads and 1 tail is $3qp^2$.

We may display these probabilities as a probability distribution.

Number of heads	Probability
0	q^3
1	$3q^2p$
2	$3qp^2$
3	p^3

Note: The expansion of the binomial expression $(q + p)^3$ is $q^3 + 3q^2p + 3qp^2 + p^3$. Each term in the expansion corresponds to a probability and the exponent of the binomial expression corresponds to the number of tosses.

Example 1.

The probability that a damaged coin comes up heads in a single toss is $\frac{4}{7}$.

a. Determine the probability that, if this coin is tossed 3 times the result will be:
   i. 0 heads. ii. 1 head. iii. 2 heads. iv. 3 heads.
   Display your answers on a probability distribution graph.

b. Determine the probability of 0 heads or 1 head or 2 heads or 3 heads.

Solution:

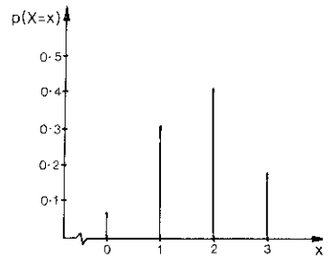
a. Let X represent "the number of heads".

Then, $P(X = 0) = (\frac{3}{7})^3 = \frac{27}{343}$

$P(X = 1) = 3.(\frac{3}{7})^2(\frac{4}{7}) = \frac{108}{343}$

$P(X = 2) = 3.(\frac{3}{7})(\frac{4}{7})^2 = \frac{144}{343}$

$P(X = 3) = (\frac{4}{7})^3 = \frac{64}{343}$



b. $P(X = 0, 1, 2 \text{ or } 3) = \frac{27}{343} + \frac{108}{343} + \frac{144}{343} + \frac{64}{343} = 1$

Example 2.

a. Write down the expansion of $(q + p)^2$. The terms of this expansion correspond to probabilities of tossing a coin n times, where p is the probability of getting a head on a single toss, and $q = 1 - p$.

Determine:

b. the number of tosses, n.
c. the probability of getting 1 head in terms of p and q.
d. the probability of getting 1 or 2 heads in terms of p and q.
e. the probability of getting 1 or 2 heads with a fair coin. Substitute for p and q.

Solution:

a. $(q + p)^2 = q^2 + 2qp + p^2$

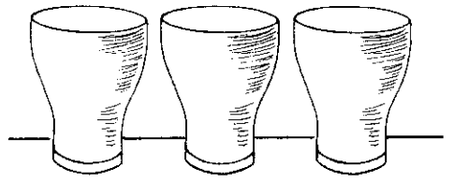
p^2 — p(2 heads)
$2qp$ — p(1 head)
q^2 — p(0 heads)

b. number of tosses, $n = 2$.
c. $P(1 \text{ head}) = 2qp$
d. $P(1 \text{ or } 2 \text{ heads}) = 2qp + p^2$
e. For a fair coin $p = \frac{1}{2}$, $q = \frac{1}{2}$

$$\therefore \quad P(1 \text{ or } 2 \text{ heads}) = (2 \times \tfrac{1}{2} \times \tfrac{1}{2}) + (\tfrac{1}{2})^2 = \tfrac{3}{4}$$

Example 3.

A brewery produces a very popular brand of lager. To expand its market it wishes to introduce an ale. It is important that the ale tastes different from the lager. In a **triangle test**, a taster is asked to sample each of three similar-looking glasses of beer, two of which are lager and one of which is ale.



a. If he guesses, what is the probability that he identifies the ale correctly?
b. If he does the triangle test three times, what is the probability that he identifies the ale correctly:
   i. twice? ii. three times?

Solution:

a. P(identifies ale correctly on a single trial) $= \frac{1}{3}$
b. For three trials, we may use the binomial expansion of $(q + p)^3$ where $p = \frac{1}{3}$

Number of successes	Probability
0	$q^3 = (\frac{2}{3})^3 = \frac{8}{27}$
1	$3q^2p = 3.(\frac{2}{3})^2(\frac{1}{3}) = \frac{4}{9}$
2	$3qp^2 = 3.(\frac{2}{3})(\frac{1}{3})^2 = \frac{2}{9}$
3	$p^3 = (\frac{1}{3})^3 = \frac{1}{27}$

i. P(identifies the ale correctly twice) $= \frac{2}{9}$
ii. P(identifies the ale correctly 3 times) $= \frac{1}{27}$

Some experiments are repetitive: for example, flipping a coin ten times or performing the triangle taste test three times.

We may be interested in the total number of heads in the coin flip or the total number of correct identifications in a triangle taste test. In each case we are concerned with the **total number of successes**. This is a random variable for these experiments.

Imagine n trials in one of these experiments. The probabilities that the random variable takes on the values 0, 1, 2, 3, 4, ..., n correspond to the terms of the expansion of the binomial, $(q + p)^n$, where p is the probability of a success on a single trial and $q = 1 - p$.

Properties of a Binomial Distribution

To guarantee that the probabilities correspond to the terms of the binomial expansion and to calculate the probabilities, the following conditions must be met:

- There must be a fixed number of trials so that n is known.
- The probability of a success on a single trial, p, must be known.
- p must not change from trial to trial. That is, the outcome of each trial is independent of previous outcomes.
- The random variable must be "the total number of successes".

Example 4.

Consider the expansion:

$(q + p)^7 = q^7 + 7q^6p + 21q^5p^2 + 35q^4p^3 + 35q^3p^4 + 21q^2p^5 + 7qp^6 + p^7$

a. Write down the term which corresponds to the probability of 3 successes in a seven-trial binomial experiment.

b. Eighty per cent of printed circuits will pass a quality-control test. A sample of seven is to be selected randomly from the next month's production. Let X represent "the number of printed circuits which will pass a quality control test". Find:

i. $P(X = 3)$ ii. $P(X < 3)$ iii. $P(X \geqslant 4)$

Solution:

a. $P(3 \text{ successes}) = 35q^4p^3$

b. X is binomially distributed with $n = 7$ and $p = 0{\cdot}8$

i. $P(X = 3) = 35q^4p^3 = 35(0{\cdot}2)^4(0{\cdot}8)^3 \approx 0{\cdot}0287$

ii. $P(X < 3) = q^7 + 7q^6p + 21q^5p^2$
$= (0{\cdot}2)^7 + 7(0{\cdot}2)^6(0{\cdot}8) + 21(0{\cdot}2)^5(0{\cdot}8)^2 \approx 0{\cdot}0047$

iii. $P(X \geqslant 4) = 1 - [P(X = 3) + P(X < 3)] \approx 1 - 0{\cdot}0334 = 0{\cdot}9666$

TWO METHODS FOR CALCULATING BINOMIAL PROBABILITIES

A. Use a formula.

The formula gives the general term of the expansion of $(q + p)^n$. In fact, it is the term corresponding to the probability of r successes in n trials.

Binomial Probability Distribution Formula

If X is a binomially distributed random variable, the probability that X takes the value r after n trials is given by:

$$P(X = r) = \binom{n}{r}p^rq^{n-r} = \binom{n}{r}p^r(1-p)^{n-r}$$

B. Use four-figure tables.

Probabilities for 0, 1, 2, 3, ..., n successes in n trials are given for various values of n and p. The tables are quicker to use than a calculator but give probabilities correct only to four decimal places. Sometimes the tables do not have the required value of n or p.

Example 5.

Two columns of individual probabilities from a typical four-figure binomial distribution table are given:

n = 4	r \ p	0·1	0·2	0·3	0·4	0·5	0·6	0·7	0·8	0·9
	0				0·1296	0·0625				
	1				0·3456	?				
	2				?	0·3750				
	3				0·1536	0·2500				
	4				0·0256	0·0625				

Use the formula to calculate the two missing entries.

Solution:

For $n = 4$, $p = 0{\cdot}4$,

$$P(X = 2) = \binom{4}{2}(0{\cdot}4)^2(0{\cdot}6)^2 = \frac{4 \times 3}{2 \times 1}(0{\cdot}4)^2(0{\cdot}6)^2 = 0{\cdot}3456$$

For $n = 4$, $p = 0{\cdot}5$,

$$P(X = 1) = \binom{4}{1}(0{\cdot}5)^1(0{\cdot}5)^3 = 4 \times (0{\cdot}5)^4 = 0{\cdot}2500$$

Example 6.

Twenty per cent of patients brought to a hospital by ambulance should be admitted to the intensive-care ward. On a particular night the ward had 6 spare beds and 15 people were brought to the hospital by ambulance.



a. What is the probability that, in the intensive-care ward, 4 spare beds were filled that night?

b. What is the probability that at least one intensive-care patient was turned away that night because all spare beds were filled?

c. How many spare beds should the ward carry so that when 20 ambulance patients are brought in, there is less than one chance in fifty of having to turn someone away from the intensive-care ward?

d. What would be the expected number of admissions on a night when 20 ambulance patients were brought in?

Solution:

Let X represent "the number of patients who should be admitted to the intensive-care ward".

Then $n = 15$; $p = 20\% = 0{\cdot}2$; and X has a binomial distribution.

a. $P(X = 4) = \binom{15}{4}(0{\cdot}2)^4(0{\cdot}8)^{11} \approx 0{\cdot}1876041 \approx 0{\cdot}1876$

This probability may be obtained directly from tables.

b. P(at least one intensive-care patient turned away)
$= $ P(more than 6 require intensive-care beds)
$= $ P(X = 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15)
$= 1 - P(0 \leqslant X \leqslant 6)$
$\approx 1 - (0{\cdot}0352 + 0{\cdot}1319 + 0{\cdot}2309 + 0{\cdot}2501 + 0{\cdot}1876 + 0{\cdot}1032 + 0{\cdot}0430)$
$= 1 - 0{\cdot}9819$
$= 0{\cdot}0181$

c. In this part, $n = 20$.

Imagine that there are 6 spare beds.

Then P(at least one intensive-care patient is turned away)

$= $ P(X = 7 or 8 or 9 or ... or 20)
$\approx 1 - 0{\cdot}9133$
$= 0{\cdot}0867$, which is greater than $0{\cdot}02$; that is, 1 chance in 50.

Imagine that there are 7 spare beds.

Then P(at least one intensive-care patient is turned away)

$= $ P(X = 8 or 9 or 10 or ... or 20)
$\approx 1 - 0{\cdot}9679$
$= 0{\cdot}0321$, which is still a little greater than 1 chance in 50.

Imagine that there are 8 spare beds.

Then P(at least one intensive-care patient is turned away)

$= $ P(X = 9 or 10 or 11 or ... or 20)
$\approx 1 - 0{\cdot}9900$
$= 0{\cdot}01$, which is less than 1 chance in 50.

Therefore, the hospital should carry a minimum of 8 intensive-care spare beds.

d. The most likely number of admissions is 20% of 20 ambulance patients.

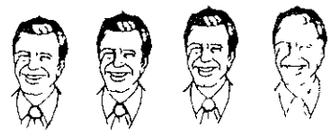
i.e., $E(X) = 0{\cdot}2 \times 20 = 4$ patients.
($0{\cdot}2$ = p, 20 = n)

Expected Value of Binomially Distributed Random Variable, X

$$E(X) = np$$

Example 7.

A photographer knows from experience that, on average, $\frac{3}{4}$ of the photographs he takes are satisfactory.



Determine the minimum number of photographs the photographer should take so as to have at least a 95% chance of producing a satisfactory photograph.

Solution:

$p = \frac{3}{4}$; $q = \frac{1}{4}$; $n = ?$

We require $q^n < 5\%$ where q is the probability of a failure.

i.e., $(\frac{1}{4})^n < 0{\cdot}05$

We can use logarithms to solve this inequality.

$$n \log(\tfrac{1}{4}) < \log 0{\cdot}05$$

i.e., $$n > \frac{\log 0{\cdot}05}{\log(\frac{1}{4})}$$ Reverse the inequality when dividing by a negative number.

i.e., $$n > 2{\cdot}16$$

Therefore, the photographer should take 3 photographs to have better than a 95% chance of a satisfactory photograph.

Exercise 9: The Binomial Distribution

1. A research chemist knows that 40% of the rats used in an experiment will die after ingesting a specified dose of a "new experimental wonder drug". Should the chemist be surprised if, from a sample of ten rats, no rats die after receiving the specified dose of the drug? Hint: Calculate the probability that no rats die.

2. A particular fullback in a hockey team is nominated to take the penalty corner hit on goals. He has a probability of scoring a goal of 0·42. Determine the probability of this fullback scoring four goals from his next ten attempts.

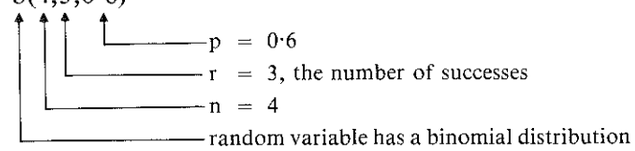
3. A local rugby team has a probability of 0·85 of winning each time it plays. If the team plays four matches, determine the probability that it will win:

a. all four matches. b. exactly two matches. c. no matches.

4. Use appropriate binomial distribution tables to determine, correct to four decimal places, the value of:

a. b(3;3,0·4) b. b(4;3,0·6) c. b(10;5,0·8)
d. b(15;8,0·5) e. b(10;2,0·6) f. b(20;10,0·1)

Note: b(4;3,0·6)



5. A fair coin is tossed six times. Use tables to determine the probability of obtaining:

a. no heads at all. b. one head only. c. at the most, one head.

6. A fair die is rolled six times. A successful roll occurs when either a 3 or a 5 occurs. Find the probability that a 3 or a 5 occurs exactly twice.

7. A kangaroo shooter brings down 80% of the animals at which he shoots. What is the probability that three of the next five kangaroos at which he shoots, will escape?

8. The bite from an Australian funnel-web spider is fatal in 70% of cases. A new anti-venene (antidote) is developed and administered to seven victims of the funnel-web spider and all of them survive. Would you say that the anti-venene is effective? What is the probability that it has no effect?

9. A touring professional soccer team has a probability of 0·75 of winning each time it plays. If the team plays eight games on its tour of Australia, what is the probability that the team will win:

a. all games? b. at least 75% of its games?

10. A clairvoyant was blindfolded and told of the ten coins that were in a row on the table in front of him. Some were showing heads and some were showing tails. Find the probability that, by chance alone, seven coins were identified correctly.

11. One-quarter of traffic tickets given out by policemen are for speeding. A traffic policeman gave out five tickets one afternoon. Determine the probability that:

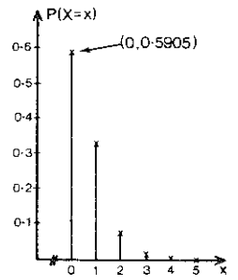
a. three were for speeding.
b. the first, the third and the fifth were for speeding but the others were not.

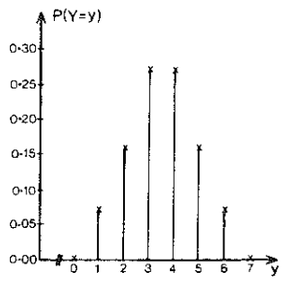
12. A card was drawn from a standard pack of 52 cards and then it was replaced. This was repeated four times. Determine the probability that:

a. exactly two spades were drawn.
b. all the cards drawn were spades.

13. If I throw four dice together, what is the chance that I obtain no sixes or just one six?

14. The random variables X and Y have binomial probability distributions. The graphs of the distributions are given. Supply the missing information. Tables would be helpful.





a. i. n = ? ii. p = ? b. i. n = ? ii. p = ?
iii. P(X > 1) = ? iv. E(X) = ? iii. P(Y = 5) = ? iv. E(Y) = ?

15. A Gallup Poll reveals that 70% of people are in favour of harsher drink-driving penalties. Five people are chosen at random. What is the probability that there will be a majority of people NOT in favour of harsher drink-driving penalties?

16. A fair die is rolled five times. Determine the probability that the number 3:

a. does not occur at all.
b. occurs only once.
c. occurs at the most once.

17. A Corgi bitch gives birth to a litter of four puppies. If the birth of a male and female are assumed to be equally likely, calculate the probability that there are at least two bitches in the litter.

18. An urn contains 4 white discs and 6 red discs. A disc is selected, the colour noted, and then replaced before the next draw is made. If a disc is drawn and replaced three times, determine the probability that:

a. only one red disc was drawn.
b. at most, one red disc was drawn.

19. A fair coin was tossed four times. Determine the probability that, at most, two heads occurred.

20. The probability that a student entering university will succeed in the first year is 0·65. What is the probability that fewer than 3 of 5 randomly selected new students will be successful in their first year at university?

21. Find the probability that, in a family of four children, there will be:

a. at least one boy. b. at least one boy and one girl.

22. Fifteen per cent of all cars produced by a major manufacturer have faulty pollution control devices. From a random sample of four cars, determine the probability that:

a. only one has a faulty pollution control device.
b. none has a faulty pollution control device.
c. at most, two have faulty pollution control devices.

23. The probability that a professional golfer hits an "under par" round of golf is 0·35. If he plays four rounds of golf, what is the probability that:

a. only one round is "under par"?
b. at least two rounds are "under par"?

24. A gardener purchases a punnet of ten Snapdragon seedlings. The chance that a seedling reaches maturity is 0·85. If three seedlings are selected at random, what is the probability that:

a. all the seedlings will reach maturity?
b. exactly two seedlings will reach maturity?
c. only one seedling will reach maturity?
d. at least two seedlings will reach maturity?

25. John and Peter decide to play ten games of snooker. Peter's chance of winning any game is 0·4. Determine the probability that Peter:

a. wins exactly four games.
b. loses only four games.
c. wins more than half the games.

26. Over a year, a restaurateur noted that 30% of all diners ordered a red wine with their meal. Seven diners were selected at random at the restaurant. What is the probability that:

a. exactly 3 had ordered a red wine?
b. a majority of those chosen had ordered a red wine?
c. at most, 2 had ordered a red wine?

27. The probability that a patient recovers from a particular disease is 0·7. Of the next ten people to enter hospital with this disease, what is the probability that:

a. at least 8 survive? b. 5, 6, 7 or 8 survive?

28. Four business executives have a luncheon meeting every Friday. To satisfy their gambling spirits, they toss to see who will pay the bill. By use of a binomial distribution, find the probability that Sally, who is one of the executives, will pay the bill:

a. only once in the next 4 weeks.
b. twice only in the next 8 weeks.
c. at least 3 times in the next 8 weeks.

29. Twelve parking bays are provided outside a local doctor's surgery. The probability that, during normal visiting hours, any particular bay is occupied is 0·6. Before you arrive at the parking area, on your next visit to the doctor, what is:

a. the probability that exactly 5 bays are occupied?
b. the most likely number of bays to be occupied?
c. the probability that at least 3 bays are vacant?
d. the probability that you will need to look for a parking area other than the one offered by the doctor?

30. A switchboard of a small business can cater for, at most, eight telephone calls at any one time. The chance that any one line is busy at any one time is 0·3. What is:

a. the chance that exactly 5 of the lines are busy at any one time?
b. the chance that no more than 3 of the lines are busy at any one time?
c. the most probable number of lines to be busy at any one time?

31. Of all students at Senior Colleges, eight out of ten enter tertiary education. Find the probability that from a group of eight Senior College students chosen at random:

a. five will enter a tertiary institution.
b. at most, five will enter a tertiary institution.
c. only the first five selected will enter a tertiary institution.

32. A multiple-choice objective test contains ten questions, each of which has five alternative answers. There is only one correct answer to each question. A student chooses all of his answers entirely by guesswork. Find:

a. the probability that he answers exactly seven questions correctly.
b. the probability that he passes the test if 50% is the pass mark.
c. the most likely number of questions he could expect to answer correctly.
d. x, given that the probability he answers at most x questions correctly is approximately $\frac{7}{8}$.

33. In the game of Yahtzee, five dice are thrown simultaneously. Find the probability that in a single roll of 5 dice:

a. exactly 3 sixes were showing.
b. at least 1 six was showing.
c. at most, 3 sixes were showing.

34. A fair die is rolled many times. Have I a better chance of throwing 2 sixes in the first 4 rolls or 3 sixes in the first 6 rolls?

35. An amateur golfer plays 18 holes with a professional player. The probability that the amateur wins any one hole is 0·4. What is the probability that the amateur wins:

a. at least five of the first 9 holes?
b. at least five of the second 9 holes?
c. at least five of the first 9 holes and at least five of the second 9 holes?
d. at least five of the first 9 holes or at least five of the second 9 holes?

36. Derek and Gaye are evenly matched at tennis. They decide to play ten sets.

a. Determine the probability that Derek wins 3 sets out of the first five and 2 sets out of the last five.
b. Is the answer to part a. the same as the probability of winning any 5 sets out of ten?

37. A boy has a handful of coins and throws them high into the air. When they fall he counts the heads. Repeating this act 99 times, he found that only six times out of 100 did he get less than five heads.

a. How many coins did he have?
b. What would be the average number of heads on a very large number of trials?
c. How many times would you have expected him to get less than five tails?

38. A multiple-choice objective test in Economics consists of fifteen questions, each with five possible answers. There is only one correct answer to each question. What is the probability that a student who has no knowledge of the subject guesses the correct answers to:

a. exactly two questions? b. not more than two questions?

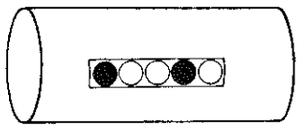
39. A multiple-choice objective test has twenty questions. There are five alternatives to each question and the candidate must choose one alternative—the correct answer.

a. What is the probability of getting 50% or more on this test just by guessing the answers? Would you suggest that the examiner makes a deduction from all of the scores for guessing?
b. What is the probability of getting 50% or more on a twenty-question **true-false** test by guessing all of the answers?
c. A candidate was sure about a particular 5 questions out of twenty in the original multiple-choice test. He had to guess the answers to the rest. What is the probability that he got 50% or more?

40. How many dice must be rolled so that there is better than an even chance of obtaining a 5?

41. A card is drawn from a standard pack of 52 cards. It is then replaced. Determine the minimum number of times a card must be drawn so that there is at least an even chance of drawing a spade.

42. A "binomial sampler" is a closed container which contains hundreds of black and white beads. The beads are identical except for colour.



There is a small window in the side of the container through which 5 beads may be seen. The container was shaken and the number of black and white beads in the window was noted. This was performed 1 000 times with the following results:

Colour	Occurrence
0 black; 5 white	2 times
1 black; 4 white	30 times
2 black; 3 white	135 times
3 black; 2 white	299 times
4 black; 1 white	362 times
5 black; 0 white	172 times

a. What proportion of black beads would you expect to find in the sampler?

All of the beads were tipped into a new "binomial sampler" with a window for 8 beads.

b. How many black beads would you most likely see in this window?
c. What should be the average number of black beads visible in this window for 100 or more shakes?

43. A paper manufacturer produces three different sizes of duplicating paper: *A4*, *A3* and *foolscap*. Of the sizes produced, 60% is *A4*, 25% is *A3* and 15% is *foolscap*. A salesman selects at random, eight reams of paper from the stock. Determine the probability that the salesman selected:

a. four reams of *A3*.
b. at least two reams of *A4*.

If the salesman did not select any reams of *A3*:

c. what is the probability that five of the eight reams were *foolscap*?

44. An operator of a game of chance tosses a coin. The operator and you put in $1 each. If the result is "heads" you win, and if "tails" the operator wins. The operator uses a biased coin which has a probability equal to 0·45 of falling heads. You decide to play this game ten times.

a. Find the probability that you have:
   i. exactly 5 wins. ii. 7 wins and 3 losses. iii. at least 4 wins.
   iv. at least 8 wins. v. 4 wins or 4 losses.
b. Find the expected return for your $10.

45. How many times should a fair die be rolled so that the probability of rolling a six once is the same as the probability of not rolling a six at all?

46. A minibus takes tourists sightseeing. It has eight passenger seats. The tour manager always accepts 10 bookings for each tour because he knows that there are likely to be last-minute cancellations. The probability that any booking will be cancelled is 0·2.

a. Determine the probability that, on the next tour, the minibus will be overbooked (that is, more than eight people will turn up).
b. Determine the probability that there will be some spare seats on the next tour.
c. What proportion of tours, for which there were no empty seats, was it necessary to turn people away because of overbooking?

47. A car hiring company has ten cars available for hire. Six are large cars which are hired out, on average, 70% of the time. The others are small cars which are hired out, on average, 60% of the time. The company manager checks the car yard on a randomly selected day.

a. What is the probability that he finds 3 large cars and 3 small cars left in the yard?
b. What is the probability that he finds that only 2 cars had been hired out?
c. If two cars had been hired out, what is the probability that they were both small cars?
d. The daily profit from hiring is \$18 per large car and \$16 per small car. What is the expected daily profit for this hiring company?

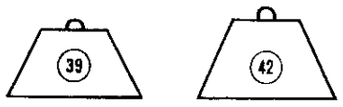
48. Out of every x people, two are in favour of a proposal. Eight people are selected at random and asked their opinions with regard to the proposal.

a. Show that the probability that five of these people are in favour of the proposal is given by: $1792(x-2)^3 \div x^8$
b. Use binomial distribution tables, or otherwise, to solve:

$1792(x-2)^3 = 0{\cdot}123863x^8$ Hint: $p = \frac{2}{x}$

49. To gain a certificate, a student must pass at least three subjects when she takes her final examinations. She estimates that she has an 80% chance of passing **each subject** if she takes three subjects; 75% chance if she takes four; and a 60% chance if she takes five. Should she take three, four, or five subjects? Why?

50. A chain is made of fifteen mass-produced links. Ninety per cent of the links will hold between 35 kg and 40 kg, but only 80% will hold between 40 kg and 45 kg.



a. Determine the probability that the chain will break if it is loaded with:

i. 39 kg. ii. 42 kg.

b. To test the chain, one of the weights is chosen at random. What is the probability that the chain will not break?
c. Given that the chain did not break, what is the probability that it was loaded with the 39 kg weight?
d. Would the chain become stronger if more mass-produced links were added? Explain.

51. A random experiment consists of two independent trials. The probability of a success in each trial is p. The probability of one or two successes in two trials is $\frac{72}{121}$. Find p.

52. The binomial expansion of $(x + y)^6$ is

$$x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6$$

A binomial experiment has six independent trials. The probability of a success in a single trial is x, and $y = 1 - x$.

a. Express in terms of x and y, the probability of 2 or 3 or 4 successes.
b. Use binomial tables to solve the simultaneous equations:

$$x^2y^2(3x^2 + 4xy + 3y^2) = 0{\cdot}02284; \quad x + y = 1$$

53. A transport survey of city workers puts them into three categories:

Private vehicles:	60%
Public transport:	30%
Others:	10%

Ten randomly chosen city workers were interviewed. Find the probability that:

a. five used private vehicles.
b. the first five interviewed used private vehicles.
c. five used private vehicles and five used public transport.

Given that the first two people interviewed used private vehicles,

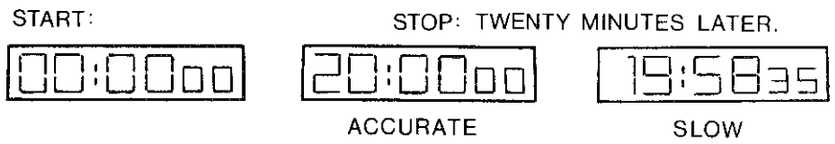
d. what is the probability that five used private vehicles?

Chapter 10:

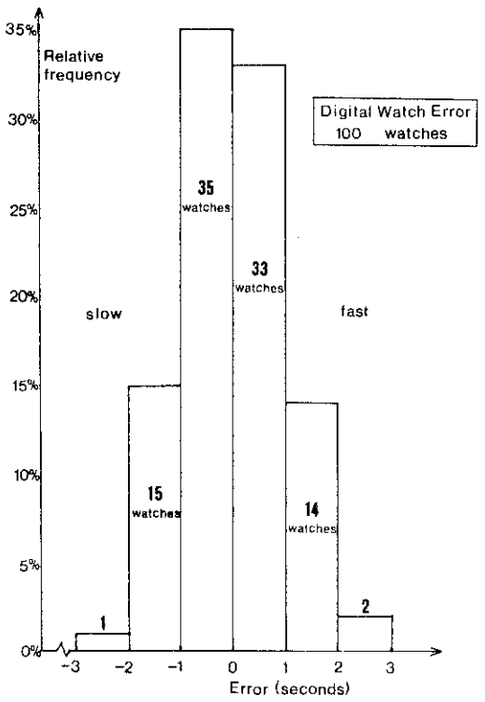
The Normal Distribution

Samples and Populations

Digital watches are very accurate. The accuracy of a stop-watch of a particular brand was checked by an experiment. A **sample** of 100 watches was used. The watches were started simultaneously by using an electronic device. Exactly 20 minutes later, all of the watches were stopped. Some of them showed the wrong time.



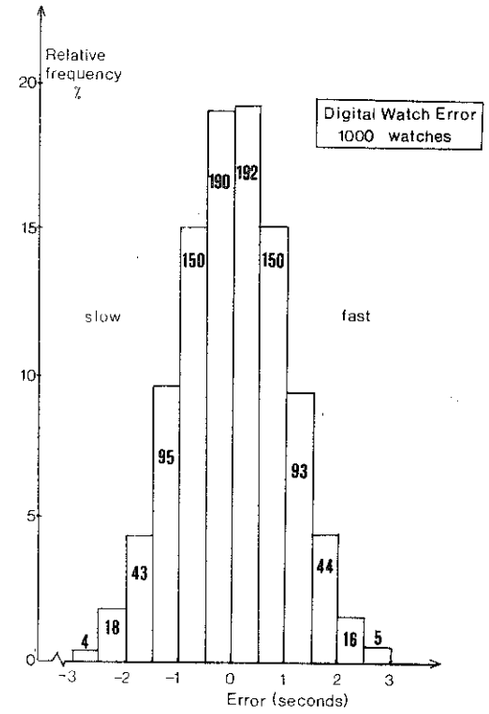
All of the displayed times were within 3 seconds of the exact time. The results are presented in the relative frequency histogram:



This histogram has several characteristics:

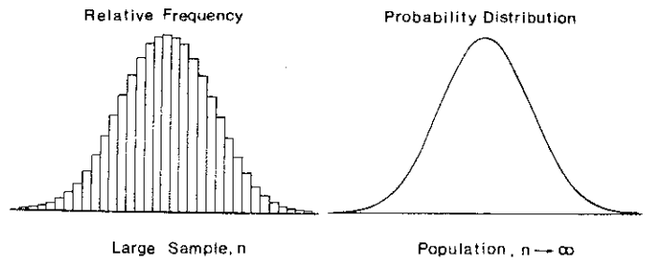
- it is almost symmetrical about the mean which is approximately zero.
- about two-thirds of the times are within one second of the correct time.

The same experiment was performed with a **sample** of 1 000 watches.



This histogram has the same characteristics as the previous one. However, the class intervals are reduced from 1 second to $\frac{1}{2}$ second.

Now consider the whole **population** of digital watches of this type. We define the population as all of the times from all of the watches, even those which have yet to be manufactured. Obviously, the population is infinite. We cannot draw a relative frequency histogram for an infinite population. Instead, we imagine a succession of histograms for larger and larger samples. It is necessary to reduce the width of the histogram columns to accommodate the increasing numbers of observations.



In the limiting case, the relative frequency histogram becomes a smooth curve called a **probability distribution** curve. This describes the population.

Populations and Samples

The **population** is the set of all scores (measurements or counts) under consideration. The population may be infinite.

A **sample** is a selection of scores from the population.

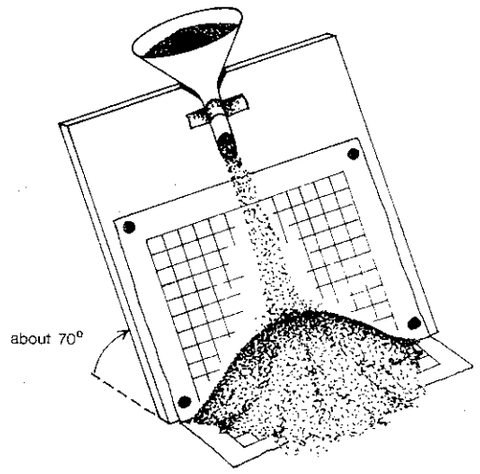
Statistics and Parameters

A **statistic** is a number which describes a sample: for example, the sample mean, $\bar{x}$, or the sample standard deviation, s.

A **parameter** is a number which describes a population. Some examples are: the population mean, μ, and the population standard deviation, σ. Parameters are estimated from sample statistics.

The Normal Curve

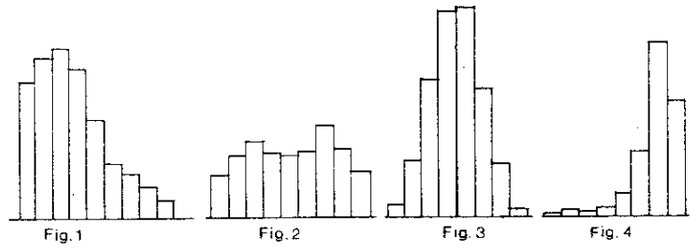
The normal curve is a probability distribution curve of a particular shape. Try this experiment: set up the apparatus as shown in the picture.



Hold your finger over the end of the funnel until it is full of sugar and then let it go. Trace around the top of the pile of sugar. You will notice that the resulting curve is the same shape as the probability distribution curve for the digital watch experiment, perhaps a little flatter. They are both **normal curves.**

Example 1.

Four frequency histograms are given without scales.



a. Consider the shape of the histograms and match them to the following random variables:

   X, the weights of students at a university;
   Y, the average weekly expenditure on bread for Australian households;
   W, the weights of 12-year-old Australian boys;
   T, the ages of Australian females at death.

b. Which random variable could be modelled by a normal probability distribution?

Solution:

a. X, the weights of students at a university is figure 2. There are two modes which correspond respectively to the mean weight for females and males.

   Y, the average weekly expenditure on bread for Australian households is figure 1. This distribution has no left-hand "tail" because it is impossible to spend less than zero on bread. There is a long right-hand "tail" corresponding to the several households which spend well-above-average on bread.

   W, the weights of 12-year-old Australian males is figure 3. Most weights are close to the mean and the distribution is symmetrical.

   T, the ages of Australian females at death is figure 4. Nearly all of the deaths occur between 70 years and 85 years. The left-hand "tail" corresponds to the few deaths of children.

b. Figure 3 (W, the weights of 12-year-old Australian males) could be modelled by a normal probability distribution.

The shape of a probability distribution curve may be described using the population mean, μ, and the population standard deviation, σ.

Description of a Normal Probability Distribution

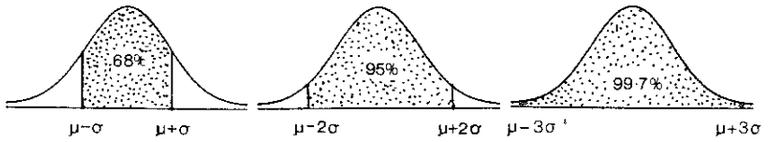
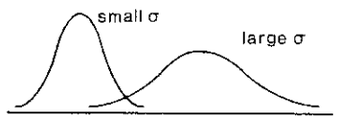


fig. 1 fig. 2 fig. 3

- Half of the population is above the mean and half is below.
- The curve is symmetrical.
- Just over two-thirds (68%) of the population is within one standard-deviation unit of the mean. See figure 1.
- Ninety-five per cent of the population is within two standard-deviation units of the mean. See figure 2.
- Nearly all (99·7%) of the population is within three standard-deviation units of the mean. See figure 3.
- Populations with large standard deviations give rise to flatter probability curves.



Standard Scores: z-Scores

Consider the population, X: 2,4,6,8,10,12,14. The mean, $\mu = 8$, and the standard deviation, $\sigma = 4$.

Each score in this population has a corresponding z-score. The z-score indicates the distance of a raw score above or below the mean in standard-deviation units. For example, a raw score $x = 12$ is above the mean by 4, which is 1 standard-deviation unit. Therefore, $z = 1$.

Also, a raw score $x = 2$ is below the mean by 6, which is $1\frac{1}{2}$ standard-deviation units. The corresponding z-score is $-1{\cdot}5$; the negative indicates that the score is below the mean.

Standardizing Scores

- For any population with mean μ and standard deviation σ, a standardized score z is given by:

$$z = \frac{x - \mu}{\sigma}$$

  where x is the raw score.
- If raw scores from populations are standardized, it allows us to compare probability distributions with different means and standard deviations.
- The mean and standard deviation of any set of standardized scores is 0 and 1, respectively.

Example 2.

A population has a mean $\mu = 7$ and a standard deviation $\sigma = 3$. Calculate the standardized scores (z-scores) corresponding to these raw scores (x-scores):

a. 10 b. 12 c. 2·3

Solution:

a. If $x = 10$ than $z = \frac{10 - 7}{3} = 1$. This indicates that a raw score of 10 is 1 standard-deviation unit above the mean.

b. If $x = 12$ then $z = \frac{12 - 7}{3} \approx 1{\cdot}67$. A raw score of 12 is $1\frac{2}{3}$ standard-deviation units above the mean.

c. If $x = 2{\cdot}3$ then $z = \frac{2{\cdot}3 - 7}{3} \approx -1{\cdot}57$. The negative sign indicates that the x-score is below the mean (approximately $1\frac{1}{2}$ standard-deviation units).

Example 3.

A population has a mean of 9 and a standard deviation of 2. Find the raw score corresponding to a standardized score of $-1{\cdot}5$.

Solution:

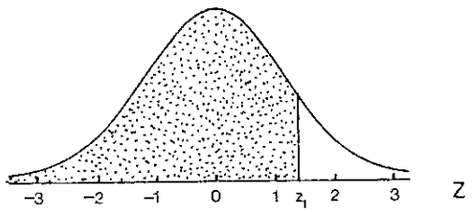
If $z = -1{\cdot}5$ then $x = 9 - (1{\cdot}5)(2) = 6$

For $z = -1{\cdot}5$, start with the mean 9, and then subtract 1·5 standard-deviation units.

Standard Normal Distribution

If a population, Z, of standardized scores has a normal distribution, it is said to have a **standard** normal distribution. The mean is 0 and the standard deviation is 1.

PROPORTIONS, PROBABILITIES AND AREAS

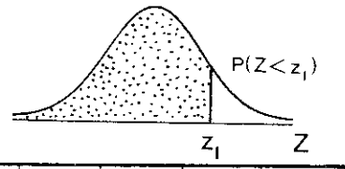


The total area under a standard normal distribution curve is 1. The area of the shaded region is equal to:

- the proportion of the population with standardized scores less than z_1; or
- the probability that a randomly selected member of the population has a standardized score less than z_1.

USING TABLES OF NORMAL PROBABILITIES

Table of Normal Probabilities



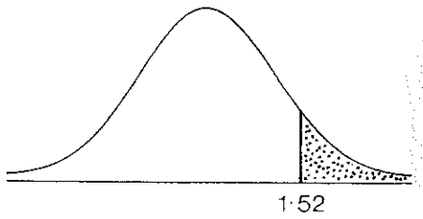
z_1	·00	·01	·02	·03	·04	·05	·06	·07	·08	·09
·0	·5000	·5040	·5080	·5120	·5160	·5199	·5239	·5279		
·1	·5398	·5438	·5478	·5517						
·2	·5793	·5832								
1·5	·9332	→	·9357			·9394				
1·6	·9452				·9495					
1·7	·9554	·9564	·9573	·9582						

The probability that a randomly selected score from a normal population has a standardized score less than 1·52 is 0·9357.

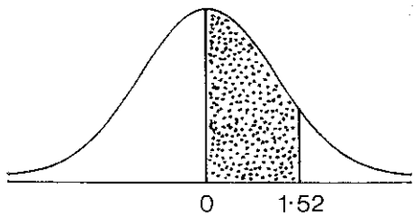
We write: $P(Z < 1.52) = 0.9357$ where Z is the random variable in standardized units.

From this single fact we may calculate probabilities such as:

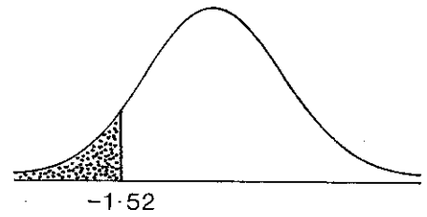
- $P(Z > 1{\cdot}52) = 1 - 0{\cdot}9357 = 0{\cdot}0643$
  (1 — the total area under the curve.)



- $P(0 < Z < 1{\cdot}52) = 0{\cdot}9357 - 0{\cdot}5 = 0{\cdot}4357$
  (0·5 — the area to the left of the mean.)



- $P(Z < -1{\cdot}52) = P(Z > 1{\cdot}52) = 0{\cdot}0643$



Example 4.

A random variable, Z, has a standard normal distribution.

a. What proportion of the population would you expect to be:
   i. greater than the mean? ii. less than 2?

b. What is the probability that a randomly selected member of the population is:
   i. between 0 and 2? ii. greater than 2·52?

Solution:

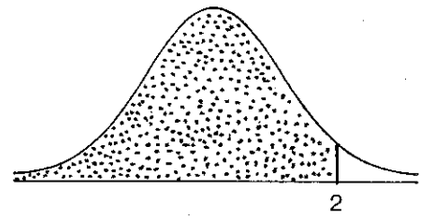
a. i. Proportion of the population greater than the mean,

$$P(X > \mu) = 0{\cdot}5$$

ii. Proportion of the population less than 2,

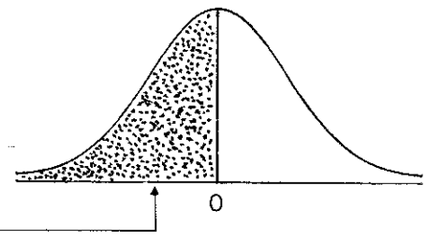
$$P(X < 2) = 0{\cdot}9772$$

(0·9772 — from tables for accuracy to 4 decimal places.)



b. i. Probability that a score is between 0 and 2,

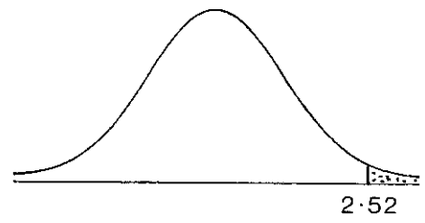
$$P(0 < X < 2) = 0{\cdot}9772 - 0{\cdot}5 = 0{\cdot}4772$$



ii. Probability that a score is greater than 2·52,

$$P(X > 2{\cdot}52) = 1 - 0{\cdot}9941 = 0{\cdot}0059$$

(1 — total area; 0·9941 — from tables)



Example 5.

The errors in digital watch times are recorded as positive if the watch is fast, but negative if the watch is slow.

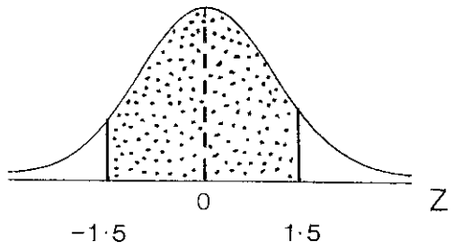


For a particular brand, the distribution of errors is normal with a mean of zero and a standard deviation of 1 second.

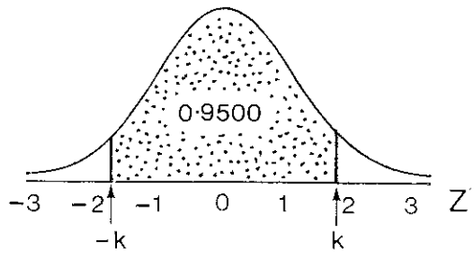
a. What proportion of watches would be within $1\frac{1}{2}$ seconds of the correct time?

b. The manufacturer wishes to identify the most accurate 95% of the watches. What is the maximum error size for this group of watches?

Solution:



a. P(watch is within $1\frac{1}{2}$ seconds of correct time)
$= P(-1{\cdot}5 < Z < 1{\cdot}5)$
$= 2[P(0 < Z < 1{\cdot}5)]$
$= 2 \times 0{\cdot}4332$
$= 0{\cdot}8664$



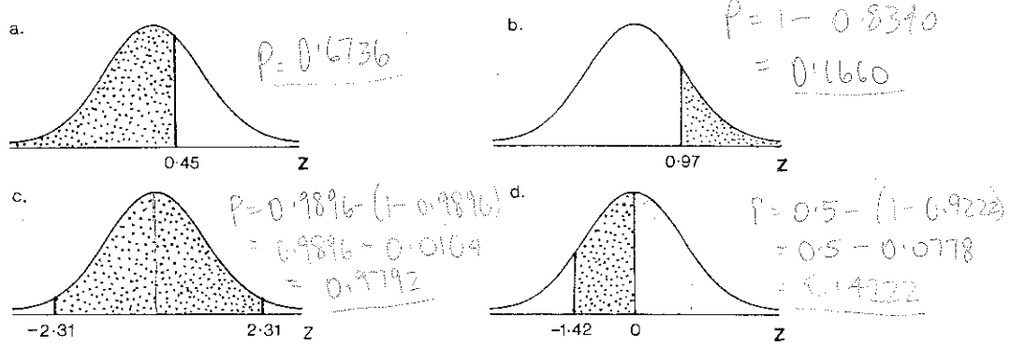
b. Let k be the maximum error. We require k, such that:

$P(-k < Z < k) = 0{\cdot}9500$
i.e., $P(0 < Z < k) = 0{\cdot}4750$
i.e., $P(Z < k) = 0{\cdot}9750$

Therefore, from tables, $k = 1{\cdot}96$. That is, the maximum error size is 1·96 seconds.

Exercise 10A: Standard Normal Distribution

1. Describe and calculate the probabilities represented by the shading in the following sketches of standard normal curves. The random variable is Z.



2. What proportion of a normally distributed population has standardized scores:

a. greater than 0·98?
b. less than $-1{\cdot}5$?
c. greater than 0·98 or less than $-1{\cdot}5$?

3. A sample of 60 adults was given a reaction-time test. The results are given in the frequency table:

Reaction time: (milliseconds)	0–49	50–99	100–149	150–199	200–249	250–299
Number of Adults:	13	27	12	6	1	1

Is this random variable, reaction time, normally distributed? Why?

4. A population is normally distributed with a standard deviation, σ. What percentage of observations differ from the mean by:

a. less than $0{\cdot}52\sigma$?
b. more than $1{\cdot}3\sigma$?

5. Determine the probability that an item selected at random from a standard normal distribution lies:

a. in the interval $[-0{\cdot}6;2{\cdot}2]$.
b. outside the interval $[-0{\cdot}6;2{\cdot}2]$.

6. What proportion of a normally distributed population would you expect to have standardized scores between:

a. 0 and 1·5?
b. $-1{\cdot}5$ and 1·5?
c. 2·15 and 2·72?
d. $-2{\cdot}15$ and 2·72?

7. Z is a population of standardized scores which are normally distributed. Calculate the following probabilities:

a. $P(0{\cdot}8 < Z < 2{\cdot}5)$
b. $P(-1{\cdot}1 < Z < 2{\cdot}5)$
c. $P(-2{\cdot}5 < Z < 2{\cdot}0)$
d. $P(Z > 1{\cdot}0 \text{ or } Z < -1{\cdot}0)$
e. $P(Z < -1{\cdot}5 \text{ or } Z > 0{\cdot}6)$
f. $P(Z < 1{\cdot}5 \text{ or } Z > -1{\cdot}5)$

8. What is the probability that a randomly selected score from a normal population is:

a. greater than the mean by less than one standard-deviation unit?

b. greater than the mean by more than one standard-deviation unit but less than two standard-deviation units?

9. Which one of the following random variables is the most likely to be normally distributed? Why did you reject the others?

A, the number of sixes when 10 fair dice are tossed.

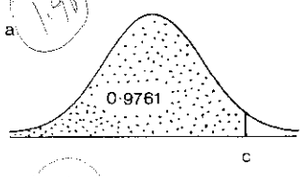
B, the weights of merino sheep.

C, the weights of merino rams.

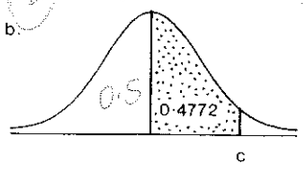
D, the heights of 13-year-old Australians.

10. Find the constant, c, in the following sketches of standard normal curves. In each case an area is given.

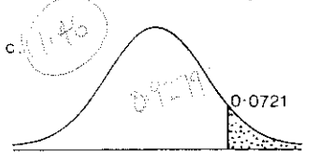
a.



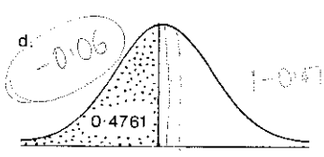
b.



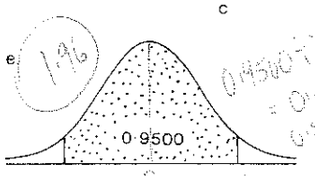
c.



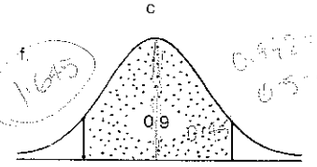
d.



e.



f.



11. The random variable Z is a standard normal variable. Find, correct to 4 decimal places:

a. $P(Z < 2{\cdot}55)$

b. $P(Z < -2{\cdot}55)$

c. $P(0 < Z < 2{\cdot}55)$

d. $P(Z < 1{\cdot}55 \mid Z < 2{\cdot}55)$

12. Let X be a random variable with a standard normal distribution. Determine:

a. $P(0 < X < 1{\cdot}2)$ b. $P(-1{\cdot}3 < X < 0{\cdot}85)$ c. $P(X > 1{\cdot}25)$

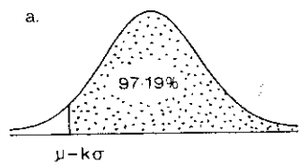
d. $P(X < -2{\cdot}5)$ e. $P(|X| < 0{\cdot}6)$ f. $P(X = 1{\cdot}2)$

13. A set of scores has a mean of 10 and a standard deviation of 3. Complete the table which shows some of the scores and corresponding standardized scores.

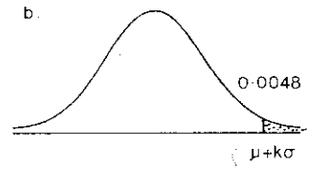
Score	Standardized score
16	2
24	?
1	?
?	1·5
?	−2·33

14. Use normal distribution tables to find a constant, k, corresponding to the given proportions:

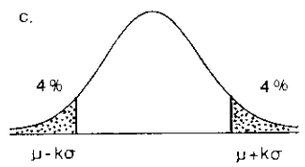
a.



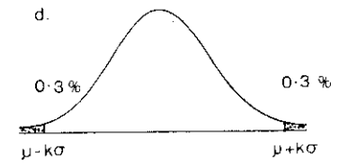
b.



c.



d.



15. A random variable Z has a normal distribution with a mean of 0 and a standard deviation of 1. Find the constant, k, such that:

a. $P(Z < k) = 0{\cdot}8315$ b. $P(Z > k) = 0{\cdot}0118$

c. $P(0 < Z < k) = 0{\cdot}4983$ d. $P(k < Z < 0) = 0{\cdot}4429$

16. A random variable, X, has a normal distribution with mean μ and standard deviation σ. Determine:

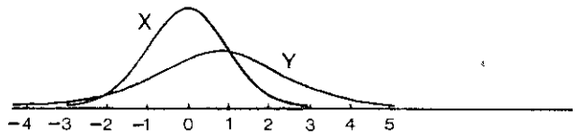
a. $P(\mu - \sigma \leqslant X \leqslant \mu + \sigma)$

b. $P(\mu - 2\sigma \leqslant X \leqslant \mu + 2\sigma)$

c. the constant, k, such that, $P(\mu - k\sigma \leqslant X \leqslant \mu + k\sigma) = 0{\cdot}95$

d. the constant, k, such that, $P(\mu - k\sigma \leqslant X \leqslant \mu + k\sigma) = 0{\cdot}90$

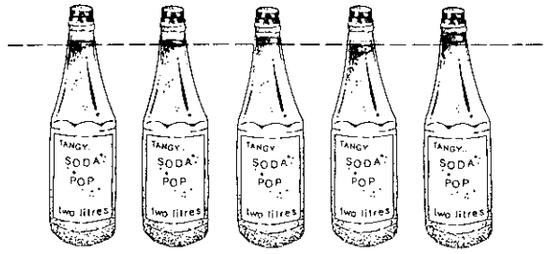
17. Consider the normally distributed populations, X and Y. The probability distributions are sketched.



a. Which population has the higher mean? Why?
b. Which population has the higher standard deviation? Why?
c. Which population has a mean of 0 and standard deviation of 1?
d. Given that $\sigma = 2$ for the population Y, find the constant "c", such that:
   i. $P(Y < c) = P(X < 0)$
   ii. $P(Y < c) = P(X < 1)$
   iii. $P(Y < c) = P(X < 2)$

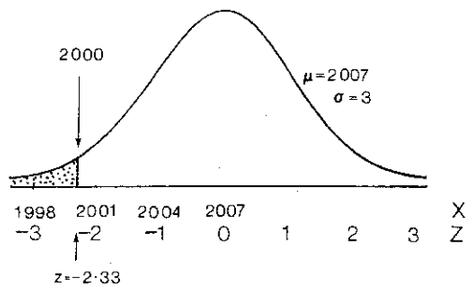
APPLYING NORMAL PROBABILITIES

If you line up some "identical" bottles of soft drink you will see that there are slightly different amounts in each bottle. The bottles are filled by an automatic machine. Because of faults in the machine, some bottles get a few millilitres more and some a few millilitres less.



It is against the law to have less than the stated amount (2 litres, in this instance) in the bottles. For this reason, the bottler sets the machine to deliver a little more to each bottle than the stated amount.

On average, a 2-litre bottle contains 2 007 millilitres. The standard deviation of the actual amounts is 3 millilitres and the distribution of the actual amounts is normal. What is the probability that a randomly selected bottle contains less than the stated 2 litres?



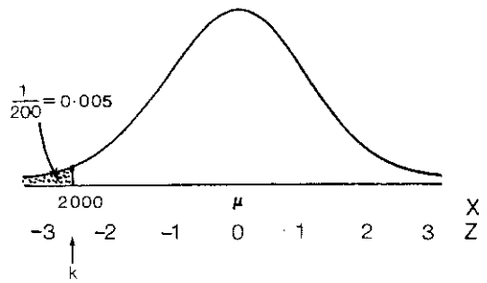
The first step is to find the standardized score corresponding to 2 litres.

$$\text{If } x = 2\,000 \text{ then } z = \frac{2\,000 - 2\,007}{3} \approx -2{\cdot}33$$

Now, P(randomly selected bottle contains less than 2 000 ml)

$= P(Z < -2{\cdot}33)$
$= 1 - 0{\cdot}9901$ (from tables)
$= 0{\cdot}0099$
$\approx 0{\cdot}01$ (i.e., one chance in 100)

There is 1 bottle in every 100 which contains less than the stated amount. The bottler wishes to reduce this to 1 bottle in 200. He decides that it is too costly to do anything about the standard deviation but decides to adjust the mean upwards. What should be the new mean?



We require a standardized score, k, such that:

$$P(Z < k) = 0{\cdot}005$$
$$\text{i.e., } P(Z \geqslant k) = 0{\cdot}9950$$
$$\therefore \quad k = -2{\cdot}575 \quad \text{(from tables)}$$

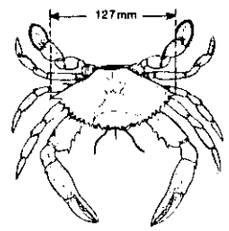
That is, 2 000 millilitres must be 2·575 standard-deviation units below the new mean, μ.

$$\therefore \quad \mu = 2\,000 + (2{\cdot}575)(3) = 2\,007{\cdot}725$$
$$\approx 2\,008$$

The new mean should be 2 008 millilitres.

Example 6.

Blue manna crabs, *Portunus pelagicus*, can be caught in the Peel Inlet, Western Australia. These crabs are very popular with amateur fishermen and the Department of Fisheries and Wildlife sets a minimum legal size.



The distance between the spines on the carapace must be at least 127 mm. In one particular season, the Department estimates the mean carapace width to be 139 mm with a standard deviation of 30 mm. Assume the widths to be normally distributed.

a. What is the probability that a randomly selected crab would have a carapace width less than the minimum?

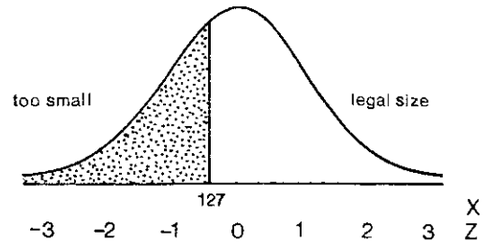
An amateur fisherman can take, at most, 36 legal-size crabs from the water per day. Any crab which is too small must be thrown back.

b. Including the crabs which were thrown back, how many crabs is a fisherman most likely to catch, to make up his bag limit of 36 legal-size crabs?

c. Of the legal-size crabs, what proportion would have carapace widths greater than 200 mm?

Solution:

a.



If $x = 27$, $z = \frac{127 - 139}{30} = -0{\cdot}4$

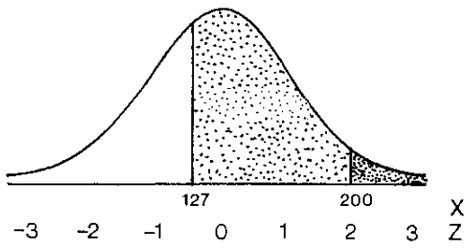
(use $z = \frac{x - \mu}{\sigma}$)

$$\begin{aligned} \text{P(less than minimum legal size)} &= P(X < 127) \\ &= P(Z < -0{\cdot}4) \\ &= 1 - 0{\cdot}6554 \\ &= 0{\cdot}3446 \end{aligned}$$

b. Thirty-six crabs represents 65·54% of the catch.

Therefore, the total catch is $\frac{36}{0{\cdot}6554} \approx 55$ crabs.

c.



If $x = 200$, $z = \frac{200 - 139}{30} \approx 2{\cdot}03$

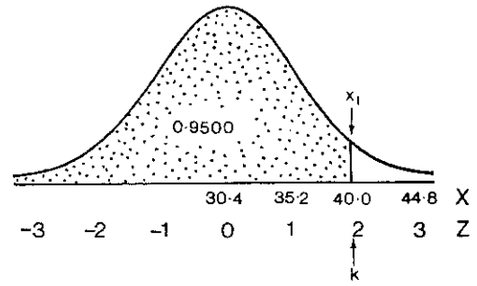
Proportion of legal-size crabs greater than 200 mm

$$\begin{aligned} &= P(X > 200 \mid X > 127) \\ &= P(Z > 2{\cdot}03 \mid Z > -0{\cdot}4) \\ &= \frac{1 - 0{\cdot}9788}{0{\cdot}6554} \\ &= 0{\cdot}0323 \end{aligned}$$

Example 7.

Pharmaceutical companies test some of their drugs on laboratory rats. The minimum lethal dose, MLD, is the smallest quantity of the drug which will kill a rat. The MLD for a particular drug is normally distributed with a mean of 30·4 mg and standard deviation of 4·8 mg. How many milligrams of the drug will kill 95% of the rats?

Solution:

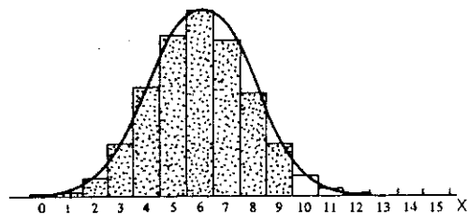


We require x_1 such that $P(X < x_1) = 0{\cdot}9500$
That is, we require k such that $P(Z < k) = 0{\cdot}9500$
Then, from tables, $k = 1{\cdot}645$
Therefore, $x_1 = 30{\cdot}4 + 1{\cdot}645(4{\cdot}8)$
$\approx 38{\cdot}3$ mg

If the rats are each given 38·3 mg of the drug, 95% of them will die.

Example 8.

The random variable, X, is "the total number of successes in 15 trials". X has a binomial distribution. The probability, p, of a success in a single trial is 0·4. The probabilities of 0,1,2,3,...,15 successes are shown in the probability distribution graph:



a. Use binomial tables to calculate the probability of 9 or less successes.

b. What is the total area enclosed by the columns associated with

$$X = 0, 1, 2, 3, 4, 5, 6, 7, 8, \text{ or } 9?$$

A normal curve with $\mu = np$ and $\sigma = \sqrt{npq}$ has been superimposed on the binomial probability histogram.

c. Use normal distribution tables to estimate the probability of 9 or less successes.

Solution:

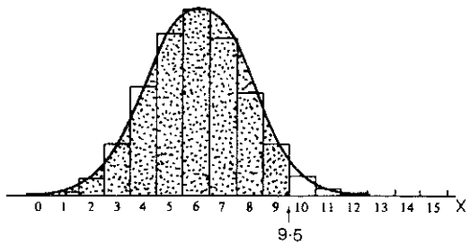
a. $P(X \leqslant 9) = 0{\cdot}9662$, from binomial tables.

b. Total area $= 0{\cdot}9662$

c. $\mu = np = 15 \times 0{\cdot}4 = 6$;

$\sigma = \sqrt{npq} = \sqrt{15 \times 0{\cdot}4 \times 0{\cdot}6} = \sqrt{3{\cdot}6} \approx 1{\cdot}90$

The area of the histogram's columns from $x = 0$ to $x = 9$ is approximated by the area under the normal curve to the left of 9·5.



If, $x = 9{\cdot}5$, $z = \dfrac{9{\cdot}5 - 6}{1{\cdot}9} \approx 1{\cdot}84$

Now, $P(X < 9{\cdot}5) \approx P(Z < 1{\cdot}84)$

$= 0{\cdot}9671$ (from normal tables)

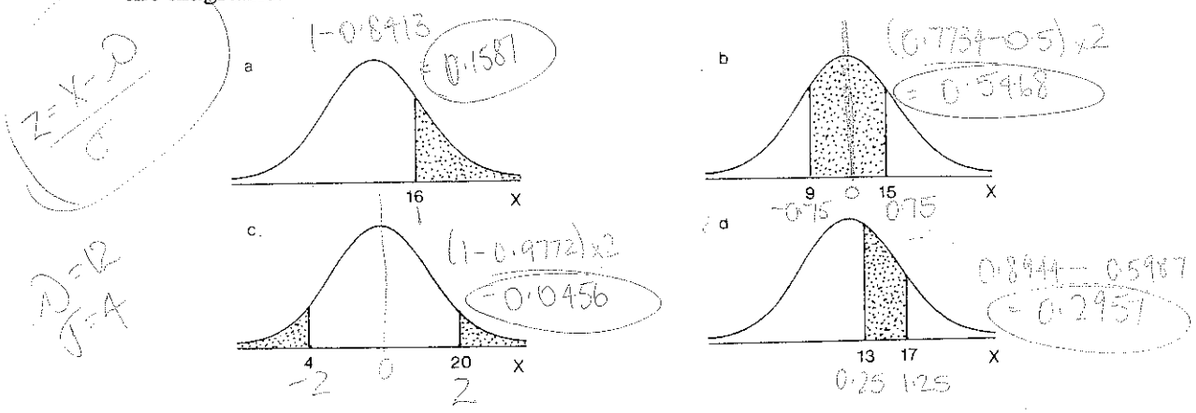
Compare this answer to the answer in part a., 0·9662

Normal Approximation to the Binomial Distribution

A normal curve with $\mu = np$ and $\sigma = \sqrt{npq}$ "fits" a binomial probability histogram provided that n is large or p is close to $\frac{1}{2}$.

Exercise 10B: Applying Normal Probabilities

1. A random variable X is normally distributed with a mean of 12 and a standard deviation of 4. Describe and find the probabilities indicated by the shaded areas in the diagrams.



2. A normally distributed population has a mean of 42 and standard deviation of 6. For this population, what is:

a. the median?

b. the approximate value of the range?

c. the variance?

3. The mathematics staff of a large secondary school prepared a measurement activity for National Mathematics Week. Every student in the school participated. Two knitting needles were placed 54 metres apart on the school oval. The distance between them was determined accurately by using surveying techniques. Each person was asked to measure the distance between the needles with a long tape. The errors in the student measurements were normally distributed with a mean of 0 and a standard deviation of 1 cm. Calculate the probability that a student selected at random from this school gave a distance which was within $\frac{1}{2}$ centimetre of 54 metres.

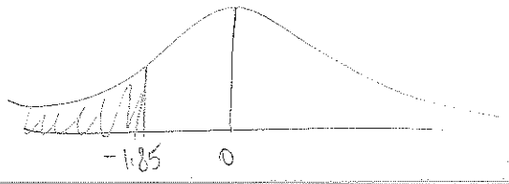
4. Gamblers at a casino lose, on average, \$83 per night. These losses are normally distributed with a standard deviation of \$28. What is the probability that a randomly selected gambler makes a profit?

5. The heights of 493 men are normally distributed with a mean of 1·740 m and standard deviation of 0·0686 m. How many of the men would you expect to have heights:

a. less than 1·613 m?

b. between 1·727 m and 1·816 m?

6. A machine produces rods in lengths normally distributed with mean 9·95 cm and standard deviation 0·1 cm. The specifications call for rods with lengths within the range $10 \pm 0{\cdot}12$ cm. What proportion of rods fail to meet the specifications?



7. The weight of swimming pool salt, in bags labelled "25 kg", is normally distributed about a mean of 26 kg and has a standard deviation of 250 grams.

a. Determine the probability that a bag, selected at random, will contain at least 25·6 kg but not more than 26·3 kg.
b. A hardware store receives a delivery of 600 bags of salt. How many bags of salt would you expect to weigh less than 25 kg?

8. A continuous variable is normally distributed with $\mu = 11{\cdot}2$ and $\sigma = 3{\cdot}1$. Find the probability that a randomly selected score will be greater than 20 or less than 5.

9. A random variable X has a normal distribution with a mean μ and a standard deviation σ.

a. What is the probability that X lies between $\mu - 1{\cdot}71\sigma$ and $\mu + 2{\cdot}31\sigma$?
b. Given that X lies between $\mu - 1{\cdot}71\sigma$ and $\mu + 2{\cdot}31\sigma$, what is the probability that it lies above the mean?

10. The mean height of men in Japan is estimated to be 1·69 m with a standard deviation of 0·06 m. The distribution of heights is normal.

a. Find the probability that a randomly chosen Japanese man will have a height between 1·60 m and 1·74 m.
b. What is the least height of the tallest 10% of Japanese men?
c. If 400 Japanese men are chosen at random, how many would you expect to have heights greater than 1·74 m?

11. The amount of juice that may be squeezed from a lime is normally distributed with a mean of 15 millilitres and a standard deviation of 1·62 millilitres. Assume that the amounts are normally distributed. What proportion of limes could be expected to deliver between 15 and 18 millilitres of juice?

12. A fishing line manufacturer tests the breaking strain of his products. He has found that, for a particular weight of line, the maximum breaking strain follows a normal distribution with a mean of 21·5 kg and a standard deviation of 1·5 kg.

a. If a line is selected at random, find the probability that it has a breaking strain of between 22 kg and 23·5 kg.
b. If the manufacturer decides to reject all lines with maximum breaking strains of 19·5 kg or less, what percentage are rejected?

13. Sixty applicants for a management position are given an aptitude test. The specifications of the test reveal that the population distribution is normal with a mean of 110 and a standard deviation of 12. Management applicants are required to have a score of at least 100.

a. How many applicants would be likely to be rejected?
b. Of the applicants who score more than 100, what proportion would be likely to score over 130?
c. Of the applicants who score over 100, how many would be likely to score over 130?

14. A soft-drink vending machine is regulated so that it delivers 210 ml per cup. The quantity of drink is normally distributed with a mean of 210 ml and a standard deviation of 15 ml.

a. What fraction of the cups would you expect to contain more than 234 ml?
b. What is the probability that a randomly selected cup of drink contains between 201 ml and 219 ml?

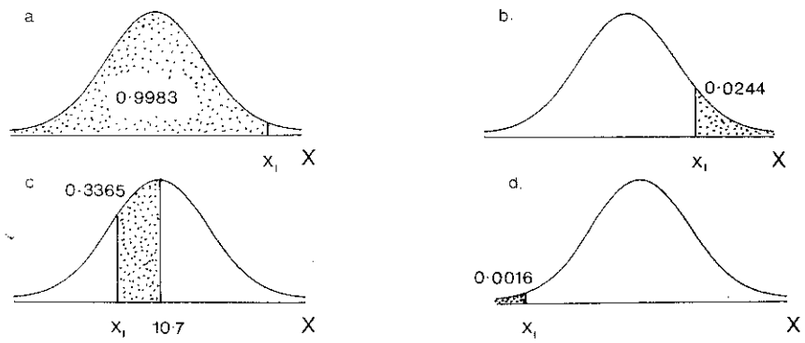
The machine uses 240 ml cups.

c. In the next 100 drinks, how many cups are likely to overflow?

15. A normal distribution has a mean of 18 and a standard deviation of 4·2. It is known that 899 scores exceed 19·5. How many scores are in the distribution?

16. In a national science competition the average score was 48·3. The scores were normally distributed with a standard deviation of 10·0. All students with scores between 39 and 57 were awarded a pass certificate. There were 2 390 students who received a pass certificate. How many entered the competition?

17. A random variable X has a normal distribution with mean, $\mu = 10{\cdot}7$, and standard deviation, $\sigma = 3{\cdot}4$. The probabilities associated with a randomly selected score x_1 are shown in the diagrams. Find x_1, correct to one decimal place.



18. Given that X is a normal random variable with mean 6·0 and standard deviation of 0·75, find q if:

a. $P(X < 7{\cdot}25) = q$
b. $P(Y > q) = 0{\cdot}2328$
c. $P(6 - q < X < 6 + q) = 0{\cdot}6$

19. Army recruits are given a test. Ten per cent of the recruits scored 112 or more. A different 10% scored 84 or less. Assume that the distribution of test scores is normal.

a. Determine the mean and standard deviation of the distribution.
b. Determine the smallest score of the top 5% of the recruits.

20. The Electricity Commission installed 3 000 experimental light bulbs in the streets throughout the city. The bulbs have an average burning life of 1 200 hours with a standard deviation of 240 hours. The burning life of the bulbs is normally distributed.

a. How many bulbs might be expected to burn out during the first 840 hours?
b. What burning period would you expect to elapse before 10% of the bulbs burn out?

21. Tomatoes have weights which are normally distributed with an average weight of 128·3 grams. The standard deviation of the weights is 11·9 grams. The tomatoes are graded according to weight.

small	20%
medium	55%
large	15%
very large	10%

Calculate the two borderline weights for the "medium" grade.

22. The lengths of nails made by a machine are normally distributed with a mean of 29·80 mm and a standard deviation of 0·54 mm.

a. If all nails less than 29 mm in length are rejected, what percentage are rejected?
b. Only those nails with lengths in the range 29·8 − L millimetres to 29·8 + L millimetres are accepted. Ninety per cent of the nails are accepted. What is the value of L, to one decimal place?

23. A company makes car batteries for $35 each and sells them for $65 each. If a battery fails in under one year, the money is refunded. The life of the battery is normally distributed with a mean of 3·25 years and a standard deviation of 0·8 years.

a. What is the probability that a battery will fail in under one year?
b. Taking into account that some batteries will fail in under one year, what is the expected net profit to the company from the sale of 10 000 batteries?

In the face of competition, the company wishes to extend its guarantee to offer a 50% refund for batteries which fail between one and two years. Note: the company still gives a full refund for batteries which fail in under one year.

c. What is the expected profit to the company from the sale of 10 000 batteries under the extended guarantee?

24. An orchard produces a particular variety of apple. The weights of the apples are normally distributed with a mean, μ, of 89 grams and a standard deviation, σ, of 12 grams. The apples are sorted according to weight:

Grade A	at least 95 grams
Grade B	between 80 grams and 95 grams
Grade C	less than 80 grams

a. What percentage of apples would be in each grade?

Apples in grade A, grade B, and grade C sell for 8 cents each, 5 cents each, and 2 cents each, respectively.

b. Calculate, to the nearest dollar, the amount the orchardist could expect to receive from the sale of 10 000 of these apples.

Apples which weigh less than 60 grams are used for fruit juice.

c. What percentage of grade C apples are used for fruit juice?

25. A normally distributed population has a standard deviation of 3. The probability that a randomly selected score exceeds 42 is 0·03. What is the population mean?

26. One hundred and seventy scores out of 1 000 lie between 45 and 46. The scores are normally distributed with a mean of 45. Find the standard deviation of the distribution.

27. The heights of a particular group of male students is normally distributed with a mean of 1·74 m. Twelve per cent of the students are at least 1·81 m in height. Calculate the standard deviation of the distribution.

28. Margarine is sold in 500-gram tubs. The true weight of a particular brand is normally distributed with a mean of 500 grams. The manufacturer faces a stiff penalty if the actual weight of a tub is less than 450 grams. He wishes the probability of a fine to be less than 1 chance in 100. What is the largest possible value of the standard deviation?

29. A random variable X is normally distributed. The probability that a randomly selected score is greater than 16 is 0·1587 but the probability that it is greater than 19 is 0·0228. Find the mean and standard deviation of the distribution.

30. A chain is made up of ten links. On average, a link will support a maximum load of 107 kilograms. However, the maximum load which individual links will support is normally distributed with a standard deviation of 8·6 kilograms. What is the probability that:

a. a randomly selected link will not break if a load of 108 kilograms is applied to it?
b. the chain will not break if a load of 108 kilograms is applied to it?

31. A poultry farmer produces chickens for the table. The weights of fifteen-week-old chickens are normally distributed with a mean of 1·40 kg and a standard deviation of 0·10 kg.

a. In a batch of 1 000 chickens, how many would you expect to weigh less than 1·32 kg?

The farmer wishes to try an improved feed on a new batch of 1 000 chickens. The feed manufacturer claims that it will increase the final weight of fifteen-week-olds by 10%.

b. For the new batch of chickens, what should be:
i. the mean weight? ii. the standard deviation of the weights?

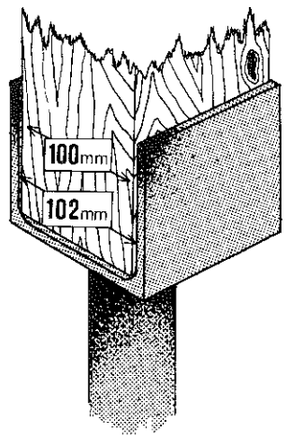
Assume that the weights of the chickens in the new batch are normally distributed.

c. How many chickens in the new batch would you expect to weigh less than 1·32 kg?

The farmer can sell chickens which weigh less than 1·32 kg for \$1.60 but if they are 1·32 kg or over he can sell them for \$2 each. However, for 1 000 chickens the improved feed costs \$30 more than the old feed.

d. Is it more profitable for the farmer to use the old feed or the improved feed? What is the margin for 1 000 chickens?

32. Metal stirrups are designed to hold wooden posts with square cross-sections.



The thickness of the posts is normally distributed with a mean of 100 mm and standard deviation of 4 mm.

a. What is the probability that a randomly selected post would not fit into a stirrup of width 102 mm?

The width of the stirrups is normally distributed also. The mean is 102 mm and the standard deviation is 1 mm.

b. What is the probability that a post with a thickness of 100 mm would not fit into a randomly selected stirrup?

A post and a stirrup are chosen independently and at random.

c. What is the probability that the thickness of the post is more than 101 mm and the width of the stirrup is less than 101 mm?
d. What is the probability that the thickness of the post is more than 101 mm or the width of the stirrup is less than 101 mm?

33. A sugar company has a machine which fills packets of sugar, each with a nominal weight of 2 kg. The actual weight of sugar in packets is normally distributed with a standard deviation of 0·01 kg. When the inspector visits the packing plant, he always chooses five packets at random and weighs the contents.

a. If the mean weight per packet is 2·015 kg, find the probability that an individual packet is under nominal weight.
b. What is the probability that the inspector will find at least one of the five packets underweight?

The risk of the inspector finding at least one underweight packet is to be reduced to 1 in 10. The machine is adjusted to alter the average weight per packet. This adjustment does not alter the standard deviation.

c. What must be the new average weight per packet?

34. Determine the probability that, out of 25 true-false questions, a student can guess correctly the answers to 15 or more.

35. A fair coin is tossed eight times. Determine the probability of getting more than two but less than seven tails. Use:

a. the binomial distribution.
b. the normal approximation to the binomial distribution.

36. A computer program will print 500 single-digit random numbers per page. The program asks for six pages of random digits. What is the probability that the digit *3* occurs between 290 and 325 times?

37. A coin is tossed 800 times. Use a normal curve approximation to find the probability of obtaining:

a. between 370 and 420 heads.
b. exactly 410 heads.
c. less than 380 or more than 420 heads.

38. A butcher orders a large number of leg hams for Christmas. The weights of the hams are approximately normally distributed about a mean of 4·2 kg. On delivery of a very large number of hams, he finds that 3% of the hams weigh less than 3·9 kg. Determine the standard deviation of the distribution of the weights of leg hams.

39. A set of random number tables contains 40 000 random digits. Determine the probability, p, that the digit *7* appears at the most 3 900 times.

40. The entire contents of a child's money box, which consists of only 2-cent coins, is tipped on to a table. Before counting the money the child separates the coins into two piles; those coins showing heads are put to the right and those showing tails, to the left. If the money box contained \$12.80, what is the probability that the pile on the right is between \$6.80 and \$7.12 in total?

41. The probability that a patient recovers from a kidney transplant operation is 0·9. What is the probability that, of the next 100 patients,

a. at least 85 survive?
b. less than 96 survive?
c. at least 85 but less than 96 survive?

Use a normal approximation.

42. A traffic engineer claims that 70% of cars travelling down a B-class road are destined for the city. A survey revealed that out of eighty cars, 69 were destined for the city. What is the probability of this happening, given the engineer's claim is true?

43. A manufacturer of diet pills claims that the pills will reduce weight for 75% of people. In an experiment, 300 individuals took a course of pills and it was decided to accept the claim if 210 or more people reduced their weight.

a. What is the probability that the claim will be rejected when the probability is, in fact, 0·75?
b. What is the probability that the claim will be accepted when the probability is as low as 0·65?

44. A pair of fair dice is rolled 192 times. We are concerned with the outcome, "score ten", which is considered as a success.

a. Estimate the population mean, μ, and standard deviation, σ, of the number of successes.
b. Determine the probability of at least 13 and at most 22 successes.
c. Calculate the approximate probability of obtaining exactly 18 successes.
d. Find n, such that the probability of obtaining more than n successes is $\frac{1}{4}$.

Answers

Chapter 1: The Multiplication Principle

Exercise 1.

1. $20.19 = 380$
2. **a.** $5.6 = 30$ **b.** $1.6 = 6$
3. **a.** $6.2 = 12$ **b.** $8.2 = 16$
4. $4.3.6 = 72$
5. $5.3.4 = 60$
6. $2.4.2 = 16$
7. $3.5.3 = 45$
8. $5.3.4 = 60$
9. $2.4.3.7.4 = 672$
10. $9.26.26 = 6084$
11. $3.5.4.3 = 180$
12. $5.2.2.6.2 = 240$
13. $9.10^6 = 9000000$
14. $24.24.24.10^4 = 13824 \times 10^4$
15. $3.3.3 = 27$
16. $10.10.10.10 = 10000$
17. $2.2.2.2.2 = 32$
18. $1.10.10.10.10 = 10000$
19. $9^8 = 43046721$
20. $10^5 = 100000$
21. **a.** $10^4 = 10000$
    b. $10^4 - 501 = 9499$
22. **a.** $5.6 = 30$
    b. $30.30 = 900$
23. **a.** $5.4 = 20$ **b.** $5.1 = 5$
    c. $5.5 = 25$
24. **a.** $6.6 = 36$ **b.** $6.1 = 6$
    c. $6.5 = 30$
25. **a.** $5.4.1 = 20$
    b. $5.4.3 = 60$
26. **a.** $5.2.3 = 30$
    b. $1.2.3 = 6$
    c. $1.2.1 = 2$
27. **a.** $8.8.9 = 576$
    b. $8.1.9 = 72$
28. **a.** $6.8.4.10.10.5.8.5 = 3840000$
    b. $4.8.10 = 320$
29. **a.** 18 **b.** 6 **c.** 3
30. **a.** $5.6 = 30$ **b.** $6.4 = 24$
    c. $1.1.4 = 4$
31. **a.** $6.4.5.4 = 480$
    b. $6.1.5.4 = 120$
32. **a.** $1.2^7 = 128$
    b. $1.1^7 = 1$
33. **a.** $100.10 = 1000$
    b. $100.10.1000 = 1000000$
34. **a.** $5.2 = 10$ **b.** $3.1 = 3$
35. **a.** $6.9.9 = 486$
    b. $6.8.7 = 336$
    c. $6.1.8 = 48$
36. **a.** $9.10.10.10 = 9000$
    b. $5.10.10.10 = 5000$
    c. $(1.10.10.5) - 1 = 499$
37. **a.** $5.4.3.2 = 120$
    b. $1.4.3.2 = 24$
    c. $4.3.2.2 = 48$
    d. none
38. $7.6.3.5.4 = 2520$
39. **a.** $2.5 = 10$
    b. **i.** 1:5 **ii.** 1:3
40. **a.** $4.3 = 12$
    b. **i.** $12.12 = 144$
    ii. $12.2.3 = 72$
41. **a.** $80.45 = 3600$
    b. $75.45.12 = 40500$
    c. $155.12 = 1860$
42. **a.** 42
    b. $85.25 = 2125$
    c. $44.25.42 = 46200$
    d. $127.44 = 5588$
43. **a.** $1.2.1.3 = 6$
    b. $1.1.1.3 = 3$
    c. $1.2.1.1 = 2$
    d. $1.2.1.1 = 2$
44. **a.** $5^{10} = 9765625$
    b. $5^5 = 3125$
    c. $3^{10} = 59049$
45. $143.126 = 18018$
46. **a.** $2^6.6.6 = 2304$
    b. $2^3.6.6 = 288$
47. **a.** $3.2.4.6.3.2.2.2.3 = 10368$
    b. $4.3.2.2.2 = 96$
48. **a.** $3.4.2.2 = 48$
    b. $1.4.2.2 = 16$
    c. $1.4.2.2 = 16$
    d. **i.** $3.2.2.3 = 36$
    ii. $3.4.2.2.3.2 = 288$
    iii. $3.4.2.2.2.1 = 96$
49. **a.** $4^6 = 4096$
    b. $1.4^5 = 1024$
    c. $7^6 = 117649$
50. **a.** $5.5 = 25$ **b.** $9 = 9$
    c. 25; 9
51. **a.** $10.10 = 100$
    b. $9.10.10 = 900$
    c. **i.** $10.4.10 = 400$
    ii. $10.1.1 = 10$
52. **a.** $10.9.8.7 = 5040$
    b. $5.4.3.2 = 120$
    c. $5.5.5.5 = 625$
    d. $5^4.2 = 1250$
    e. $2(5.4.8.7) = 2240$
53. **a.** $9.8 = 72$
    b. $(9.8.7) \div 2 = 252$
    c. $(9.8.7.6) \div 4 = 756$
    d. $(8.6.5) \div 2 = 120$

Chapter 2: Permutations

Exercise 2A.

1. **a.** 6 **b.** 39916800
   c. 24 **d.** 720 **e.** 35280
   f. 66 **g.** 8 **h.** 151170
   i. 120 **j.** 2880
2. **a.** $3!$ **b.** $5!$ **c.** $\frac{12!}{8!}$
   d. $\frac{17!}{14!}$ **e.** $7!$ **f.** $(5!)^2$
   g. $\frac{n!}{(n-4)!}$ **h.** $\frac{8!}{5!3!}$
   i. $\frac{52!}{47!5!}$ **j.** $\frac{(n+2)!}{(n+1)!}$
3. **a.** 5.4.3.2.1 **b.** 8.7.6.5
   c. 12.11.10
   d. $n(n-1)(n-2)$
   e. $r(r+1)$ **f.** 5.4.3.2.1
4. **a.** $\frac{10!}{10} = 362880$
   b. $11.10! = 39916800$
5. $7! = 5040$
6. 720
7. $15! = 1307674400000$
8. 120
9. 6
10. 24

Exercise 2B.

1. 120
2. 24
3. 6;
   XYZ, XZY, YXZ, YZX, ZXY, ZYX
4. 24
5. 720
6. **a.** 840 **b.** 30 **c.** 6300
   d. 60 **e.** 2652
7. 24
8. 40320
9. 6.5.4.3 = 360
10. 8.7.6 = 336
11. **a.** 12! = 479001600
    b. CONSIDERABLY
12. 60
13. 25! − 1
14. 60
15. 3024
16. **a.** Paris, Rome, Lisbon, Washington, Singapore
    b. **i.** 5! − 1 = 119
    ii. 8! − 1 = 40319
    iii. 6! − 1 = 719
17. 4! × 10c = $2.40
18. **a.** $^{24}P_{16} = \frac{24!}{8!}$
    b. $^{20}P_{16} = \frac{20!}{4!}$
    c. 16!
19. **a.** 120 **b.** 24
20. **a.** $^{20}P_5$ = 1860480
    b. $^{19}P_4$ = 93024
21. **a.** 9! = 362880
    b. **i.** 362880
    ii. 9.8.7 = 504
22. 3^5 = 243
23. **a.** 6 **b.** 6 **c.** 3^4 = 81
24. **a.** 5^4 = 625
    b. 5.4.3.2 = 120
25. **a.** 24 **b.** 4 **c.** 12
    d. 40
26. **a.** $^{10}P_8$ = 1814400
    b. $^{10}P_7$ = 604800
28. **a.** 61 **b.** 5 **c.** 12
    d. 10 **e.** 5
    f. n = 17, r = 8
    g. n = 5, r = 2
    h. n = 11, r = 3

Exercise 2C.

1. 2.3.2.1 = 12
2. 9.9.8.7.6 = 27216
3. **a.** 8! = 40320
   b. 2.6!1 = 1440
4. **a.** 8.7.6.5 = 1680
   b. 7.6.5.1 = 210
   c. 8.7.6.5.4 = 6720
5. **a.** $^{12}P_7$ = 3991680
   b. $^{11}P_6$ = 332640
6. 11! = 39916800;
   3!8! = 241920
7. **a.** 8.7.4 = 224
   b. 9.9.4 = 324
8. 8.7.6.5.4.1 = 6720
9. **a.** 6135, 6315, 6513, 6153, 6351, 6531
   b. 4! − 6 = 18
10. 5! = 120
11. **a.** $\frac{9!}{2!}$ = 181440
    b. MISERABLE
12. 7.6.5.4.3 = 2520; 6! = 720
13. **a.** 4! = 24
    b. 2.3!1 = 12
14. **a.** $^{15}P_6$ = 3603600
    b. $^{15}P_{11} = \frac{15!}{4!}$
15. 2.3!3! = 72
16. **a.** 1 **b.** 21.5.20. = 2100
    c. 10! = 3628800
    d. first and last
17. **a.** 6! = 720
    b. 1.5! = 120
    c. 2.5! = 240
18. **a.** **i.** I82X4T **ii.** 720
    b. **i.** I82X **ii.** 24
    c. 6.5.4.3 = 360
    d. **i.** UC5MTX2
    ii. 6! = 720
19. 7!6! = 3628800
20. **a.** 11! = 39916800
    b. 10! = 3628800
    c. 10!2 = 7257600
21. **a.** 5040 **b.** 5040
    c. 2!5! = 240
22. **a.** 10! = 3628800
    b. 13! = 6227020800
    c. 6! = 720
    d. 1.8!1 = 40320
23. **a.** 5040
    b. 3!5! = 6! = 720
24. **a.** 8.7.6.5 = 1680
    b. 7.6.5.1 = 210
    c. 1680 − 210 = 1470
    d. 4.4.3.3.2.2.1.1 = 576
25. **a.** 6! = 720
    b. $^8P_6$ = 20160
26. 5(4.3.2.1) = 120; x = 3
27. **a.** 5.4.3.2 = 120
    b. 120 ÷ 2 = 60
28. **a.** 720 **b.** 3!2.2.2 = 48
    c. 3!2 = 12
29. **a.** 362880
    b. 4!3!2! = 288
    c. 4!3!2!3! = 1728
30. **a.** 720 **b.** $3!2^3$ = 48
    c. 2.3!3! = 72
    d. 4.3!3! = 144
31. **a.** 40320 **b.** 1.7! = 5040
    c. 5.7! = 25200
    d. 5.4.6! = 14400
    e. 6!1 = 720
    f. 6!2 = 1440
    g. 2.7! = 10080
32. **a.** 12! = 479001600
    b. 1.10!1 = 3628800
    c. 1.1.10! = 3628800
    d. 11! = 39916800
    e. 2.11! = 79833600
33. **a.** 1.7!1 = 5040
    b. 8! = 40320
    c. 2.8! = 80640
    d. 2.5!4! = 5760
    e. 5.5!4! = 14400
    f. 1.1.5!2 = 240
    g. 0
34. **a.** 5040 **b.** 7.6.5.4 = 840
    c. 840 **d.** 840 **e.** 4200
    f. 1.6.5.4.1 = 120
35. **a.** 5! = 120
    b. 1.4! = 24
    c. 5! − 4! = 96
    d. 4! = 24
    e. 5! − 2(4!) = 72
    f. .....
36. **a.** 4^6 = 4096 **b.** 4^n
    c. $4^9 < 1000000$, minimum number is 10.
37. **a.** 60 **b.** 6
    c. **i.** n(n − 1)(n − 2) = $n^3 - 3n^2 + 2n$, n ⩾ 3 and integral
    ii. n(n − 1)(n − 2) = 990 = 11.10.9, n = 11
39. **a.** $\frac{10!}{(8-r)!}$ **b.** r = 6
40. **a.** 362880 **b.** 15120
41. **a.** **i.** 6 **ii.** 4 **iii.** 24
    b. **i.** 4! = 24
    ii. $4!6^4.4^4$ = 7962624
42. **a.** 6! = 720 **b.** 5 units
    c. 2.6 = 12

Exercise 2D.

1. $\frac{7!}{4!2!}$ = 105; $\frac{2.6!}{4!2!}$ = 30
2. $\frac{7!}{5!}$ = 42
3. $\frac{11!}{3!3!}$ = 1108800
4. 19958400
5. 60
6. **a.** 50400 **b.** 5040
   c. 1260
7. $\frac{20!}{10!5!5!}$ = 46558512
8. **a.** $\frac{10!}{4!}$ = 151200
   b. *SANTA MARIA*
9. $\frac{8!}{2!}$ = 20160
10. **a.** $\frac{9!}{4!2!3!}$ = 1260
    b. $\frac{5!}{2!3!}$ = 10
11. **a.** $\frac{11!}{5!2!3!}$ = 27720
    b. $\frac{11!}{5!2!4!}$ = 6930
12. **a.** 12! = 479001600
    b. 1
    c. $\frac{12!}{6!6!}$ = 924
13. **a.** 11550
    b. 11! = 39916800
14. $3!\frac{3!}{2!}$ = 18
15. **a.** $\frac{12!}{4!4!4!}$ = 34650
    b. $\frac{12!}{4!4!4!}3!$ = 207900
16. **a.** $\frac{11!}{2!2!2!}$ = 4989600
    b. **i.** $\frac{1.9!1}{2!2!2!}$ = 45360
    ii. $\frac{10!}{2!2!2!}$ = 453600
17. **a.** $\overline{WWW}$ WWWW
    b. **i.** $\frac{5!}{3!2!}$ = 10
    ii. $\frac{6!}{3!3!}$ = 20
18. Three $1 and seven 10c

Exercise 2E.

1. $\frac{12!}{12}$ = 39916800
2. 8! = 40320
3. 10! = 3628800
4. 5! = 120
5. **a.** 5! = 120
   b. (n − 1)! = 6, n = 4
6. **a.** 12! = 479001600
   b. 1.12! = 479001600
7. 11! ÷ 2 = 19958400
8. **a.** The second: opposite faces of a regular die add up to 7
   b. **i.** 3 **ii.** 2 **iii.** 2
9. **a.** 2.6! = 1440
   b. 7! − 2.6! = 3600
10. **a.** 6!3! = 4320
    b. $\frac{6!3!}{6}$ = 720
11. **a.** 4!2 = 48
    b. 4!4 = 96
    c. $\frac{4!4}{8}$ = 12
12. **a.** 4! = 24 **b.** 1 **c.** 10
13. $\frac{2.4!}{2}$ = 24
14. **a.** 15! **b.** 2 **c.** 3!
15. $\frac{6!}{6}2^6$ = 7680

Chapter 3: Combinations

Exercise 3A.

1. **a.** $\frac{5!}{2!3!}$ **b.** $\frac{10!}{8!2!}$
   c. $\frac{20!}{9!11!}$ **d.** $\frac{15!}{3!12!}$
2. $^{20}C_5$ = 15504
3. $^{12}C_3$ = 220
4. $^{15}C_3$ = 455
5. $^9C_2$ = 36
6. **a.** 45 **b.** 45 **c.** 220
   d. 220 **e.** 4 **f.** 1 **g.** 1
   h. 2598960
7. $^9C_6$ = 84
8. $^{100}C_4$ = 3921225
9. $^{11}C_8$ = 165
10. $^8C_2$ = 28
11. 495
12. 126
13. 21
14. **a.** $^{52}C_5$ = 2598960
    b. 270725
15. **a.** $^8C_2$ = 28 **b.** 56
    c. 70
17. **a.** $\binom{6}{2}$ = 15 **b.** $\binom{6}{4}$
18. **a.** 21 **b.** 21
19. **a.** 10 **b.** 10
20. **a.** $^6C_3$ = 20
    b. Yes, $^6C_4 < 20$
21. **a.** 28 **b.** 28
22. **a.** 0,5 **b.** 3,9
    c. 2,(n − 2) **d.** 3,8
    e. 2,10 **f.** 4,6
23. **a.** 35 **b.** 10 **c.** 20
    d. 165
24. **a.** $^{13}C_6$ = 1716
    b. $^1C_1.^{12}C_5$ = 792
25. $\frac{52!}{13!39!}$; $\frac{36!}{13!23!}$
26. **a.** $^3C_1.^4C_1.^3C_1$ = 36
    b. $^1C_1.^4C_1.^3C_1$ = 12
27. **a.** $^{13}C_5$ = 1287
    b. $^{26}C_5$ = 65780
    c. $^{40}C_5$ = 658008
28. **a.** 126 **b.** 56 **c.** 35
29. **a.** 5 **b.** 1 **c.** 15
    d. 0
30. **a.** 220 **b.** 10 **c.** 1
    d. 4
31. **a.** $\binom{54}{25}$ **b.** $\binom{80}{25}$
    c. $\binom{99}{25}$ **d.** $\binom{38}{25}$
32. **a.** **i.** $^5C_3$ = 10
    ii. $^5C_4$ = 5
    b. 2^5 = 32
33. **a.** $\binom{16}{10}$ = 8008
    b. $\binom{16}{8}$ = 12870
    c. $\binom{16}{16}$ = 1
34. **a.** $^6C_3$ = 20
    b. 6.5.4 = 120
35. **a.** 11.10.9.8.7 = 55440
    b. $^{11}C_5$ = 462

Exercise 3A—*continued*

36. $\binom{7}{2} - 1 = 20$

37. **a.** 220 **b.** 1320

38. **a.** 12 **b.** 7 **c.** 12
d. 17 **e.** 52 **f.** 11

39. 10

40. 7

41. 8

42. $\binom{n}{2} - n$

Exercise 3B.

1. **a.** 35 **b.** 90 **c.** 290004
d. 1863225 **e.** 10
f. 3920

2. $\binom{3}{2}\binom{6}{3} = 60$

3. $\binom{5}{3}\binom{6}{3} = 200$

4. $\binom{5}{2}\binom{20}{3} = 11400$

5. $\binom{7}{2}\binom{7}{1} = 147$

6. $\binom{7}{3}\binom{4}{2} = 210$

7. $\binom{5}{2}\binom{7}{2}\binom{12}{2} = 13860$

8. $\binom{9}{2}\binom{13}{3} = 10296$

9. $\binom{52}{2}\binom{14}{1} = 18564$

10. **a.** $\binom{30}{12} = 86493225$

b. $\binom{2}{2}\binom{28}{10} = 13123110$

11. 22308

12. **a.** $\binom{8}{5} = 56$

b. $\binom{8}{5}\binom{4}{2} = 336$

13. $\binom{5}{2}\binom{5}{2} = 100$

14. **a.** 2520 **b.** 3150

15. **a.** 45 **b.** 25 **c.** 5

16. **a.** 1 **b.** 36 **c.** 16

17. $\binom{5}{3}\binom{6}{3} = 200$

18. 553644

19. $\binom{7}{4}^4 = 1500625$

20. 5280

21. $\binom{2}{2}\binom{5}{2}\binom{3}{2} = 30$

22. $\binom{13}{2}\binom{13}{2}\binom{6}{1} = 36504$

23. 60060

24. $\binom{2}{1}\binom{4}{2}\binom{6}{3}\binom{8}{4} = 16800$

25. $\binom{7}{2}\binom{4}{2}\binom{6}{1}\binom{5}{2} = 7560$

26. **a.** 38760 **b.** 210
c. 136800

27. **a.** $\binom{4}{2}\binom{5}{2}\binom{7}{2} = 1260$

b. $\binom{5}{4}\binom{7}{2} = 105$

c. $\binom{4}{3}\binom{12}{3} = 880$

28. **a.** $\binom{13}{2}\binom{13}{3} = 22308$

b. $\binom{4}{3}\binom{4}{2} = 24$

c. $\binom{13}{2}\binom{13}{2}\binom{13}{1} = 79092$

d. $\binom{4}{1}\binom{4}{1}\binom{4}{1}\binom{4}{1}\binom{4}{1}$
$= 1024$

e. $\binom{26}{2}\binom{26}{3} = 845000$

f. $\binom{1}{1}\binom{48}{4} = 194580$

g. $\binom{4}{2}\binom{4}{2}\binom{4}{1} = 144$

29. 1102500

30. $\binom{5}{2}\binom{4}{1}\binom{2}{1} = 80$

31. **a.** 792 **b.** 171600

32. **a.** 12376

b. **i.** $\binom{8}{5}\binom{4}{3}\binom{3}{2}\binom{2}{1}$
$= 1344$

ii. $\binom{8}{6}\binom{4}{2}\binom{3}{2}\binom{2}{1}$
$= 1008$

33. $\binom{13}{4}\binom{13}{3}\binom{13}{5}\binom{13}{1}$
$\approx 3{\cdot}4213 \times 10^9$

34. **a.** 24

b. $\binom{4}{3}\binom{4}{2}\binom{12}{1} = 288$

c. $\binom{13}{1}\binom{4}{3}\binom{12}{1}\binom{4}{2}$
$= 3744$

35. **a.** $\binom{20}{4}\binom{41}{4}\binom{28}{4}$

b. $\binom{75}{12}$

c. $\binom{55}{6}\binom{34}{6}$

36. **a.** $\binom{13}{9}\binom{13}{3}\binom{13}{1}$
$= 2658370$

b. **i.** $\binom{39}{13} \approx 8{\cdot}1224 \times 10^9$

ii. $\binom{13}{10}\binom{13}{3} = 81796$

iii. $\binom{13}{10}\binom{39}{3}$
$= 2613754$

37. **a.** $\binom{15}{6} = 5005$

b. $\binom{4}{2}\binom{5}{2}\binom{6}{2} = 900$

38. **a.** $\binom{1}{1}\binom{7}{3}\binom{6}{2} = 525$

b. $\binom{6}{3}\binom{6}{3} = 400$

39. **a.** $\binom{6}{2}\binom{8}{2} = 420$

b. **i.** $\binom{5}{2}\binom{8}{2} = 280$

ii. $\binom{4}{2}\binom{8}{2} = 168$

40. **a.** $\binom{10}{5}\binom{5}{5} = 252$

b. $\binom{6}{3}\binom{4}{2} = 120$

41. $\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2} = 2520$

42. $\binom{30}{15}\binom{15}{13} \approx 1{\cdot}629 \times 10^{10}$

43. **a.** $\binom{16}{8} = 12870$

b. $\binom{8}{4}\binom{8}{4} = 4900$

44. **a.** **i.** $\binom{10}{2}\binom{8}{8} = 45$

ii. $\binom{10}{5}\binom{5}{5} = 252$

b. 5, because there are more ways of getting 5 correct.

47. **a.** $\binom{6}{3} = 20$

b. $\binom{6}{3} - 7 = 13$

48. **a.** $\binom{9}{5}\binom{4}{4} = 126$

b. $\binom{9}{3}\binom{6}{3}\binom{3}{3} = 1680$

c. $\binom{9}{4}\binom{5}{3}\binom{2}{2} = 1260$

49. **a.** {(AB,CD,EF),(AB,CE,DF), (AB,CF,DE)}

b. 5 **c.** 15

50. **a.** 4 **b.** $\binom{12}{2} - 4 = 62$

51. $\binom{12}{2}\binom{10}{2}\binom{8}{2}\binom{6}{2}$
$= 1247400$

52. **a.** $\binom{12}{3} = 220$

b. $\binom{1}{1}\binom{11}{2} = 55$

c. $\binom{1}{1}\binom{8}{2} = 28$

d. $\binom{1}{1}\binom{3}{2} = 3$

e. $\binom{8}{1}\binom{1}{1}\binom{3}{1} = 24$

Chapter 4: The Addition Principle and Miscellaneous Counting

Exercise 4A.

1. 5.4.3 + 3.2.1 = 66

2. 7.6.5 + 7.6.5.4 = 1050

3. 5.4.3.2 + 5.4.3.2.1 = 240

4. **a.** 256, 265, 52, 56, 526, 562, 62, 65, 625, 652
b. 3 + 3.2 + 3.2 = 15

5. **a.** 5.4 + 5.4.3.2 = 140
b. 5.5 + 5.5.5.5 = 650

6. **a.** 4.4.4.3 = 192
b. 1.2.1.1 + 1.2.1.1 = 4
c. 18 **d.** 15

7. 5.4 + 5.4.3 + 5.4.3.2 + 5.4.3.2.1 = 320

8. **a.** 6! = 720
b. 5!1 = 120
c. 2.5! = 240
d. 5! = 120
e. 15.4! = 360

9. **a.** 720
b. 6 + 6.5 + 6.5.4 + 6.5.4.3 + 6.5.4.3.2 + 6! = 1956

10. **a.** 3
b. **i.** 3.1 + 3.1 = 6
ii. 3.2 + 3.2 = 12
c. 18
d. **i.** 3 **ii.** 6
iii. 2(1.1.3) + 2(1.1.3) + 2(1.1.3) = 18
e. 3 + 6 + 18 = 27
f. 3 + 18 + 27 = 48

11. **a.** 18 **b.** 324 **c.** 294

12. 999100

13. 4.5.6 + 3.5.6 = 210

14. 8.7.6.5 + 8.7.6.5.4 + 8.7.6.5.4.3 + 8! + 8! = 109200

15. 1.3.6.10 + 9.6.10 = 720

16. **a.** 4.3.2 = 24
b. 4.3.2 + 4.3.2.1 = 48
c. 3.2.1.1 + 3.2.1.1 = 12
d. 1.2.1.1 + 1.2.1.1 + 1.2.1.2. = 8

17. 6! = 720;
a. 5!1 = 120
b. 1.5! = 120
c. 1.5! + 5!1 = 240
d. 1.5! + 5!1 − 1.4!1 = 216

18. **a.** 45
b. 12 + 18 = 30
c. 3.1.2 + 30 = 36

19. **a.** 6 **b.** 4.3 = 12

20. 2.8!1 + 1.8!1 = 120960

21. **a.** **i.** 120 **ii.** 400
b. 1240

22. **a.** (43??), (4?3?), (4??3), (?43?), (??43), (?4?3)
b. 1.10.10.1 = 100
c. 6.1.10.10.1 = 600

23. **a.** 10.9.8 + 10.9 = 810
b. (1.9.8 + 1.9) + (9.1.8 + 9.1) = 162

24. 6!(8.7.6.5) + (6.5.4.3).(8.7.6.5.4.3) = 8467200

25. **a.** **i.** 5040
ii. 3!2 = 12
b. **i.** 3.5! + 2.5! = 600
ii. 2.3! + 3.3!2.2 = 84

26. 270

27. **a.** 9.5 + 9.10.5 = 495
b. 9.2 + 9.10.2 = 198
c. 9.9 + 9.9.8 = 729
d. 9.1 + 8.1 + 9.8.1 + 8.8.1 = 153
e. 8.5 + 8.8.5 = 360

28. **a.** 3! = 6
b. 6 + 5.4.3.2 = 126
c. 5.4.4.4 + 3.2.2 = 332

29. 6 + 6.5 + 6.5.4 + (1.5.4.3 + 1.5.4.3) = 276

30. 243

31. **a.** 3.3.2 = 18
b. 2.2.2 + 1.1.2 + 1.2.2 = 14
c. (3 + 3 + 2) + (3.3 + 3.2 + 3.2) + 3.3.2 = 47

Exercise 4B.

1. $\binom{12}{4} + \binom{12}{5} + \binom{12}{6}$
$= 2211$

2. $\binom{5}{1} + \binom{5}{2} + \binom{5}{3}$
$+ \binom{5}{4} + \binom{5}{5} = 31$

3. $\binom{6}{1} + \binom{6}{2} + \binom{6}{3}$
$+ \binom{6}{4} + \binom{6}{5} + \binom{6}{6}$
$= 2^6 - 1 = 63$

4. $\binom{7}{1} + \binom{7}{2} + \binom{7}{3}$
$+ \binom{7}{4} + \binom{7}{5} + \binom{7}{6}$
$+ \binom{7}{7} = 127$

5. 127

6. **a.** $\binom{5}{1} + \binom{5}{2} + \binom{5}{3}$
$+ \binom{5}{4} + \binom{5}{5} = 31$

b. 31 − 3 = 28

7. $\binom{8}{2}\binom{6}{1} + \binom{6}{2}\binom{8}{1} = 288$

8. $\binom{5}{3}\binom{6}{3} + \binom{5}{4}\binom{6}{2} + \binom{5}{5}\binom{6}{1}$
$= 281$

9. **a.** $\binom{13}{6} = 1716$

b. $\binom{8}{4}\binom{5}{2} = 700$

c. $\binom{5}{2}\binom{8}{4} + \binom{5}{3}\binom{8}{3}$
$+ \binom{5}{4}\binom{8}{2} + \binom{5}{5}\binom{8}{1}$
$= 1408$

10. **a.** 50388 **b.** 17640
c. 6546

11. **a.** 2970 **b.** 924
c. 3916

12. **a.** $\binom{7}{3}\binom{5}{3}\binom{4}{3} = 1400$

b. $\binom{4}{3}\binom{12}{6} + \binom{4}{4}\binom{12}{5}$
$= 4488$

c. $\binom{7}{0}\binom{9}{9} + \binom{7}{1}\binom{9}{8}$
$+ \binom{7}{2}\binom{9}{7} = 820$

13. **a.** 4368 **b.** 1568
c. 147 **d.** 4312

14. **a.** $\binom{8}{5}\binom{16}{11} = 244608$

b. $\binom{8}{6}\binom{16}{10} + \binom{8}{7}\binom{16}{9}$
$+ \binom{8}{8}\binom{16}{8} = 328614$

15. **a.** $\binom{10}{3} = 120$

b. $\binom{3}{3}\binom{7}{0} = 1$

c. $\binom{3}{2}\binom{7}{1} + \binom{3}{3}\binom{7}{0} = 22$

d. $\binom{3}{0}\binom{7}{3} + \binom{3}{1}\binom{7}{2} = 98$

16. **a.** $\binom{4}{3}\binom{48}{2} = 4512$

b. $\binom{4}{0}\binom{48}{5} + \binom{4}{1}\binom{48}{4}$
$+ \binom{4}{2}\binom{48}{3} = 2594400$

18. **a.** 45 **b.** 21 **c.** 35

19. **a.** 210 **b.** 420

20. **a.** 495 **b.** 1 **c.** 224
d. 201 **e.** 294

21. **a.** 15 **b.** 11

Exercise 4B—*continued*

22. **a.** 42 **b.** 84 **c.** 130
23. **a.** 38760 **b.** 5320
c. 21120
24. **a.** 210 **b.** 90 **c.** 80
d. 84 **e.** 182
25. **a.** 28 **b.** 16 **c.** 13
26. **a.** $\binom{10}{4} = 210$
b. $\binom{2}{2}\binom{8}{2} = 28$
c. $\binom{10}{4} - \binom{2}{2}\binom{8}{2} = 182$
27. **a.** $\binom{2}{2}\binom{10}{3} + \binom{10}{5} = 372$
b. $2\binom{1}{1}\binom{1}{0}\binom{10}{4} + \binom{10}{5}$ $= 672$
28. **a.** 9 **b.** 126 **c.** 108
d. 252
e. $\binom{3}{2}\binom{9}{2} + \binom{3}{3}\binom{9}{1}$ $= 117$
29. **a.** $\binom{16}{11} = 4368$
b. $\binom{1}{1}\binom{12}{10} = 66$
c. $\binom{1}{1}\binom{3}{1}\binom{12}{9}$ $+ \binom{1}{1}\binom{3}{2}\binom{12}{8}$ $+ \binom{1}{1}\binom{3}{3}\binom{12}{7} = 2937$
d. $\binom{12}{11} = 12$
30. $\binom{25}{4} - \binom{10}{2}\binom{15}{2} = 7925$
31. $\binom{13}{3} - \binom{7}{3} - \binom{4}{3} = 247$
32. $\binom{1}{1}\binom{17}{15} + \binom{1}{1}\binom{17}{7}$ $+ \binom{1}{1}\binom{17}{3} = 20264$
33. **a.** $\binom{6}{3}\binom{3}{3} = 20$
b. $\binom{6}{4}\binom{2}{2} + \binom{6}{3}\binom{3}{3} = 35$
34. $\binom{8}{7}\binom{1}{1} + \binom{8}{6}\binom{2}{2}$ $+ \binom{8}{5}\binom{3}{3} = 92$
35. $\binom{9}{3}\binom{3}{1} + \binom{9}{6}\binom{3}{2}$ $+ \binom{9}{9}\binom{3}{3} = 505$
36. **a.** $\binom{6}{2}\binom{6}{2} = 225$
b. $1 + 4 + 9 + 16 + 25$ $= 55$
c. $225 - 55 = 170$
37. **a.** $\binom{4}{1} = 4$
b. $\binom{4}{1} = 4$
c. $\binom{4}{1}\binom{3}{1} = 12$
d. $4 + 4 + 12 = 20$

Exercise 4C.

1. $6 \times 20 \times 2 = 240$
2. 34
3. 200
4. **a.** 1275 **b.** 500500
c. 676 **d.** 729
e. 250500

Exercise 4D.

1. **a.** C **b.** C **c.** C **d.** C
e. P **f.** P
2. ${}^{42}P_3 = 68880$
3. ${}^{23}C_4 = 8855$
4. $\binom{12}{3} = 220$
5. ${}^{7}P_7 = 5040$
6. ${}^{6}P_3 = 120$
7. **a.** 2822400 **b.** 576
8. $\binom{7}{2} = 21$
9. **a.** $12.12.12 = 1728$
b. $12.11.10 = 1320$
c. ${}^{12}C_3 = 220$
10. **a.** $10^3 = 1000$
b. $10.9.8 = 720$
11. **a.** ${}^{15}P_{12} \approx 2{\cdot}1795 \times 10^{11}$
b. ${}^{15}C_3 = 455$
12. **a.** ${}^{12}C_4 = 495$
b. ${}^{10}C_4 = 210$
13. **a.** 12 **b.** 144
14. 210
15. **a.** $\binom{12}{5} = 792$
b. $\binom{1}{1}\binom{11}{4} = 330$
c. **i.** 0
ii. $\binom{1}{1}\binom{8}{4} = 70$
iii. $\binom{1}{1}\binom{3}{2}\binom{8}{2} = 84$
16. $9.8.7.4.6.5.4 = 241920$
17. 10
18. $26.26.10.10 = 67600$
19. **a.** $5! = 120$
b. $\frac{9!}{3!2!} = 30240$
20. **a.** 90 **b.** $\binom{9}{1} = 9$
21. **a.** 10080 **b.** 720
22. **a.** 36 **b.** 45
c. 1620
23. Neither; $\binom{10}{8} = \binom{10}{2}$
24. **a.** 24 **b.** 8 **c.** 8
25. **a.** 70 **b.** 1680
26. $\binom{20}{5}\binom{15}{5}\binom{10}{5}\binom{5}{5}4!$ $\approx 2{\cdot}8159 \times 10^{11}$
27. **a.** $6.3.10 = 180$
b. $6.10 = 160$
28. **a.** 9555 **b.** 588
29. 3150
30. **a.** 1 **b.** $3! - 1 = 5$
31. **a.** $4^3 = 64$ **b.** $4.4 = 16$
c. $4.3.2 = 24$
d. $1.4.4 = 16$
32. **a.** $8^8 = 16777200$
b. $8! = 40320$
33. **a.** 2560 **b.** 1050
34. 22308
35. $\binom{4}{1}\binom{13}{3} = 1144$
36. **a.** 336 **b.** 42
37. $\binom{7}{4}\binom{3}{2}\binom{1}{1} = 105$
38. **a.** $\frac{9!}{2!2!2!2!} = 22680$
b. $\binom{9}{3}\binom{6}{1} = 504$
39. **a.** 362880 **b.** 40320
40. **a.** 120 **b.** 720
c. $\frac{6!}{3!3!} = 20$
41. **a.** $\binom{4}{2}\binom{48}{3} = 103776$
b. $\binom{4}{2}\binom{4}{1}\binom{44}{2} = 22704$
c. $\binom{13}{3}\binom{13}{1}\binom{26}{1} = 96668$
42. 15400
43. **a.** 5040 **b.** $3!4! = 144$
c. $4.3!3! = 144$
44. $1{\cdot}1404 \times 10^{10}$
45. 40
46. **a.** 4 **b.** 10
47. **a.** $\binom{8}{2}\binom{10}{2}\binom{14}{2}\binom{12}{2}$ $= 7567560$
b. $\binom{4}{2}\binom{5}{2}\binom{7}{2}\binom{12}{2}$ $= 83160$
48. **a.** $2^3 = 8$ **b.** 4
49. **a.** 7 **b.** 13

Exercise 4D—*continued*

50. **a.** $4.3.2 = 24$
b. $4! = 24$
c. $4! + 4! = 48$
51. $\binom{4}{1}\binom{6}{1} + \binom{2}{1}\binom{3}{1} = 30$
52. **a.** $1.3.1 = 3$
b. $1.4.3 + 4.3.1 - 1.3.1$ $= 21$
53. **a.** $7.6.5.4.3 = 2520$
b. $5.4.3.3.3 = 540$
54. 720
56. **a.** 1330 **b.** 165
57. 20
58. $\binom{n}{2} = 780$; $n = 40$
59. $\binom{9}{3}\binom{5}{2}\binom{3}{1}6!$ $= 1814400$
60. $(x! - 6)(x! + 2) = 0$; $x = 3$
61. 95040
62. **a.** 35 **b.** 20 **c.** 10
d. 10 **e.** 5
63. **a.** $\frac{6!}{3} = 240$ **b.** $5! = 120$
64. $4.3 + 4.3.2 + 4! = 60$
65. $4 + 2.4.3 + \binom{4}{2} + 1 = 35$
66. $\binom{4}{2}\binom{5}{2}4! = 1440$
67. The middle number must be 5; $1.8.2 = 16$
68. **a.** 15 **b.** $2.2.2.3! = 48$
69. **a.** $\binom{8}{4}\binom{5}{2}6! = 504000$
b. $\binom{8}{4}\binom{5}{2}3.4!2! = 100800$
c. $\binom{8}{4}\binom{5}{2}5! = 84000$
70. **a.** 210
b. $\binom{10}{1}\binom{9}{1}\binom{8}{4} = 6300$
71. **a.** $\binom{8}{3} = 56$
b. $\binom{8}{3}\binom{5}{2} = 560$
c. $\binom{8}{3}\binom{5}{2}3! = 3360$
72. **a.** $5! = 120$ **b.** $3! = 6$
c. **i.** $3!3! - 1 = 35$
ii. $2.3!3! - 1 = 71$
73. 98
74. **a.** $1.8.6.4.2 = 384$
b. $\frac{384}{8} = 48$
75. $\binom{10}{2} + 10 = 55$
76. $\frac{\binom{12}{4}\binom{8}{4}\binom{4}{4}}{3!}$ $= 5775$
77. **a.** $\binom{7}{3} = 35$
b. $3! = 6$
c. $35 \times 6 = 210$
d. $\binom{2}{2}\binom{5}{1} = 5$
78. $\frac{\binom{10}{2}\binom{8}{2}\binom{6}{2}\binom{4}{2}\binom{2}{2}}{5!}$ $= 945$
79. **a.** **i** $5! = 120$ **ii.** 0
iii. $5! + \binom{5}{2}\binom{2}{2}4!$ $= 360$
b. $1 + \frac{4!}{2!2!} = 7$
80. 4
81. **a.** $\binom{100}{18} = \frac{100!}{18!82!}$
b. $\binom{9}{3} = 84$
c. $\binom{9}{1} + \binom{9}{2} + \binom{9}{3}$ $= 129$
d. $\binom{10}{1} + \binom{10}{2} + \binom{10}{3}$ $= 175$
e. 129×175^9
82. **a.** **i.** $\binom{15}{10}\binom{5}{5} = 3003$
ii. $2 \times 3003 = 6006$
b. $\binom{15}{9}\binom{6}{4}\binom{2}{2}3!$ $= 450450$
83. **a.** $r - 10 + r = 20$; $r = 15$
b. 455 **c.** 816
84. $\frac{1}{2!}\binom{10}{5}\binom{5}{5} - 1 = 125$
85. **a.** 4 **b.** 12

Chapter 5: Beginning Probability

Exercise 5A.

1. **a.** $\frac{38}{120} \approx 0{\cdot}317$
b. $\frac{34}{120} \approx 0{\cdot}283$
c. $\frac{108}{120} = 0{\cdot}9$
2. No; too few trials.
3. Invalid; maybe not random outcomes.
4. No; too few trials.
5. **a.** $\frac{209}{288} \approx 0{\cdot}726$
b. $\frac{15}{288} \approx 0{\cdot}052$
c. $\frac{33}{288} \approx 0{\cdot}115$
6. **a.** $\frac{36}{200} = 0{\cdot}18$
b. $\frac{98}{200} = 0{\cdot}49$
c. $\frac{102}{200} = 0{\cdot}51$
d. $\frac{91}{200} = 0{\cdot}455$
7. **a.** $\frac{31}{300} \approx 0{\cdot}103$
b. $\frac{27}{300} = 0{\cdot}09$
c. $\frac{86}{300} \approx 0{\cdot}287$
d. $\frac{176}{300} \approx 0{\cdot}587$
e. $\frac{147}{300} = 0{\cdot}49$
f. $\frac{153}{300} = 0{\cdot}51$

Exercise 5A—*continued*

8. a. i. $\frac{101}{440} \approx 0{\cdot}230$

ii. $\frac{118}{440} \approx 0{\cdot}268$

iii. $\frac{109}{440} \approx 0{\cdot}248$

iv. $\frac{112}{440} \approx 0{\cdot}255$

v. $\frac{221}{440} \approx 0{\cdot}502$

vi. $\frac{219}{440} \approx 0{\cdot}498$

b. Hearts; less proportion than other suits.

9. a. $\frac{47}{300} \approx 0{\cdot}157$

b. $\frac{52}{300} \approx 0{\cdot}173$

c. $\frac{151}{300} \approx 0{\cdot}503$

d. $\frac{97}{300} \approx 0{\cdot}323$

10. a. $\frac{109}{150} \approx 0{\cdot}727$

b. $\frac{129}{150} = 0{\cdot}86$

c. $\frac{145}{150} \approx 0{\cdot}967$

11. a. $\frac{66}{526} \approx 0{\cdot}125$

b. $\frac{59}{526} \approx 0{\cdot}112$

c. $\frac{208}{526} \approx 0{\cdot}395$

d. $\frac{193}{526} \approx 0{\cdot}367$

12. a. $\frac{12}{50} = 0{\cdot}24$

b. $\frac{14}{50} = 0{\cdot}28$

c. $\frac{19}{50} = 0{\cdot}38$

d. $\frac{2}{50} = 0{\cdot}04$

e. $\frac{47}{50} = 0{\cdot}94$

13. a. $\frac{8}{50} = 0{\cdot}16$

b. $\frac{14}{50} = 0{\cdot}28$

14. a. P(D) $\approx 0{\cdot}26$

b. Not equally likely; P(Down) $< \frac{1}{2}$

15. a. $\frac{50}{230} \approx 0{\cdot}217$

b. $\frac{209}{230} \approx 0{\cdot}909$

c. $\frac{21}{230} \approx 0{\cdot}091$

d. $\frac{38}{230} \approx 0{\cdot}165$

e. $\frac{192}{230} \approx 0{\cdot}835$

16. a. $\frac{18}{80} \approx 0{\cdot}225$

b. $\frac{8}{80} = 0{\cdot}1$

c. $\frac{77}{80} \approx 0{\cdot}962$

17. (A, 1), (B, 3), (C, 4), (D, 2)

18. a. (I, B), (II, A), (III, D), (IV, C)

b. 1

19. a. $\frac{201}{300} = 0{\cdot}67$

b. Yes; $\frac{2}{\pi} \approx 0{\cdot}64 \approx 0{\cdot}67$

20. a. B; P(H on B) is not close to 0·5.

b. C

c. **i.** 1 **ii.** 0 **iii.** 1 **iv.** 0

21. a. $\frac{118}{190} \approx 0{\cdot}621$

b. $\frac{106}{190} \approx 0{\cdot}558$

c. $\frac{72}{190} \approx 0{\cdot}379$

d. $\frac{27}{190} \approx 0{\cdot}142$

e. $\frac{96}{190} \approx 0{\cdot}505$

22. a. $\frac{329}{426} \approx 0{\cdot}772$

b. $\frac{\pi}{4} \approx 0{\cdot}785$

c. At random.

d. $\pi \approx \frac{4 \times \text{no. landing on shading}}{\text{no. of flies landing}}$

23. More likely not to break because, 0·7 = P(N) > P(Y) = 0·3

24. a. $\frac{13\,250}{200\,000} \approx 0{\cdot}066$

b. $\frac{170\,500}{200\,000} \approx 0{\cdot}852$

25. a. (1, 0·200), (2, 0·173), (3, 0·143), (4, 0·160), (5, 0·163), (6, 0·160)

b. No; P(6) not significantly different.

Exercise 5B.

1. $\frac{1}{1\,000}$

2. $\frac{52}{88}$

3. a. $\frac{3}{20}$ **b.** $\frac{7}{10}$ **c.** $\frac{3}{10}$

4. a. $\frac{1}{7}$ **b.** $\frac{2}{7}$ **c.** $\frac{6}{7}$

d. $\frac{2}{7}$ **e.** 0

5. a. $\frac{3}{20}$ **b.** $\frac{3}{10}$ **c.** $\frac{11}{20}$

d. $\frac{9}{20}$ **e.** $\frac{17}{20}$

6. a. $\frac{1}{4}$ **b.** $\frac{3}{4}$ **c.** $\frac{1}{2}$

d. 0

7. a. i. $\frac{1}{8}$ ii. $\frac{3}{4}$ iii. $\frac{1}{2}$

b. Outcomes are random and equally likely.

8. a. $\frac{1}{5}$ **b.** $\frac{3}{10}$ **c.** $\frac{1}{10}$

d. $\frac{2}{5}$

Exercise 5B—*continued*

9. $\frac{1}{12}$

10. a. $\frac{1}{26}$ **b.** $\frac{21}{26}$ **c.** $\frac{3}{26}$

11. Wrong; amber on for less time, hence not equally likely outcomes.

12. a. $\frac{1}{10}$ **b.** $\frac{2}{5}$ **c.** $\frac{1}{2}$

d. $\frac{3}{10}$ **e.** $\frac{1}{5}$

13. a. $\frac{1}{2}$ **b.** $\frac{1}{2}$

14. a. $\frac{1}{75}$ **b.** $\frac{11}{75}$

15. a. $\frac{1}{4}$ **b.** $\frac{1}{4}$ **c.** $\frac{1}{52}$

d. $\frac{1}{52}$ **e.** $\frac{1}{13}$ **f.** $\frac{1}{13}$

16. a. $\frac{1}{11}$ **b.** $\frac{1}{11}$ **c.** $\frac{1}{11}$

d. $\frac{1}{11}$ **e.** $\frac{2}{11}$ **f.** $\frac{2}{11}$

17. a. $\frac{1}{26}$ **b.** $\frac{1}{26}$ **c.** $\frac{1}{13}$

d. $\frac{25}{26}$ **e.** 1

18. $\frac{1}{8}$

19. a. $\frac{1}{7}$ **b.** $\frac{1}{7}$ **c.** $\frac{5}{7}$

d. $\frac{2}{7}$

20. a. $\frac{18}{37}$ **b.** $\frac{1}{37}$ **c.** $\frac{18}{37}$

d. $\frac{1}{37}$ **e.** $\frac{36}{37}$

21. $\frac{1}{7}$ if he took first one touched, but not if he could feel them first.

22. a. $\frac{1}{2}$ **b.** $\frac{1}{4}$

23. Odd number.

24. Red; P(R) > P(Y)

25. No; not equally likely outcomes.

26. a. $\frac{7}{13}$ **b.** $\frac{1}{2}$ **c.** $\frac{6}{11}$

27. a. Neither; red and black are equally likely.

b. Diamond; P(diamond) > P(Jack)

28. a. $\frac{1}{5}$ **b.** $\frac{4}{5}$ **c.** $\frac{2}{5}$

d. 1

29. a. $\frac{1}{3}$ **b.** $\frac{1}{2}$

30. $\frac{1}{12}$

31. a. i. $\frac{1}{9}$

ii. Random choice; all squares equally likely of being chosen.

b. $\frac{3}{8}$

32. a. 1 or 2—left, 3 or 4—middle, 5 or 6—right

b. i. $\frac{2}{3}$ ii. $\frac{1}{3}$

33. a. $\frac{1}{6}$ **b.** $\frac{1}{6}$ **c.** $\frac{1}{3}$

34. (A, 1), (B, 0·94), (C, 0·5), (D, 0·004), (E, 0)

35. E_2; $P(E_2) > P(E_1)$

36. P(R) : P(G) : P(B)

$= \frac{1}{2} : \frac{1}{3} : \frac{1}{6}$

$= 6 : 4 : 2$

$\therefore$ 12 beads

37. Ace of spades.

38. a. A **b.** A or B **c.** C **d.** A or C **e.** A **f.** C

39. a. i. $\frac{1}{25}$ ii. $\frac{4}{25}$ iii. $\frac{1}{5}$ iv. $\frac{9}{25}$

b. Cannot be assigned; not random, the child took aim.

40. 8

41. a. i. $\frac{1}{6}$ ii. $\frac{1}{3}$ iii. $\frac{2}{3}$ iv. $\frac{5}{6}$ v. $\frac{1}{6}$ vi. $\frac{1}{2}$

b. Assign each person to a number on a die.

42. a. $\frac{1}{8}$ **b.** $\frac{7}{8}$ **c.** $\frac{1}{2}$

d. $\frac{1}{4}$

43. a. $\frac{1}{3}$ **b.** $\frac{1}{3}$ **c.** $\frac{2}{3}$

d. 1; every female is someone's daughter.

44. a. $\frac{1}{4}$ **b.** $\frac{3}{8}$ **c.** $\frac{3}{4}$

d. $\frac{1}{4}$ **e.** 1

Exercise 5C.

1. $\frac{1}{20}$

2. $\frac{1}{225}$

3. a. 0·045 **b.** 0·955

4. 0·3

5. $\frac{10 + 10}{40 + 25 + 25} = \frac{2}{9}$

6. $\frac{1}{3}$

7. 0·881

8. a. $\frac{1}{10}$ **b.** $\frac{1}{4}$

9. a. i. $\frac{1}{3}$ ii. $\frac{2}{3}$

b. $\frac{1}{2}$

10. $\frac{1}{4}$

11. $\frac{1}{8}$

12. $\frac{12 + 13 + 10 + 13 + 12}{24 + 24 + 10 + 10 + 39} = \frac{60}{107}$

Exercise 5D.

1. a. $\frac{1}{6}$ b. $\frac{1}{3}$ c. $\frac{1}{6}$
2. a. {it, is, ii, ti, ts, tt, si, st, ss}
   b. $\frac{4}{9}$
3. $\frac{1}{6}$
4. Wrong; regions are of unequal areas and shooting is not random.
5. Wrong; winning and losing not equally likely events.
6. a. i. $\frac{1}{8}$ ii. $\frac{1}{8}$ iii. $\frac{1}{8}$
   b. i. $\frac{1}{8}$ ii. $\frac{1}{8}$ iii. $\frac{3}{8}$
7. a. $\frac{1}{4}$ b. $\frac{1}{2}$ c. $\frac{1}{3}$
   d. $\frac{1}{2}$
8. a. $\frac{1}{8}$ b. $\frac{1}{4}$ c. $\frac{1}{4}$
9. a. $(\frac{1}{2}, \frac{3}{4}, \frac{1}{3})$, $(\frac{1}{2}, \frac{1}{3}, \frac{3}{4})$, $(\frac{3}{4}, \frac{1}{2}, \frac{1}{3})$, $(\frac{3}{4}, \frac{1}{3}, \frac{1}{2})$, $(\frac{1}{3}, \frac{3}{4}, \frac{1}{2})$, $(\frac{1}{3}, \frac{1}{2}, \frac{3}{4})$
   b. i. $\frac{1}{6}$ ii. $\frac{1}{6}$ c. $\frac{2}{3}$
10. a. $\frac{1}{3}$ b. $\frac{2}{9}$ c. $\frac{5}{6}$
    d. 0 e. No; $\frac{35}{90} < \frac{1}{2}$
11. a. (V,F,?), (F,V,?), (V,?,F), (F,?,V), (?,V,F), (?,F,V)
    b. i. $\frac{1}{3}$ ii. $\frac{2}{3}$ iii. $\frac{1}{6}$
    iv. $\frac{2}{3}$ v. $\frac{2}{3}$
12. a. $\frac{14}{25}$ b. $\frac{9}{25}$ c. $\frac{17}{50}$
13. a. $\frac{1}{3}$ b. $\frac{1}{3}$ c. $\frac{2}{3}$
14. a. (O,H), (O,O), (O,L), (H,O), (H,L), (H,H), (L,O), (L,H), (L,L)
    b. i. $\frac{1}{9}$ ii. $\frac{8}{9}$ iii. $\frac{2}{3}$
15. a. i. $\frac{1}{6}$ ii. $\frac{1}{6}$
    b. i. $\frac{2}{3}$ ii. $\frac{1}{3}$
16. a. (1,3), (1,4), (2,3), (2,4), (3,1), (4,1), (3,2), (4,2)
    b. i. $\frac{1}{4}$ ii. $\frac{1}{4}$ iii. $\frac{1}{8}$
17. a. $\frac{1}{3}$ b. $\frac{1}{6}$ c. $\frac{2}{3}$
    d. $\frac{1}{2}$
18. a. {(3,4), (3,5), (3,6), (4,5),(4,6), (5,6)}
    b. $\frac{2}{3}$
19. a. $\frac{1}{6}$ b. $\frac{1}{3}$ c. $\frac{1}{3}$
20. a. $\frac{1}{6}$ b. i. $\frac{1}{3}$ ii. $\frac{2}{3}$
21. a. (HPJ), (HJP), (PHJ), (PJH), (JHP), (JPH)
    b. i. $\frac{1}{3}$ ii. $\frac{1}{3}$ iii. $\frac{2}{3}$
22. a. (O,O,O), (O,O,C), (O,C,O), (O,C,C), (C,O,O), (C,O,C), (C,C,O), (C,C,C)
    b. i. $\frac{1}{8}$ ii. $\frac{3}{8}$ iii. $\frac{3}{8}$
23. a. $\frac{152}{800} = 0{\cdot}19$
    b. $\frac{380}{800} = 0{\cdot}475$
    c. $\frac{88}{800} = 0{\cdot}11$
    d. $\frac{180}{800} = 0{\cdot}225$
24. 0·3
25. a. $\frac{1}{6}$ b. $\frac{1}{3}$ c. $\frac{1}{3}$
26. a. (14,55,44), (41,55,44), (14,44,55), (41,44,55), (55,14,44), (55,41,44), (44,14,55), (44,41,55), (55,44,14), (55,44,41), (44,55,14), (44,55,41)
    b. i. $\frac{1}{3}$ ii. $\frac{1}{6}$ iii. $\frac{1}{12}$
27. a. 12
    b. i. $\frac{1}{4}$ ii. $\frac{1}{2}$ iii. $\frac{1}{3}$
28. a. $\frac{1}{9}$ b. $\frac{2}{9}$
29. a. $\frac{1}{6}$ b. $\frac{1}{3}$ c. $\frac{1}{3}$
    d. $\frac{1}{3}$ e. $\frac{1}{2}$
30. a. $\frac{1}{8}$ b. $\frac{3}{8}$ c. $\frac{1}{2}$
    d. $\frac{1}{4}$ e. $\frac{1}{2}$
31. a. 0·3 b. 0·1 c. 0·6
    d. 0·7 e. 0
32. a. $\frac{1}{3}$ b. $\frac{1}{9}$ c. $\frac{4}{9}$
    d. $\frac{2}{9}$
33. a. $\frac{11}{200} = 0{\cdot}055$
    b. $\frac{24}{200} = 0{\cdot}12$
    c. $\frac{15}{200} = 0{\cdot}075$
    d. $\frac{17}{200} = 0{\cdot}085$
    e. $\frac{61}{200} = 0{\cdot}305$
34. $\frac{1}{4}$
35. a. $\frac{16}{148} \approx 0{\cdot}108$
    b. $\frac{20}{148} \approx 0{\cdot}135$
    c. $\frac{56}{148} \approx 0{\cdot}378$
    d. $\frac{28}{148} \approx 0{\cdot}189$
36. a. $\frac{1}{6}$ b. $\frac{1}{4}$ c. $\frac{1}{3}$
    d. $\frac{1}{6}$ e. $\frac{5}{12}$ f. $\frac{11}{36}$
37. a. (UUU), (UUD), (UDU), (UDD), (DUU), (DUD), (DDU), (DDD)
    b. $\frac{1}{8}$ c. $\frac{3}{8}$
38. a. $\frac{1}{12}$ b. $\frac{1}{2}$ c. $\frac{1}{2}$
39. a. (B_1B_2), (B_1Y_1), (B_1Y_2), (B_2B_1), (B_2Y_1), (B_2Y_2), (Y_1B_1), (Y_1B_2), (Y_1Y_2), (Y_2B_1), (Y_2B_2), (Y_2Y_1)
    b. $\frac{1}{3}$
40. a. $\frac{4}{9}$ b. $\frac{1}{3}$ c. $\frac{2}{3}$
    d. Even; $P(E) \geq P(O)$

Chapter 6: Applying Probability: Expectation and Odds

Exercise 6A.

1. a. $\frac{5}{6}$ b. $\frac{1}{6}$
2. a. $\frac{1}{8}$ b. $\frac{7}{8}$
3. a. $\frac{1}{4}$ b. $\frac{3}{4}$
4. $1 - \frac{12}{36} = \frac{2}{3}$
5. a. $\frac{7}{12}$ b. $\frac{5}{12}$ c. $\frac{7}{12}$
   d. $\frac{5}{12}$
6. a. $\frac{1}{2}$ b. $\frac{12}{13}$ c. $\frac{51}{52}$
7. a. $\frac{35}{36}$ b. $\frac{25}{36}$
8. a. $\frac{1}{5}$ b. $\frac{7}{15}$ c. $\frac{1}{3}$
   d. $\frac{2}{3}$ e. $\frac{8}{15}$ f. $\frac{4}{5}$
9. a. $\frac{1}{4}$ b. $\frac{3}{4}$
10. D
11. $\frac{1}{3}$

Exercise 6B.

1. \$1340
2. \$28400
3. Yes; fair bet is \$7
4. a. Yes; E(return) = 32c
   b. Yes; E(return) = 2c
5. No; E(return) = −50c
6. \$740
7. 2·42 − 1 = 1·42
8. Yes; E(return) = 0
9. Yes; E(return) = 17c
10. House; E(return) ≈ −17c
11. a. 1·5 b. E(return) = 0
12. 6·8
13. −\$1
14. a. 0·729
    b. E(defectives) ≈ 0·3
    c. $\frac{100}{0{\cdot}3} = 333\frac{1}{3}$ min.
    ≈ 6 hours
15. \$23.70
16. a. \$31.16 b. \$62.33
17. 7
18. a. $\frac{20}{80}(\$10) + \frac{60}{80}(-\$5)$
    = −\$1.25
    b. \$3.75
    c. 500 × \$1.25 = \$625
19. a. (0, 0·1), (1, 0·3), (2, 0·3), (3, 0·2), (4, 0·1)
    b. 1·9
20. a. −5c b. −3c
    c. Betting on 23 because you lose more slowly.
21. a. 2·67 ≈ 3 b. 2·724

Exercise 6C.

1. 1:5
2. 3:1
3. $\frac{3}{5}$
4. 4:1
5. a. 20% b. 4:1 c. 1:4
6. a. $\frac{4}{5}$
   b. 1:4
   c. 4:1
7. a. 1:5
   b. 7:5
8. a. 2:3
   b. p:1 − p
9. W:B = 1k:2k;
   1k = 2k − 5; n = 15
10. a. $\frac{7}{10}$ b. $\frac{3}{10}$ c. 3:7
11. 1:7
12. a. $\frac{1}{6}$ b. $\frac{5}{6}$
13. a. 7:13 b. 17:23
14. 3:5
15. a. 2:7 b. 4:5 c. 5:4
16. a. 5:7 b. 5:7
    c. 1:5; for a "fair" game the odds should be 5 for 1.
17. a. $\frac{1}{4}$ b. 3:1 c. You
18. a. 1:3 b. $\frac{1}{2}$
19. a. $\frac{3}{10}$ b. $\frac{3}{5}$ c. $\frac{1}{10}$
20. a. 6R, 9W, 18G b. 8:3

Chapter 7: Conditional Probability: Independence and Laws

Exercise 7A.

1. a. $\frac{5}{14}$ b. $\frac{5}{14}$ c. $\frac{5}{14}$
2. $\frac{1}{3}$
3. a. $\frac{2}{3}$ b. $\frac{9}{14}$ c. $\frac{5}{13}$
4. $\frac{1}{5}$
5. a. $\frac{1}{6}$ b. $\frac{1}{7}$
6. $\frac{1}{150}$
7. a. No b. Yes c. Yes
   d. Yes e. No f. Yes
   g. Yes h. No

Exercise 7A—*continued*

8. a. P(fatality | accident) = 0·09; P(alcohol | fatality) = 0·80
b. P(live to 65 + | aged 60) = 0·72
c. P(injury | no warm-up) = 1; P(injury | warm-up) = $\frac{5}{124}$
d. P(strong drugs | use marijuana) = 0·28; P(strong drugs | no marijuana) = 0·01
e. P(lower birth weight | tobacco) = 5 [P(lower birth weight | no tobacco)]

9. a. 0 **b.** $\frac{1}{5}$

10. a. $\frac{1}{3}$ **b.** $\frac{1}{2}$

11. a. $\frac{1}{2}$ **b.** $\frac{1}{2}$ **c.** $\frac{1}{2}$

12. a. $\frac{1}{3}$ **b.** $\frac{1}{4}$ **c.** 0

13. $\frac{1}{39}$

14. a. i. $\frac{1}{4}$ **ii.** $\frac{1}{4}$ **iii.** $\frac{1}{4}$ **iv.** $\frac{1}{4}$
b. i. $\frac{1}{3}$ **ii.** $\frac{1}{3}$ **iii.** 0 **iv.** $\frac{1}{3}$

15. a. $\frac{12}{25}$ **b.** $\frac{28}{55}$ **c.** $\frac{3}{5}$

16. a. i. 0·6 **ii.** 0·8
b. Yes; P(A) = P(A | B)

17. a. $\frac{1}{13}$ **b.** $\frac{1}{12}$ **c.** $\frac{1}{25}$

18. a. $\frac{1}{100\,000}$ **b.** $\frac{1}{99\,999}$
c. $\frac{1}{99\,998}$

19. a. $\frac{1}{38}$ **b.** $\frac{1}{38}$

20. a. i. $\frac{1}{13}$ **ii.** $\frac{1}{17}$ **iii.** $\frac{1}{13}$ **iv.** $\frac{4}{51}$
b. i. $\frac{1}{13}$ **ii.** $\frac{1}{13}$ **iii.** $\frac{1}{13}$ **iv.** $\frac{1}{13}$

21. a. $\frac{1}{3}$ **b.** $\frac{2}{3}$

22. a. Probability that I pass my next mathematics exam.
b. Probability that I pass my next mathematics exam if I do all of my homework.
c. Probability that I fail my next mathematics exam if I do not do all of my homework.
d. No; P(M | H) ≠ P(M)

23. a. $\frac{1}{4}$ **b.** $\frac{1}{4}$ **c.** $\frac{1}{2}$ **d.** $\frac{3}{4}$

24. a. Yes; P(A | B) = P(A) = $\frac{1}{6}$
b. Yes **c.** Yes

25. a. $\frac{1}{3}$ **b.** $\frac{1}{3}$ **c.** $\frac{2}{3}$

26. (A,B) not; P(A | B) = $\frac{1}{3}$ but P(A) = $\frac{5}{12}$
(B,C) not; P(B | C) = $\frac{2}{5}$ but P(B) = $\frac{1}{2}$
(A,C) not; P(A | C) = $\frac{1}{2}$ but P(A) = $\frac{5}{12}$

27. a. i. P(TB) = 0·7 **ii.** P(TA) = 0·3
b. i. P(TB | F) = 0·7 **ii.** P(TA | F) = 0·3
c. Preference is independent of sex; P(TB | F) = P(TB)

28. a. $\frac{4}{9}$ **b.** $\frac{1}{4}$

29. a. $\frac{61}{70}$ **b.** $\frac{32}{35}$
c. Yes; Probabilities approximately equal.
d. $\frac{36}{37}$
e. No; Answer to part d. is not equal to answer in part a.

30. a. $\frac{3}{8}$ **b.** $\frac{5}{12}$

31. a. $\frac{2}{3}$ **b.** $\frac{1}{3}$

32. $\frac{1}{4}$

33. a. 0·35 **b.** 0·02 **c.** 0·06 **d.** 0·44

Exercise 7B.

1. 0·6 × 0·4 = 0·24

2. a. i. $\frac{1}{6}$ **ii.** $\frac{5}{36}$ **iii.** $\frac{1}{36}$
b. No; P(T).P(R) ≠ P(T ∩ R)

3. $\frac{12}{30} \times 0{\cdot}7 = 0{\cdot}28$

4. $\frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216}$

5. a. 0·2 × 0·85 = 0·17
b. 0·2 × 0·15 = 0·03
c. 0·8 × 0·85 = 0·68
d. 0·8 × 0·15 = 0·12

6. a. 0·6 × 0·25 = 0·15
b. 0·4 × 0·25 = 0·10
c. 0·6 × 0·75 = 0·45
d. 0·4 × 0·75 = 0·30

7. a. i. 0·3 × 0·8 = 0·24
ii. 0·7 × 0·8 = 0·56
iii. 0·3 × 0·2 = 0·06
iv. 0·7 × 0·2 = 0·14
b. 0·24 + 0·56 + 0·06 + 0·14 = 1

8. a. 0·85 × 0·65 = 0·5525
b. 0·15 × 0·35 = 0·0525
c. 0·35 × 0·85 = 0·2975
d. 0·65 × 0·15 = 0·0975

9. 0·4, 0·3

10. $\frac{9}{10} \cdot \frac{3}{4} \cdot \frac{2}{3} = \frac{9}{20}$

11. a. 0·075 **b.** 0·175 **c.** 0·525 **d.** 0·475

12. a. 0·688 **b.** 0·028

13. a. 0·8 × 0·7 × 0·75 = 0·42
b. 0·8 × 0·7 × 0·25 = 0·14
c. 0·2 × 0·3 × 0·75 = 0·045

14. a. $\frac{1}{5}$ **b.** $\frac{1}{4}$ **c.** $\frac{1}{20}$

15. 0·096

16. 0·0375

17. A; P(A) = $(0{\cdot}99)^3$ which is greater than P(B) = $(0{\cdot}99)^4$

18. a. 0·038 **b.** 0·446

19. a. i. $\frac{4}{15}$ **ii.** $\frac{2}{5}$
iii. $\frac{4}{15} \cdot \frac{2}{5} = \frac{8}{75}$
b. $\frac{11}{15} \cdot \frac{9}{15} = \frac{11}{25}$

Exercise 7B—*continued*

20. a. 0·13 **b.** 0·44

21. a. i. 0·7 **ii.** 0·8
b. Not independent.

22. a. 0·95 × 0·75 × 0·35 ≈ 0·249
b. 0·95 × 0·25 × 0·35 ≈ 0·083
c. 0·05 × 0·35 × 0·25 ≈ 0·004
d. 0·95 × 0·65 × 0·75 ≈ 0·463
e. 1 − 0·004 ≈ 0·996

23. $\frac{1}{5} \cdot \frac{1}{3} = \frac{1}{15}$

24. a. $\frac{3}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{4}$
b. No difference; p = $\frac{1}{4}$ for all.
c. $\frac{1}{4} \cdot \frac{2}{3} \cdot \frac{1}{2} = \frac{1}{12}$

25. a. $(0{\cdot}8)^3(0{\cdot}4)^3 \approx 0{\cdot}0328$
b. $(0{\cdot}2)^3(0{\cdot}6)^3 \approx 0{\cdot}00173$
c. (1 − 0·00173)(1 − 0·0328) ≈ 0·966

26. a. (4,0·30745), (5,0·3767), (6,0·439) and (8,0·5456)
i. 0·30745 **ii.** 0·376705 **iii.** 0·4390345
b. 0·4543821 **c.** Seven.
d. $p_{k-1} < 1$; $0{\cdot}9p_{k-1} < 0{\cdot}9$
e. $q_k = 1 - (0{\cdot}1 - 0{\cdot}9p_{k-1})$

27. a. $(0{\cdot}92)^n < 0{\cdot}05$; 36 is minimum number.
b. Yes; $(0{\cdot}92)^{36} = 0{\cdot}0497$, slightly less than 5%.

28. a. i. 0·4 **ii.** 0·42 **iii.** 0·8 **iv.** 0·08
b. (A,B), (A,$\bar{B}$)

Exercise 7C.

1. $\frac{5}{9} + \frac{4}{9} - \frac{2}{9} = \frac{7}{9}$

2. a. $\frac{1}{2} + \frac{1}{6} - \frac{1}{12} = \frac{7}{12}$
b. $\frac{1}{2} + \frac{1}{6} - \frac{1}{12} = \frac{7}{12}$

3. a. $\frac{2}{7}$ **b.** $\frac{6}{7}$

4. a. P(A) = P(A ∪ B) − P(B) + P(A ∩ B) = 0·78
b. P(B | A) = $\frac{0{\cdot}32}{0{\cdot}78} \approx 0{\cdot}41$

5. a. $\frac{1}{2} \cdot \frac{4}{52} = \frac{1}{26}$
b. $\frac{1}{2} + \frac{4}{52} - \frac{1}{26} = \frac{7}{13}$

6. a. 0·84 **b.** 0 **c.** 0·52 **d.** 0·64 **e.** 0·48

7. a. $\frac{4}{25}$
b. $\frac{12}{25} + \frac{8}{25} - \frac{4}{25} = \frac{16}{25}$

8. $\frac{18}{30} + \frac{10}{30} - \frac{6}{30} = \frac{11}{15}$

9. a. $\frac{1}{2}$ **b.** $\frac{3}{4}$ **c.** $\frac{7}{12}$

10. a. $\frac{1}{4}$ **b.** $\frac{13}{32}$ **c.** $\frac{7}{32}$
d. $\frac{7}{8} = \frac{28}{32}$

11. $\frac{7}{16}$

12. a. 0·1 **b.** $\frac{1}{3}$ **c.** $\frac{1}{7}$

13. a. $\frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{7}{13}$
b. $\frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{4}{13}$
c. $\frac{12}{52} + \frac{26}{52} - \frac{6}{52} = \frac{8}{13}$

14. P(M ∪ N) = P(M) + P(N) − P(M)P(N);
0·95 = 0·85 + P(N) − 0·85P(N);
P(N) = $\frac{0{\cdot}1}{0{\cdot}15} = \frac{2}{3}$

15. a. $\frac{1}{36}$ **b.** $\frac{1}{18}$

16. a. $\frac{16}{52} + \frac{13}{52} - \frac{4}{52} = \frac{25}{52}$
b. $1 - \frac{25}{52} = \frac{27}{52}$

17. a. $\frac{4}{52} \cdot \frac{4}{51} = \frac{4}{663}$
b. $\frac{8}{663}$

18. a. $\frac{1}{3}$ **b.** $\frac{1}{12}$ **c.** $\frac{1}{6}$ **d.** $\frac{2}{9}$ **e.** $\frac{7}{12}$ **f.** $\frac{5}{18}$ **g.** $\frac{2}{9}$ **h.** $\frac{13}{18}$

19. 0·1904

20. a. i. $\frac{5}{36}$ **ii.** $\frac{13}{36}$
b. $\left(\frac{13}{36}\right)^2 = \frac{169}{1296}$

21. a. $\frac{1}{10} \cdot \frac{1}{4} = \frac{1}{40}$
b. $\frac{\frac{9}{10} \cdot \frac{1}{9}}{\frac{1}{40} + \frac{1}{10}} = \frac{4}{5}$
c. $\frac{\frac{1}{10} \cdot \frac{3}{4}}{\frac{3}{40} + \frac{3}{10}} = \frac{3}{35}$

22. $\frac{1}{2} \cdot \frac{1}{8} + \frac{1}{2} \cdot \frac{1}{12} = \frac{5}{48}$

23. a. 0·75(0·85) = 0·6375
b. 0·75 + 0·85 − 0·6375 = 0·9625

24. a. $\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$
b. $\frac{3}{5} \cdot \frac{1}{5} \cdot \frac{1}{5} = \frac{3}{125}$
c. $\binom{3}{1}\left(\frac{3}{5}\right)\left(\frac{1}{5}\right)\left(\frac{1}{5}\right) = \frac{9}{125}$

25. a. 0·34 **b.** 0·28
c. 0·66 + 0·70 − 0·42 = 0·94

26. a. $\frac{19}{30}$ **b.** $\frac{13}{30}$
c. $\frac{70 + 80 - 30}{300} = \frac{4}{10}$
d. $\frac{40 + 90 - 10}{300} = \frac{4}{10}$
e. $1 - \frac{130}{300} = \frac{17}{30}$

Exercise 7C—*continued*

27. a. P(Perth) = x,
P(O.S.) = 10x,
P(E.S.) = 30x;
$41x = 1$; $x = \frac{1}{41}$

b. $\frac{1}{41} + \frac{30}{41} = \frac{31}{41}$

c. $\frac{10}{41} \div \frac{40}{41} = \frac{1}{4}$

28. $\frac{1}{50} + \frac{49}{50}\cdot\frac{1}{35} + \frac{49}{50}\cdot\frac{34}{35}\cdot\frac{1}{17}$
$= 0{\cdot}104$

29. $\frac{1}{2}\cdot\frac{1}{3} + \frac{1}{2}\cdot\frac{3}{5} = \frac{7}{15}$

30. $0{\cdot}95 \times 0{\cdot}9 \times 0{\cdot}2$
$+ 0{\cdot}95 \times 0{\cdot}1 + 0{\cdot}05$
$= 0{\cdot}316$

31. a. $\frac{1}{4}$ **b.** $\frac{1}{3}$ **c.** $\frac{1}{12}$

d. $\frac{1}{2}$ **e.** $\frac{1}{4}\cdot\frac{1}{3} \times 2 = \frac{1}{6}$

f. $\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{12}$
$= \frac{7}{12}$

g. $\frac{\frac{1}{12}}{\frac{1}{12} + \frac{1}{12}} = \frac{1}{2}$

32. A. $\left(\frac{1}{6}\right)^4 = \frac{1}{1296}$

B. $4!\left(\frac{1}{6}\right)^4 = \frac{1}{54}$

C. $\binom{4}{1}\left(\frac{1}{6}\right)^4 = \frac{1}{1296}$

D. $\binom{6}{1}\left(\frac{1}{6}\right)^4 = \frac{1}{216}$

E. $4\left(\frac{1}{6}\right)^4 = \frac{1}{324}$

F. $6.5.4.\left(\frac{1}{6}\right)^4 = \frac{5}{54}$

33. a. $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$

b. $2\left(\frac{1}{2}\right)^3 = \frac{1}{4}$

c. $3\left(\frac{1}{2}\right)^4 = \frac{3}{16}$

d. $(n-1)\left(\frac{1}{2}\right)^n$

34. $\frac{4}{12} \div \frac{6}{12} = \frac{2}{3}$

35. a. $\frac{3}{50}$ **b.** $\frac{5}{8}$

c. Not independent.

d. Not independent.

36. a. $\frac{1}{2}\cdot\frac{1}{3} = \frac{1}{6}$

b. $\frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8}$

c. $\frac{1}{6} + \frac{1}{8} = \frac{7}{24}$

d. $\frac{1}{6}$

37. a. $(0{\cdot}7)(0{\cdot}9) = 0{\cdot}63$

b. $(0{\cdot}7)(0{\cdot}1) + (0{\cdot}3)(0{\cdot}9)$
$+ 0{\cdot}63 = 0{\cdot}97$

c. $\frac{0{\cdot}56}{0{\cdot}7} = 0{\cdot}8$

38. a. 0·5 **b.** $(0{\cdot}1)(0{\cdot}2) = 0{\cdot}02$

c. $0{\cdot}1 + 0{\cdot}2 - 0{\cdot}02 = 0{\cdot}28$

d. $(0{\cdot}1)(0{\cdot}2) = 0{\cdot}02$

e. $(0{\cdot}1)(0{\cdot}1) + (0{\cdot}3)(0{\cdot}5)$
$= 0{\cdot}16$

39. a. 0·7

b. $2(0{\cdot}7)(0{\cdot}3) = 0{\cdot}42$

40. a. $\frac{3.2.1 + 2.2.2}{3.3.2} = \frac{7}{9}$

b. $1 - \frac{10}{18} = \frac{4}{9}$

c. $\frac{5}{9} + \frac{3}{9} - \frac{1}{9} = \frac{7}{9}$

41. a. $(0{\cdot}7)(0{\cdot}8) = 0{\cdot}56$

b. $0{\cdot}2 + 0{\cdot}1 = 0{\cdot}3$

c. $(0{\cdot}7)(0{\cdot}2) = 0{\cdot}14$

d. $(0{\cdot}1)(0{\cdot}2) + (0{\cdot}2)(0{\cdot}5)$
$+ (0{\cdot}1)(0{\cdot}9) = 0{\cdot}33$

e. $\frac{0{\cdot}1 + 0{\cdot}01}{0{\cdot}67} = \frac{11}{67}$

42. a. $\frac{1}{16}$ **b.** $\frac{3}{16}$ **c.** 0

43. a. i. $\frac{1}{3}\cdot\frac{6}{12} = \frac{1}{6}$

ii. $\frac{1}{3}\cdot\frac{1}{6} + \frac{1}{3}\cdot\frac{5}{9}$
$+ \frac{1}{3}\cdot\frac{3}{8} \approx 0{\cdot}3657$

b. $\frac{1}{3}\left(\frac{6}{12}\cdot\frac{5}{11} + \frac{2}{9}\cdot\frac{1}{8}\right.$
$\left. + \frac{3}{8}\cdot\frac{2}{7}\right) \approx 0{\cdot}1207$

c. $\frac{1}{2}\cdot\frac{2}{9} + \frac{1}{3}\cdot\frac{2}{9} \approx 0{\cdot}1852$

44. a. P(A) + P(E) = 1·1;
P(A).P(E) = 0·3;
i. P(A) = 0·6
ii. P(E) = 0·5

b. $\frac{0{\cdot}2}{0{\cdot}5} = 0{\cdot}4$

45. 0·0588

46. a. $\frac{1}{3}$

b. $\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right) = \frac{2}{243}$

c. $\binom{5}{1}\left(\frac{1}{3}\right)^4\left(\frac{2}{3}\right) = \frac{10}{243}$

d. **i.** $\frac{1}{3}$ **ii.** $\frac{2}{9}$ **iii.** $\frac{4}{27}$
iv. $\frac{8}{81}$ **v.** $\frac{65}{81}$

47. a. $\frac{18}{31}$ **b.** $\frac{324}{961}$ **c.** $\frac{13}{31}$

d. $\frac{169}{961}$ **e.** $\frac{792}{961}$

f. $\binom{4}{1}\left(\frac{18}{31}\right)\left(\frac{13}{31}\right)^3 \approx 0{\cdot}171$

48. a. $(0{\cdot}6)(0{\cdot}65) = 0{\cdot}39$

b. $(0{\cdot}39)(0{\cdot}7) = 0{\cdot}273$

c. 0·527

49. a. $\frac{1}{3}$

b. $\frac{1}{3}\cdot\frac{1}{3} = \frac{1}{9}$

c. $\binom{2}{1}\frac{1}{3}\cdot\frac{1}{3} = \frac{2}{9}$

d. $\left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^3 = \frac{2}{27}$

e. $1 - \left(\frac{2}{3}\right)^4 = \frac{65}{81}$

f. 1

g. {AB, AC, AD, ~~BA~~,
~~BC~~, BD, ~~DA~~, ~~DC~~,
~~DB~~, ~~CA~~, CB, CD}

h. $\left(\frac{1}{6}\right)^2 = \frac{1}{36}$

50. AB, CD; AC, BD; AD, BC

a. $\frac{1}{3}$

b. $\frac{1}{3}\cdot\frac{1}{2} = \frac{1}{6}$

c. $\frac{1}{3}\cdot\frac{1}{2} + \frac{1}{3}\cdot\frac{1}{10} + \frac{1}{3}\cdot\frac{4}{10}$
$= \frac{1}{3}$

d. $\frac{1}{3}(0{\cdot}5)(0{\cdot}7)(0{\cdot}1)$
$+ \frac{1}{3}(0{\cdot}4)(0{\cdot}7)(0{\cdot}1)$
$= 0{\cdot}021$

Chapter 8: Probability Using Counting Techniques

Exercise 8.

1. $\frac{1}{4!} = \frac{1}{24}$

2. $\frac{1}{5!} = \frac{1}{120}$

3. $\frac{23}{24}$

4. $\frac{119}{120}$

5. $\frac{2 \times 3!}{5!} = \frac{1}{10}$

6. a. $\frac{4}{\binom{4}{2}} = \frac{2}{3}$ **b.** $\frac{1}{3}$

7. $\frac{2 \times 3!}{5!} = \frac{1}{10}$

8. a. $\binom{7}{2} = 21$ **b.** $\frac{1}{21}$

c. $\frac{2}{21}$

9. $\frac{8! \times 7!}{15!} = \frac{1}{6\,435}$

10. a. $6! = 720$

b. $\frac{1 \times 4! \times 1}{6!} = \frac{1}{30}$

c. $\frac{5!}{6!} = \frac{1}{6}$

11. a. $\frac{\binom{10}{6}}{\binom{40}{6}} = \frac{210}{\binom{40}{6}}$
$= 210 \times \frac{1}{\binom{40}{6}}$

b. $210 \times 25c = \$52.50$

12. $\frac{6}{\binom{8}{2}} = \frac{6}{28} = \frac{3}{14}$

13. a. $6 \times 5 \times 4 \times 3 = 360$

b. i. $\frac{1}{360}$

ii. $\frac{5 \times 4 \times 3}{360} = \frac{1}{6}$

iii. $\frac{4!}{360} = \frac{1}{15}$

14. a. $\frac{\binom{5}{2}}{\binom{26}{2}} = \frac{2}{65}$

b. $\frac{\binom{21}{2}}{\binom{26}{2}} = \frac{42}{65}$

c. $\frac{\binom{5}{1}\binom{21}{1}}{\binom{26}{2}} = \frac{21}{65}$

d. $\frac{\binom{1}{1}\binom{5}{1}}{\binom{26}{2}} = \frac{1}{65}$

15. a. $\frac{1}{4!} = \frac{1}{24}$ **b.** $\frac{1}{3}$

16. $\frac{2 \times 7!}{8!} = \frac{1}{4}$

17. $\frac{1 \times 4 \times 3 \times 1}{6 \times 5 \times 4 \times 3} = \frac{1}{30}$

18. a. $\frac{\binom{5}{2}\binom{7}{2}\binom{12}{2}}{\binom{24}{6}} = \frac{45}{437}$

b. $\frac{\binom{12}{3}\binom{12}{3}}{\binom{24}{6}} = \frac{1\,100}{3\,059}$

c. $\frac{\binom{8}{3}\binom{16}{3}}{\binom{24}{6}} = \frac{1\,120}{4\,807}$

19. a. $\frac{\binom{11}{6}\binom{19}{6}}{\binom{30}{12}}$

b. $\frac{\binom{2}{2}\binom{28}{10}}{\binom{30}{12}}$

c. $\frac{\binom{11}{0}\binom{19}{12}}{\binom{30}{12}}$

d. $\frac{\binom{2}{2}\binom{9}{4}\binom{19}{6}}{\binom{11}{6}\binom{19}{6}}$

20. a. $\frac{3!2^3}{6!} = \frac{1}{15}$

b. $\frac{2 \times 3! \times 3!}{6!} = \frac{1}{10}$

c. $\frac{3!4!}{6!} = \frac{1}{5}$

21. a. $\frac{\binom{2}{2}\binom{3}{1}\binom{6}{2}}{\binom{5}{3}\binom{6}{2}} = \frac{3}{10}$

b. $\frac{\binom{5}{3}\binom{5}{2}}{\binom{5}{3}\binom{6}{2}} = \frac{2}{3}$

22. a. $\binom{7}{3} = 35$ **b. i.** $\frac{1}{35}$

ii. $\frac{34}{35}$ **iii.** $\frac{\binom{6}{3}}{35} = \frac{4}{7}$

23. a. i. 13! **ii.** 9!4!

b. $\binom{13}{4} = 715$

c. i. $\frac{\binom{9}{4}}{715} = \frac{126}{715} \approx 0{\cdot}176$

ii. $\frac{\binom{4}{1}\binom{9}{3}}{715} = \frac{336}{715}$
$\approx 0{\cdot}470$

24. a. $\binom{8}{3} = 56$ **b.** $\frac{1}{56}$

c. $\frac{\binom{1}{1}\binom{7}{2}}{56} = \frac{3}{8}$

d. $\frac{9}{14}$

25. a. $\frac{\binom{5}{3}\binom{3}{3}\binom{4}{2}\binom{3}{1}\binom{1}{1}}{\binom{16}{10}}$
$= \frac{45}{2\,002}$

b. $\frac{\binom{10}{10}}{\binom{16}{10}} = \frac{1}{8\,008}$

26. a. $8 \times 7 \times 6 \times 5 \times 2$
$= 3\,360$

b. $\frac{1}{3\,360}$

c. $\frac{1}{4!2} = \frac{1}{48}$

27. a. $12! = 479\,001\,600$

b. $9!3! = 2\,177\,280$

c. $\frac{2\,177\,280}{479\,001\,600} = \frac{1}{220}$

d. $\frac{\binom{4}{2}}{\binom{6}{2}} = \frac{2}{5}$

e. $\frac{\binom{2}{2}}{\binom{6}{2}} = \frac{1}{15}$ **f.** $\frac{3}{5}$

28. a. $\frac{10!}{\frac{11!}{2!}} = \frac{2}{11}$ **b.** $\frac{9}{11}$

Exercise 8—*continued*

29. a. $\frac{\binom{13}{2}\binom{13}{1}}{\binom{52}{3}} \approx 0{\cdot}046$

b. $\frac{\binom{4}{1}\binom{13}{3}}{\binom{52}{3}} \approx 0{\cdot}052$

c. $\frac{\binom{4}{2}\binom{4}{1}}{\binom{52}{3}} \approx 0{\cdot}00106$

d. $\frac{\binom{4}{2}\binom{48}{1} + \binom{4}{3}}{\binom{52}{3}}$

$\approx 0{\cdot}013$

30. a. $\frac{\binom{5}{3}}{\binom{18}{3}} \approx 0{\cdot}0123$

b. $\frac{\binom{12}{3}}{\binom{18}{3}} \approx 0{\cdot}270$

c. $\frac{\binom{12}{2}}{\binom{18}{2}} \approx 0{\cdot}431$

31. a. i. $\frac{4}{\binom{8}{2}} = \frac{1}{7}$ **ii.** $\frac{1}{7}$

iii. $\frac{5}{7}$ **b.** 1

32. a. $\binom{8}{4} = 70$

b. i. $\binom{1}{1}\binom{7}{3} = 35$ **ii.** $\frac{1}{2}$

c. $\frac{\binom{2}{2}\binom{6}{2}}{70} = \frac{3}{14}$

d. $\frac{1}{2} + \frac{1}{2} - \frac{3}{14} = \frac{11}{14}$

e. $\frac{\frac{3}{14}}{\frac{1}{2}} = \frac{3}{7}$ **f.** $4! = 24$

33. $\frac{\binom{4}{3}\binom{48}{2} + \binom{4}{4}\binom{48}{1}}{\binom{52}{5}}$

$\approx 0{\cdot}00175$

34. a. $\frac{\binom{1}{1}\binom{6}{2}}{\binom{7}{3}} = \frac{3}{7}$

b. $\frac{1}{35}$ **c.** $\frac{\binom{5}{3}}{35} = \frac{2}{7}$

d. $\frac{3}{7} + \frac{3}{7} - \frac{\binom{2}{2}\binom{5}{1}}{35}$

$= \frac{5}{7}$

35. a. $\frac{1}{\binom{4}{2}\binom{6}{2}} = \frac{1}{90}$

b. $\frac{1 + 1}{90} = \frac{1}{45}$

36. $\frac{\binom{13}{1}\binom{12}{1}\binom{4}{3}\binom{4}{2}}{\binom{52}{5}}$

$\approx \frac{1}{694}$; 693:1

37. a. $\frac{4.6.5}{7.6.5} = \frac{4}{7}$

b. $\frac{4.6.5}{7.6.5} = \frac{4}{7}$

c. $\frac{1.1.5 + 1.6.5}{7.6.5} = \frac{1}{6}$

d. $\frac{1.3.5 + 2.6.5}{7.6.5} = \frac{5}{14}$

e. 0

38. a. $\frac{\binom{8}{5}\binom{7}{0}}{\binom{15}{5}} = \frac{8}{429}$

b. $\frac{\binom{8}{0}\binom{7}{5}}{\binom{15}{5}} = \frac{1}{143}$

c. $1 - \left(\frac{8}{429} + \frac{1}{143}\right)$

$= \frac{418}{429}$

d. $\frac{\binom{7}{3}\binom{8}{2}}{\binom{15}{5}} = \frac{140}{429}$

e. $\frac{\binom{7}{3}\binom{8}{2} + \binom{7}{4}\binom{8}{1} + \binom{7}{5}\binom{8}{0}}{\binom{15}{5}}$

$= \frac{61}{143}$

39. a. $\frac{\binom{26}{0}\binom{26}{13} + \binom{26}{13}\binom{26}{0}}{\binom{52}{13}}$

$\approx 3{\cdot}276 \times 10^{-5}$;
i.e., 30 527:1

b. $\frac{\binom{26}{0}\binom{26}{13} + \binom{13}{13}\binom{39}{0}}{\binom{52}{13}}$

$\approx 1{\cdot}638 \times 10^{-5}$;
i.e., 61 054:1

40. $\frac{\binom{3}{2}\binom{5}{2} + \binom{3}{3}\binom{5}{1}}{\binom{8}{4}}$

$= \frac{1}{2}$

41. $\frac{6}{12.11.10} \approx 0{\cdot}00454$; 219:1

42. a. $\frac{5}{6}$

b. $\frac{5}{6} \times \frac{1}{5} = \frac{1}{6}$

43. a. $8! = 40\,320$

b. $\frac{2 \times 7!}{8!} = \frac{1}{4}$

c. $\frac{1}{4}$

d. $\frac{2 \times 2 \times 6!}{8!} = \frac{1}{14}$

e. $\frac{1}{4} + \frac{1}{4} - \frac{1}{14} = \frac{3}{7}$

f. $\frac{\frac{1}{14}}{\frac{1}{4}} = \frac{2}{7}$

44. a. $7.6.5.4 = 840$

b. $\frac{1.6.5.4}{840} = \frac{1}{7}$

c. $\frac{6.5.4.3}{840} = \frac{3}{7}$

d. $\frac{2}{7}$

e. $\frac{\frac{2}{7}}{\frac{4}{7}} = \frac{1}{2}$

45. a. $\binom{5}{3} = 10$

b. $5 \times 4 \times 3 = 60$ **c.** $\frac{1}{60}$

d. $\frac{1.1.3}{60} = \frac{1}{20}$

e. $\frac{\binom{4}{3}}{10} = \frac{2}{5}$

f. $\frac{4 \times 3 \times 1 + 4 \times 3 \times 2}{60}$

$= \frac{3}{5}$

46. a. $\binom{70}{10}$ **b.** $\frac{7^{10}}{\binom{70}{10}}$

47. a. $\frac{\binom{4}{4}\binom{48}{1}}{\binom{52}{5}} = \frac{1}{54\,145}$;

54 144:1

b. $\frac{\binom{13}{1}\binom{4}{4}\binom{48}{1}}{\binom{52}{5}}$

$= \frac{1}{4\,165}$; 4 164:1

Exercise 8—*continued*

48. a. $\frac{\binom{3}{2}\binom{5}{0}}{\binom{8}{2}} = \frac{3}{28}$

b. $\frac{\binom{4}{2}\binom{4}{0}}{\binom{8}{2}} = \frac{3}{14}$

c. $\frac{1 \times 4}{8 \times 7} = \frac{1}{14}$

d. $\frac{4 \times 1}{8 \times 7} = \frac{1}{14}$

49. a. $\frac{\binom{6}{6}\binom{6}{3}}{\binom{12}{9}} = \frac{1}{11}$

b. $\frac{1}{11}$

c. $\frac{\binom{6}{4}\binom{6}{5}}{\binom{12}{9}} = \frac{9}{22}$

d. $\frac{\binom{1}{1}\binom{1}{1}\binom{10}{7}}{220} = \frac{6}{11}$

e. $\frac{5!4!}{9!} = \frac{1}{126}$

f. $\frac{5!4!}{9!} = \frac{1}{126}$

50. a. $\frac{\binom{3}{3}\binom{36}{10}}{\binom{39}{13}} = \frac{22}{703}$

$\approx 0{\cdot}0313$

b. $\frac{2 \times \binom{4}{4}\binom{22}{9}}{\binom{26}{13}} = \frac{11}{115}$

$\approx 0{\cdot}0957$

51. a. $\frac{\binom{4}{1}\binom{3}{1}\binom{5}{1}}{\binom{12}{3}} = \frac{3}{11}$

b. $\frac{\binom{4}{3} + \binom{3}{3} + \binom{5}{3}}{220}$

$= \frac{3}{44}$

c. $\frac{\binom{1}{1}\binom{11}{2}}{220} = \frac{1}{4}$

52. a. $4 \times 3 \times 5 = 60$

b. $\binom{4}{2}\binom{8}{1} = 48$

c. $\binom{3}{2}\binom{9}{1} = 27$

d. $\binom{5}{2}\binom{7}{1} = 70$

e. $60 + 48 + 27 + 70 = 205$

f. $\frac{60}{205} = \frac{12}{41}$

g. $\frac{\binom{3}{2} \times \binom{5}{1} + \binom{5}{2} \times \binom{3}{1}}{205}$

$= \frac{9}{41}$

h. $\frac{\frac{15}{145}}{\frac{45}{145}} = \frac{1}{3}$

53. a. $\frac{\binom{5}{3}\binom{12}{9}}{\binom{17}{12}} = \frac{550}{1547}$

$\approx 0{\cdot}356$

b. $\frac{\binom{5}{2}\binom{12}{10} + \binom{5}{1}\binom{12}{11} + \binom{5}{0}\binom{12}{12}}{\binom{17}{12}}$

$= \frac{103}{884} \approx 0{\cdot}117$

54. $\frac{2 \times 4! \times 4!}{8!} = \frac{1}{35}$

$\approx 0{\cdot}0286$

55. a. $10! = 3\,628\,800$

b. i. $\frac{8!}{10!} = \frac{1}{90}$

ii. $\frac{8 \times 8! \times 2}{10!} = \frac{8}{45}$

iii. $1 - \left(\frac{1}{90} + \frac{8}{45}\right) = \frac{73}{90}$

56. $\frac{\binom{6}{1}\binom{5}{1}}{6^6} = \frac{5}{7776}$

$\approx 0{\cdot}000643$

57. a. $\frac{\binom{16}{4}\binom{4}{0}}{\binom{20}{4}} = \frac{364}{969}$

$\approx 0{\cdot}376$

b. $1 - \frac{364}{969} = \frac{605}{969}$

$\approx 0{\cdot}624$

c. $\frac{\frac{364}{969}}{\frac{16}{20}} = \frac{455}{969}$

$\approx 0{\cdot}470$

58. a. $\frac{\binom{10}{6}\binom{8}{0}}{\binom{18}{6}} = \frac{5}{442}$

$\approx 0{\cdot}0113$

b. $\frac{\binom{10}{3}\binom{8}{3}}{18\,564} = \frac{80}{221}$

$\approx 0{\cdot}362$

c. $\frac{18\,564 - 28 - \binom{10}{6}\binom{8}{0}}{18\,564}$

$= \frac{77}{78} \approx 0{\cdot}987$

d. $1 - (0{\cdot}8)^3(0{\cdot}75)^3 = 0{\cdot}784$

e. $(1 - 0{\cdot}75^3)(1 - 0{\cdot}8^3)$
$\approx 0{\cdot}282$

f. $(1 - 0{\cdot}75^n)(1 - 0{\cdot}8^n)$
$> 0{\cdot}95$; minimum n is 15;
30 children.

Exercise 8—*continued*

59. a. $\dfrac{\binom{5}{3}\binom{7}{0}}{\binom{12}{3}} = \dfrac{1}{22}$

b. $\dfrac{\binom{12}{3} - \binom{5}{3}\binom{7}{0} - \binom{5}{0}\binom{7}{3}}{\binom{12}{3}}$

$= \dfrac{35}{44}$

c. $1 - \dfrac{35}{44} = \dfrac{9}{44}$

d. $\dfrac{\frac{7}{12}\cdot\frac{6}{11}\cdot\frac{5}{10} + \frac{5}{12}\cdot\frac{4}{11}\cdot\frac{3}{10}}{\frac{7}{12}\cdot\frac{6}{11} + \frac{5}{12}\cdot\frac{4}{11}}$

$= \dfrac{27}{62}$

60. a. $\dfrac{\binom{4}{3}}{\binom{11}{3}} = \dfrac{4}{165}$

b. $\dfrac{4}{165}$

c. $\dfrac{\binom{3}{1}\binom{4}{1}\binom{4}{1}}{\binom{11}{3}} = \dfrac{16}{55}$

d. $1 - \dfrac{4}{165} = \dfrac{161}{165}$

e. $\dfrac{\frac{1}{165}}{\frac{161}{165}} = \dfrac{1}{161}$

61. a. $\dfrac{2}{\binom{16}{4}} = \dfrac{1}{910}$

b. $\dfrac{9}{1820}$

c. $1 - \dfrac{\binom{12}{4}}{1820} = \dfrac{265}{364}$

d. $1 - \dfrac{10}{1820} = \dfrac{181}{182}$

e. $\dfrac{4 \times 3 \times 2 \times 1}{1820}$

$= \dfrac{6}{455}$

f. $\dfrac{\binom{12}{3}}{\binom{15}{3}} = \dfrac{44}{91}$

62. a. i. $(0{\cdot}6)^{10} \approx 0{\cdot}00605$

ii. $\binom{10}{2}(0{\cdot}6)^2(0{\cdot}4)^8$
$\approx 0{\cdot}0106$

iii. $1 - (0{\cdot}4)^{10} \approx 0{\cdot}9999$

iv. $1 - [(0{\cdot}6)^{10} + (0{\cdot}3)^{10} + (0{\cdot}1)^{10}]$
$\approx 0{\cdot}994$

b. E(B) = 30% of 10 = 3

c. $(0{\cdot}9)^n < 0{\cdot}1$;
minimum n is 22.

d. $\binom{10}{4}(0{\cdot}1)^4(0{\cdot}9)^6$
$= 0{\cdot}0112 \approx 1\%$;
Yes, there is a 1% chance that he is wrong.

63. a. $12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 = 79\,833\,600$

b. $\dfrac{1.1.10.9.8.7.6.5.4}{12.11.10.9.8.7.6.5.4}$

$= \dfrac{1}{132} \approx 0{\cdot}00768$

c. $\dfrac{4}{12} \times \dfrac{3}{11} = \dfrac{1}{11}$

d. $\dfrac{\binom{4}{4}\binom{8}{5}}{\binom{12}{9}} = \dfrac{14}{55}$
$\approx 0{\cdot}255$

e. $\dfrac{4}{9} \times \dfrac{3}{8} \times \dfrac{2}{7} \times \dfrac{1}{6}$

$= \dfrac{1}{126} \approx 0{\cdot}00794$

Chapter 9: The Binomial Distribution

Exercise 9.

1. $\binom{10}{10}(0{\cdot}6)^{10}(0{\cdot}4)^0 \approx 0{\cdot}00605$; Yes, the probability is very small.

2. $\binom{10}{4}(0{\cdot}42)^4(0{\cdot}58)^6 \approx 0{\cdot}2488$

3. a. $\binom{4}{4}(0{\cdot}85)^4(0{\cdot}15)^0 \approx 0{\cdot}5220$

b. $\binom{4}{2}(0{\cdot}85)^2(0{\cdot}15)^2 \approx 0{\cdot}0975$

c. $\binom{4}{0}(0{\cdot}85)^0(0{\cdot}15)^4 \approx 0{\cdot}0005$

4. a. 0·0640 **b.** 0·3456
c. 0·0264 **d.** 0·1964
e. 0·0106 **f.** 0·0000

5. a. 0·0156 **b.** 0·0938
c. 0·1094

6. $\binom{6}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^4 \approx 0{\cdot}3292$

7. $\binom{5}{3}(0{\cdot}2)^3(0{\cdot}8)^2 \approx 0{\cdot}0512$

8. Yes, but P(no effect) = 0·0002

9. a. 0·1001 **b.** 0·6785

10. $\binom{10}{7}\left(\frac{1}{2}\right)^7\left(\frac{1}{2}\right) \approx 0{\cdot}1172$

11. a. $\binom{5}{3}\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^2 \approx 0{\cdot}0879$

b. $\frac{1}{4} \times \frac{3}{4} \times \frac{1}{4} \times \frac{3}{4} \times \frac{1}{4}$
$= 0{\cdot}00879$

12. a. 0·2109 **b.** 0·0039

13. $\left(\frac{5}{6}\right)^4 + \binom{4}{1}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^3$
$\approx 0{\cdot}8681$

14. a. i. 5 **ii.** 0·1 **iii.** 0·0815 **iv.** 0·5

b. i. 7 **ii.** 0·5, symmetry **iii.** 0·1641 **iv.** 3·5

15. 0·1631

16. a. 0·4019 **b.** 0·4019
c. 0·8038

17. $\frac{11}{16}$

18. a. 0·2880 **b.** 0·352

19. 0·6875

20. 0·2352

21. a. 0·9375
b. 1 − (0·0625 + 0·0625) = 0·8750

22. a. 0·3685 **b.** 0·5220
c. 0·9880

23. a. 0·3845 **b.** 0·4370

24. a. 0·6141 **b.** 0·3251
c. 0·0574 **d.** 0·9392

25. a. 0·2508 **b.** 0·1115
c. 0·1662

26. a. 0·2269 **b.** 0·1260
c. 0·6472

Exercise 9—*continued*

27. a. 0·3828 **b.** 0·8034

28. a. 0·4219 **b.** 0·3115
c. 0·3214

29. a. 0·1009 **b.** $12 \times 0{\cdot}6 \approx 7$
c. 0·9165
d. $\binom{12}{12}(0{\cdot}6)^{12}(0{\cdot}4)^0 \approx 0{\cdot}0022$

30. a. 0·0467 **b.** 0·8059
c. $8 \times 0{\cdot}3 \approx 2$

31. a. 0·1468 **b.** 0·2031
c. $(0{\cdot}8)^5(0{\cdot}2)^3 \approx 0{\cdot}0026$

32. a. 0·0008 **b.** 0·0328
c. 2, from tables.
d. x = 3

33. a. 0·0322 **b.** 0·5981
c. 0·9967

34. $\binom{4}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^2 \approx 0{\cdot}1157$;
$\binom{6}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^3 \approx 0{\cdot}0536$;
Two sixes from 4 rolls is more likely.

35. a. 0·2666 **b.** 0·2666
c. $(0{\cdot}2666)^2 \approx 0{\cdot}0711$
d. 0·2666 + 0·2666 − 0·0711 = 0·4621

36. a. $\binom{5}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2$
$\times \binom{5}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^3 \approx 0{\cdot}0977$

b. No, $\binom{10}{5}\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^5$
$\approx 0{\cdot}2461$

37. a. n = 15, from tables.
b. $15 \times 0{\cdot}5 = 7{\cdot}5$
c. 6 in 100

38. a. 0·2309 **b.** 0·3980

39. a. 0·0026; No, very small chance of passing
b. 0·5881 **c.** 0·1642

40. $\left(\frac{5}{6}\right)^n < 0{\cdot}5$; $n > 3{\cdot}8$;
4 dice needed.

41. $\left(\frac{3}{4}\right)^n < 0{\cdot}5$; $n > 2{\cdot}4$;
minimum is 3.

42. a. 0·7, from tables
b. $8 \times 0{\cdot}7 \approx 6$ **c.** 5·6

43. a. 0·0865 **b.** 0·9915
c. $\binom{8}{5}(0{\cdot}15)^5(0{\cdot}6)^3 \approx 0{\cdot}00092$

44. a. i. 0·2340 **ii.** 0·0746
iii. 0·7340 **iv.** 0·0274
v. 0·3979 **b.** $1 loss

45. $\binom{n}{1}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{n-1} = \left(\frac{5}{6}\right)^n$;
n = 5

46. a. 0·3758 **b.** 0·3222
c. $\dfrac{0{\cdot}3758}{1 - 0{\cdot}3222} \approx 0{\cdot}5544$

47. a. $\binom{6}{3}(0{\cdot}7)^3(0{\cdot}3)^3$
$\times \binom{4}{1}(0{\cdot}6)(0{\cdot}4)^3 \approx 0{\cdot}0284$
b. 0·00333
c. $\dfrac{0{\cdot}0007 \times 0{\cdot}3456}{0{\cdot}00333}$
$\approx 0{\cdot}0726$
d. $(0{\cdot}7 \times 6 \times \$18)$ + $(0{\cdot}6 \times 4 \times \$16)$ = $114

48. a. $\binom{8}{5}\left(\frac{2}{x}\right)^5\left(\frac{x-2}{x}\right)^3$
b. $p = 0{\cdot}4 = \frac{2}{x}$; x = 5

49. Probability of passing at least three is, respectively, 0·512, 0·7383, 0·6826 if she takes 3, 4, or 5 subjects; take 4.

50. a. i. 0·7941 **ii.** 0·9648
b. $\frac{1}{2}(0{\cdot}2059 + 0{\cdot}0352)$
$= 0{\cdot}12055$
c. $\dfrac{(0{\cdot}5)(0{\cdot}2059)}{0{\cdot}12055} \approx 0{\cdot}854$
d. No, $1 - (0{\cdot}9)^n$ and $1 - (0{\cdot}8)^n$ both tend to 1 as n increases.

51. $2pq + p^2 = \frac{72}{121}$; $p = \frac{4}{11}$

52. a. $15x^4y^2 + 20x^3y^3 + 15x^2y^4$
b. x = 0·9, y = 0·1 or x = 0·1, y = 0·9

53. a. 0·2007
b. $(0{\cdot}6)^5 \approx 0{\cdot}0778$
c. $\binom{10}{5}(0{\cdot}6)^5(0{\cdot}3)^5$
$\approx 0{\cdot}0476$
d. $\binom{8}{3}(0{\cdot}6)^3(0{\cdot}4)^5$
$\approx 0{\cdot}1239$

Chapter 10: The Normal Distribution

Exercise 10A.

1. a. $P(Z < 0{\cdot}45) = 0{\cdot}6736$
b. $P(Z > 0{\cdot}97) = 0{\cdot}1660$
c. $P(-2{\cdot}31 < Z < 2{\cdot}31) = 0{\cdot}9792$
d. $P(-1{\cdot}42 < Z < 0) = 0{\cdot}4222$

2. a. 0·1635 **b.** 0·0668
c. 0·2303

3. No, frequency distribution is skewed to the right.

4. a. 39·7% **b.** 19·36%

5. a. 0·7118 **b.** 0·2882

6. a. 0·4332 **b.** 0·8664
c. 0·0125 **d.** 0·9809

7. a. 0·2057 **b.** 0·8581
c. 0·9710 **d.** 0·3174
e. 0·3411 **f.** 1

8. a. $P(0 < Z < 1) = 0{\cdot}3413$
b. $P(1 < Z < 2) = 0{\cdot}1359$

9. A, Binomial, positive skew; B, Bimodal, rams are heavier than ewes; C, Normal; D, Bimodal, girls are taller than boys.

10. a. 1·98 **b.** 2·00 **c.** 1·46
d. −0·06 **e.** 1·96 **f.** 1·645

11. a. 0·9946 **b.** 0·0054
c. 0·4946
d. $\dfrac{0{\cdot}9394}{0{\cdot}9946} \approx 0{\cdot}9445$

12. a. 0·3849 **b.** 0·7055
c. 0·1056 **d.** 0·0062
e. $2 \times 0{\cdot}2257 = 0{\cdot}4514$
f. 0

13. (24,4·67), (1, −3), (14·5,1·5), (3, −2·33)

14. a. −1·91 **b.** 2·59 **c.** 1·75
d. 2·75

15. a. 0·96 **b.** 2·26 **c.** 2·93
d. −1·58

16. a. 0·6826 **b.** 0·9544
c. 1·96 **d.** 1·645

17. a. Y, $\mu_y = 1 > \mu_x = 0$
b. Y, more spread out
c. X **d. i.** 1 **ii.** 3 **iii.** 5

Exercise 10B.

1. **a.** $P(X > 16) = 0{\cdot}1587$
b. $P(9 < X < 15) = 0{\cdot}5468$
c. $P(X < 4 \text{ or } X > 20) = 0{\cdot}0456$
d. $P(13 < X < 17) = 0{\cdot}2957$
2. **a.** 42 **b.** 36 **c.** $6^2 = 36$
3. 0·3830
4. 0·0015
5. **a.** $0{\cdot}0322 \times 493 \approx 16$
b. $0{\cdot}4418 \times 493 \approx 218$
6. 28·66%
7. **a.** 0·8301
b. $P(X < 25) = P(Z < -4) \approx 0{\cdot}0000$; none
8. 0·0251
9. **a.** 0·9460
b. $\frac{0{\cdot}4896}{0{\cdot}9460} \approx 0{\cdot}5175$
10. **a.** 0·7299 **b.** 1·77 m
c. $0{\cdot}2033 \times 400 \approx 81$
11. 0·4678
12. **a.** 0·2789 **b.** 9·18%
13. **a.** 12 **b.** 0·0596
c. $0{\cdot}0475 \times 60 \approx 3$
14. **a.** 0·0548 **b.** 0·4514
c. $0{\cdot}0228 \times 100 \approx 2$
15. 2500
16. 3784
17. **a.** 20·7 **b.** 17·4 **c.** 7·4
d. 0·7
18. **a.** 0·9525 **b.** 6·55 **c.** 0·63
19. **a.** 98, 10·94 **b.** 116
20. **a.** 200 **b.** 893 h
21. 118·3 g, 136·3 g
22. **a.** 6·94% **b.** 0·9
23. **a.** 0·0025 **b.** \$298 375
c. \$298 375 − 569(\$32.50) ≈ \$279 882.50
24. **a.** 30·85%, 46·49%, 22·66%
b. \$525 **c.** 3·44%
25. 36·36
26. 2·27
27. 0·059
28. 21·5 g
29. 13, 3
30. **a.** 0·4522
b. $(0{\cdot}4522)^{10} \approx 0{\cdot}0004$
31. **a.** 212
b. **i.** $1{\cdot}40 \times 1{\cdot}1 = 1{\cdot}54$ kg
ii. $0{\cdot}1 \times 1{\cdot}1 = 0{\cdot}11$ kg
c. 23
d. Improved feed; Margin is \$45.60
32. **a.** 0·3085 **b.** 0·0228
c. $0{\cdot}4013 \times 0{\cdot}1587 \approx 0{\cdot}0637$
d. 0·4963
33. **a.** 0·0668
b. $1 - (0{\cdot}9332)^5 \approx 0{\cdot}2923$
c. 2·02 kg
34. $\mu = 12{\cdot}5$, $\sigma = 2{\cdot}5$; 0·2119
35. **a.** 0·8203
b. $\mu = 4$, $\sigma = 1{\cdot}4142$; 0·8170
36. $\mu = 300$, $\sigma \approx 16{\cdot}432$; $P(289{\cdot}5 \leqslant X \leqslant 325{\cdot}5) = 0{\cdot}6783$
37. **a.** 0·9111 **b.** 0·0218
c. 0·1470
38. $P(Z > k) = 0{\cdot}97$, $k \approx 1{\cdot}88$; 0·16 kg
39. 0·0485
40. 0·0599
41. **a.** 0·9664 **b.** 0·9664
c. 0·9328
42. $P(68{\cdot}5 < X < 69{\cdot}5) = 0{\cdot}0006$
43. **a.** 0·0192 **b.** 0·0392
44. **a.** 16, 3·83 **b.** 0·7740
c. 0·0905 **d.** 19